

DeskPRO Admin Manual

October 30th, 2014 • Revision #119

DeskPRO Admin Manual	1
1. Introducing DeskPRO	7
1.1. DeskPRO features and benefits	8
1.2. DeskPRO basic concepts	9
1.3. Account types	11
User accounts	11
Agent accounts	11
Admin accounts	11
1.4. DeskPRO interfaces	12
2. Creating Your Helpdesk	14
2.1. Helpdesk domains	15
2.2. Cloud sign-up	16
2.3. Installing DeskPRO Download	17
Requirements	17
Installation process	17
2.4. Configuring your new helpdesk	19
Entering basic details	19
Configuring a custom domain with Cloud	19
Disabling the helpdesk	20
Next steps to setting up your helpdesk	20
3. Agents	22
3.1. Creating agents	23
Agent properties	23
Copying agent settings	24
Bulk adding agents	25
Agent licensing	26
3.2. Agent permissions	28
Agent zones	28
Agent ticket permissions	28
Agent department permissions	30
Agent people permissions	30
Agent chat permissions	31
Agent publish permissions	32
Agent profile permissions	32
Agent notifications	32
Agent task permissions	33
3.3. Permission groups	34
Permission overrides	35
3.4. Logging in as an agent	38

3.5. Monitoring agents	39
3.6. Managing agent passwords	40
3.7. Editing agent profiles	41
3.8. Deleting an agent	42
3.9. Admin access log	43
3.10. IP whitelisting	44
3.11. Agent teams	45
Creating and editing teams	45
Setting primary team of an agent	46
3.12. Satisfaction survey	48
4. Departments	50
4.1. Creating and editing ticket departments	51
Department properties	51
Built-in department triggers	52
Nested departments	52
4.2. Using departments	54
4.3. Department permissions	55
Agent department permissions	55
User department permissions	55
4.4. Ticket form layouts	59
5. Email Accounts	63
5.1. Adding ticket email accounts	64
Default account for non-ticket email	64
DeskPRO Cloud email configuration	64
Sending email using DeskPRO Cloud email servers	66
DeskPRO Download email configuration	67
5.2. Email account New Ticket triggers	70
5.3. Linking a department to an email account	71
Linking email accounts to departments	71
Linking non-email tickets to email accounts	71
5.4. Email handling options	73
Adding agents as followers with cc:	73
Priority of email headers	73
Ticket forwarding: inbound	73
Ticket forwarding: outbound	75
5.5. Running an email-only helpdesk	76
5.6. Monitoring email accounts	77
Incoming email log	77
Outgoing email log	78
6. Ticket Fields	79
6.1. Categories	80
6.2. Priorities	82
6.3. Workflows	83
6.4. Products	84
6.5. Custom fields	85
Custom field types	85
6.6. Field data validation	88

Text validation with regular expressions	88
Date validation options	89
7. Automating the Helpdesk	90
7.1. Ticket triggers	91
Trigger event filtering	92
Trigger order and trigger control	92
Department and Email Account triggers	94
Default triggers	95
Custom triggers	97
Triggers example: automatically assign agents to tickets they reply to by email	98
Triggers example: assigning out-of-hours tickets to a special team	99
Triggers example: filtering bounce messages	100
Triggers example: tickets from organization managers are more urgent	101
Triggers example: emailing users when their ticket is assigned	102
Triggers example: sending a different auto-reply to new users	104
7.2. SLAs	108
Creating an SLA	108
7.3. Macros	110
Creating a macro	110
7.4. Ticket escalations	111
Creating an escalation	111
7.5. Round robins	113
7.6. Using urgency to prioritize tickets	115
7.7. Displaying IDs for items	116
7.8. Sending SMS using automatic actions	117
7.9. Actions guide	120
Ticket assignment	120
Ticket properties	120
Ticket SLAs	121
Ticket actions	121
Send email	122
Trigger control actions	123
Create task	124
Custom ticket fields	124
Custom person (user) fields	124
8. CRM	125
8.1. User permissions	126
Everyone usergroup	126
Registered usergroup	126
Portal access and usergroups	126
User permission details	127
8.2. User registration and authentication	129
User registration settings	129
Restricting access to part of the portal	129
CAPTCHAs	130
Authentication apps	130
8.3. Managing user passwords	131

8.4. Custom user and organization fields	132
Adding custom fields to ticket forms	132
Per-user fields	136
Editing per-user fields from the portal	139
Per-organization fields	140
9. Portal	141
9.1. Basic portal customization	142
9.2. Portal branding	143
Color chooser	143
Custom favicon	143
Adding custom HTML	144
Adding a logo	144
9.3. Changing portal design and content with templates	146
9.4. Customizing portal content	147
Rearranging portal sections	147
Hiding and displaying portal sections	147
Ticket deflection	148
Portal contact form layout	149
Showing your Twitter account on the portal	149
Custom portal sidebars	149
Custom ticket form	150
9.5. Website integration	151
Embedding the Feedback & Support tab	151
Embedding the chat widget	153
Using a custom button with the chat widget	155
Embedding a contact form	155
Embedding the whole portal	156
9.6. Ticket ref codes	158
10. Editing Templates	160
10.1. Introducing templates	161
10.2. Twig syntax basics	164
Variables	164
Tags	164
Comments	164
10.3. Customizing the email templates	166
Templates example: custom email auto-response by department	166
10.4. Customizing the portal templates	170
10.5. Useful template variables	171
11. Localization	174
11.1. Date & time settings	175
11.2. Languages	177
Installing new languages	178
Translating names of custom values	178
Editing user interface phrases	179
Changing default language	180
Agent interface language setting	180
Crowdsourced translation	180

12. Extending DeskPRO	181
12.1. Installing the ready-made apps	182
Gravatar	182
Share Widget	183
Google Analytics	184
Highrise	185
HipChat	186
Microsoft Translator	188
JIRA	190
Salesforce	192
Twilio SMS	193
Clickatell SMS	194
Authentication apps	194
12.2. Using web hook actions	195
Using custom headers and data	196
12.3. Using the DeskPRO API	199
Managing DeskPRO API keys	199
12.4. Creating your own widgets and apps	201
13. Chat	202
13.1. Chat departments	203
14. Labels	204
14.1. Managing labels and controlling agent label creation	205
14.2. Setting label colors	206
14.3. Merging labels	207
15. Agent Interface Options	208
15.1. Automatic ticket locking	209
15.2. Setting ticket defaults for agents	210
15.3. Time log and billing	212
Modify billing/logs agent permission	212
Billing users	213
15.4. Custom filters	214
15.5. Ticket archiving	215
15.6. Ticket purging	216
16. Importing User Data	217
16.1. Importing user data in CSV format	218
Importing into custom fields	221
Importing contact data	221
16.2. Importing data from legacy systems	222
Importing data from Zendesk	222
Importing data with the DeskPRO API	222
17. Billing and licensing	223
17.1. Deleting agents to free up agent seats	225
18. Managing DeskPRO Download	226
18.1. Setting account credentials from the command line	227
Token login method	227
18.2. Backing up or moving a DeskPRO server	228
18.3. Updating DeskPRO	229

Command-line updater	230
Manually installing updates	230
Updating multiple servers	231
18.4. Proxy servers and load-balancing	232
18.5. DeskPRO Download performance tuning	233
18.6. Storing binary data	234
Switching to filesystem storage	234
Filesystem storage with multiple web servers	234
Switching to Amazon S3 storage	235
S3 security and reliability	236
Server configuration and file size	237
18.7. Error reporting	238
18.8. Elasticsearch	239
Installing Elasticsearch	239
18.9. Securing DeskPRO	240
19. DeskPRO Terminology	241

1. Introducing DeskPRO

Welcome to DeskPRO, the helpdesk software platform.

This manual explains the details of installing, configuring and managing a DeskPRO helpdesk as an **administrator**.

For a quick overview of how to get a DeskPRO helpdesk up and running, see the [Admin Quick Launch Guide](#).

There are separate manuals for two advanced admin topics:

- the [Reports Manual](#) covers DeskPRO's Reports interface, which can generate detailed custom reports about the helpdesk
- the [Apps Developer Manual](#) explains how to modify and extend the functionality of DeskPRO by developing apps and widgets.

If you are looking for information about using DeskPRO as an **agent** (i.e. to help customers/users), see the [Agent Quickstart Guide](#) and [Agent Manual](#).

1.1. DeskPRO features and benefits

We've worked hard to make DeskPRO **powerful yet simple**. We understand that to be accepted by your agents, our software has to be intuitive and pleasant to use, but powerful enough that it enhances their productivity rather than holding them back.

We've carried the principle of "powerful yet simple" through to the administration of the platform. You can set up a working DeskPRO helpdesk in minutes, but the platform is flexible enough so you can configure it to match your business processes instead of changing the way you work.

Here are some of the key benefits of DeskPRO:

Flexible configuration: You can organize your helpdesk using teams and departments to reflect how your business processes work. You can add extra information to tickets to make sure all the information you need is captured, and use fine-grained access permissions to control what agents and users can see.

Scalability: Major corporations have used DeskPRO to handle millions of tickets in a single helpdesk.

Multiple channels: DeskPRO enables users to contact you via email, real-time chat, through a web portal or your own site. All the information about an issue is combined into a single ticket, so you can see the whole history of the user's communication with the helpdesk at a glance.

Integrated publishing: Your web portal contains a knowledgebase which allows your agents to create and manage articles offering solutions to common problems. The knowledgebase can be used to deflect tickets before they're submitted, by showing relevant help content as the user describes their problem. The portal can also host file downloads and news posts, all created and managed from within DeskPRO.

Automation: DeskPRO supports powerful automatic actions, which you can set up with a simple web interface. This means that you can automate repetitive or routine tasks, freeing up your agents to be more productive. You can customize the default automatic behavior of the helpdesk using the same system. Agents can speed up their own workflow using time-saving features like macros, custom filters and snippets.

Monitoring and reporting: You can ensure that tickets don't spend too long unanswered or unresolved using automatic Service Level Agreements and escalations. You can get detailed information about agent performance using the powerful Reports interface and built-in user satisfaction survey.

Branding: You can fully customize the web portal and embed it into your website, or integrate DeskPRO features like live chat and ticket submission forms into your website.

Extensibility: We refer to DeskPRO as a platform because it can be extended using DeskPRO apps. We offer pre-made apps to integrate your helpdesk with third-party services, and to enable single sign-on. You can customize the agent interface by writing simple widgets using just HTML and Javascript, or write your own full apps. DeskPRO has a full REST API to enable you to access helpdesk data and actions.

Truly multi-lingual: We've designed DeskPRO with international usage in mind. You can quickly install new language packs to offer a translated user experience for most major languages. If you need a language that we don't yet offer, we have a crowdsourcing translation system so you can provide or commission a new language pack.

Simple pricing structure: We don't lock useful features until you upgrade to a more expensive pricing plan. You just pay a simple per-agent fee and get access to all the power of the DeskPRO platform.

1.2. DeskPRO basic concepts

Here are a few key terms and concepts we'll be using throughout this manual.

Tickets: every issue (like a complaint or question) becomes a ticket in DeskPRO; a ticket gives you a unified view of all the communication about an issue.

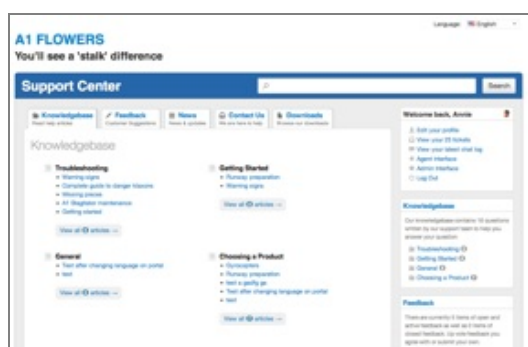
Users: people who need help and support from your helpdesk, whether that is external customers or others within your enterprise (for example if you're running an IT helpdesk). Every ticket belongs to a user.

Agent: someone within your enterprise who uses DeskPRO to help users, such as an IT support worker or customer service agent.

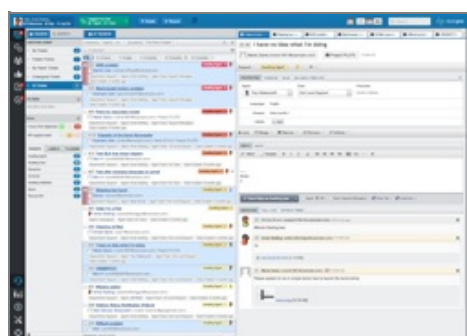
Admin: an admin is an agent who can configure the helpdesk by accessing the DeskPRO admin interface.

Department: departments are a way to divide up the helpdesk and control access. They are typically used to reflect the internal organization of your enterprise e.g. you could have Sales, Support and Research departments. Every ticket always has a department.

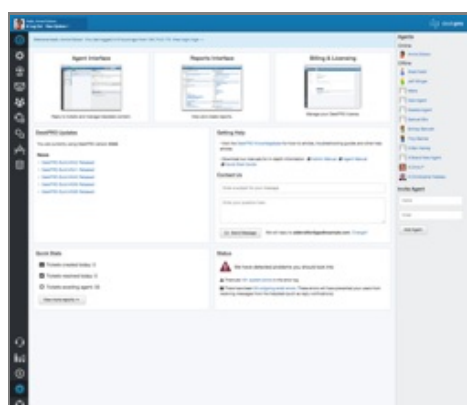
Web portal this is an optional way to allow users to interact with the helpdesk through the web; depending on how you choose to configure it, users can view their tickets, submit feedback, see knowledgebase articles, etc. If enabled, it will be accessible to users at http://YOUR_HELPDESK/.



Agent interface: the DeskPRO interface that **agents** use to respond to tickets. When you have set up your helpdesk, it will be at http://YOUR_HELPDESK/agent.



Admin interface: the DeskPRO interface that you will use to configure and maintain the helpdesk. It will be at http://YOUR_HELPDESK/admin.



The actual web address that's represented above by `YOUR_HELPDESK` will depend on your configuration.

1.3. Account types

There are three different account types in DeskPRO: user accounts, agent accounts and admin accounts.

User accounts

User accounts govern how your users interact with the helpdesk.

By default, an account is created as soon as a user emails the helpdesk or submits a ticket. You can change the requirements for creating a user account with by changing your helpdesk [registration](#) settings.

What a user can do on the helpdesk depends on their [usergroup permissions](#). Examples of user actions include:

- Creating a ticket via the web portal.
- Creating a ticket by sending an email.
- Rating web portal content and leaving comments.
- Initiating live chat with agents.

You can configure your helpdesk to control which of these functions are available to users who aren't logged in (guest users).

See the [User permissions](#) section for more details.

Agent accounts

Agent accounts are user accounts, with the additional permission to log in to the agent interface. They are designed for people within your organization who will be using the helpdesk to help users, but don't need to change helpdesk settings.

Individual agents have permissions settings which determine what they can do from the agent interface. You can also assign permissions to multiple agents at a time using **agent permissions groups**.

Agents may or may not have permission to access the **reports** interface to generate detailed, custom tables and graphs about helpdesk activity (see the [DeskPRO Reports Manual](#) for details).

See the [Agent permissions](#) section for more details.

Admin accounts

Admin accounts are agent accounts, with the additional permission to log in to the admin interface.

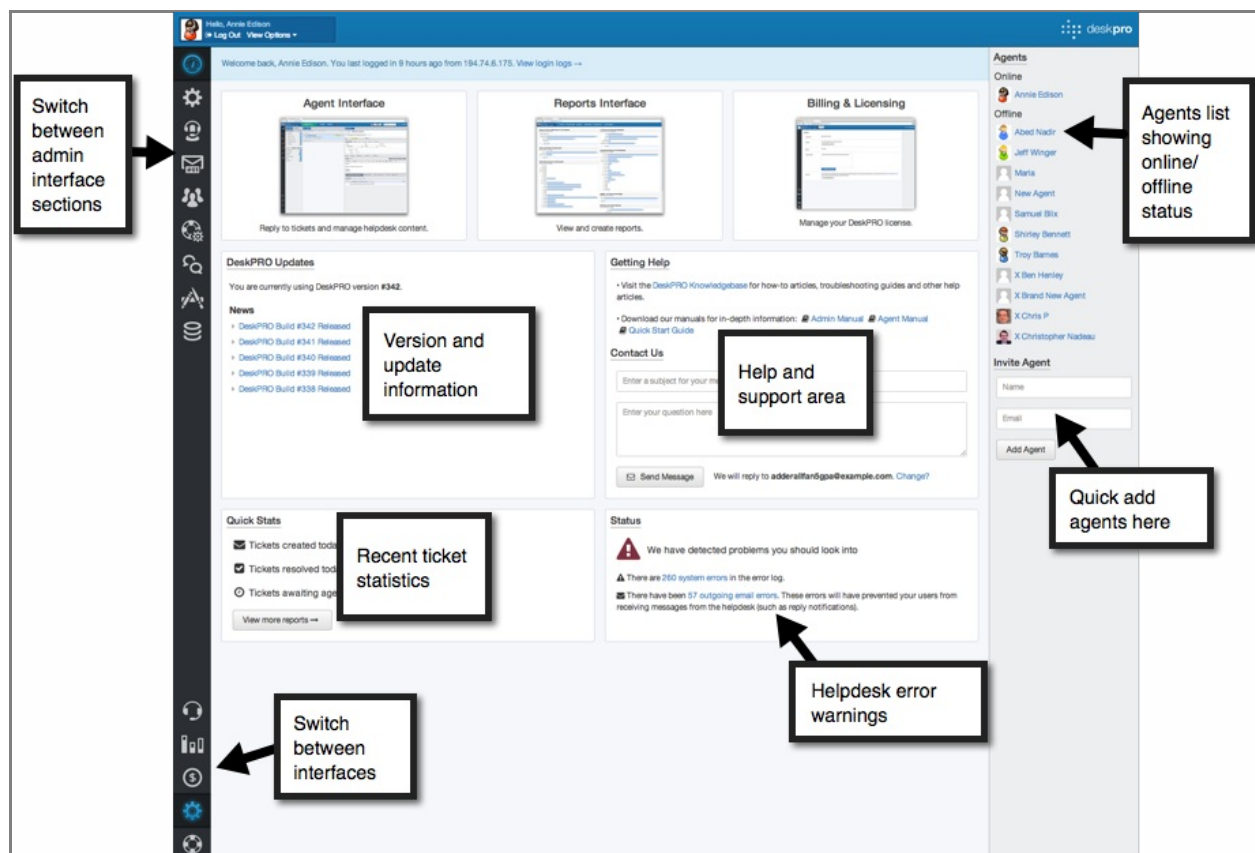
Admins can change all the helpdesk settings, and edit the permissions of all agent and admin accounts, including converting an agent to or from an admin.

Note that all admin accounts have the same abilities; DeskPRO has no 'master' admin account, and there is no difference between the initial admin account you use to create the helpdesk and any subsequent admins you create.

1.4. DeskPRO interfaces

Once your helpdesk has been created, you will be able to log in to the **admin** interface.

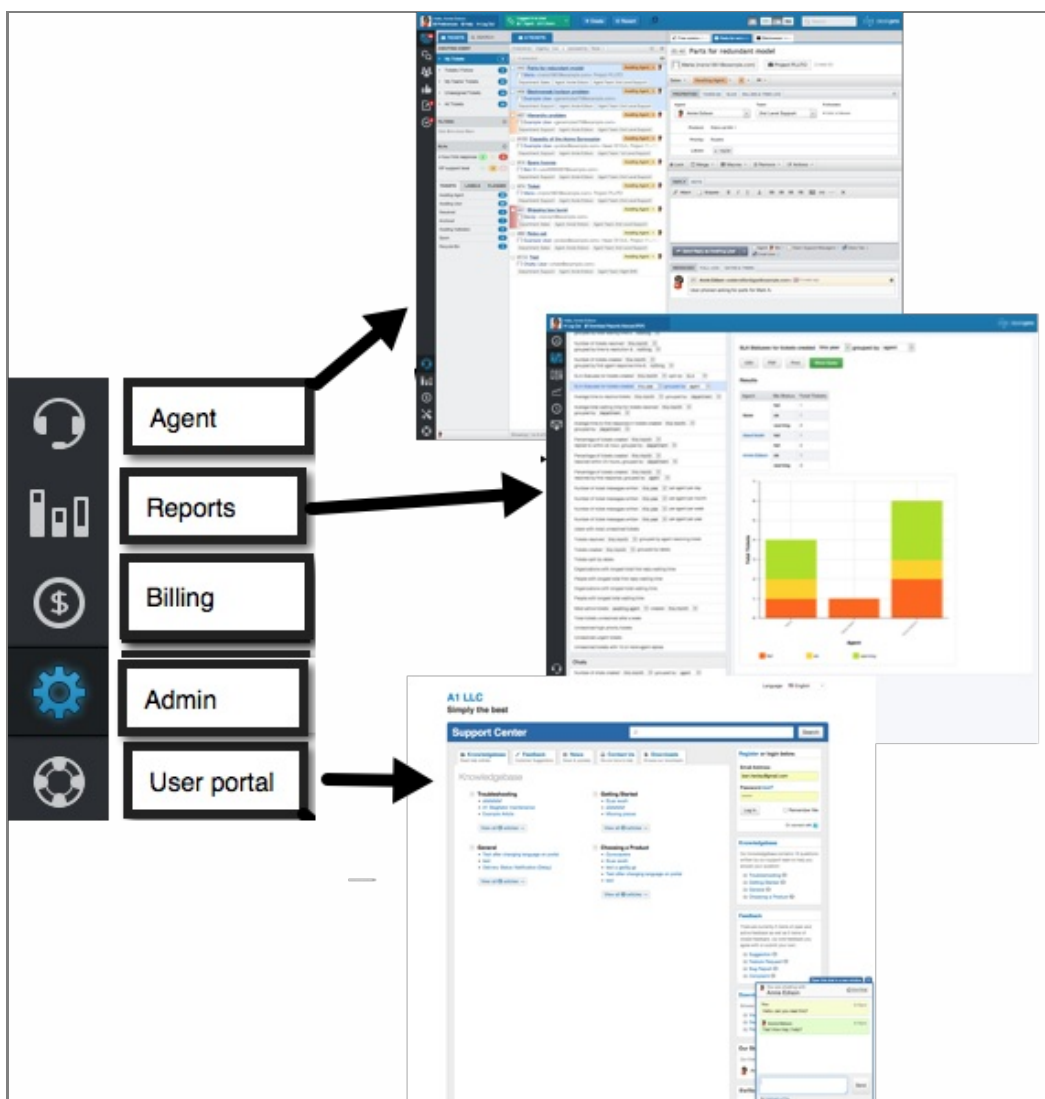
The admin home page gives you an overview of the status of your DeskPRO installation.



You can access different sections of the admin interface using the upper icons in the bar at far left.

	Setup	Basic configuration, languages
	Agents	Manage agents, teams, agent permissions and view audit log
	Tickets	Configure tickets, departments, email accounts, automation...
	CRM	User-related settings: registration, usergroups...
	Portal	Configure portal, embedding, and Publish content settings
	Chat	Chat settings, embed widget
	Apps	Manage apps to extend & integrate DeskPRO, API keys
	Server	Server settings (DeskPRO Download only)

You can switch between interfaces (including the web portal for users) with the icons at the lower left.



- The **agent** interface is where your agents will reply to tickets, chat with users, create portal content etc.. See the [Agent Manual](#) for details.
- The **reports** interface enables you to generate detailed tables and graphs about your helpdesk.
- The **billing** interface is where you pay for and manage your license to use DeskPRO.
- The **user portal** is where users can browse Knowledgebase articles and other content managed in the agent Publish app, submit tickets and chat with agents.

Note that as an admin, you can log into any of these interfaces using your account.

2. Creating Your Helpdesk

When you set up your helpdesk, you have two deployment options:

- **DeskPRO Cloud:** where the software runs on our servers, and we manage the technical details of hardware, upgrades, off-site backups, network and bandwidth. This is the quickest and easiest method, and it's what we recommend for most customers. See [Cloud sign-up](#) for the simple procedure to get a Cloud helpdesk.
- **DeskPRO Download:** where you download and install DeskPRO on your own servers. This option gives you more control over hosting and access: for example, you can run an internal helpdesk that is only accessible from your intranet. You will have full access to the DeskPRO source code. You will need the technical skills to be able to configure and maintain a webserver and database, and you will have to arrange hosting and bandwidth yourself. See [Installing DeskPRO Download](#) for setup details.

Note that you can *always* migrate between these two options: you can evaluate DeskPRO using Cloud, then move your helpdesk to your own servers; if you decide that you no longer want to self-host, you can move your helpdesk into Cloud.

2.1. Helpdesk domains

When you first set up a Cloud helpdesk, it is hosted on a sub-domain like `yourcompany.deskpro.com` ; agents log in at `yourcompany.deskpro.com/agent/` and the admin interface is at `http://yourcompany.deskpro.com/admin/`

You can configure a Cloud helpdesk so it's accessible through your own custom sub-domain such as `help.yourcompany.com` or `support.yourcompany.com` , where `yourcompany.com` is your company's domain.

You could even use a dedicated domain just for your helpdesk e.g. `yourcompanyhelp.com` .

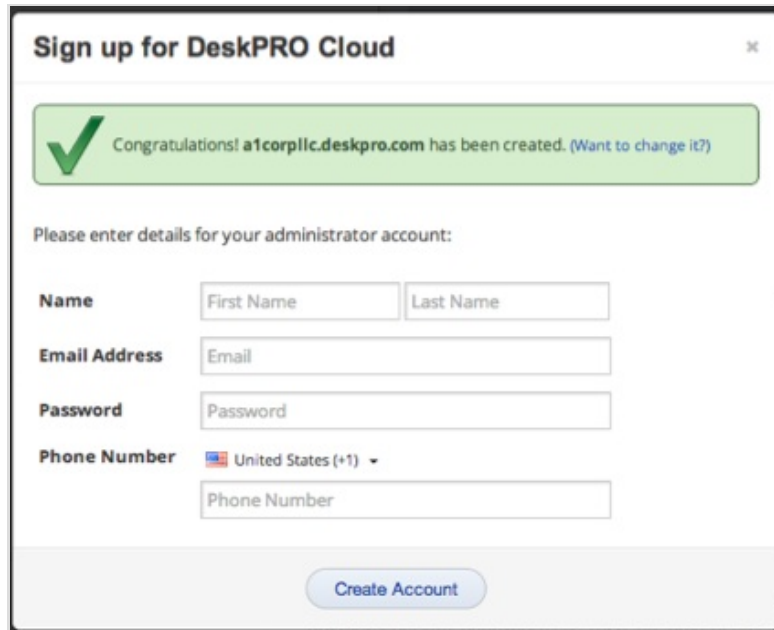
See the section [Configuring a custom domain with Cloud](#) for details.

If you're using DeskPRO Download, you configure the domain or IP address where your helpdesk is hosted.

2.2. Cloud sign-up

Signing up for a Cloud helpdesk is easy:

- 1 Go to <https://www.deskpro.com/>, go to the Cloud sign-up area and enter a name for your helpdesk, then click **Create My Free Helpdesk**
- 2 The **Sign up for DeskPRO Cloud** pop-up opens. Choose the details for your admin account.



The screenshot shows a pop-up window titled "Sign up for DeskPRO Cloud" with a close button (X) in the top right corner. Inside the window, there is a green success message: "Congratulations! a1corp1lc.deskpro.com has been created. (Want to change it?)" with a green checkmark icon. Below this, it says "Please enter details for your administrator account:". The form fields are: "Name" (split into "First Name" and "Last Name" boxes), "Email Address" (labeled "Email"), "Password" (labeled "Password"), and "Phone Number" (with a dropdown menu showing "United States (+1)" and a "Phone Number" box). At the bottom right is a blue "Create Account" button.

- 3 You can now log in with the email address and password you provided.

2.3. Installing DeskPRO Download

DeskPRO is web-based software written in the PHP language. It is designed to be independent of the underlying operating system and web server you use. Setting up a DeskPRO Download helpdesk is fairly simple, but does require server and database administration skills on your platform of choice.

The installation instructions we have provided in this manual are written to give a general outline of the process, applicable to all platforms.

You may be able to find more detailed installation advice for your platform in our knowledgebase at <https://support.deskpro.com/kb>. We also offer an [installation service](#) for an additional fee.

Requirements

DeskPRO will run on any recent version of Windows, Linux or Mac OS X.

Installing DeskPRO requires the following:

A webserver: any webserver that works well with PHP should be able to run DeskPRO, such as a recent version of Apache, nginx/PHP-FPM or Microsoft IIS.

PHP: minimum supported version is v5.3.2, but we recommend always using the latest version.

DeskPRO will also need access to the command line version of PHP (often called php or php-cli on Linux or php-win.exe on Windows). This is generally installed by default as part of the PHP installation process.

PHP extensions that you need enabled are:

- PDO with the MySQL driver installed
- mbstring or iconv
- json
- session
- ctype
- dom
- tokenizer
- GD or Imagick or Gmagick
- OpenSSL (recommended)

If you are using a version of PHP before 5.5, it is recommended to increase performance that you install an **opcode cache** such as [Alternative PHP Cache](#) or [Windows Cache Extension](#), which can dramatically increase performance. Versions above 5.5 come with a built-in opcode cache (OPCache).

mySQL database: The minimum supported version is 5.0; we recommend always using the latest 5.x.x. version.

cron (on a Unix-based OS) or **Scheduled Tasks** (on Windows) to run regular processes such as sending email

An email server (optional): DeskPRO can send email via an SMTP server, using the PHP mail() function or via a Gmail/Google Apps account. It can receive email from a POP3, IMAP or Exchange server or a Gmail/Google Apps account.

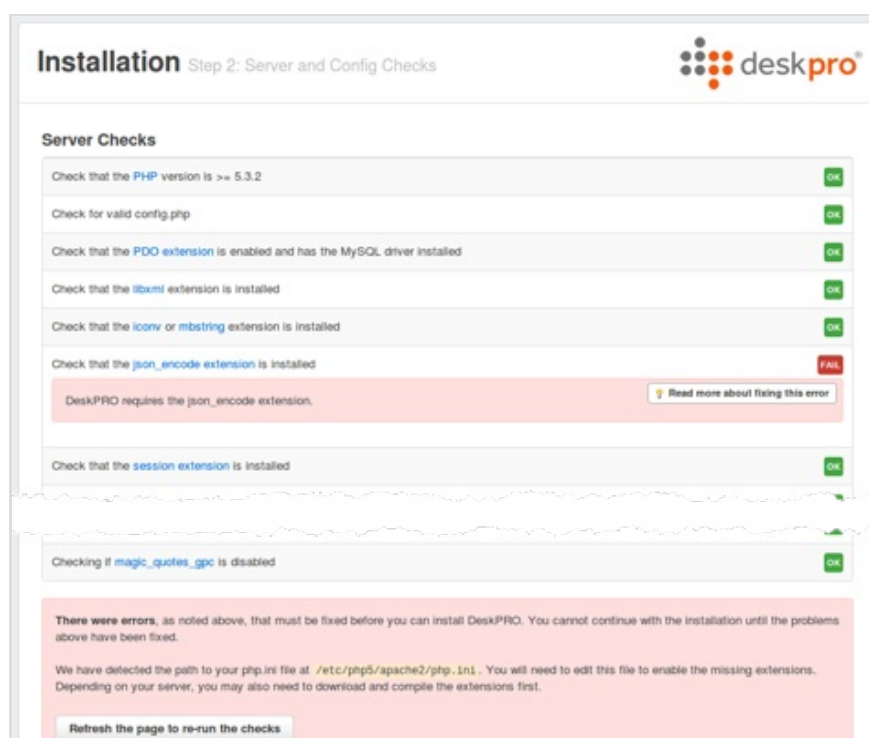
Hardware: DeskPRO runs well on modest hardware, and can easily run from a standard virtual machine or even virtual hosting account. If you are expecting a very busy helpdesk, the limiting factor for speed is generally the database, which can be helped by increasing the amount of RAM and using fast disks for storage. You can run all the required software on one machine; however, you may wish to separate the web server from the database server.

Installation process

Once you've set up the basic PHP/MySQL environment, DeskPRO's installation wizard will guide you through the rest of the process.

The installation process will of course vary depending on your chosen platform, but the general steps are:

- 1 Install a web server and enable PHP support and URL rewriting.
- 2 Install PHP, with the required PHP extensions listed in [Requirements](#).
- 3 Install MySQL, and create a 'deskpro' user account.
- 4 Download the latest version of DeskPRO from <https://www.deskpro.com/downloads/deskpro.zip>
 - remove the default page (e.g. `index.html`) from your web server's base directory, and unzip `deskpro.zip` into it
 - copy the resulting `config.new.php` file as `config.php`
 - set a cron job or Scheduled Task to run `cron.php` once per minute
 - edit `config.php` to add your email address and MySQL database credentials
- 5 Ensure the web server is running and go to the URL (or IP address) where it is serving DeskPRO.
- 6 The wizard will take you through the steps required to complete installation. It can advise you on how to fix common configuration problems and missed steps.



- 7 Once installation is complete, you can log on with the account details you gave during installation.

There is more platform-specific installation guidance available in the [DeskPRO Knowledgebase](#).

2.4. Configuring your new helpdesk

Entering basic details

The first step when configuring your new helpdesk is to enter the details of your organization.

Go to the **Setup** section by clicking the gear icon in the black bar at upper left.



Enter the details requested under **Your Helpdesk** and **Your Website**.

Settings

Your Helpdesk

Helpdesk Name
This is the name of your helpdesk. The name is displayed to users in their browser window title, and is also

Helpdesk URL
The primary URL for your helpdesk. This URL is used whenever the system needs to link to your helpdesk.

Your Website

Website Name
Enter the name of your company or company website.

Website URL
Enter the URL to your company website.

For DeskPRO Download, you will see a **Helpdesk URL** entry; this should be the web/intranet address where users can access your web portal.

While you're on this page, check that the **Default Timezone** at the top of the **Dates and Times** section is set correctly.

Configuring a custom domain with Cloud

If you are using DeskPRO Cloud, your helpdesk will initially be hosted at the default URL you selected when you created it: e.g. `http://yourcompany.deskpro.com`.

You can use a sub-domain for the helpdesk, e.g. `http://support.yourcompany.com`, or even use a dedicated domain like `http://www.yourcompanyhelp.com`.

To set this up:

- 1 Go to **Setup > Domain / URL Setup**
- 2 Click **Use a custom URL**. Enter the domain name you want to use.
- 3 Edit the DNS settings for your domain to create a CNAME record mapping `yourcompany.deskpro.com` to your custom domain:

NAME	TYPE	VALUE

support.yourcompany.com.	CNAME	yourcompany.deskpro.com.

Note that the admin interface will always be accessible through the original `yourcompany.deskpro.com/admin/` address. This is useful if you run into problems with your custom domain (e.g. it expires).

You can also select whether to enable HTTPS security. Note that using HTTPS with a custom domain requires manual setup by one of our agents. Contact support@deskpro.com to arrange this.

Disabling the helpdesk

If you need to turn off the user-facing parts of the helpdesk while you configure it, you can disable the helpdesk from **Setup > Settings**

☒ **Disable the helpdesk**

This message will be displayed when the helpdesk is offline or when you are installing updates. HTML is allowed.

The helpdesk is undergoing routine maintenance. We will be back in about 10 minutes.

When you select this option:

- the user portal is disabled and replaced with the message you specify
- any [website integrations](#) you have set up stop working
- email processing and sending is stopped

The agent and admin interfaces will still be available while the helpdesk is disabled.

Next steps to setting up your helpdesk

The steps you need to take now will depend on how you want to use DeskPRO.

Consult the [Quick Launch Guide](#) for a checklist of essential decisions and steps you'll need to get a new helpdesk ready for use.

Here are some sections of this manual relating to the common setup tasks that most helpdesks will need:

- 1 Configure [departments](#) to divide up the helpdesk.
- 2 Add more [agents](#) - create an agent account for everyone who will work on the helpdesk.
- 3 Set up ticket [email accounts](#).
- 4 Customize the [portal](#) or [integrate it with your website](#) using embeds.
- 5 Customize [ticket fields](#) to store extra information about tickets.

Not all of these steps will apply to all helpdesks.

For example, you may prefer to disable the portal and interact with users by email only. See the section [Running an email-only helpdesk](#) for details on how to do this.

3. Agents

As an admin, you can create and manage all agent accounts.

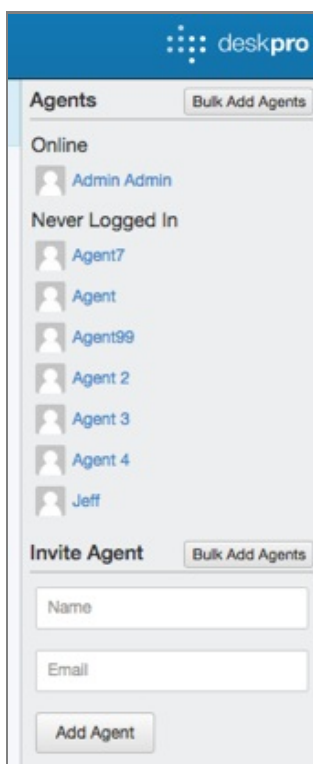
Use [agent permissions](#) to control what each agent is allowed to do within your helpdesk. There are many fine-grained settings, so to simplify management, you can add agents to predefined [permission groups](#).

You can organize agents into [teams](#), which allow tickets to be assigned to groups of agents.

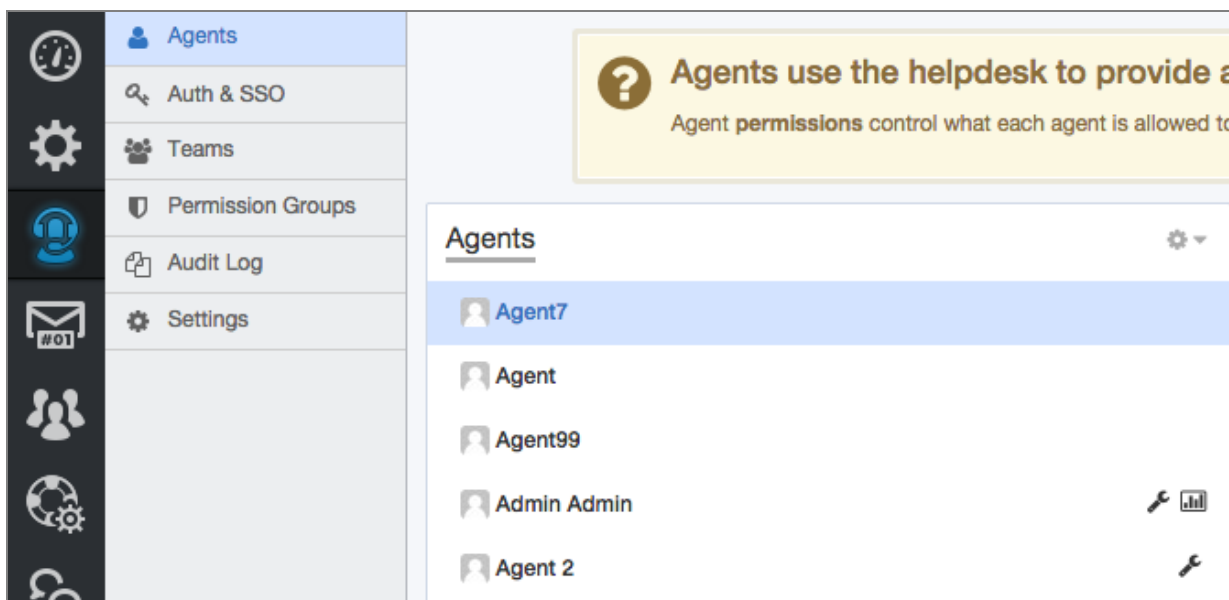
3.1. Creating agents

You can see a list of agents/admins in the system at the top right of the admin Home page.

You can quick-add an agent using the **Invite Agent** form on the Home page - note that this gives the agent All Permissions (in other words they will be able to use all agent interface functions).



For more detailed administration of agents, go to the **Agents** section of the admin interface.



Adding a new agent from this section using the **Add** button gives you more control of details like their account properties and [permissions](#).

When you create a new agent account for someone, they receive an email notification with their login details and the [Agent Quick Start Guide](#) (a brief overview of how the agent interface works) attached as a PDF.

Agent properties

You can define the following properties for each agent from the **Agents** section:

- **Name** (and an optional override name that will be shown to users) *
- **Email Addresses** *
- **Phone Number** (this is used only for [sending SMS notifications](#) so you should enter a mobile/cell number) *
- **Zones** (whether the agent has access to admin and/or the reports interface)
- **Teams** (set membership of [agent teams](#))
- **Permission Groups** (set membership of [agent permission groups](#))

Properties marked with * can also be edited by the agent themselves from the **Preferences** section in the agent interface.

The screenshot shows the 'Properties' tab of the agent settings interface. It features a top navigation bar with tabs: 'Properties' (selected), 'Permissions', 'Departments', 'Ticket Notifications', and 'Other Notifications'. A gear icon is on the right. The main content area has several sections: 'Name' with a text input field and a checkbox 'Display a different name to end-users'; 'Email Addresses' with a text input field containing 'agent7@example.net' and a note 'Separate multiple email addresses with a comma.'; 'Phone Number' with a dropdown menu showing a flag icon; 'Zones' with checkboxes for 'Admin Interface' and 'Reports Interface'; 'Teams' with checkboxes for '1st Level Support', '2nd Level Support', and 'Support Managers'; and 'Permission Groups' with checkboxes for 'All Permissions' and 'All Non-Destructive Permissions'. A note at the bottom of the 'Permission Groups' section says 'Switch to the Permissions or Departments tab to manage permission overrides.' At the bottom of the form are 'Save' and 'Delete' buttons.

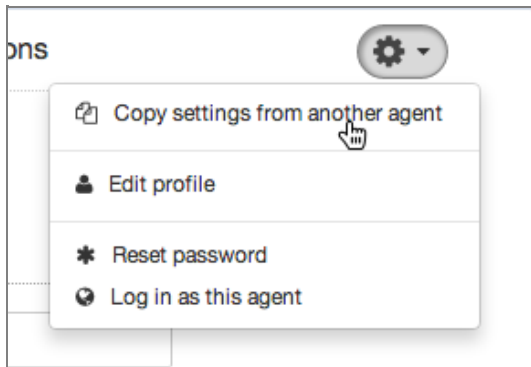
In other tabs, you can set:

- **Permissions** (individual [agent permissions](#) for the agent)
- **Departments** (set which [departments](#) the agent can use; see the section on [agent department permissions](#))
- **Ticket Notifications** - control which email/browser notifications the agent receives about tickets; by default, agents can edit their notifications settings from the **Preferences** section, although as an admin you can disable this ability for each agent
- **Other Notifications** - control which email/browser notifications the agent receives about other helpdesk events like [chats](#) and tasks; agents can always edit their own settings for these notifications

Copying agent settings

Once you've configured one agent's account, you may want to apply the same settings to other agents on the helpdesk.

When viewing an individual agent's details, you can copy settings from another agent using the gear icon at the top right.

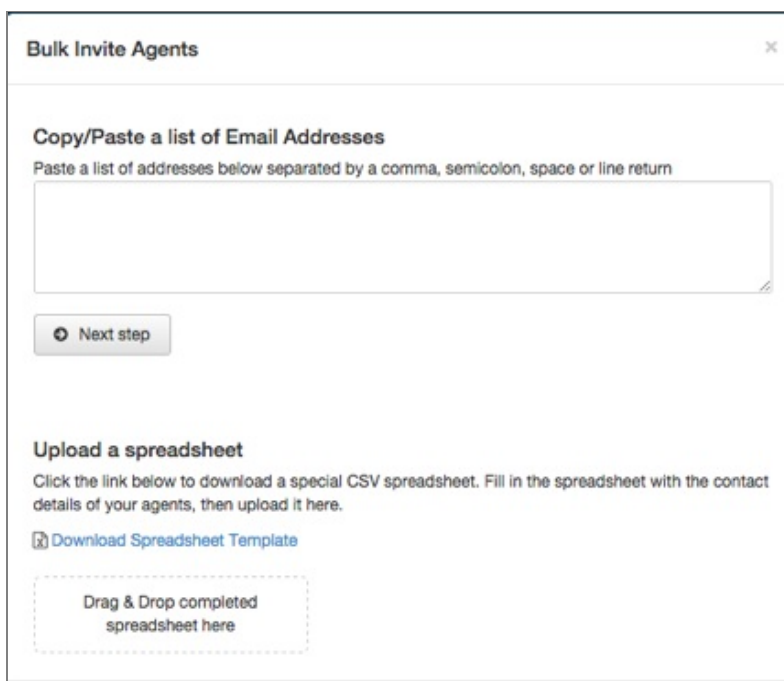


You can select which settings to copy.

A screenshot of a 'Copy Settings' dialog box. At the top, it says 'Copy Settings' with a close button (X). Below this, there is a label 'Copy settings from' followed by a dropdown menu showing 'Maria Cama'. Underneath, the question 'Which settings do you want to copy?' is followed by a list of six settings, each with a checked checkbox: 'Zone Access', 'Teams', 'Permission Groups', 'Permission Overrides', 'Ticket Notification Preferences', and 'Other Notification Preferences'. A note at the bottom states: 'Note that settings will be applied to the edit form and you will need to click the "Save" button to actually save any changes.' At the very bottom, there is a blue button labeled 'Apply settings to form'.

Bulk adding agents

You can create multiple agents at a time using the **Bulk Add Agents** button. You can either supply a list of email addresses to invite, or a CSV spreadsheet of agent data.



The image shows a 'Bulk Invite Agents' dialog box with a close button (X) in the top right corner. It contains two main sections. The first section, 'Copy/Paste a list of Email Addresses', includes a text area for pasting addresses and a 'Next step' button. The second section, 'Upload a spreadsheet', includes a link to 'Download Spreadsheet Template' and a dashed box for dragging and dropping a completed spreadsheet.

Bulk Invite Agents [X]

Copy/Paste a list of Email Addresses
Paste a list of addresses below separated by a comma, semicolon, space or line return

[Text Area]

[Next step]

Upload a spreadsheet
Click the link below to download a special CSV spreadsheet. Fill in the spreadsheet with the contact details of your agents, then upload it here.

[Download Spreadsheet Template](#)

Drag & Drop completed spreadsheet here

Using the spreadsheet option enables you to specify the following information for each agent account you create:

- Email (required)
- Name (required)
- Permission Groups (required, specify by ID number, where 1 means All Permissions and 2 means All Non Destructive Permissions)
- Teams (specified by ID number)
- Admin access (Yes or No)
- Reports access (Yes or No)
- Notifications (Yes to enable default notifications, No to enable none)
- Signature (HTML format)

Agent licensing

Your DeskPRO license entitles you to a certain number of agent accounts at any one time.

If you attempt to create more agents than your license allows, you will be prompted to upgrade your license.

License Limit Reached

You are already using all of the agent seats that your license allows.

Increase your license from 5 to agents to create a new agent.

Upgrade now for **\$6100 USD**

[Download PDF Invoice](#) • [View Details](#)

☒ Use your existing card

 xxxx xxxx xxxx 4008 Exp: 11 / 15

☐ Enter a new card

Upgrade license Now

Your payment information is being sent securely over HTTPS.

Either upgrade your license, or delete an agent account that is no longer in use.

Note that the limit is on the number of agent accounts you have, not the total number you have created. In other words, deleting an agent and then adding another does not "use up" an agent license.

3.2. Agent permissions

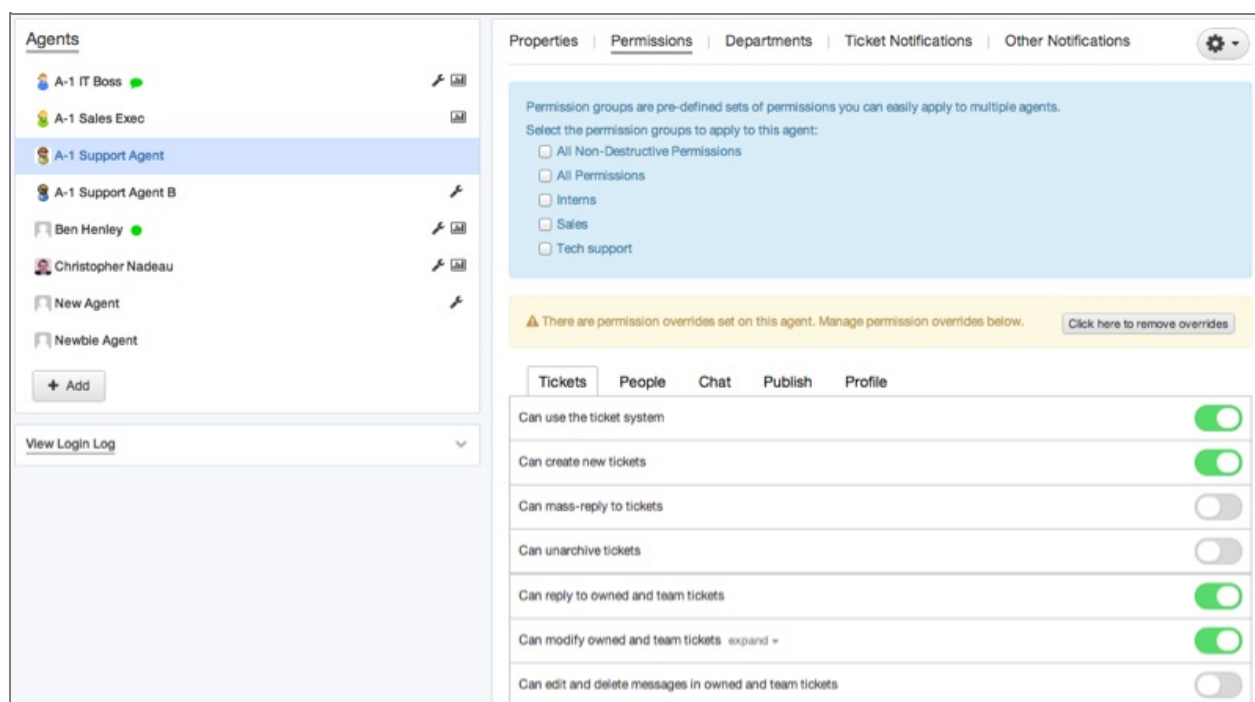
Agent permission settings control what an agent is allowed to see and do on the helpdesk. You can use this to limit agent accounts so they can only carry out the actions that are required by their role.

For example, you may want to limit a trainee's account so that they can't reply to tickets, but can still read them and assign them to other agents.

There are a large number of permissions available, so you can restrict what an agent does at a high level (e.g. preventing them from using an app like Tickets or Chat altogether), or at a fine-grained level (for example, prevent them from using a single ticket function).

You can use [department permissions](#) to limit an agents access based on departments.

To set individual permissions for an agent, go to **Agents** and click on the agent's name, then the **Permissions** tab.



You can use the toggle controls at right to enable and disable the agent's ability to carry out a wide variety of actions. Click the **People**, **Chat**, **Publish** or **Profile** tabs to set permissions for other parts of the helpdesk.

To avoid having to set up permissions for each agent individually, you can use permission groups to quickly apply stored sets of permissions to multiple agents.

See the section on [Permission Groups](#) for more details.

Agent zones

In the **Properties** tab, you can select what **zones** the agent can use.

Granting **admin** zone access means the agent is an admin and can use the admin interface. Note that there is no way to customize admin permissions; admin access is all or nothing.

Granting **report** zone access enables the agent to use the Reports interface to view statistics and charts about your helpdesk's tickets, agents' performance, billing, etc.

Agent permissions for tickets and chats are also limited by departments: see the section on [Department Permissions](#) for details.

Agent ticket permissions

You can limit agents' access to tickets based on assignment/following status.

For example, if you disable an agent's **view unassigned tickets** permission, they can't view unassigned tickets.

If you disable both **view unassigned tickets** and **view tickets assigned to others** permissions, the agent can only see tickets that are assigned to them (or one of their teams).

A ticket being assigned to one of an agent's teams grants the same access as if it was assigned to the agent directly.

Note that agents can *always* view tickets which are assigned to them/their team *or* that they are following. So, for example, if an agent has **can view unassigned tickets** disabled, but is added to an unassigned ticket as a follower, they will still be able to see the ticket.

These are the ticket settings available for each agent. Note that an agent's 'owned' tickets means the tickets that are assigned directly to that agent.

Tickets	People	Chat	Publish	Profile	Tasks
Can use the ticket system					☒
Can create new tickets					☒
Can mass-reply to tickets					☒
Can unarchive tickets					☒
Can modify billing and time log records					☒
Can reply to owned and team tickets					☒
Can modify owned and team tickets expand ▾					☒
Can edit and delete messages in owned and team tickets					☒
Can delete and spam owned and team tickets					☒
Can reply to followed tickets					☒
Can modify followed tickets expand ▾					☒
Can edit and delete messages in followed tickets					☒
Can delete and spam followed tickets					☒
Can view unassigned tickets					☒
Can reply to unassigned tickets					☒
Can modify unassigned tickets expand ▾					☒
Can edit and delete messages in unassigned tickets					☒
Can delete and spam followed tickets					☒
Can view tickets assigned to others					☒
Can reply to tickets assigned to others					☒
Can modify tickets assigned to others expand ▾					☒
Can edit and delete messages in tickets assigned to others					☒
Can delete and spam tickets assigned to others					☒

The "modify tickets" permissions (shown collapsed above) are:

- Modify department
- Modify fields
- Can assign agents
- Can assign teams
- Can assign tickets to self
- Can modify CCs
- Can merge this ticket
- Can modify labels
- Can modify SLAs
- Can add notes
- Can set on-hold
- Can set status to awaiting user
- Can set status to awaiting agent
- Can set status to resolved
- Can unresolve ticket

You can't grant an agent the permission to mark a ticket as spam but not the permission to delete it. This is by design, since marking a ticket as spam is effectively a different way to delete it; spam tickets are purged after a set period of time. See the [Ticket purging](#) section for details.

Agent department permissions

For each department, an agents can have either **Full**, **Assign** or no access.

Note that the ticket permission settings interact with the ticket department permissions: to do something to a ticket, an agent must have permission for the relevant action *and* the required level of access to the relevant department.

For example, if an agent had the **view unassigned tickets** ticket permission enabled, but only had **Assign** access to the Sales department, they would still not be able to view any Sales tickets.

See the section on [department permissions](#) for details.

Agent people permissions

These control what agents can do with users and organizations.

Tickets	People	Chat	Publish	Profile
Can use the CRM system				
Can create new user profiles				
Can edit user profiles				
Can validate users and user email addresses				
Can edit users email addresses				
Can reset user passwords				
Can add notes to user profiles				
Can completely delete users				
Can disable users				
Can log in as users				
Can merge users				
Can create organizations				
Can edit organizations				
Can add notes to organizations				
Can delete organizations				

Agent chat permissions

These control what agents can do with the user chat system.

Tickets	People	Chat	Publish	Profile
Can use chat				
Can view old chat transcripts				
Can view unassigned chats				
Can view chats assigned to others				
Can delete chat transcripts				

These settings apply only to the user-facing chat system. Agent IM is always enabled for all agents; agents are logged in to IM whenever they are logged in to the agent interface.

Note that the agent must also have the correct chat department permission to do anything within a chat department.

Agent publish permissions

These control what the agent can do in the agent interface Publish app which is used to publish content on the portal.

Tickets	People	Chat	Publish	Profile
Can create content				
Can delete content				
Can edit content				
Can validate content and comments (including own)				

Agent profile permissions

These control whether the agent is allowed to set their own profile picture (displayed in the agent interface and on the portal) and signature (appended to messages to users).

Agent notifications

You can set the details of the notifications each agent receives, both for ticket notifications and other notifications (e.g. for chat, tasks, account logins).

[Properties](#) |
 [Permissions](#) |
 [Departments](#) |
 [Ticket Notifications](#) |
 [Other Notifications](#)

[Email Notifications](#) |
 [Browser Alerts](#)

☒ Do not allow this agent to change these notification settings

	New Ticket	Assignment	User Message	Agent Message	Agent Note	Property Change
My Tickets						
My Team's Tickets						
Tickets I Follow						
Unassigned Tickets						
All Tickets						

Filters

	New Ticket	Filter Match	User Message	Agent Message	Agent Note	Property Change
Tickets on hold from Company X, urgency over 6						

By default, each agent can change the ticket notification settings for their own account using **Preferences** in the agent interface. You can disable an agent's ability to do this by checking the **Do not allow this agent to change these notification settings** checkbox.

Agents with custom filters can set notifications for tickets matching that filter. This is done in the **Filters** section which appears below the other ticket notifications if the agent has custom filters. You and your agents can use this to create very precise custom notifications.

Agent task permissions

You can control agent access to the Tasks app using the **Tasks** section of the admin interface.

You can enable/disable the Task application for the whole helpdesk.

You can also grant the ability to use tasks to individual agents, or to permission groups.

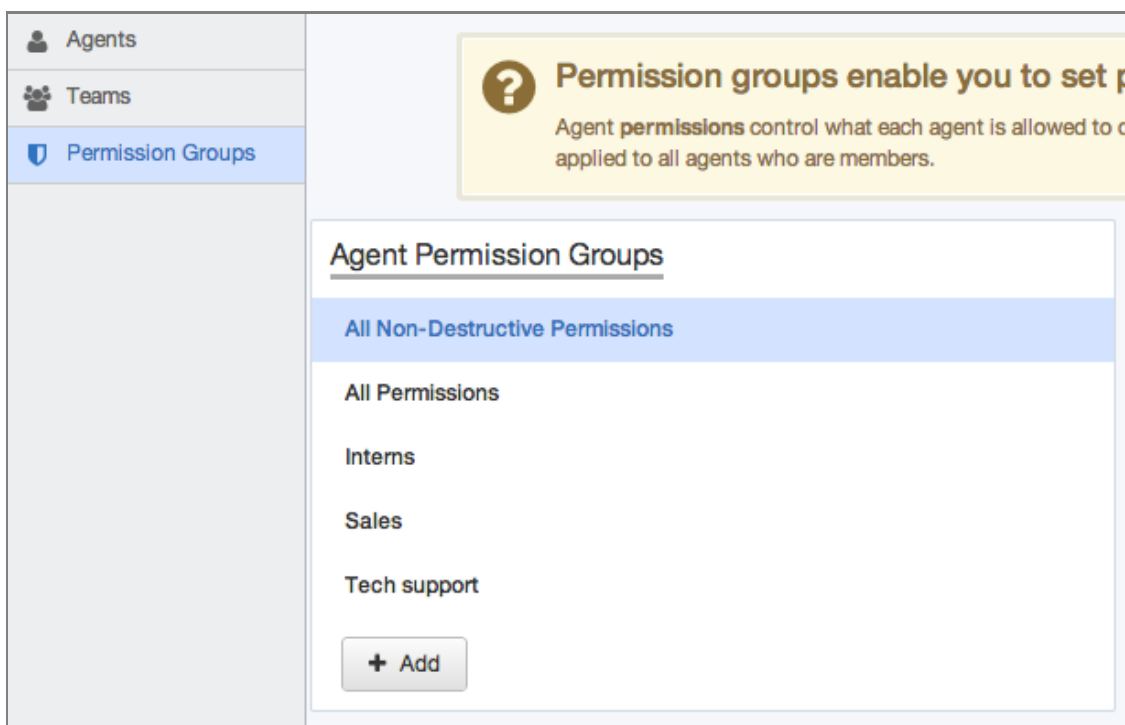
If an agent is granted permission to use tasks through one of their permission groups, it is indicated with a gray tick to show that you cannot remove it from the individual agent.

You can also set the time of day that task reminders are sent out. This affects the timing of **A task assigned to me is due today** notifications. If the agent does not have this enabled (in **Agent > Preferences > Notifications > Tasks**) they will not receive the reminder.

	Email Notifications	Browser Notifications
A task is assigned to me	☒	☒
A task is assigned to my team	☐	☐
A task I delegated is completed	☐	☐
A task assigned to me is due today	☒	☒

3.3. Permission groups

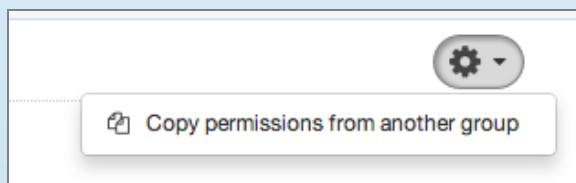
Instead of making decisions about each of the dozens of possible actions for every one of your agents, you can use permission groups to store sets of permissions which you can quickly apply to multiple agents. These are managed from **Agents > Permission Groups**



The default groups when you first create your helpdesk are "All Permissions" and "All Non-Destructive Permissions". The latter allows the agent to use most DeskPRO features, except ones which involve deleting information permanently.

To create a new permission group, click the **Add** button.

After you click **Add**, you can copy the permission settings from an existing group with the gear icon at the top right. This can save you time compared to starting from scratch.



In the **Properties** tab, enter a **Group Name**. Then click on the **Permissions** tab and set permissions as for an individual agent. You can also set [department permissions](#) for the group.

Agents can belong to more than one permission group. If an action is allowed by *any* of an agent's groups, the agent can do it.

You can select permission groups for an agent from their **Properties** tab:

Permission Groups

☒ All Non-Destructive Permissions

☐ All Permissions

☐ Interns

☒ Sales

☒ Tech support

Switch to the **Permissions** tab to manage permission overrides.

Save

or add agents to a permission group from the group's **Properties**:

Properties | Permissions | Departments

Group Name

All Non-Destructive Permissions

Members

☒ A-1 IT Boss

☒ A-1 Sales Exec

☐ A-1 Support Agent

☒ A-1 Support Agent B

☐ New Agent

☐ Newbie Agent

Save

Permission overrides

You can combine individual permission settings and permission groups.

For example, this agent is a member of "All Non-Destructive Permissions".

Properties | Permissions | Departments | Ticket Notifications | Other Notifications

Permission groups are pre-defined sets of permissions you can easily apply to multiple agents.
Select the permission groups to apply to this agent:
☒ All Non-Destructive Permissions
☐ All Permissions
☐ Interns
☐ Sales
☐ Tech support

Tickets | People | Chat | Publish | Profile

Can use the ticket system	
Can create new tickets	
Can mass-reply to tickets	
Can unarchive tickets	
Can reply to owned and team tickets	

Suppose you need to grant the agent permission to mass-reply to tickets.

You can just click the **Can mass-reply to tickets** toggle to add that permission.

There are permission overrides set on this agent. Manage permission overrides below. [Click here to remove overrides](#)

Tickets | People | Chat | Publish | Profile

Can use the ticket system	
Can create new tickets	
Can mass-reply to tickets	
Can unarchive tickets	
Can reply to owned and team tickets	

A permission that's not granted through a permission group is considered an **override**. To make it easier to work out which agents have which permissions, DeskPRO alerts you if there are overrides, both in the agent's **Permissions** tab as you can see above, and in the agent **Properties** tab:

Permission Groups

- ☒ All Non-Destructive Permissions 
- ☐ All Permissions 
- ☐ Interns 
- ☐ Sales 
- ☐ Tech support 

  This agent has permission overrides on the **Permissions** tab.

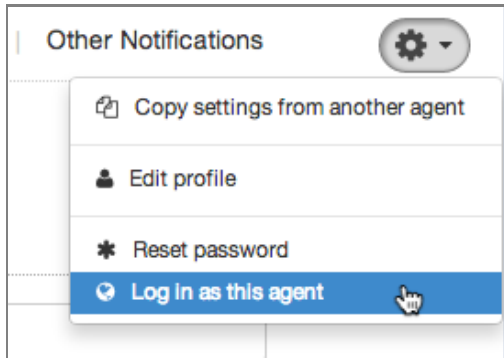
Note that you can't *remove* permissions which are granted by a permission group; this is indicated by the lock symbols on the other toggle controls.

Agent permissions are always additive - an agent only needs to be granted a permission by a single group or an override to be able to use it, even if none of their other groups include that permission.

3.4. Logging in as an agent

As an admin, you can log in to the agent interface as any agent (including other admins). This can be useful for troubleshooting purposes.

- 1 | Navigate to the **Agents** settings page, then select the agent you want to log in as.
- 2 | Click **Log in as this agent**



Note that when you are logged in as an agent, it's exactly as if you have signed in with their account and password. For example, if the agent doesn't have permission to alter their own notifications, you won't be able to alter them either. If you send a response to a user, it will appear to come from the agent who's account you're using.

3.5. Monitoring agents

To help monitor how your agents are using the helpdesk, you can view a **Login log** below the **Agents** list.

For more detailed information about agent activity, you can use the **Reports** interface, accessible from the chart icon at the lower left.

There are three sections of the Reports interface that are useful for checking agent performance:



Agent Activity shows you a log of all the actions agents have taken.



Agent Hours shows you the times when agents were actively using the agent interface (writing a reply, clicking something, moving the mouse). Breaks do not necessarily indicate that the agent interface was closed or that the agent was logged out, just that the interface was not being actively used.



Ticket Satisfaction shows you the results of user ratings of agent replies from the [satisfaction survey](#).

3.6. Managing agent passwords

Go to **Agents > Settings** to set the requirements for creating agent account passwords.

By default, DeskPRO requires only that passwords are 5 characters long. You can use the **Custom Password Policy** option if you want to set a more restrictive policy for security reasons.

☐ Standard Password Policy

☒ Custom Password Policy
Specify custom password requirements.

Password Policy

Minimum password length: characters
The minimum number of characters the password has to be.

Maximum password age: days
How many days a password can remain unchanged until DeskPRO requires the agent to change it.

☐ Forbid password reuse
When enabled, agents will not be able to change their password to a password they have used in the past.

Require this many lower-case characters:
Define the minimum number of lower-case (a-z) characters that must be used in the password.

Require this many upper-case characters:
Define the minimum number of upper-case (A-Z) characters that must be used in the password.

Require this many numbers:
Define the minimum number of numbers (0-9) that must be used in the password.

Require this many symbols:
Define the minimum number of symbols (!, @, \$, % etc) that must be used in the password.

The policy will only apply the next time each agent sets their password. When you set a new password policy, agents whose passwords do not meet the policy will not be forced to change them (unless you set a maximum password age).

You can reset agent passwords from the admin interface. Go to **Agents**, select the agent you want to reset, then select **Reset password** from the gear menu at top right.

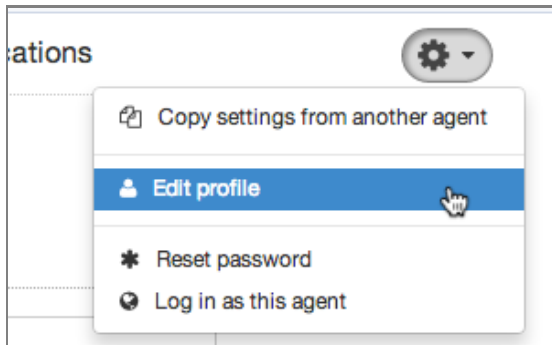
You choose whether to randomly generate a new password, or enter one yourself. The new password will be emailed to the agent.

If you are using DeskPRO Download, you can also set agent passwords and admin access from the command line. See the section on [Setting account credentials from the command line](#) for details.

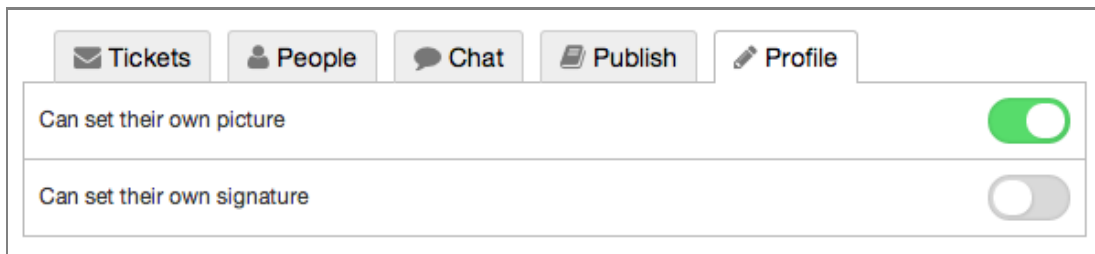
3.7. Editing agent profiles

In the agent interface **Preferences** agents can set a picture (to represent them in the agent interface), timezone and an HTML signature for ticket messages.

You can edit an agent's profile from the admin interface using the **Edit profile** option on the gear icon.



Note that you can disable the ability for an agent to set their own picture and/or signature using the **Profile** permissions.



3.8. Deleting an agent

You can delete any agent using the **Delete** button displayed at the lower right when viewing the agent's details.

When you delete an agent, you can choose to either convert their account to a user account, or mark the account deleted.

- **Convert to user**
 - Turns the agent into a user.
 - Immediately unassigns all tickets currently assigned to the agent.
 - You can turn the user back into an agent later, but some settings (such as permissions) will be lost.
 - The person will still be able to use the account to interact with the helpdesk e.g. submit tickets as a user, log in to the portal.
- **Mark as deleted**
 - Stops the person being able to log in to either the portal or agent interfaces.
 - Tickets assigned to the agent will remain assigned until they become **awaiting agent**, then become unassigned.
 - You can fully restore the agent later.
 - Use this option if you want to keep reports for the agent based on their assigned tickets.

Either method means that the agent no longer counts towards the [licensing limit](#) on the number of agents on your helpdesk.

Of course, if you want to convert a user back to an agent, or restore a deleted agent, you need to have a spare agent seat, or buy more agents.

3.9. Admin access log

Admins have the power to lock out or delete other users, and make major changes to your helpdesk data that can't be undone.

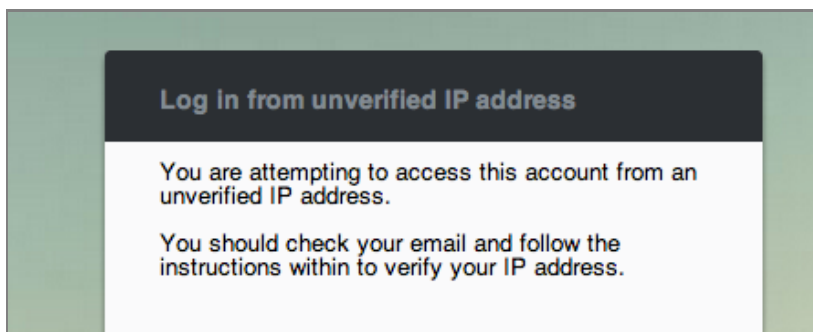
You can view a log of all admin interactions with the helpdesk under **Agents > Audit Log**. This can be useful if you are trying to investigate an unexpected change to your helpdesk, or if you suspect that an admin account has been abused.

140	A1 Admin <ben.henley@deskpro.com>	is_agent_group:	UPDATE usergroups#1	2014-05-12 17:14:50
139	A1 Admin <ben.henley@deskpro.com>	is_agent_group: 1	UPDATE usergroups#1	2014-05-12 17:12:00
138	Christopher Nadeau <chris.nadeau@deskpro.com>	permissions: REMOVE: Full for people#2 -- A1 Admin <ben.henley@deskpro.com>	UPDATE departments#2	2014-05-12 16:44:35
137	Christopher Nadeau <chris.nadeau@deskpro.com>	permissions: ADD: Full for usergroups#1 -- Everyone	UPDATE departments#1	2014-05-12 16:43:07
136		date_last_login: 2014-05-12 16:42:36	UPDATE people#1	2014-05-12 16:42:36
135	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#12	2014-05-12 15:26:53
134	A1 Admin <ben.henley@deskpro.com>		DELETE usergroups#11	2014-05-12 15:26:29
133		date_last_login: 2014-05-12 15:18:06	UPDATE people#2	2014-05-12 15:10:06
132	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#11	2014-05-12 15:05:41
131	A1 Admin <ben.henley@deskpro.com>	first_name: A1 last_name: Admin name: A1 Admin	UPDATE people#2	2014-05-12 13:58:39
130		date_last_login: 2014-05-12 12:42:22	UPDATE people#2	2014-05-12 12:42:22
129		date_last_login: 2014-05-12 12:31:17	UPDATE people#2	2014-05-12 12:31:17
128		date_last_login: 2014-05-12 08:25:27	UPDATE people#2	2014-05-12 08:25:27
127	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#10	2014-05-09 16:32:05
126		date_last_login: 2014-05-09 15:31:02	UPDATE people#2	2014-05-09 15:31:02
125		date_last_login: 2014-05-09 11:55:33	UPDATE people#2	2014-05-09 11:55:33
124	A1 Admin <ben.henley@deskpro.com>	title: Sales	UPDATE usergroups#6	2014-05-09 11:18:51
123	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#9	2014-05-09 11:18:40
122	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#8	2014-05-09 11:17:41
121	A1 Admin <ben.henley@deskpro.com>	title: Sales team	UPDATE usergroups#6	2014-05-09 11:16:15
120	A1 Admin <ben.henley@deskpro.com>		DELETE usergroups#7	2014-05-09 10:50:01
119	A1 Admin <ben.henley@deskpro.com>	permissions: ADD: Full for usergroups#7 -- YSP	UPDATE departments#1	2014-05-09 10:24:49
118	A1 Admin <ben.henley@deskpro.com>		CREATE usergroups#7	2014-05-09 10:23:52
117	A1 Admin <ben.henley@deskpro.com>	parent: null	UPDATE departments#2	2014-05-09 10:06:05

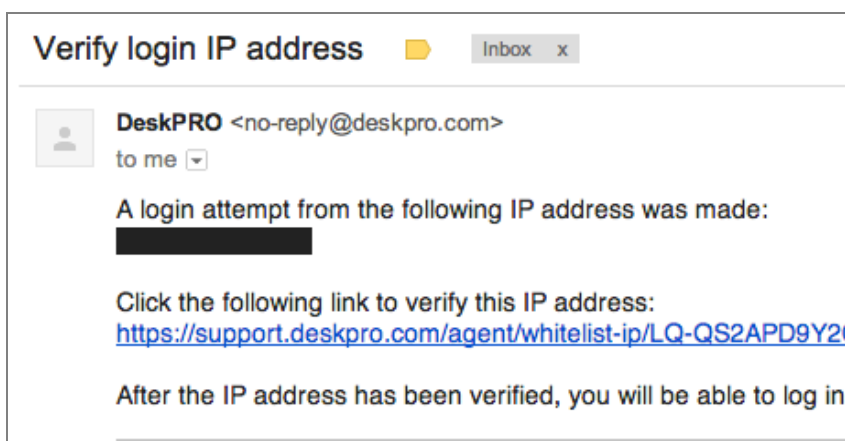
3.10. IP whitelisting

IP whitelisting is a security measure that makes it harder for an unauthorized user to log into an agent account, even if they know the account password.

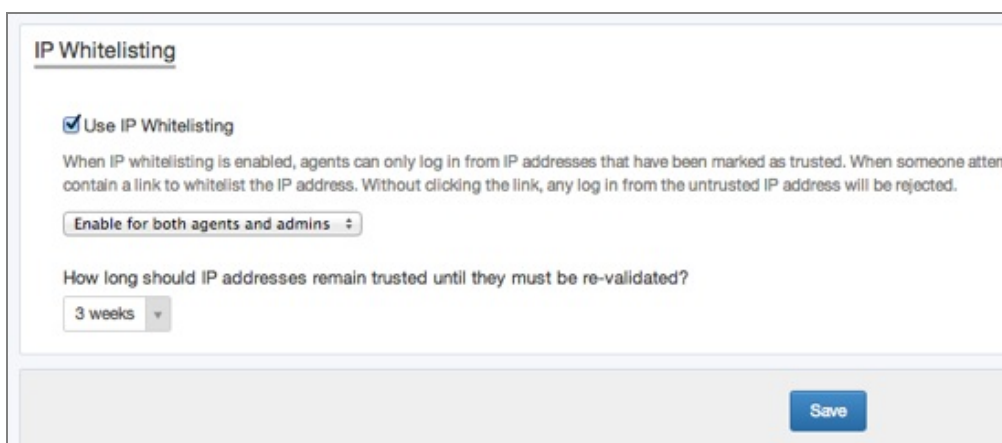
When enabled, agents are only allowed to log in from trusted IP addresses. When an agent attempts to log in from an untrusted address, they will see a message like this:



An email is then sent asking the agent to verify that the IP address is one that's in use by your organization by clicking a link:



You can enable IP whitelisting from **Agents > Settings**



You can choose whether IP whitelisting applies to both agents and admins, or just to admins.

You can also select how long the helpdesk will remember a trusted IP address before the agent has to validate it again; this can be from one day to two years.

Make sure that outgoing email is working before you enable IP whitelisting; otherwise you will be not be able to verify your IP, and will be locked out.

3.11. Agent teams

Agent **teams** are a way to allow group ticket assignment. Assigning a ticket to a team brings it to the attention of all the members of that team: it will show up in the **My Teams' Tickets** filter in the agent interface.

An agent can belong to more than one team.

For ticket permissions purposes, assigning a ticket to a team to which an agent belongs is equivalent to assigning it to the agent directly.

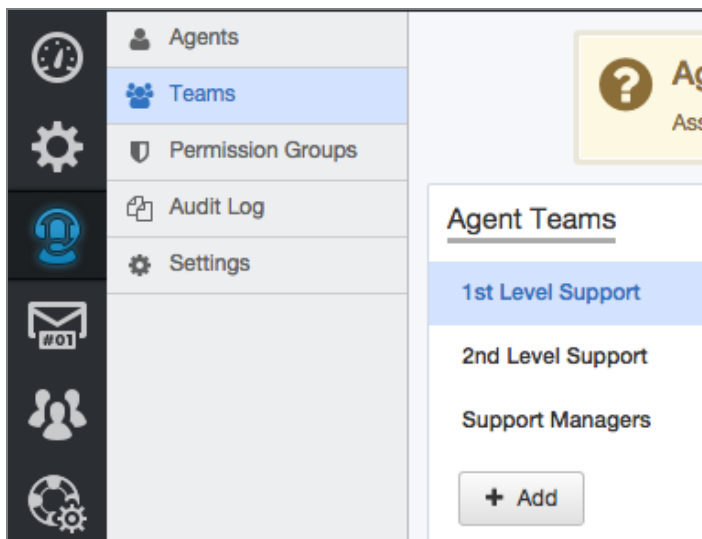
Teams are useful if you want to divide your agents up within a department, or across departments.

For example, if the agents who handle tickets in your Support department are divided into first-tier, second-tier and third-tier support, you could arrange them into three teams.

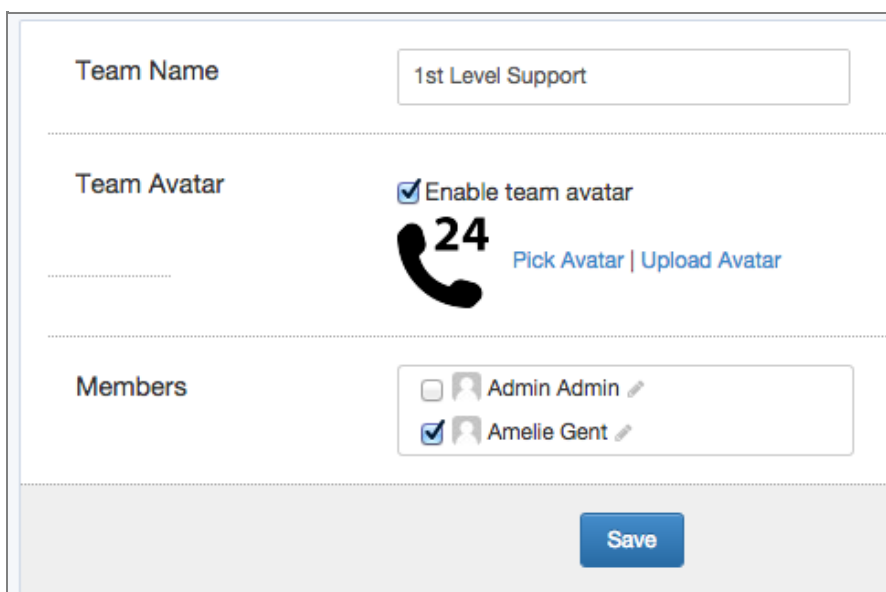
You can also use teams to assign tickets to groups of agents who usually work in different departments. For example, you could have a team for senior agents across different parts of your organization, a team for agents who all handle a certain type of user, etc.

Creating and editing teams

You can create and edit teams under **Agents > Teams**. Click on an existing team to edit its details.



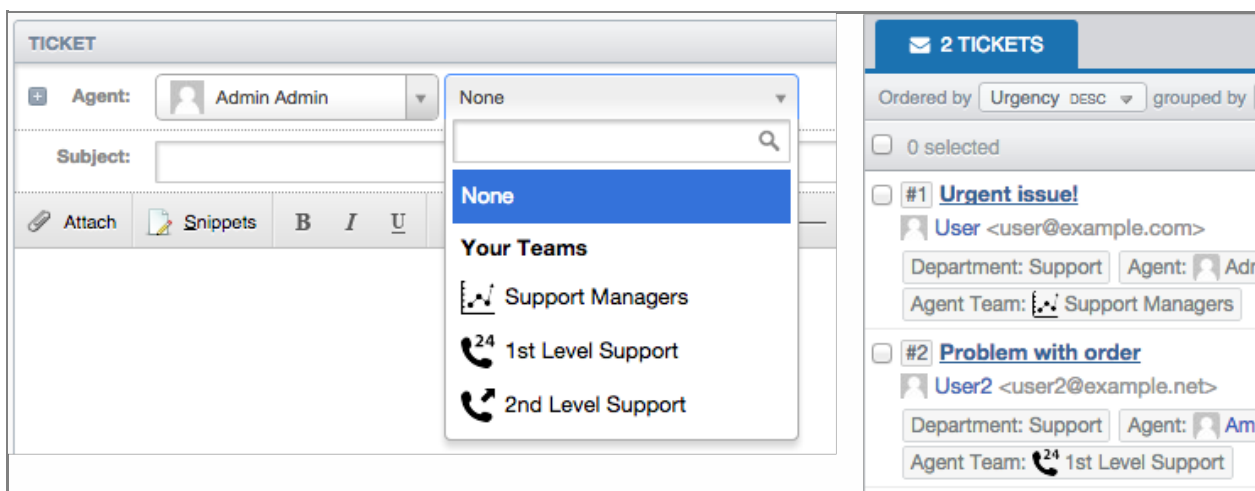
Click **+Add** to create a new team, enter its name, and select which agents are members using the checkboxes.



You can also assign a team **avatar**: an icon which is shown in various parts of the agent interface.

Select **Enable team avatar**, then click **Pick Avatar** to select an icon for the department from a predefined set, or click **Upload Avatar** to upload your own.

Agents see the team avatar when a team is shown in the agent interface:



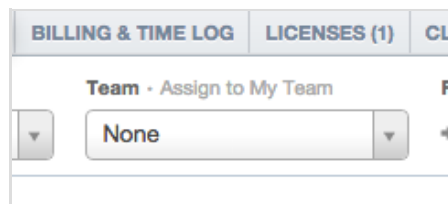
Setting primary team of an agent

If an agent has more than one team, you can select a **primary team** which you can set on the agent's profile in **Agents**.

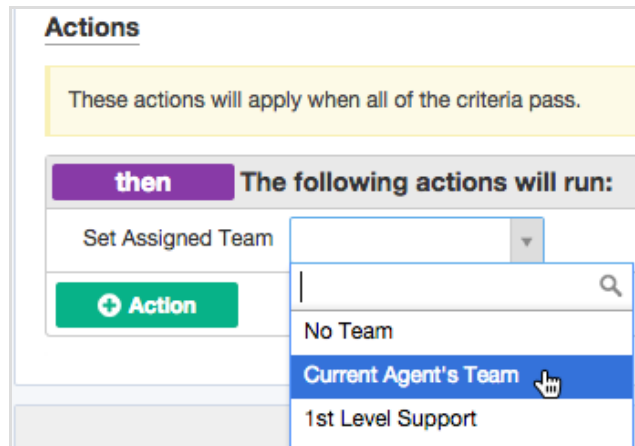


The primary team has two effects:

- 1 It is the team used when the agent selects **Assign To My Team** from the individual ticket view in the agent interface.



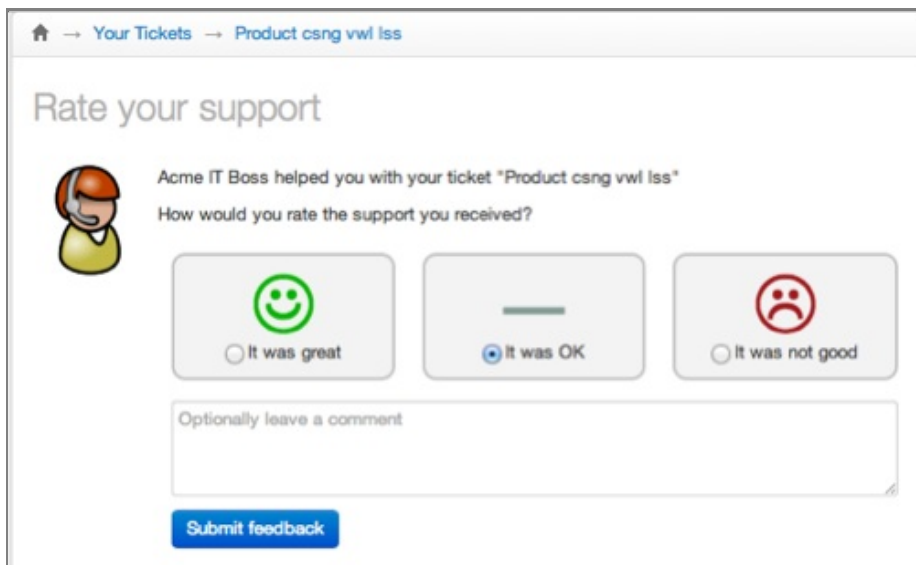
- 2 Some [automated actions](#) have an option to assign a ticket to the 'current agent's team'; this is the primary team.



3.12. Satisfaction survey

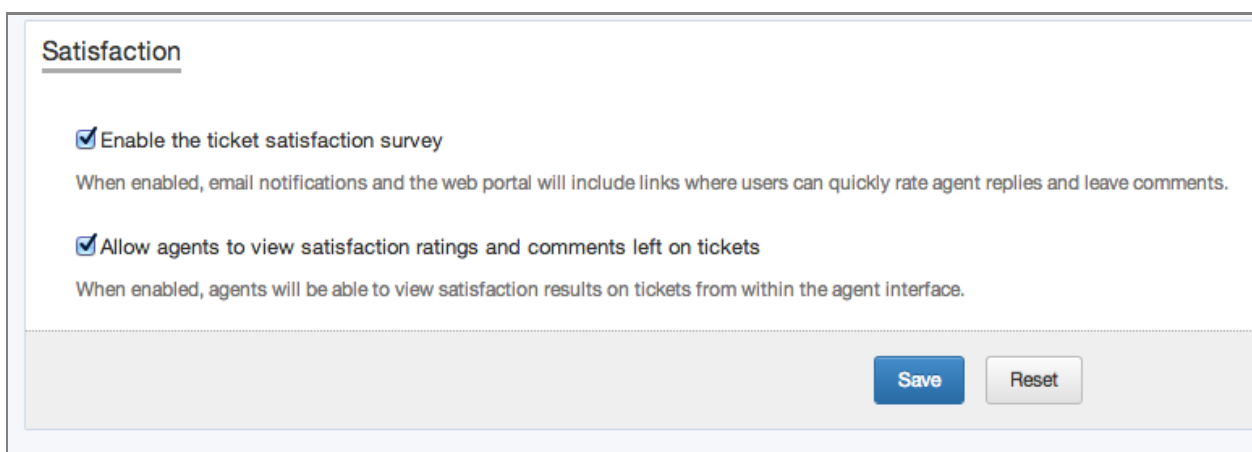
To help measure agent performance, you can enable a **satisfaction survey**.

This offers users the option to submit a brief rating and comment when agents



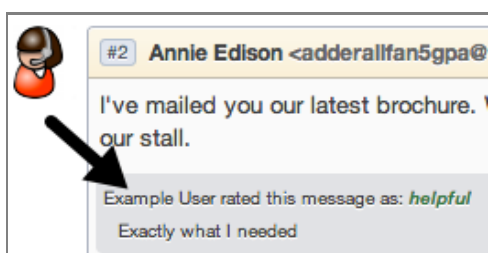
The screenshot shows a web interface for a satisfaction survey. At the top, there's a breadcrumb trail: Home → Your Tickets → Product csng vwl lss. The main heading is "Rate your support". Below it, a cartoon agent icon is shown next to the text: "Acme IT Boss helped you with your ticket 'Product csng vwl lss'". The question is "How would you rate the support you received?". There are three radio button options: "It was great" (with a green smiley face), "It was OK" (with a neutral face), and "It was not good" (with a red frowny face). Below these is a text input field labeled "Optionally leave a comment". At the bottom is a blue "Submit feedback" button.

Go to **Tickets > Satisfaction** to enable the survey.



The screenshot shows the "Satisfaction" settings page. It has a title "Satisfaction" with a link icon. There are two checkboxes, both of which are checked: "Enable the ticket satisfaction survey" and "Allow agents to view satisfaction ratings and comments left on tickets". Below each checkbox is a descriptive text. At the bottom right are "Save" and "Reset" buttons.

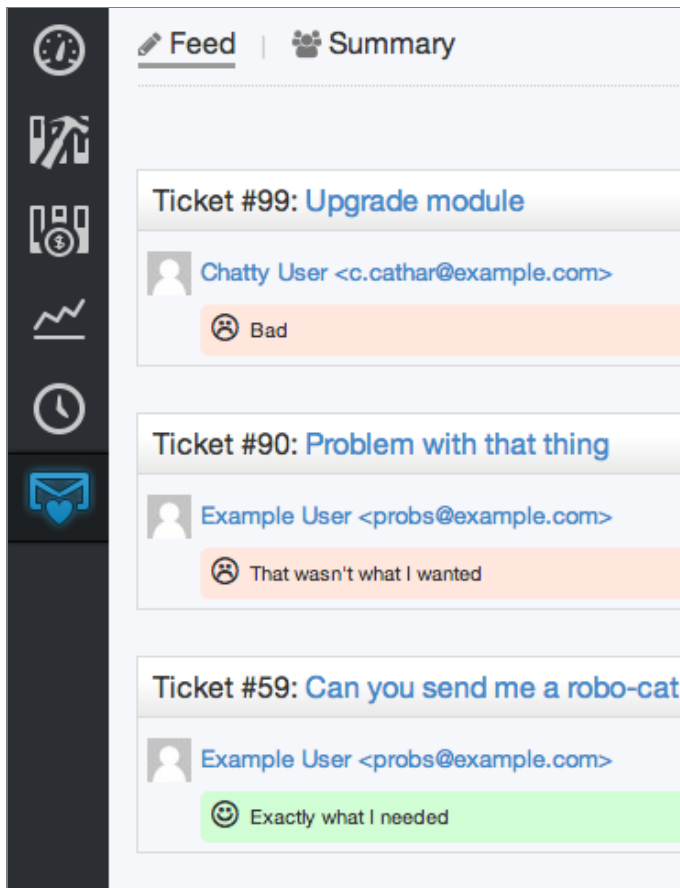
You can choose whether agents can view satisfaction ratings and comments on their own tickets:



The screenshot shows a ticket conversation. On the left is an agent icon. The ticket number is "#2" and the customer name is "Annie Edison <adderallfan5gpa@...>". The message says: "I've mailed you our latest brochure. V our stall." Below the message, there's a grey box showing a rating: "Example User rated this message as: **helpful**" and a comment: "Exactly what I needed". An arrow points from the agent icon to the rating box.

You can use the satisfaction rating in the criteria for [triggers](#), SLAS, escalations and other automation - for example, you might want to automatically assign tickets with a negative rating to a quality assurance team.

You can access satisfaction data through the **Reports** interface:



The **Feed** view shows you details of recent ratings, while the **Summary** view shows you the total of each agent's positive, neutral and negative ratings for the current month.

4. Departments

Departments are an important concept within DeskPRO. They are a way to divide up the helpdesk and control access.

Departments are often used to reflect different parts of an organization: for example, you might have "Support", "Delivery" and "Sales" departments, or "US", "France" and "UK" departments.

There are **ticket departments** and **chat departments**. The two types of department are not linked; you configure departments separately for tickets and chats.

The focus of this chapter is on ticket departments, since they are more frequently used and more powerful than chat departments. These are some important properties of ticket departments:

- A ticket must always have a department.
- Each ticket email account is linked to a department by an [email account trigger](#). For example, if your `help@` account is linked to your Support department, tickets created from emails to `help@` will be assigned to that department. Note that one department can be linked to several email accounts e.g. your `info@` account could also send tickets to Support.
- Each department is linked to an email account by a [department trigger](#). This determines which email account is used for messages relating to tickets that are submitted through the portal. For example, if your Recruitment department is linked to the `jobs@yourcompany.com` account, a Recruitment department ticket created from the web portal will have related emails sent from the `jobs@` account by default.
- Access to departments can be limited by agent and user permissions, so if you want to let people access certain tickets but not others, you can implement this with departments.
- The form to create a ticket through the portal or agent interface (including which fields are required) can be different depending on the department.
- You can give each department a user-facing name; e.g. a department that your agents see as "Terms of Service Violations" could be shown to users as "Moderation".

4.1. Creating and editing ticket departments

To manage ticket departments, go to **Tickets > Departments**. At left, you'll see the list of departments currently in your helpdesk. The defaults for a new helpdesk are "Support" and "Sales", but you are free to edit these.

You can change the order of departments by dragging them with the handles at left. This changes the order in which they're displayed to users and agents.

You can have an unlimited number of departments. Most enterprises will have fewer than ten departments, but you can structure your departments as you feel suitable.

If your helpdesk is very small, you may decide that you do not need to use departments. You need to keep one department in your helpdesk, but if you only have a single department, users are not shown the department field on the ticket submission forms.

Department properties

To make a new department, you need to enter information in the **Properties** tab.

The screenshot shows the 'Properties' tab of the department configuration interface. It includes the following sections:

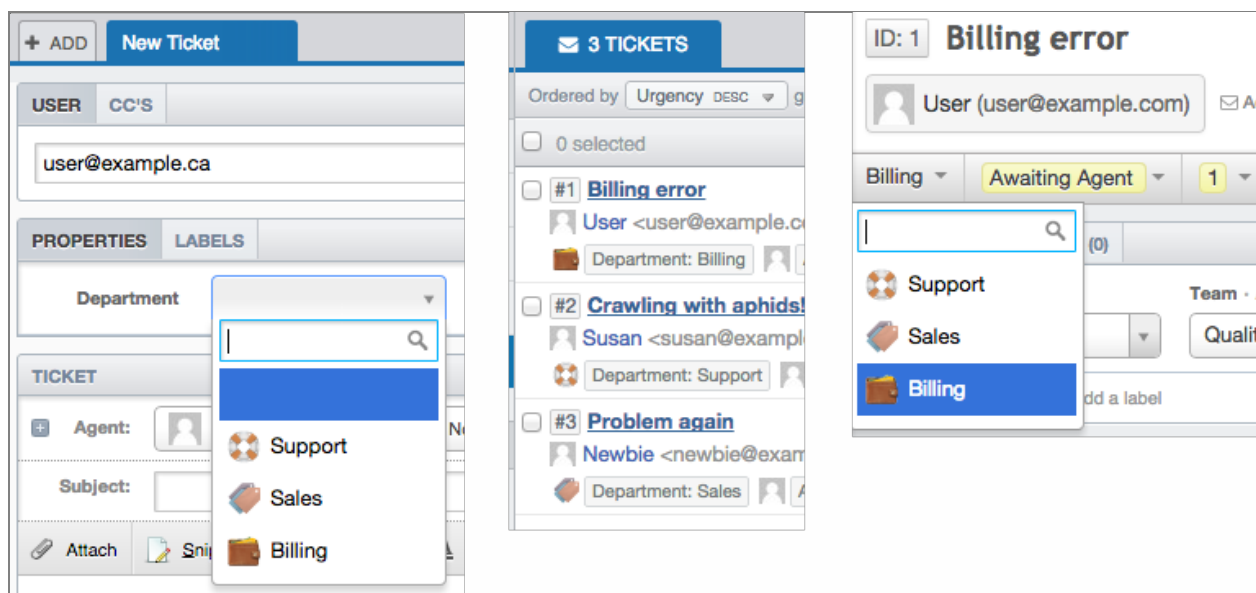
- Title ***: A text input field containing 'Billing'. Below it is a note: 'This is the title as it will appear throughout the agent and user interfaces.' There is also a checkbox labeled 'Show a different title to end-users'.
- Department Avatar**: A section with a checked checkbox 'Enable department avatar'. Below it is a file picker icon and the text 'Pick Avatar | Upload Avatar'.
- Parent**: A dropdown menu set to 'No Parent'. Below it is a note: 'By setting a parent department, this department becomes a child. This is just an organizational feature that helps you create department structures that are easier to use.'
- New Ticket Trigger**: A section titled 'When a new ticket is submitted through the portal in this department:'. It contains two checkboxes: 'Set Email Account' (checked) with a dropdown showing 'support@site19.deskprodemo.com', and 'Send User Email' (unchecked). There is a green 'Action' button.
- Ticket Changed Trigger**: A section titled 'When a ticket department is changed to this department:'. It contains one checkbox: 'Set Email Account' (unchecked). There is a green 'Action' button.

At the bottom of the form are three buttons: 'Save' (blue), 'Reset' (grey), and 'Delete' (red).

Edit the **Title** field to change the department's name. You can select **Show a different title to end-users** if you want to enter an alternative department name to be displayed to users.

If you have multiple languages enabled, you will need to enter a translation for department titles for each language: go to **Settings > Languages**, click **Edit Phrases** and enter translations for **Ticket Departments**.

You can set a **Department Avatar** - an icon for the department that will be shown in various parts of the agent interface:



Select **Enable department avatar**, then click **Pick Avatar** to select an icon for the department from a predefined set, or click **Upload Avatar** to upload your own.

Built-in department triggers

The **New Ticket Trigger** defines an automated action that runs when a ticket in this department is submitted through the portal.

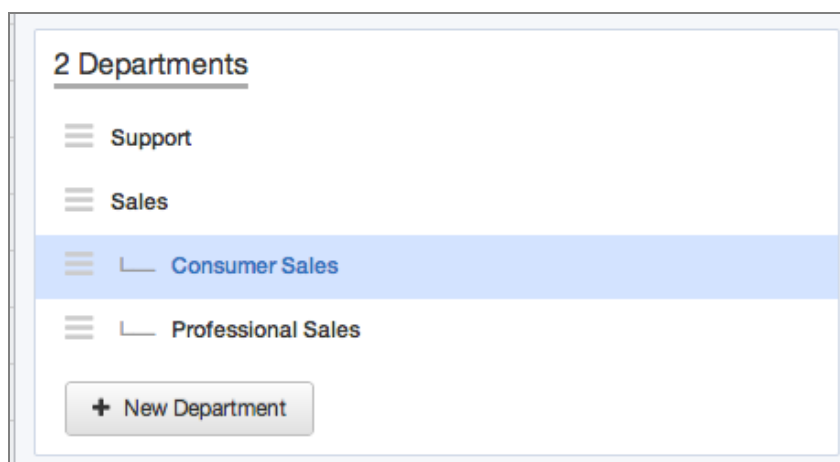
The **Ticket Changed Trigger** defines an automated action that runs when a ticket is assigned to the department.

Besides these simple, built-in department triggers, there are more built-in triggers. You can also create your own triggers, as well as other kinds of automation. See the section on [Ticket Triggers](#) and the chapter on [Automating the Helpdesk](#) for more details.

Nested departments

You can group similar departments by making them sub-departments of a parent department. This can help organize your departments if you have a lot of them.

To create a sub-department, create a new department and then select its **Parent**. For example, you could create a **Consumer Sales** department, then make its parent **Sales**.



Note that if a department has sub-departments, you can only assign tickets to the sub-departments. As a result, there is no point having only a single sub-department within a department.

If you make a department with existing tickets into a parent, its tickets are moved to the sub-department.

You can only have sub-departments one layer deep: for example, you can have **Sales > Consumer Sales** and **Sales > Professional Sales** sub-departments, but not **Sales > Consumer Sales > Budget Range**

4.2. Using departments

Departments usually reflect different divisions of your enterprise with different requirements. Here are some common scenarios where you would need to use departments:

- You want to have separate contact forms on your website for support requests and media requests, with different custom fields. You would create Support and Media departments, create custom [ticket form layouts](#) for each department, then [embed the forms in your website](#).
- You want to configure your helpdesk so that your manufacturing staff only see manufacturing tickets, and your retail staff only see retail tickets, while your research staff can see everything. You would create Manufacturing, Retail and Research departments, then control access with [agent permissions](#).
- You want anyone to be able to contact your sales department, but for only your subscribers to be allowed to create support tickets. You would use the [user permissions system](#) to remove access to the Support department from all users, then create a usergroup with full access to Support and make sure subscribers are added to it.

4.3. Department permissions

Click the **Permissions** tab for a ticket or chat department to edit the department permissions.

Agent department permissions

You can limit agent access to tickets based on ticket department.

This is useful if you want to ensure that different groups of agents work on different types of tickets.

In the **Agent Permissions** section of a department's **Permissions** tab, you can select which [agent permission groups](#) and individual agents have access to that department.

Agent / Agent Group	Assign	Full Access
 All Non-Destructive Permissions		
 All Permissions		
 Interns		
 Sales		
 Tech support		
A-1 IT Boss		
A-1 Sales Exec		

There are two levels of agent permission for departments:

- **Assign** means that the agent can assign tickets to the department.
- **Full access** means that the agent can view and work with tickets in the department as normal.

An agent who only has **assign** permission can assign a ticket to a department, but will not be able to view it afterwards.

Note that department permissions are overridden by ticket assignment and following. If a ticket is assigned to an agent or one of the agent's teams, or the agent is added as a follower, the agent can view and work with it regardless of department.

You can also manage these settings from **Agents > Agent Permission Groups**

In the screenshot above, you will see that many of the permissions are on and *locked* - you can't remove the permission. This happens for one of two reasons:

- Granting **full access** includes **assign**, so if the agent/group has been granted **full access**, **assign** is locked on.
- If an agent is a member of an agent permission group which already grants them a permission, it will be locked on.

Chat department permissions for agents work like those for ticket departments, except they control which chats agents can accept. Unlike the two permission levels of **assign/full access** for ticket departments, chat department access is either on or off.

User department permissions

In the **User Group Permissions** section of a department's **Permissions** tab, you can select which [usergroups](#) have permission to create tickets in that department. This enables you to limit access to certain departments to particular usergroups: for example, you might want to limit Support department access to paying customers only, but allow Sales department access to everyone.

User Group Permissions

User Group permissions control which of your end-users can select a department when submitting or editing their tickets from the web portal.

Note that it is still possible for agents to change the department to one that a user would not normally have access to. Triggers can also be used to assign any department regardless of user permissions. User group permissions are therefore very much about what the user can select themselves.

User Group	Use
 Everyone	☐
 Registered	☒
 VIP	☒

Note that this permission setting only affects *creating* tickets. If a ticket is assigned to the user, they will be able to see it on the portal, get replies etc, regardless of department permissions; and similarly, if the user is CC'd into the ticket, they will receive ticket messages.

If the user only has permission to use one department, the department field is not shown at all on the contact form.

For example, suppose you have two departments, "Support" and "Sales".

You set the permissions for the groups like this:

User Group		Use
 Everyone	Sales	☒
 Registered		☒

User Group		Use
 Everyone	Support	☐
 Registered		☒

On your portal contact form, a user who is logged in will see this:

Contact Us

Please complete this form and one of our agents will reply to you by

Your Email Address
[Change email address](#)
propellerhead@example.com

Your Name *
Sue Advanced User

Department *
Support
Sales

Subject *
Subject

whereas a user who isn't logged in will see this:

Contact Us

Please complete this form and one of our agents will

Your Email Address * | This is where our agent replies

If you are a registered user, please [Log In](#).

Your Name *

Priority
Priority

Subject *
Subject

What is your question? *
Message

The user registration settings under **CRM > Registration** can override department permissions; for example, if the **Everyone** usergroup has permission to create tickets in a department, but registration is required, guest users who aren't logged in will not see the contact form at all.

Chat department permissions for users work as for ticket departments, except that they control which departments users can select when they initiate a chat.

are here to help

Open this chat in a new window

Start a Chat with an Agent

Your Name
Sue Advanced User

Your Email Address
propellerhead@example.com ([Log Out](#))

Choose department

- ✓ Troubleshooting
- Sales
- Billing**
- Setting up products

4.4. Ticket form layouts

You can configure the fields which appear in the forms used to submit, and edit forms separately for each department.

These forms are:

- For users, the Contact Us form in the web portal, and any [embedded contact forms](#).

The left screenshot shows the 'Support Center' 'Contact Us' form. It has a header with links to Knowledgebase, Downloads, Feedback, Contact Us, and News. Below the header, there's a 'New Ticket' link and a 'Contact Us' section. The form includes fields for 'Your Name *', 'Your Email Address *' (with a note that replies will be sent here), a 'Department *' dropdown, 'Subject *', and 'What is your question? *'. There's also an 'Attachments' section with a file upload button. The right screenshot shows a 'Welcome to A1 Corp' form. It has a header with a welcome message and a 'Your Email Address *' field. Below that is a 'Subject *' field, a 'What is your question? *' field, a 'Priority' dropdown (set to 'Routine'), and a 'Submit' button.

- For agents, the ticket form in the agent interface.

The screenshot shows the 'New Ticket' form in the agent interface. At the top, there's a header with user information (Hello, A1 Admin), navigation links (Preferences, Help, Log Out), and status (Logged in to chat, 1 Agent, 0 Users). Below the header, there are buttons for '+ Create' and 'Recent'. The form itself has tabs for 'USER' and 'CC'S'. Under 'USER', there's a dropdown to 'Choose a person to create this ticket for' and a 'Create a new person' button. Below that are 'PROPERTIES' and 'LABELS' tabs. Under 'PROPERTIES', there's a 'Department' dropdown. Under 'TICKET', there's an 'Agent' dropdown (set to 'A1 Admin') and a 'Subject' field. Below the subject field is a rich text editor with various formatting options (Attach, Snippets, Bold, Italic, Underline, Text color, Background color, Bulleted list, Numbered list, Indent, Outdent, Link, Unlink, etc.). At the bottom, there are buttons for 'Send Reply as Awaiting User', 'Email User', and 'Open Tab'.

You can have different layouts depending on the ticket department, and for agents and users within each department.

The default value of a field, and the data validation rules for what values to accept, are properties of the field, not the form, but you can have different validation rules for users and agents.

To edit a form layout, go to **Tickets > Departments** and click on the **Layout** tab.

By default, all your forms use the same default layout; editing the default will affect the forms for all all your departments. The buttons at the top of the **Layout** tab enable you to switch between editing the default, or to edit the form just for the current department.

The screenshot shows the 'Layout' tab in DeskPRO. At the top, there are two buttons: 'Use Default Form Layout' and 'Use Custom Form Layout for "Support"'. A callout box points to these buttons with the text: 'Select whether to edit all departments or just this one'. Below these buttons is a blue banner that says 'All of your departments are using the default layout. Click here to make changes to this layout, and all departments too.' A callout box points to this banner with the text: 'Select whether to edit the user or agent form using tabs'. Below the banner are two tabs: 'User Form' and 'Agent Form'. A callout box points to these tabs with the text: 'Select whether to edit the user or agent form using tabs'. The main area shows a list of fields currently in the form: Department, Subject, Message, User Email, Priority, Workflow, Category, CC, Product, CAPTCHA, Attachments, and User confirms T&Cs. A callout box points to this list with the text: 'Fields currently in the form'. To the right of the list are two panels: 'Ticket Fields · Manage Fields' and 'User Fields · Manage Fields'. The 'Ticket Fields' panel contains a button 'User confirms T&Cs'. The 'User Fields' panel contains buttons 'Name', 'Timezone', 'Language', and 'Level'. A callout box points to these panels with the text: 'Unused fields'. Below the 'Product' field, there is a callout box 'Remove field from form' pointing to a gear icon. Below the 'Attachments' field, there is a callout box 'Edit field details' pointing to a gear icon. Below the 'User confirms T&Cs' field, there is a callout box 'Custom field' pointing to a gear icon.

On the left is the list of fields currently in the form; the blue tabs at right are fields which are available for adding to the form (including custom fields for both tickets and users).

Drag and drop the fields you want onto the form. You can also set the order by dragging the fields on the form up and down.

Note that you can add both **ticket** fields and **user** fields. This enables you to have the user submit their identifying data, such as their account number, from the ticket form. See the section [Adding custom user fields to ticket forms](#) for details.

For all but the compulsory fields, you can click the small gear icon to set the details of how the field should behave:

Options

☒ Show this field during ticket creation

☒ Show the field to agents when viewing the ticket

☒ Only show when there is a value

☒ Allow agents to edit the field

☐ Only show this field based on custom criteria

Done

The options for a field are:

- Show this field during ticket creation
- Let the user view this field after the ticket is created/Show the field to agents when viewing the ticket
 - Only show when there is a value - if enabled, when a user views an existing ticket, they will not see the field if it's empty
- Let the user edit this field after the ticket is created/Allow agents to edit the field
- Only show this field based on custom criteria

So you can, for example, have users set a priority for a ticket when they initially submit it, which they can't then change; or have a field that users can see but only agents can edit.

☒ Only show this field based on custom criteria

Show when all of the following criteria match:

- Department
- Product
- Category
- Priority
- Workflow**

Only show this field based on custom criteria lets you show a field selectively if it matches **all** or **any** of your specified:

- Department
- Product
- Category
- Priority
- Workflow

For example, suppose your company's "Robo-cat" hardware products have a unique serial number which is useful for debugging, but your other products don't. You want to have the user form ask for a serial number for Robo-cat products only, and only for tickets to the Support department (since users submitting Sales tickets don't have the product yet).

The screenshot shows a configuration panel for a field. At the top, there is a green toggle switch labeled 'Only show this field based on custom criteria'. Below it, the text 'Show when' is followed by a dropdown menu set to 'all' and the text 'of the following criteria match:'. There are two criteria listed: 'Product is' followed by a box containing 'Robo-cat Mk 1' and 'Robo-cat Mk 2', and 'Department is' followed by a box containing 'Support'. Each criterion has a small 'x' icon to its right. At the bottom of the criteria list is a green button with a plus icon and the text 'Criteria'.

You can also set **validation** rules about the field values that the user/agent is allowed/required to enter. You can do this using the individual field settings in **Tickets > Fields**.

The screenshot shows a 'Validation' section with two checkboxes. The first checkbox is checked and labeled 'Require users to select a value for this field'. The second checkbox is unchecked and labeled 'Require agents to select a value for this field'.

The validation rules you set only apply if the field is being displayed.

See the section on [Field data validation](#) for more details.

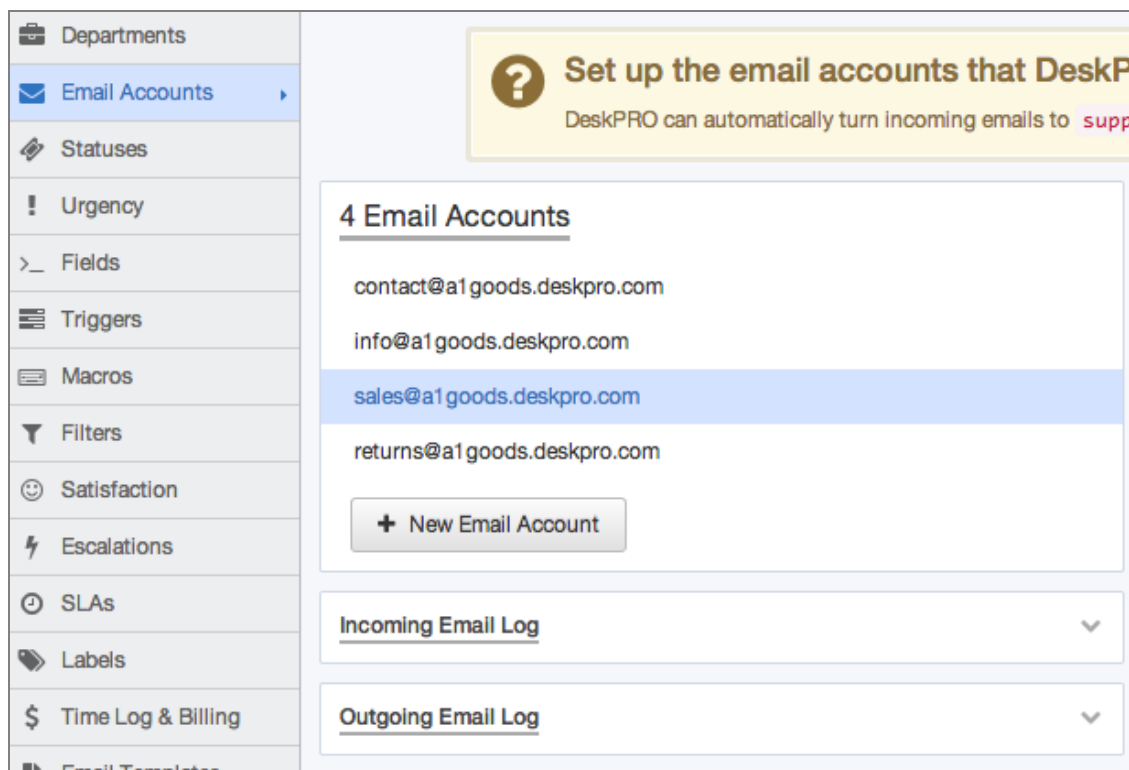
5. Email Accounts

DeskPRO can take incoming user messages to a ticket email account, and convert them to tickets.

Replies from agents (either through the agent interface, or email replies to a notification) are automatically converted into emails to the user.

5.1. Adding ticket email accounts

Under **Tickets > Email Accounts**, you can configure the email accounts that will be used to accept tickets and send out .



Any new message to one of these accounts is automatically converted to a ticket within your helpdesk. If the message is part of an ongoing conversation (i.e. it is in response to a message about an existing ticket), it will be shown as a reply in the agent interface.

A ticket email account can't also be used for a DeskPRO user/agent account. This is necessary to prevent email loops, where DeskPRO could end up replying to its own emails endlessly. It's best to add agents using individual email accounts (e.g. `john.smith@yourcompany.com`), rather than with user-facing addresses like `support@` .

Note that email account configuration differs between the Cloud and Download versions of DeskPRO.

Default account for non-ticket email

Some of the emails that DeskPRO sends are non-ticket related email such as password reset links, welcome emails, or login alerts. You can specify which account is used to send these in **Admin > Settings** in the **Default Email Account** section.

If you don't want a ticket to be created when a user replies to a password reset message, you may want to create a separate "no-reply" email account and set that as the default email account.

DeskPRO Cloud email configuration

DeskPRO Cloud offers cloud-hosted email accounts with addresses ending `@yourcompany.deskpro.com` . By default, you will have a `contact@` account linked to the Support department; you can edit this or add more accounts.

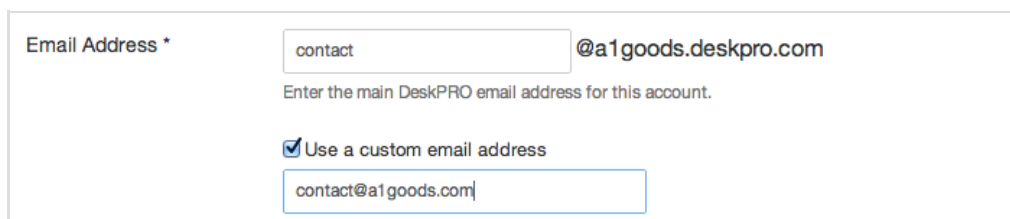
The screenshot shows the '1 Email Account' configuration page in DeskPRO. On the left, there's a sidebar with '1 Email Account' (selected), 'Incoming Email Log', and 'Outgoing Email Log'. The main area is for configuring the email account 'contact@a1goods.deskpro.com'. It includes a 'New Email Account' button, a 'Email Address *' field with the address '@a1goods.deskpro.com', and checkboxes for 'Use a custom email address' and 'Add custom email aliases'. Below this is the 'New Ticket Trigger' section with a dropdown for 'When a user submits a new ticket by emailing this address:', a checked checkbox for 'Set Department' with a dropdown set to 'Support', and an unchecked checkbox for 'Send User Email'. An 'Action' button is at the bottom of this section. At the very bottom of the main area are 'Save', 'Reset', and 'Delete' buttons.

You could ask users to submit tickets to these `deskpro.com` accounts, but you would probably prefer to use addresses with your own domain, like `help@yourcompany.com`.

You can configure DeskPRO to use an address like this (a custom email address). Users will only see the custom email address you set up: e.g. they would send email to `help@yourcompany.com`, and receive email from `help@yourcompany.com`, without ever seeing the `@yourcompany.deskpro.com` address.

To set up a custom email address:

- 1 In **Tickets > Email Accounts**, select the cloud account you want to use and check **Use a custom email address**. Enter the custom address you want to use.

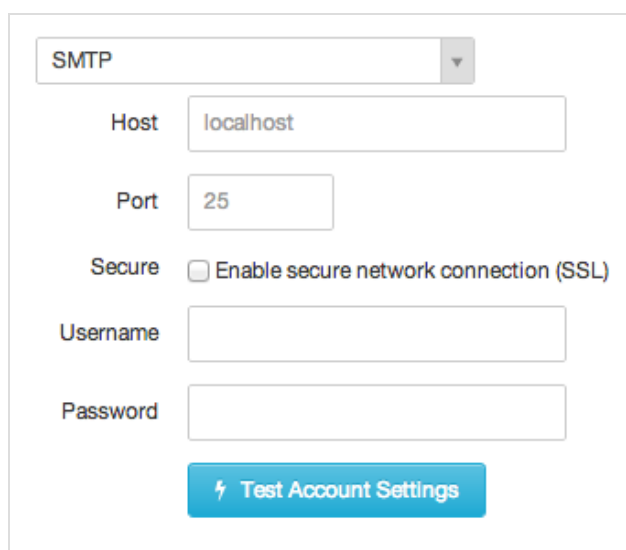


- 2 Configure your email system to forward all mail sent to the custom address to the DeskPRO Cloud account: e.g. forward from `contact@yourcompany.com` to `contact@yourcompany.deskpro.com`. Consult the documentation for the email system you use (e.g. Exchange, Gmail) to find out how to do this.

- 3 Set up how outgoing email will be sent; i.e. mail *from* the custom address. There are three options:

- **Send mail through DeskPRO:** sends email from our cloud servers; you will need to make some DNS changes to ensure email isn't marked as spam. See the section [Sending email using DeskPRO Cloud email servers](#) for details.
- **SMTP** sends email using your SMTP server.

You will need to provide some SMTP settings:



If you don't know these, you will probably be able to find them by looking at the settings of your email client.

- **Gmail or Google Apps** just enter the account password. If you have 2-step verification enabled on the Google account, will you need to create an [application-specific password](#) and enter that instead.

Sending email using DeskPRO Cloud email servers

If you use a custom email address and you want to send outgoing email from the DeskPRO servers, it's important that you take some additional steps to avoid your helpdesk email messages being wrongly flagged as spam.

Internet providers use anti-spam systems called SPF and DKIM, which compare the server that sends a message to the From: address. You need to modify your DNS to indicate that DeskPRO's email delivery provider (SendGrid) is authorized to send email on your behalf.

Make the following changes to the DNS record of your domain, replacing `yourcompany.com` with your actual domain:

- 1 Create a CNAME for `dpmail.yourcompany.com` to `sendgrid.net`, for example:
`dpmail.yourcompany.com CNAME sendgrid.net.`
- 2 Add a DNS [TXT record](#) for SPF, or modify your existing SPF TXT record, to whitelist SendGrid as a server allowed to send mail from your domain. Consult this [SPF Record Syntax guide](#) for full details. Here are some examples:
 - Basic example: `v=spf1 a mx include:sendgrid.net ~all` - doesn't whitelist a particular server, just means all servers are considered neutral (neither spam nor not spam)
 - If you use Google Apps for email:
`v=spf1 a mx include:_spf.google.com include:sendgrid.net ~all`
 - To whitelist an SMTP server:
`v=spf1 a mx include:smtp.yourcompany.com include:sendgrid.net ~all`
- 3 Create the following CNAME records for DKIM:
 - `smtpapi._domainkey.example.com CNAME dkim.sendgrid.net.`
 - `smtpapi._domainkey.dpmail.example.com CNAME dkim.sendgrid.net.`

If you don't know how to change your DNS configuration or don't have the right access to do so, consider using a different Outgoing Email setting.

If your company doesn't have an SMTP server, an easy alternative is to set up a Gmail or Google Apps email account that is only for helpdesk messages.

DeskPRO Download email configuration

Here's how to add a ticket email account in DeskPRO Download:

- 1 Go to **Tickets > Email Accounts** and click **New Email Account**.

Email Address *
Enter the main email address for this account.

☐ This account has additional email addresses associated with it (aliases)

Enabled ☒ Enable this email account
When enabled, new emails will be processed from this account and outgoing emails may be sent out from this address.

Type Ticket Email Account ▼
Ticket accounts are email accounts that DeskPRO will download emails from and process into tickets and replies.

Account Details * POP3 ▼

Host

Port

Secure ☐ Enable secure network connection

Username

Password

[↗ Test Account Settings](#)

Outgoing Email * Send mail using PHP mail() ▼

PHP Mail Send mail through the server `mail()` function.

[↗ Test Account Settings](#)

New Ticket Trigger

When a user submits a new ticket by emailing this address:

☒ Set Department Support ▼

☐ Send User Email

[➕ Action](#)

[Save](#) [Reset](#)

- 2 **Email Address** is simply the user-facing email address you want to use for support e.g. `support@yourcompany.com`
- 3 Leave **Type** as **Ticket Email Account**
- 4 Select the type of email account you are using (IMAP, POP3, Gmail/Google Apps, or MS Exchange), and enter the server details requested.

If there are details you don't know, check the settings/documentation of your email program or mail provider. Note that **Host** means the address of the IMAP or POP3 (incoming mail) server.

If you are using two-factor authentication with Gmail/Google Apps, you will need to create an application specific password and use that instead of your normal password.
- 5 Specify how DeskPRO sends outgoing email.

- The default **Send mail using PHP mail()** option sends emails directly from your DeskPRO installation with a PHP function - this will usually work without extra configuration, but you may need to configure PHP further.
- For more reliability, we recommend sending email via a separate SMTP server. You must ensure that your server is configured to relay mail from your DeskPRO installation: consult the documentation for the SMTP server on how to do this.
- You can also use a Gmail or Google Apps account to send mail.

6

Click **Test Account Settings** to try sending an email to yourself and confirm that DeskPRO can send email. If you get a connection error, check the details you entered. You may also need to change your network configuration to open up the port mentioned in the error.

5.2. Email account New Ticket triggers

Every email account you create has two simple **New Ticket triggers**. These specify what happens automatically when a new ticket is created from an email sent to the account.

The **Set Department** trigger sets the department of the newly-created ticket. Selecting a department for this trigger is compulsory, since tickets must always have a department. See [linking a department to an email account](#) for more details.

The **Send User Email** trigger enables you to send an email to the user who is associated with the ticket. You can use the **New Ticket Auto-Response** template for a simple acknowledgement email.

You could also use a custom template, for example if you wanted to have a different auto-response for each department. See the section [Customizing the email templates](#) for more details.

You can choose whether to **Include CC's**: if selected, this takes any non-agents who were cc'd in to the original email message, and adds them as CCs to the ticket. This means they will receive copies of any emails sent by DeskPRO about this ticket, even though they are not the ticket owner.

The **From Name** setting sets the name the user will see the email as coming from:

- **Helpdesk Name** is the helpdesk name you have set in **Setup > Settings**
- Alternatively you can set a **custom name** that you define.

The **From Email** controls which email address the response email comes from. You can either choose:

- **The account set on the ticket**.g. if the user emailed help@yourcompany.com to create the ticket, the reply is sent from `help@`.
- A **specific account** - for example, you might prefer to have auto-responses sent from `info@` or `noreply@`.

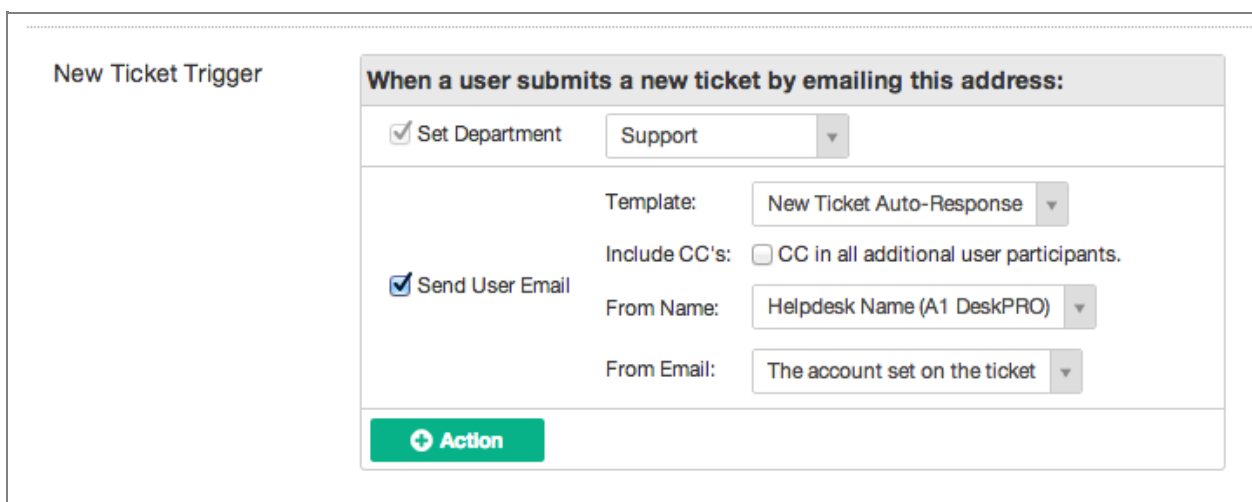
5.3. Linking a department to an email account

The link between a department and an email account is made using built-in [triggers](#).

Linking email accounts to departments

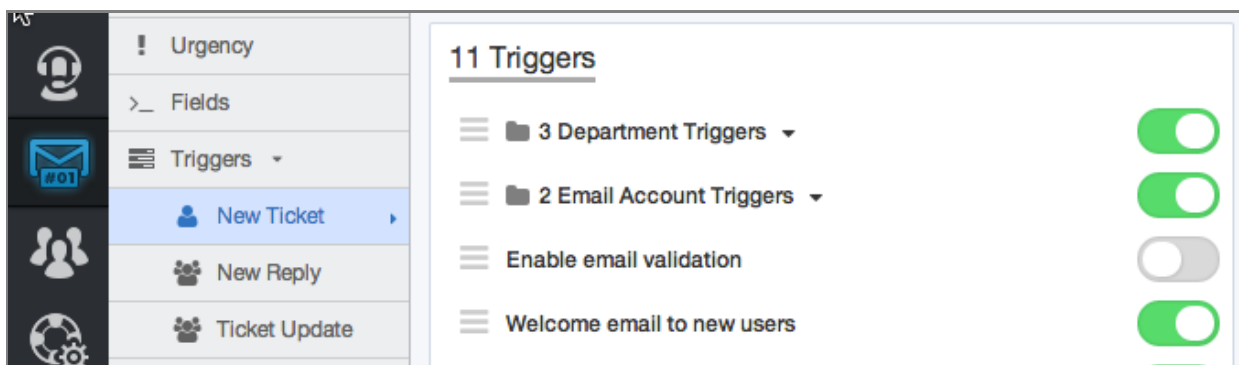
Each email account has a **New Ticket** trigger which assigns tickets submitted to that email account to a specified department. For example, you could have all tickets created by emailing `support@yourcompany.com` assigned to the Support department.

You can view and edit the trigger by going to **Tickets > Email Accounts** and selecting the relevant email account.



The screenshot shows the 'New Ticket Trigger' configuration form. It has a title 'New Ticket Trigger' on the left. The main form area is titled 'When a user submits a new ticket by emailing this address:'. It contains several settings: 'Set Department' is checked with a dropdown set to 'Support'; 'Send User Email' is checked; 'Template' is set to 'New Ticket Auto-Response'; 'Include CC's' is unchecked with the text 'CC in all additional user participants.'; 'From Name' is set to 'Helpdesk Name (A1 DeskPRO)'; and 'From Email' is set to 'The account set on the ticket'. At the bottom is a green 'Action' button with a plus icon.

You can also view and edit these triggers for each email account in **Email Account Triggers** under **Tickets > Triggers > New Ticket**.



Linking non-email tickets to email accounts

The **New Ticket** trigger in a department's properties screen sets the email account to use for tickets not submitted through email - for example, via the portal an embedded form, or the API - in that department. The selected account will be used to send messages related to the ticket. For example, a ticket submitted to the Support department via the web would have its emails sent from `support@yourcompany.com`.

New Ticket Trigger	When a new ticket is submitted through the portal in this department: ☒ Set Email Account helpdeska766@gmail.com Template: New Ticket Auto-Response Include CC's: ☐ CC in all additional user participants. ☒ Send User Email From Name: Helpdesk Name (A1 DeskPRO) From Email: The account set on the ticket + Action
Ticket Changed Trigger	When a ticket department is changed to this department: ☐ Set Email Account + Action

You can also view and edit these triggers for each department in **Department Triggers** under **Tickets > Triggers > New Ticket** and **Tickets > Triggers > Ticket Update**

! Urgency >_ Fields ☰ Triggers ▾ 👤 New Ticket ▶ 👥 New Reply 👥 Ticket Update ☰ Macros 🔍 Filters	11 Triggers ☰ 3 Department Triggers ▲ ☒ Use these triggers if you want to perform actions based on the department a user selects when submitting a ticket from the web interface. Support Sales Delivery
--	--

The **Ticket Changed** trigger sets the email account when an existing ticket is changed to the department.

5.4. Email handling options

Adding agents as followers with cc:

By default, if the user cc's an agent into a ticket email, the agent will be added as a follower of the ticket.

You may not want users to be able to set agents as followers; bear in mind that adding an agent as a follower means they can always see a ticket, overriding other permissions.

You can disable this feature from **Tickets > Settings**:

Email Settings

☒ Process agents that are in the CC line of incoming emails

When enabled, any agent email addresses that are in the CC line of incoming emails will be added as followers on tickets. This means, for example, an end-user could add an agent to their ticket by CC'ing them.

Priority of email headers

DeskPRO looks at the email headers to determine who the user is. Some email server configurations do not send the correct user information in the **From:** header. This can happen if you are automatically forwarding emails or using distribution lists.

Sometimes, the correct user address is retained in the **Reply-To** or **X-Original-From** header, even if **From:** is wrong.

You can set the order that DeskPRO looks at headers in **Admin > Tickets Settings** in the **Email Settings** section. Use the handles at left to drag the headers into the order you want, putting the most reliable header at the top.

Priority of "From Email Address" headers

DeskPRO reads these headers in all email messages to determine who the end-user is. Drag and drop the headers to prioritize them. The first valid header in an email will be used.

≡ From
≡ Reply-To
≡ X-Original-From

Ticket forwarding: inbound

Agents may sometimes receive an email to their individual address that would be better handled through your helpdesk. For example, the agent the user has emailed may not be the best person to address the issue, may not be available to respond, or just doesn't want support correspondence cluttering up their inbox.

Inbound ticket forwarding lets agents forward an email to one of the helpdesk's ticket email accounts, and have DeskPRO process it as a ticket from the original user, not the agent.

For example, `user@example.com` emails `carol.smith@yourcompany.com`, who forwards the message to `help@yourcompany.com`. If inbound ticket forwarding is enabled, your helpdesk will create a ticket with the user `user@domain.com`.

Agents can add a reply to the top of the message they forward, and DeskPRO will deliver it to the user.

To enable or disable this function, go to **Tickets > Forward Settings** and use the **Enable smart-parsing of emails forwarded into the helpdesk by agents** checkbox. It is enabled by default.

This is the process DeskPRO uses to determine if an email has been forwarded:

- 1 **Subject match:** The helpdesk looks for a string like 'FW:', 'FWD:' etc. at the beginning of the subject line of an email from an agent's email account. It takes this to mean that the agent is trying to create a ticket for a user. (Emails forwarded by non-agents will be treated as normal tickets).
- 2 **Sender match:** The helpdesk will then try and find the details for the user who originally sent the email. Different email programs will include this information in different ways; this is a typical example:

-----Original Message-----

From: Victoria Johnson [mailto:user@domain.com]

Sent: 05 November 2012 14:38

To: agent_email@yourcompany.com

Subject: Support Request

In this case, the helpdesk would match the user as Victoria Johnson, with the user email being `user@domain.com`.

In some cases, DeskPRO will not be able to find the user's details. This is typically because the format from your mail client is unknown to DeskPRO, you are using an unsupported language, or your email system has modified the headers in a way that prevents DeskPRO determining the original sender. In this case an error message will be sent to back to the agent, including the original email.
- 3 **Reply match:** DeskPRO will now look to see if the agent added a reply. If a reply is found, this response will be emailed to the end-user.
- 4 **Processing:** DeskPRO will now process the relevant triggers and create the ticket.

You can configure how subject matching works with the **Use a custom Subject regular expression for detecting forwarded emails** option in **Tickets > Forward Settings**

If you enable the option, you can modify the regular expression (search pattern) used to detect forwarded emails. This is useful if your email software uses a non-standard prefix and DeskPRO is not processing forwards correctly.

☒ **Use a custom Subject regular expression for detecting forwarded emails**

`/^(FW|FWD|VL|WG|FS|VB|RV|VS):/i`

Enter a regular expression (PCRE) that will be run against email subjects to determine if it is a forwarded message. E
When an email subject from an agent matches this expression, DeskPRO will process it as a forward.

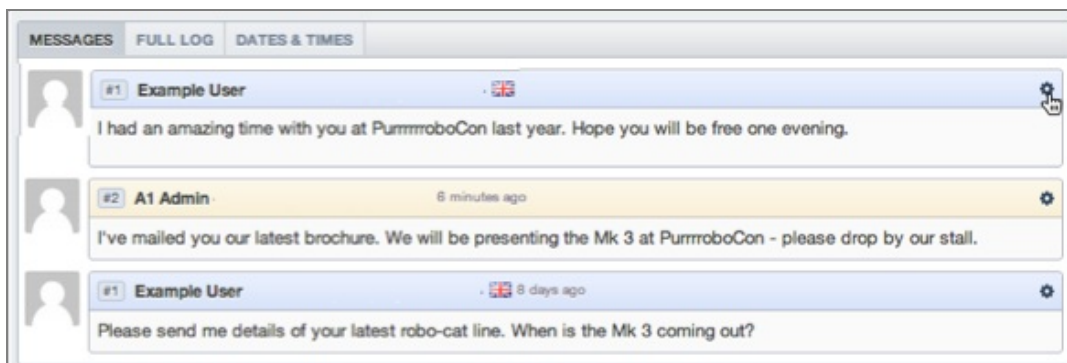
The default pattern will match subject lines that start with any of the letter combinations in brackets, followed by a colon, e.g.:

FW:
fw:
Fwd:
FS:

The regular expression syntax used is [PHP Perl Compatible Regular Expressions](#).

Ticket forwarding: outbound

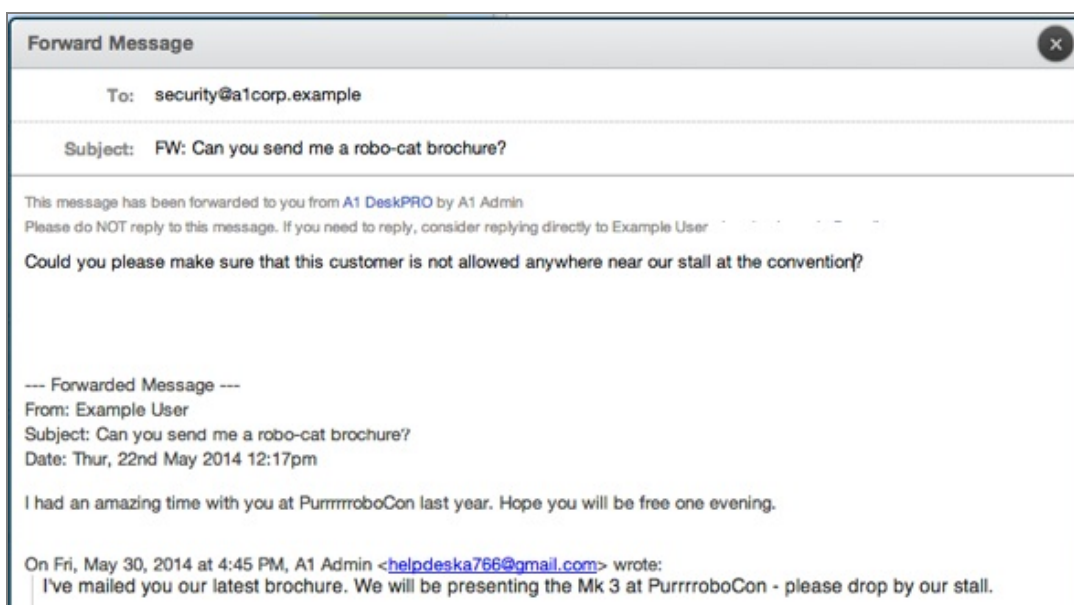
Agents can forward individual ticket messages via email, using the **Forward message** option accessible from the gear icon to the right of each message.



This is useful for forwarding messages outside of the helpdesk if they would better be dealt with as a private email conversation.

Under **Tickets > Forward Settings**, under **Ticket Forwarding - Outbound** you can change which email account is used to send the emails.

By default, the forwarded email will appear to come from ticket email account's address, but you can choose to use the agent's address in the From: field instead. For this to work, you may need to configure your email server to accept messages with any From: address.



5.5. Running an email-only helpdesk

You may want to offer support by email only, without users being aware of the web portal.

To set this up:

- 1 Make sure users don't have to validate their accounts. In **CRM > Registration**, select **Enable Registration** and disable all the registration and validation requirements. Enable the **Everyone** group.
- 2 Optionally, completely disable the portal from **Portal > Portal Editor** using the **Disable Portal** button.
- 3 If you want to leave the portal enabled (perhaps because you want it to be available to a certain group of users), you can remove links to the portal from email notifications by editing the templates in **Tickets > Email Templates**

- 1 **Layout > Footer** - remove this code:

```
{% if app.isPortalEnabled() %}
<a href="{{ deskpro_url }}">{{ deskpro_url }}</a>{% endif %}
```

- 2 **User Email Templates > Ticket Emails**- in "New Ticket Confirmation" template, remove the section referencing `ticket.link` and save the template:

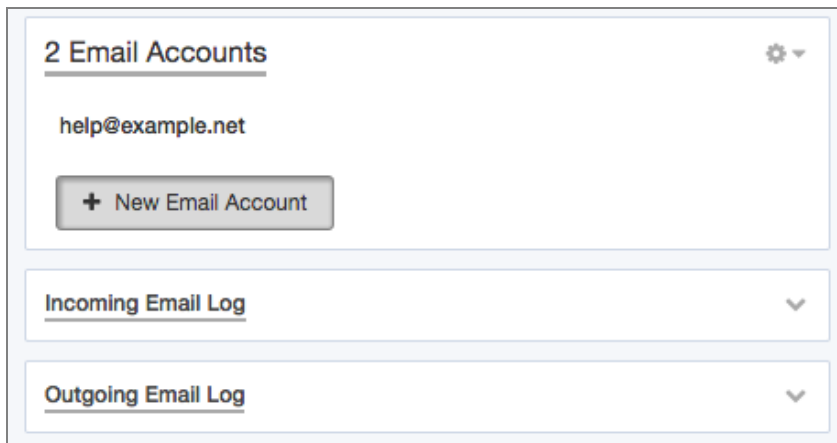
```
{% if app.isPortalEnabled() %}
<br /><br />

View and manage this ticket online:
<a href="{{ ticket.link }}">{{ ticket.link }}</a>
{% endif %}
```

- 3 Repeat above step for "New Ticket Created by Agent", "New Agent Reply" and "User Reply Auto-Response".

5.6. Monitoring email accounts

You can view logs of the status of incoming and outgoing email messages under **Tickets > Email Accounts**.



Viewing these logs can be useful when troubleshooting email problems.

Bear in mind that email processing does not happen instantly. There will be a slight delay between an email arriving in the inbox of a ticket account and DeskPRO checking the account and processing it into a ticket; similarly, outgoing emails are queued up and sent out in batches.

You can filter either log to find particular emails using the **Filter** button.

Click on an email's ID number to see full details.

You can apply actions to an email using the gear icon when viewing its details (see the [Incoming email log](#) and [Outgoing email log](#) sections below for details).

You can carry out actions on multiple emails by selecting the checkboxes at the left of the log.

Incoming Email

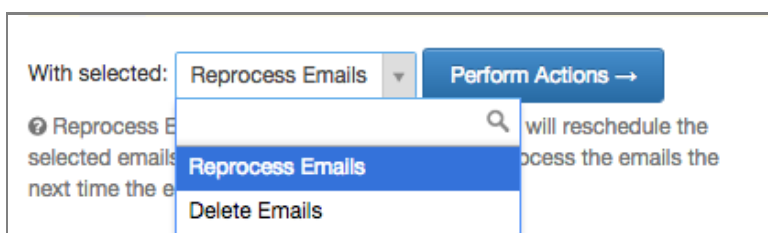
Showing all Incoming emails. Filter

☐	ID	Date	From	To	Account	Subject	Object	Status
☐	11	Sep 23, 2014 @ 11:40	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	14	retry
☒	10	Sep 23, 2014 @ 11:40	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	13	complete
☐	9	Sep 23, 2014 @ 11:38	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	12	complete
☐	8	Sep 23, 2014 @ 11:38	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	11	rejected (duplicate_message)
☒	7	Sep 23, 2014 @ 11:36	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	10	complete
☐	6	Sep 23, 2014 @ 11:36	MAILER-DAEMO...	support@site96...	support@site96.des...	Undelivered Mail Returned to ...	9	complete

Incoming email log

The incoming email log shows emails that have been received by the email accounts you have configured to receive tickets.

These are the actions you can apply to emails from the incoming log:



- **Reprocess Emails:** this will reschedule the selected emails for processing.

Use this option if email has not been correctly converted into a ticket or ticket message and you want to try again. Note that DeskPRO will detect if this will result in a duplicate message.

- **Delete emails:** use this to remove the email from the system.

Note that this deletes the raw email source and the processing log from the system, but if the deleted email has been processed, any resulting ticket or message will not be deleted.

If the deleted email has not been processed, it never will be.

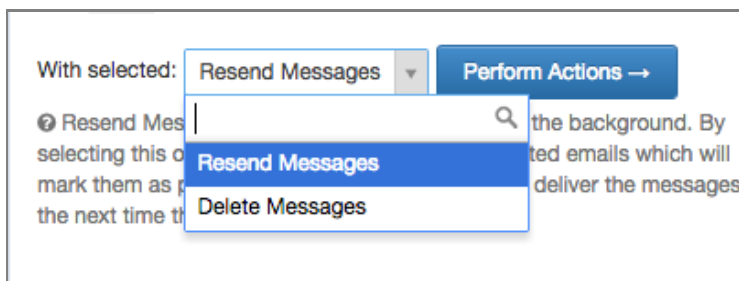
Outgoing email log

The outgoing email log displays pending messages that have not yet been sent (or messages that could not be sent due to a problem with the outgoing mail settings for one of your ticket accounts).

Messages that have been sent successfully will not be displayed on this log.

Note that if a message has been sent but not delivered due to a problem at the recipient's end, you will likely see a bounce message in the *incoming* email log.

These are the actions you can apply to emails from the outgoing log:



- **Resend Messages:** selecting this option reschedules the selected emails for delivery, marking them as pending. The system will attempt to deliver the messages the next time the message queue is processed. This option is useful if you want to resend failed messages once you have corrected a problem.
- **Delete Messages:** this deletes the message, and all records of it will be removed from the system.

6. Ticket Fields

You can add to the information stored on a ticket using optional built-in fields and custom fields.

[Categories](#) are a general-purpose field, often used to handle grouping tickets within a department.

[Priorities](#) are used to describe how important a ticket is. Priority values are a ranked series of text values e.g. "Minor"/"Routine"/"Urgent"/"Critical".

[Workflows](#) are used to record the ticket's progress through your support process - for example "Triage"/"Troubleshooting"/"Follow-Up".

[Products](#) usually denote a product or service that your helpdesk supports. If enabled, products can apply to items other than tickets, such as feedback and articles.

[Custom fields](#) are the most flexible way to add extra information to tickets. You define the name of the field as well as the possible values (text string, multiple choice, etc.).

For any of these fields, you can set a default value. You can also use **validation** rules to require users and/or agents to select a value for the field.

Note that for these fields to be visible, they must be included in the relevant ticket form layout under **Tickets > Departments**.

6.1. Categories

Categories are a general-purpose grouping field.

The main way to group tickets in DeskPRO is using departments (and sub-departments); categories can be used to break down types of tickets within departments.

For example, you could use categories to classify tickets as 'Product Defect'/'Other'; that would then enable you to get reports about the number of tickets about product defects between different departments, assign product defect tickets differently, etc.

Category choices can apply to all tickets or can be linked to tickets in specific departments.

The category field is a good choice if it applies to tickets across more than one department; if you need a field that is only relevant to tickets a single department, you will find it easier to create a custom field and link that to the department in the form layout editor.

To create categories:

- 1 Go to **Tickets > Fields** and make sure that the **Categories** switch control is on.
- 2 Click **Categories**, fill in the name of the category you want to create in the **Enter a title...** box, and click **Add**.
- 3 Enter your desired categories, then click **Save**.

The screenshot shows the 'Categories' configuration form in DeskPRO. It has three main sections: 'Categories', 'Default', and 'Validation'. The 'Categories' section has a text input 'Enter a title...', a dropdown menu 'No parent', and an 'Add' button. The 'Default' section has a dropdown menu 'No Default' and a note: 'Choose the category selected by default on the new ticket form (for both agents and users). This default will also be used in the email gateway.' The 'Validation' section has two checkboxes: 'Require users to select a value for this field' and 'Require agents to select a value for this field'. At the bottom is a blue 'Save' button.

Categories can have subcategories one layer deep. For example, you could divide the 'Questions' category into 'Basic' and 'Advanced'.

To create a 'child' category, enter a new category as normal, then change the drop-down menu to the right from **No parent** to the desired parent category. For example, this would create the subcategory 'Basic' within 'Questions':

The screenshot shows the 'Categories' configuration form with a list of existing categories: 'Product Defects', 'Questions', and 'Other'. Below the list, the 'Enter a title...' input contains 'Basic', and the dropdown menu is set to 'Questions'. An 'Add' button is visible, with a mouse cursor hovering over it.

If categories are enabled, users and agents can select a category when they create a ticket, through the portal Contact Us form or the agent interface, respectively.

Setting a **Default** category prefills that category in the user and agent forms. The user/agent can change the category; they can clear it unless validation is enabled. Tickets submitted via email will be assigned to the default category.

Under **Validation** you can require users and/or agents to select a category when they submit a ticket through the web. This does not apply to tickets submitted via email; so if it is important that all tickets are assigned a category, you should set a default category.

6.2. Priorities

Priorities are an alternative way to record how important it is to answer a certain ticket.

Priorities differ from urgency in the following ways:

- Priorities are an ordered set of named values defined by you, e.g. 'Routine', 'Urgent', 'Critical', whereas urgency is a numerical score from 1 to 10.
- Actions from triggers, SLAs, macros and escalations cannot *increase* or *decrease* priority; they can only set a particular priority value. In contrast, you can have an action that increases or decreases urgency by a set amount.
- Priority can be seen and set by users, whereas urgency is only ever visible to agents.

Priorities can be useful when the service you offer has different tiers. For example, suppose you offer a service plan where you allow users to declare a certain number of 'Critical' tickets, which you guarantee to answer faster than normal tickets.

You could implement this in DeskPRO by setting 'Critical' as a priority, allowing users to select it when they submit a ticket, and then defining an SLA that is applied to tickets with 'Critical' priority.

To create priorities:

- 1 Go to **Tickets > Fields** and make sure that the **Priorities** switch control is on.
- 2 Click **Priorities**, fill in the name of the priority you want to create in the **Enter a title...** box, and click **Add**.
- 3 Repeat until you have made all the priorities you want.
- 4 Drag priorities by the handles at left to order them, from the least important at the top to the most important at the bottom.

The screenshot shows the 'Priorities' configuration page in DeskPRO. At the top, there is a list of existing priorities: 'Routine', 'Urgent', and 'Critical', each with a handle icon on the left and a delete icon on the right. Below this list is a form to add a new priority, consisting of a text input field labeled 'Enter a title...' and an 'Add' button. Underneath the add form, there is a 'Default' section with a dropdown menu currently set to 'Routine'. A note below the dropdown states: 'Choose the category selected by default on the new ticket form (for both agents and users). This default will also be used in the email gateway.' Below the default section is a 'Validation' section with two checkboxes: 'Require users to select a value for this field' (unchecked) and 'Require agents to select a value for this field' (checked). At the bottom of the form is a blue 'Save' button.

The order of priorities is used:

- when displaying them to users/agents
- to enable agents to order tickets by ascending or descending priority in the list pane

Setting defaults and validation for priorities works just as with categories.

6.3. Workflows

DeskPRO admins often want to be able to add extra ticket statuses; for example, to record that a ticket is brand new and needs to be assigned to the right team, or that it is ready to be resolved and should be reviewed by quality control.

However, the core function of ticket status is to record whether an agent or user is due to reply to a ticket. Allowing extra custom status values would disrupt this, so we don't allow it.

Instead, we provide **workflows** as a way to record the stage of your support process that a ticket has reached.

For example, you could have workflow values of 'Triage', 'Support' and 'Quality Control', then use a trigger to set all new, unassigned tickets to 'Triage'.

To create workflows:

- 1 Go to **Tickets > Fields** and make sure that the **Workflows** switch control is on.
- 2 Click **Workflows**, fill in the name of the workflow value you want to create in the **Enter a title...** box, and click **Add**.
- 3 Repeat until you have made all the workflow values you want.

The screenshot shows the 'Workflows' configuration page in DeskPRO. At the top, there is a list of existing workflows: 'Triage', 'Support', and 'QC'. Each item has a handle (three horizontal lines) on the left and a delete icon (an 'x' in a circle) on the right. Below this list is a form to add a new workflow, consisting of a text input field labeled 'Enter a title...' and an 'Add' button. The page is divided into sections by dashed lines. The 'Default' section contains a dropdown menu currently set to 'Triage' and a descriptive text: 'Choose the category selected by default on the new ticket form (for both agents and users). This default will also be used in the email gateway.' The 'Validation' section contains two checkboxes: 'Require users to select a value for this field' and 'Require agents to select a value for this field'. At the bottom of the form is a blue 'Save' button.

You can drag workflows by the handles at left to order them. This only affects the order of display to agents.

6.4. Products

Products are a way to link tickets to products or services that your helpdesk supports. For example, a company which makes software might use products to track different versions and platforms.

Note that, unlike the other built-in fields, products can apply to items other than tickets, such as feedback and Knowledgebase articles.

Products can be nested into sub-products. These work in the same way as nested categories.

To create products:

- 1 Go to **Tickets > Fields** and make sure that the **Products** switch control is on.
- 2 Click **Products**, fill in the name of the product you want to create in the **Enter a title...** box, and click **Add**.
- 3 Repeat until you have made all the products you want.

The screenshot shows the 'Products' configuration page in DeskPRO. At the top, there is a list of existing products with handles on the left and delete icons on the right: 'Robo-cat', 'Robo-cat Mk 1', 'Robo-cat Mk 2', 'Drivers', 'Windows', and 'Mac'. Below this list is a form to add a new product, consisting of a text input field labeled 'Enter a title...', a dropdown menu currently showing 'Drivers', and an 'Add' button. The form is divided into sections by dashed lines. The 'Default' section has a dropdown menu set to 'No Default' and a descriptive text: 'Choose the category selected by default on the new ticket form (for both agents and users). This default will also be used in the email gateway.' The 'Validation' section contains two checked checkboxes: 'Require users to select a value for this field' and 'Require agents to select a value for this field'. At the bottom of the form is a blue 'Save' button.

You can drag products by the handles at the left to order them. This only affects the order of display to users/agents.

Setting defaults and validation for product works as with categories.

6.5. Custom fields

Custom ticket fields are the most flexible way to add extra information to a ticket. You choose both the name of the field, and the type of values it can store.

You can also add custom fields to users, organizations, and chats. These are all separate systems; even if you have a custom ticket field and a custom user field with the same name, you'll need to set their properties separately.

Note that you can include both custom ticket fields and custom *user* fields in the ticket form. See [Ticket Form Layouts](#) for details.

To create a custom field, go to **Tickets > Fields**, then click the **+ Add** button in the Custom Field section.

Select the type of field you want. There are a number of different types you can use, depending on the kind of data you want to store.

Enter a **Title** and **Description**. Both of these can be displayed if the field is included in a form: make sure they explain the purpose of the field in a way that will make sense to the people who will be using it.

Remember that a custom field must be added to a form layout under **Tickets > Departments** for it be visible to users/agents.

Custom field types

- Text inputs, with an optional default value (pre-filled text):
 - **Text**: a single-line text input box that users/agents can type into; accepts up to 136 characters

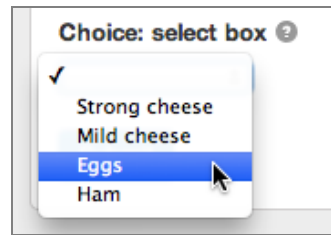


- **Textarea**: a larger multi-line text input area that users/agents can type into



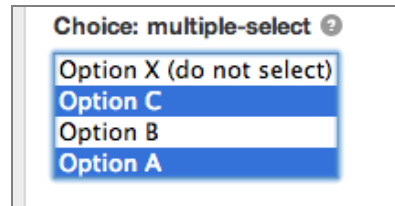
- **Choice**: a set of options you define for users/agents to pick from, with the option to set pre-selected default(s). Can be presented in the following ways:

- **Select box**



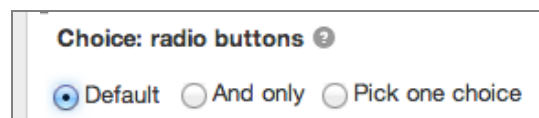
Only one can be selected at a time

- **Multiple-select box**



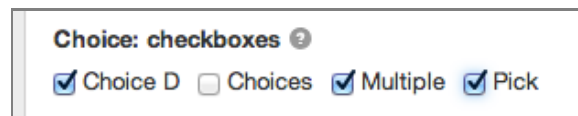
Allows multiple selections

- **Radio buttons**



Only one can be selected at a time

- **Checkboxes**

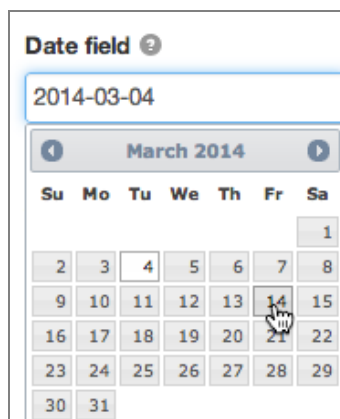


Allows multiple selections

- **Toggle:** a simple checkbox; can either be on or off by default; can require user or agent to check it (for example, to confirm they accept your terms and conditions)



- **Date:** stores a calendar date; users/agents can select from a calendar, or enter a date in text in YYYY-MM-DD format.



- **Display:** this type of field is just used to display HTML, and cannot take input; you could use it for warnings or disclaimers or design customization.



- **Hidden:** a special field that's never visible to users, which can take information from cookies or parameters in the URL; you can use this to capture information about the user's browsing session which agents can then see as part of the ticket.

6.6. Field data validation

You can set **validation** requirements for field types which take input: this enables you enforce rules when users/agents enter data.

For example, all input field types let you specify if entering a value is required for users and for agents:

User Validation	☐ No user validation
	☒ Require the user to provide a value
Agent Validation	☒ No agent validation
	☐ Require the agent to provide a value

With the setting above, the user won't be able to submit a ticket until they've entered a value for the field.

If you enable a new field and require a value, agents will be prompted to select or enter a value for the missing field before replying to the ticket.

You can control when a field is *displayed* and *editable* from the ticket form layout editor in the **Departments Layout** tab. The validation rules you set will only apply to tickets where the field is being displayed. See the [Ticket form layouts](#) section for details.

Options

☒ Show this field during ticket creation

☒ Show the field to agents when viewing the ticket
☒ Only show when there is a value

☒ Allow agents to edit the field

☐ Only show this field based on custom criteria

Done

Text validation with regular expressions

The **text** and **textarea** custom field types allow validation of the content using **regular expressions**; instead of just requiring a value, you can require the user/agent to enter a value that matches a certain pattern.

For example, suppose you need to know users' account numbers to help them. When a user submits a ticket via the web, you could use a regular expression to make sure they have entered an account number in the correct format.

Here are some examples:

```
/^[d]{8}$/ - require an 8 digit account number
```

```
/^[a-zA-Z0-9]{6,32}$/ - require an alphanumeric username 6 to 32 characters long
```

```
/(\d{5}-\d{4})|(\d{5})/ - require a valid US zip code
```

The regular expression system DeskPRO uses is [PHP Perl Compatible Regular Expressions](http://www.php.net/manual/en/regexp.reference.pcre.php). We suggest using <http://www.regexr.com/> to write and test regular expressions.

Date validation options

You can also specify additional **date validation** options:

Default value	☒ Use date at time of submission ☐ Specific date
Valid week days	☒ Monday ☒ Tuesday ☒ Wednesday ☒ Thursday ☒ Friday ☐ Saturday ☐ Sunday
Valid dates	☐ Any date is valid ☐ Date range ☒ Days before/after Date selected must be between 1 and 365 days from the time of submission

7. Automating the Helpdesk

DeskPRO provides a number of ways to automatically apply actions to tickets.

You can use these to enforce business rules, avoid your agents needing to carry out repetitive steps, and automatically prioritize tickets based on their properties.

- [Triggers](#) run actions when a ticket is created, replied to or updated. Many of the default processes of the helpdesk, like sending out notification emails, are controlled by triggers which you can edit to change their behavior.
- [SLAs](#) track your helpdesk's response time goals, highlight tickets to agents when a goal is failed or close to failing, and can run actions.
- [Macros](#) carry out a series of actions on a ticket when applied by an agent, automating repetitive tasks. Agents can create their own macros without admin access.
- [Escalations](#) run actions on tickets based on the elapsed time the ticket has been in a certain state.
- [Round robins](#) automatically assign incoming tickets to a queue of agents.

7.1. Ticket triggers

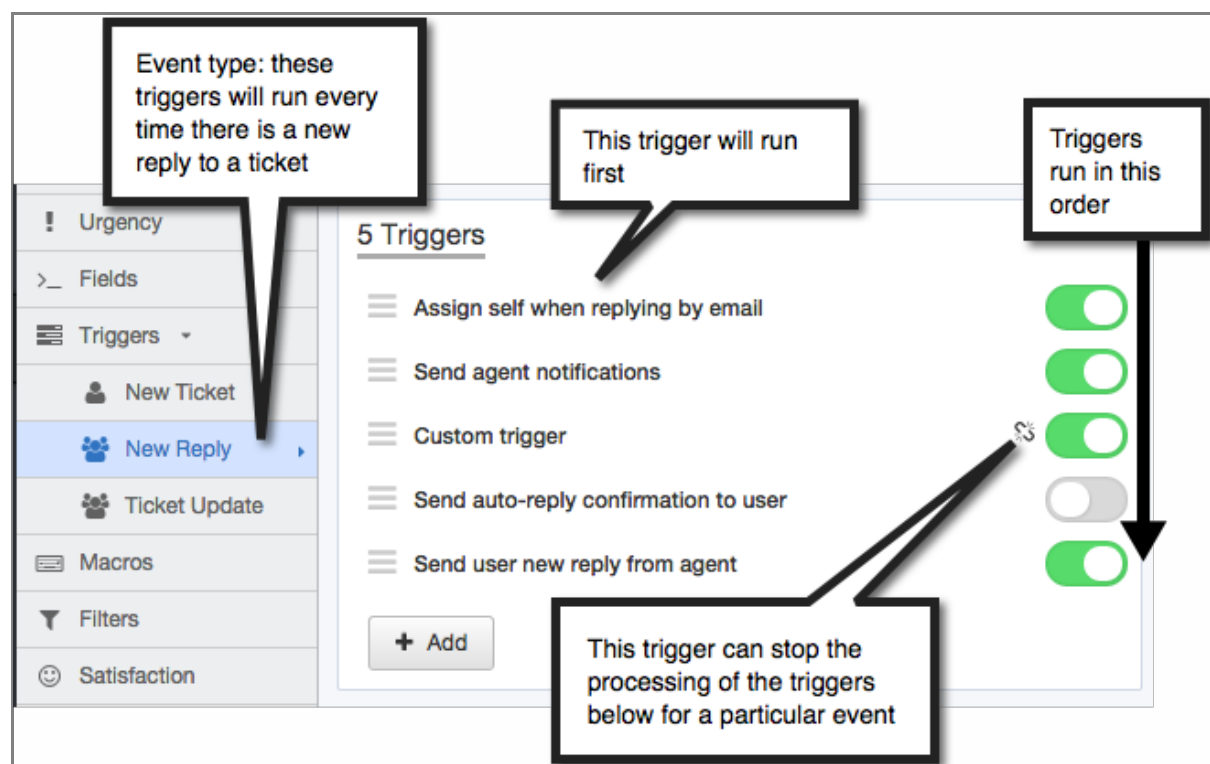
Triggers are automatic processes that perform **actions** in response to ticket **events**.

A given trigger can respond to one of these three event types:

- **New Ticket** - a new ticket is created
- **New Reply** - an agent/user replies to a ticket
- **Ticket Update** - a ticket property changes

You can see your helpdesk's triggers at **Tickets > Triggers** - there is a separate subsection for triggers with each event type.

When an event happens, the triggers for that event type run in order, from the top down.



Before a trigger performs its actions, it checks its **criteria**. The trigger's actions will only be carried out *if* the criteria are met.

The screenshot shows the 'Criteria' configuration interface. It includes a yellow informational box stating: 'The criteria section is a list of terms that must match before the actions are applied to the ticket.' Below this, there are two main sections for defining conditions. The first section is labeled 'when' and 'The following conditions are met:'. It contains a condition: 'Department is Support'. The second section is labeled 'or' and 'The following conditions are met:'. It contains two conditions: 'Urgency > 7' and 'Organization Name is Acme'. Each condition has a green '+ Criteria' button next to it.

Most of the automatic processes that the helpdesk carries out, such as sending out notification emails, are controlled by triggers and can be customized.

Trigger event filtering

A trigger can be set to respond only to certain events within its event type, based on:

- whether the event was carried out by a user or agent.
- which interface they were using.

Event

When a new ticket is created

☒ By a user

– ☒ via the web ⚙

☒ Portal ☒ Feedback and Support Tab

☒ Contact Form

– ☒ via email

– ☐ via the API

☒ By an agent

– ☒ via the agent interface

– ☐ via email

– ☐ via the API

Click the gear icon next to **By a user** > **via the web** to further refine by the particular web interface.

When a new ticket is created

☒ By a user

– ☒ via the web ⚙

☒ Portal ☒ Widget ☒ Ticket Form

– ☒ via email

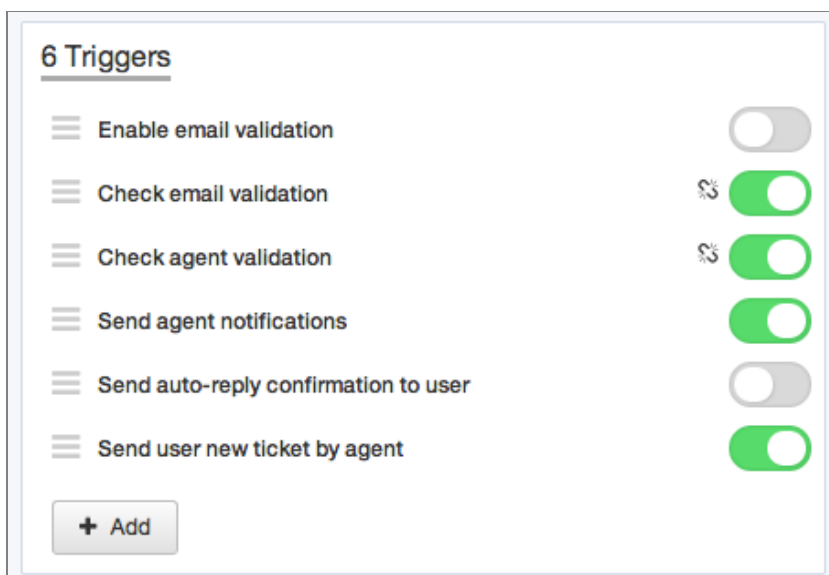
– ☐ via the API

- **Portal** means the user web portal
- **Widget** means the Feedback and Support Tab widget (see under **Portal** > **Embed**)
- **Ticket form** means the embeddable Contact Form (see under **Portal** > **Embed**)

The **via the API** setting applies to events carried out by software that has been integrated with DeskPRO. The API may carry out actions as a specific account. (See the [API documentation](#) for details).

Trigger order and trigger control

The triggers are checked in the order they are listed, with the top trigger first. You can drag triggers by the handles at the left to change the order.



In the example above, the **Check email validation** trigger runs, then **Check agent validation**, and so on.

There are **trigger control** criteria and actions which you can use to affect the usual trigger order.

For example, the **Check email validation** trigger runs the **Stop processing triggers** action if the user's email hasn't been validated, and this stops any triggers lower down from running.

The **Trigger control** actions available are:

- **Stop processing triggers**
- **Prevent emails to user**
- **Prevent emails to agents**
- **Force agent email subscriptions**
- **Set trigger variable**

The **Trigger control** criteria available are:

- **Check if user was emailed**
- **Check if agents were emailed**
- **Check trigger variable** (check the variable set by a 'set trigger variable' action)
- **Check current agent** (this checks if the agent(s) specified caused the event through the agent interface, portal or via email)
- **Check performer email** (this checks the email address of the person - user or agent - who caused the trigger event)

Note that the value of a trigger variable is only stored for the duration of the event.

Triggers with a **stop processing triggers** action are marked with a 'broken chain' icon, to remind you they can break the usual trigger order.

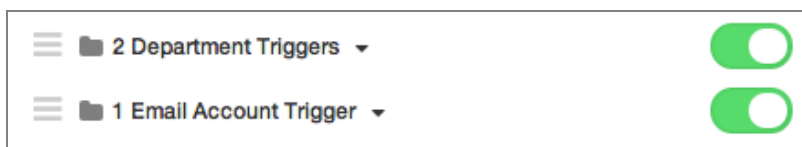


Bear in mind that the trigger control actions only apply to the *current* event. For example, suppose a ticket is updated, and the first **Ticket Update** trigger which runs applies a **Prevent emails to user** action; no emails will be sent as a result of that particular ticket update event, but that will not stop emails being sent

to the user as the result of future events.

Department and Email Account triggers

DeskPRO has built-in Department and Email Account triggers.



These simple triggers have fixed **criteria**, which you can't edit.

- **Department Triggers:** each department on your helpdesk has a New Ticket trigger which runs when a ticket is created through any means *except email*: either through the agent interface, the web portal, or an embedded form or widget.

These triggers link tickets submitted through the web to an email account; for example, tickets submitted to your Sales department have their messages sent from your `sales@` account because of a department trigger.

"Support" Department Trigger

This is a department trigger. It will apply when a user submits a new ticket from the web interface and selects this department.

Criteria

when The following conditions are met:

- if** Department is "Support"
- and** Ticket is created by a user from the web interface

Actions

then The followings actions will run:

- ☒ Set Email Account
- ☐ Send User Email

Action

There is also a **Ticket Changed Trigger** that applies when a ticket is moved into each department. This has no actions by default. If you want the email account used to send messages about a ticket to update when the ticket's department is changed, you can use this trigger.

For example, suppose you have a `support@` account linked to the Support department; by default, if a user submits a ticket to `support@`, then an agent changes the department of the ticket to Sales, the agent will continue to receive ticket emails from your `support@` address. If you wanted the user in this situation to start receiving messages from your `sales@` account, you could accomplish this with the Sales department **Ticket Changed Trigger**:

Ticket Changed Trigger

When a ticket department is changed to this department:

- ☒ Set Email Account

Action

You can view and edit the simple Department triggers from **Tickets > Departments** as well as **Tickets > Triggers**.

- **Email Account triggers:** each account you have configured to receive tickets has a simple trigger which runs when a new ticket is created *via email*.

These triggers link your email accounts to departments by assigning all new tickets created via an account to a specified department. For example, emails sent to your `sales@` account create tickets in the Sales department because of an email trigger.

"helpdeska766@gmail.com" Email Account Trigger

This is a email account trigger. It will apply when a user submits a new ticket by sending an email to this email address.

Criteria

when The following conditions are met:

if The user created a new ticket by emailing "helpdeska766@gmail.com"

Actions

then The followings actions will run:

☒ Set Department Support

☒ Send User Email

Template: New Ticket Auto-Response

Include CC's: ☒ CC in all additional user participants.

From Name: Helpdesk Name (A1 DeskPRO)

From Email: The account set on the ticket

➕ Action

You can view and edit the simple Email Account triggers from **Tickets > Email Accounts** as well as **Tickets > Triggers**.

By default, a new helpdesk doesn't send an auto-response email to the user to acknowledge a new ticket.

You can quickly enable an auto-response email to the user using the **Send User Email** action from a Department or Email Account trigger. The default New Ticket Auto-Response email template sends the user a simple acknowledgement message.

Actions

then The following actions will run:

☒ Set Department Support

☒ Send User Email

Template: New Ticket Auto-Response

Include CC's: ☐ CC in all additional user participants.

From Name: Helpdesk Name (DeskPRO)

From Email: The account set on the ticket

➕ Action

Default triggers

Aside from the built-in Department and Email Account triggers, your helpdesk comes with a set of default triggers.

You can edit these to change the behavior of your helpdesk.

For reference, here are the default triggers and what they do in the helpdesk:

New ticket:

- **Enable email validation** (off by default)

Requires the user to validate their email address by clicking a link in an email. Note that this works separately from the settings under [CRM > Registration](#). If you want email validation for all users, you can just enable it from the registration settings, but by using the trigger you can require validation

- **Check email validation** (on by default)

Sends out the email that a user must click to validate their email address before their ticket is created. Disabling it will prevent users from validating if it is required elsewhere. Uses the **New ticket requires validation (auto-response)** template. By default, only applies to user-created tickets.

- **Check agent validation** (on by default)

Checks if the user has been manually validated by an agent, if that option is enabled under **CRM > Registration**. By default, only applies to user-created tickets.

- **Send agent notifications** (on by default)

Sends out the email notifications for agents who have a notification preference set up matching the new ticket (**Subscribed agents**). Uses the **New Ticket Notification** template by default. You can also add extra notifications, and customize the name and account used to send the email.

then The following actions will run:

Template: New Ticket Notification

Agents:

- ☒ Subscribed agents
- ☐ Ben Henley
- ☐ Assigned Agent
- ☐ Agents of Assigned Team
- ☐ Ticket Followers
- ☐ Teams of Ticket Followers
- ☐ Agent Performer
- ☐ Teams of Agent Performer

Send Agent Email

From Name: Name of the person who initiated the action

From Email: The account set on the ticket

- **Send auto-reply confirmation to user**(off by default)

Sends out an auto-response acknowledging that the user's ticket has been received. You can customize the name and account used to send the email.

- **Send user new ticket by agent**(on by default)

Emails a user to inform them when an agent has created a ticket on their behalf. Uses the **New Ticket by Agent** template by default. You can customize the name and account used to send the email.

New reply:

- **Assign self when replying by email**(on by default)

When an agent replies to a ticket via email and the ticket is unassigned, assign the ticket to that agent. Note the action used to achieve this:

In the **Set Assigned Agent** action, **Current Agent** means the agent who triggered the event - in this case by replying to the ticket.

- **Send agent notifications** (on by default)

Sends out the ticket reply email notifications for agents with matching notification preferences. The equivalent of the New Ticket "Send agent notifications" trigger, but for replies.

- **Send auto-reply confirmation to user**(off by default)

Sends out an auto-response acknowledging that the user's reply has been received. You can customize the name and account used to send the email.

- **Send user new reply from agent**(on by default)

Sends the email with an agent reply to a ticket.

Ticket update:

- **Send agent notifications** (on by default)

Sends the ticket updated email notifications for agents with matching notification preferences.

Custom triggers

To customize trigger behavior, you can either modify the settings of the built-in triggers, or create your own new triggers.

When you set up a custom trigger, don't forget to change any default triggers that do the same thing. For example, suppose you create a new ticket trigger for your `sales@` account which sends different auto-replies based on whether the user is new or registered. If you leave the simple **Email Account Trigger** for `sales@` unchanged, users will get two sets of auto-reply emails.

To create a new trigger, go to **Tickets > Triggers** then either **> New Ticket**, **> New Reply** or **> Ticket Update**, depending on the type of trigger you want to make.

- 1 Specify a **Title** for the trigger that describes what it does.
- 2 You can further refine when a trigger runs using the **Event** section.
- 3 Set **criteria** which a ticket must meet before the trigger runs. These can be properties of the email, ticket, attachments, user, user's organization or the current time. You can also check [trigger control](#) criteria.

The screenshot shows the 'Criteria' section of a trigger configuration. It has a title bar 'Criteria' and a yellow informational box stating 'The criteria section is a list of terms that must match before the actions are applied to the ticket.' Below this, there are two criteria groups separated by 'or' logic. The first group is titled 'The following conditions are met:' and contains one criterion: 'User is new'. The second group is also titled 'The following conditions are met:' and contains one criterion: 'Department is Sales'. Each criterion has a green '+ Criteria' button to its left and a blue 'x' button to its right. At the bottom of each group is a green '+ Criteria' button.

- 4 Set the **actions** which the trigger will run. Actions can be used to notify agents and users, or change the state of a ticket, or even trigger an API call to a web-based service.

The screenshot shows the 'Actions' section of a trigger configuration. It has a title bar 'Actions' and a yellow informational box stating 'These actions will apply when all of the criteria pass.' Below this, there is one action group titled 'The followings actions will run:' with a 'then' logic connector. The action is 'Set Assigned Agent' with a dropdown menu showing 'A-1 Sales Exec'. There is a green '+ Action' button to the left of the action and a blue 'x' button to the right. At the bottom of the group is a green '+ Action' button.

See the [Actions guide](#) section for full details of the available actions.

Triggers example: automatically assign agents to tickets they reply to by email

In some helpdesks, agents answer users only with their email clients rather than using the agent interface.

In this situation, it would be useful to automatically assign tickets to the first agent to reply.

You can easily do this by adding a **New Reply** trigger with everything in **Event** unchecked except **By an agent, via email**.

For Criteria, put **Agent is Unassigned**.

Under Actions, add **Set Assigned Agent: Current Agent**.

Title * Assign agent to ticket if replies via email
This title will be used throughout the admin interface to refer to this trigger.

Event When a new reply is submitted
☐ By a user ☒ By an agent
 ☐ via the agent interface
 ☒ via email
 ☐ via the API

Criteria ?
when The following conditions are met:
Agent is
+ Criteria

Actions ?
then The followings actions will run:
Set Assigned Agent
+ Action

Triggers example: assigning out-of-hours tickets to a special team

Suppose you want to treat incoming tickets differently outside working hours. For example, assigning them to a night shift team who you know will be checking their email.

This is easy to accomplish with triggers:

Title * New out-of-hours tickets to night shift
This title will be used throughout the admin interface to refer to this trigger.

Event When a new ticket is created
☒ By a user ☒ By an agent
 ☒ via the web ☒ via the agent interface
 ☒ via email ☒ via email
 ☒ via the API ☒ via the API

Criteria ?
when The following conditions are met:

☒ Default working hours ☐ Set custom working hours
+ Criteria

or The following conditions are met:
+ Criteria

Actions ?
then The followings actions will run:
Set Assigned Team
+ Action

Note that the working hours here are taken from the **Default Working Hours** in **Tickets > Settings** - but you could specify custom hours just for this trigger.

You'd probably want to send out-of-hours replies from users to the same team:

The screenshot shows the configuration interface for a trigger in DeskPRO. It is divided into three main sections: Title, Event, and Criteria/Actions.

- Title:** A text box contains "Out-of-hours replies". Below it, a note states: "This title will be used throughout the admin interface to refer to this trigger."
- Event:** A section titled "When a new reply is submitted". It has two main options:
 - ☒ **By a user**: This is expanded to show three sub-options, all of which are checked:
 - via the web
 - via email
 - via the API
 - ☐ **By an agent**
- Criteria:** A section with a "when" tab. It says "The following conditions are met:". There is a dropdown menu showing "Is outside of working hours". Below this are two radio buttons:
 - ☒ **Default working hours**
 - ☐ **Set custom working hours** There are two green buttons labeled "Criteria" with a plus icon.
- Actions:** A section with a "then" tab. It says "The followings actions will run:". There is a dropdown menu showing "Set Assigned Team". Below this is a green button labeled "Action" with a plus icon.

Triggers example: filtering bounce messages

Suppose one of your ticket email accounts is used as the reply address when you send out your company newsletter. As a result, incoming notifications about undelivered emails are turned into tickets.

You could create a filter to match these tickets and delete them instantly.

Title *

Filter email bounces

This title will be used throughout the admin interface to refer to this trigger.

Event

When a new ticket is created

☒ By a user

☐ By an agent

☐ via the web

☒ via email

☐ via the API

Criteria ?

when

The following conditions are met:

Email Subject

contains

Undelivered Mail Returned To Sender

Criteria

or

The following conditions are met:

Criteria

Actions ?

then

The followings actions will run:

Set Status

Delete

Action

Triggers example: tickets from organization managers are more urgent

If your agents are using the CRM app to record which users are managers of their organizations, you could use this information to increase the urgency of managers' tickets:

Title *

RHIP

This title will be used throughout the admin interface to refer to this trigger.

Event

When a new ticket is created

☒ By a user

☒ By an agent

☒ via the web

☒ via the agent interface

☒ via email

☒ via email

☒ via the API

☒ via the API

Criteria ?

when

The following conditions are met:

Is manager of organization

+ Criteria

or

The following conditions are met:

Usergroup is

+ Criteria

Actions ?

then

The followings actions will run:

Set Priority

Set Urgency

+ Action

Save

Triggers example: emailing users when their ticket is assigned

Suppose you wanted to send an email to the user when their ticket is assigned, so they know their issue is being dealt with.

Here's how you could implement this:

- 1 Create a new trigger, and set the event to be **When a property on a ticket is changed/By an agent**.
- 2 You probably don't want the user to get a notification if their ticket was assigned and replied to at the same time, so deselect **Also run when properties have changed during a reply**.
- 3 Add the Criteria: **Agent / changed from/ Unassigned**.
- 4 Add the Action: **Send User Email > Create new template** An editor opens where you define the email template.
Here's an example template:
Subject:

```
RE: {{ ticket.subject }}
```


Body:

```
{{ phrase('user.emails.greeting') }} <br /><br />
Your ticket has been assigned to {{ ticket.agent.display_name }}. <br /><br />
{{ phrase('user.emails.ticket_access_ticket_online') }}
<a href="{{ ticket.link }}">{{ ticket.link }}</a>
```
- 5 Save the template and trigger. Make the trigger active with the toggle control next to its name.

Title *

This title will be used throughout the admin interface to refer to this trigger.

Event When a property on a ticket is changed

☐ By a user ☒ By an agent

— ☒ via the agent interface

— ☒ via email

— ☒ via the API

Options ☐ Also run when properties have changed during a reply

When this option is enabled, this trigger will run if an agent has made updates at the same time as replying. ?

Criteria ?

when The following conditions are met:

Agent changed from

Has been assigned is not

Criteria

Actions ?

then The followings actions will run:

Send User Email

Template:

Include CC's: ☐ CC in all additional user participants.

From Name:

From Email:

Has been assigned Yes

Action

Note that this trigger doesn't run when the assigned agent changes, only when the ticket moves from unassigned to assigned. So in most cases, the user would only get the custom email once, but the trigger will also run if the ticket is moved back to unassigned, then assigned to an agent again.

You could refine the trigger so that it only runs the first time the ticket is assigned. One way to implement this would be by using a custom field:

- 1 Create a toggle field "Has been assigned", make it agent-only and have it unchecked by default.
- 2 Add an action to the custom trigger that sets the field as checked when the trigger runs.
- 3 Add to the criteria of the trigger a check that 'Has been assigned' is false. So the second time that the ticket is assigned, the trigger won't run.

Triggers example: sending a different auto-reply to new users

Suppose the sales team want to customize the email auto-response when tickets are submitted to the Sales department. They want to send existing users the normal response, but they want new users to get a more elaborate welcome email explaining your product range.

We'll assume you start out with your `sales@` account linked to the Sales department with simple **new ticket** triggers, configured to send the default auto-response.

This **Email Account Trigger** sets tickets sent via the `sales@` account to the Sales department, then sends the auto-response.

6 Triggers

- Enable email validation ☒
- Check email validation ☒
- Check agent validation ☒
- 1 Email Account Trigger ☒

Use these triggers if you want to perform actions based on the email address a user sends a new ticket to.

sales@a1corp.example

2 Department Triggers ☒

- Send agent notifications ☒
- Send user new ticket by agent ☒
- Assign self when replying by email ☒

1 Email Account Trigger

"sales@a1corp.example" Email Account Trigger

This is a email account trigger. It will apply when a user submits a new ticket by sending an email to this email address.

Criteria

when The following conditions are met:

if The user created a new ticket by emailing "sales@a1corp.example"

Actions

then The followings actions will run:

- ☒ Set Department Sales
- ☒ Send User Email
 - Template: New Ticket Auto-Response
 - Include CC's: ☐ CC in all additional user participants.
 - From Name: Site Name (A1 Inc)
 - From Email: The account set on the ticket

Action

This **Department Trigger** sets tickets submitted via the web where the user has selected the Sales department to the sales@ email account, then sends the auto-response.

6 Triggers

- Enable email validation ☒
- Check email validation ☒
- Check agent validation ☒
- 2 Department Triggers ☒

Use these triggers if you want to perform actions based on the department a user selects when submitting a ticket from the web interface.

Support

Sales

1 Email Account Trigger ☒

- Send agent notifications ☒
- Send user new ticket by agent ☒
- Assign self when replying by email ☒

2 Department Triggers

"Sales" Department Trigger

This is a department trigger. It will apply when a user submits a new ticket from the web interface and selects

Criteria

when The following conditions are met:

if Department is "Sales"

and Ticket is created by a user from the web interface

Actions

then The followings actions will run:

- ☒ Set Email Account sales@a1corp.example
- ☒ Send User Email
 - Template: New Ticket Auto-Response
 - Include CC's: ☐ CC in all additional user participants.
 - From Name: Site Name (A1 Inc)
 - From Email: The account set on the ticket

Action

The simple triggers can't do what the sales team wants, because their criteria are fixed, so there's no way to specify that you only want to do something to new users.

Instead, you would create a new trigger:

Title *

Welcome email to new users

This title will be used throughout the admin interface to refer to this trigger.

Event

When a new ticket is created

- ☒ By a user
 - ☒ via the web
 - ☒ via email
 - ☐ via the API
- ☐ By an agent

This trigger is checked when a user or agent submits a ticket via email or web. (We'll assume that you don't want to send acknowledgements for tickets created using the API).

The criteria you would use are:

Criteria ?

when The following conditions are met:

Department is Sales

User is new

+ Criteria

That means the new trigger will only run its actions when the department is Sales, and the user is new. We're going to rely on the **Email Account Trigger** to set the department to Sales. Make sure the new trigger is below the simple triggers.

The action you want to run is sending a custom email. You would create a custom email template (see the chapter on [Editing Templates](#) for details), then run it like this:

Actions ?

then The followings actions will run:

Send User Email

Template: new-user-sales-welcome.html

Include CC's: ☐ CC in all additional user participants.

From Name: Site Name (A1 Inc)

From Email: The account set on the ticket

+ Action

You don't want users receiving more than one auto-response email when they submit a new ticket. So you would disable the **Send User Email** action for the simple triggers:

Actions

then The followings actions will run:

☒ Set Email Account sales@a1corp.example

☐ Send User Email

+ Action

Actions

then The followings actions will run:

☒ Set Department Sales

☐ Send User Email

+ Action

You also want users who *don't* receive the custom email to receive the default auto-response email. Since the simple triggers aren't sending that any more, another new trigger is needed to do the job.

The simplest way to avoid a user getting both the emails is to put in a check to see if the custom email has already been sent, and make sure the default auto-response trigger is below the one that sends the custom response.

Criteria ?

when The following conditions are met:

The user has NOT been emailed ☐ a specific email template new-user-sales-welcome.html

Criteria

Actions

These actions will apply when all of the criteria pass.

then The followings actions will run:

Send User Email

Template: New Ticket Auto-Response

Include CC's: ☐ CC in all additional user participants.

From Name: Site Name (A1 Inc)

From Email: The account set on the ticket

Action

Here's the completed triggers in the right order:

 2 Department Triggers ▼

 1 Email Account Trigger ▼

 Welcome email to new users

 Send default auto-response






7.2. SLAs

SLAs (Service Level Agreements) enable you to set goals for ticket response time. These goals might represent a formal SLA with a client who is paying for support, or just an internal quality standard.

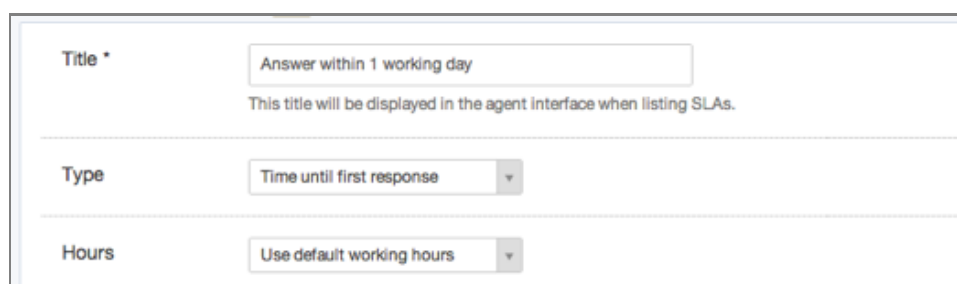
Tickets that are close to failing the goal (you define what is considered 'close') enter a **warning** condition, and tickets that exceed the goal have **failed**.

There are two effects of SLAs:

- Tickets close to or failing an SLA are highlighted in the agent interface: tickets on warning are edged amber, tickets that have failed are edged red.
- SLAs can run automatic **actions** to modify the ticket on warning or failure.

Creating an SLA

Create a new SLA by going to **Tickets > SLAs** and clicking **Add**.



Set **Type** to decide whether the SLA tracks:

- time until first response
- time until ticket is resolved
- user waiting time until ticket resolution (total)

You can use **Hours** to set whether time should be counted **24x7** (continuously), during default working hours (as defined in **Tickets > Settings > Default Working Hours**), or during custom hours that you define.

For example, suppose your helpdesk's default working hours are set as 9am to 6pm Monday to Friday, and you have an SLA that requires you to make a first response within 8 hours. A ticket arrives at 5pm on Monday and doesn't get answered. If you have set **Hours** as **24x7**, the elapsed time will be counted all night, and the ticket will fail the SLA 8 hours later at 1am. But if you set the SLA to count **Default Working Hours**, elapsed time will only be counted until 6pm. Then the clock will stop until 9am the next day, when the remaining 7 hours will start counting down again.

SLA Warning

After hours the ticket is close to failing and the warning status is applied.

then The followings actions will run:

Set Urgency

then Set Assigned Team

+ Action

SLA Failure

After hours the ticket is close to failing and the failed status is applied.

then The followings actions will run:

Set Urgency ☒ Only set if urgency is lower

then Set Assigned Team

+ Action

Define an **SLA Warning** time (when the ticket is considered close to failing) and a **SLA Failure** time.

Note the value you enter in SLA Failure is the total time elapsed on the ticket, *not* the time since warning.

You can define actions that run automatically at warning and at failure. For example, you might want to increase the **urgency** of a ticket that is about to fail an SLA, or assign the ticket to a more experienced team.

Click the **+ Action** button under **SLA Warning** or **SLA Failure** to add **actions** that will run on warning.

SLA Application

Save

Use **SLA Application** to set which tickets are affected by the SLA:

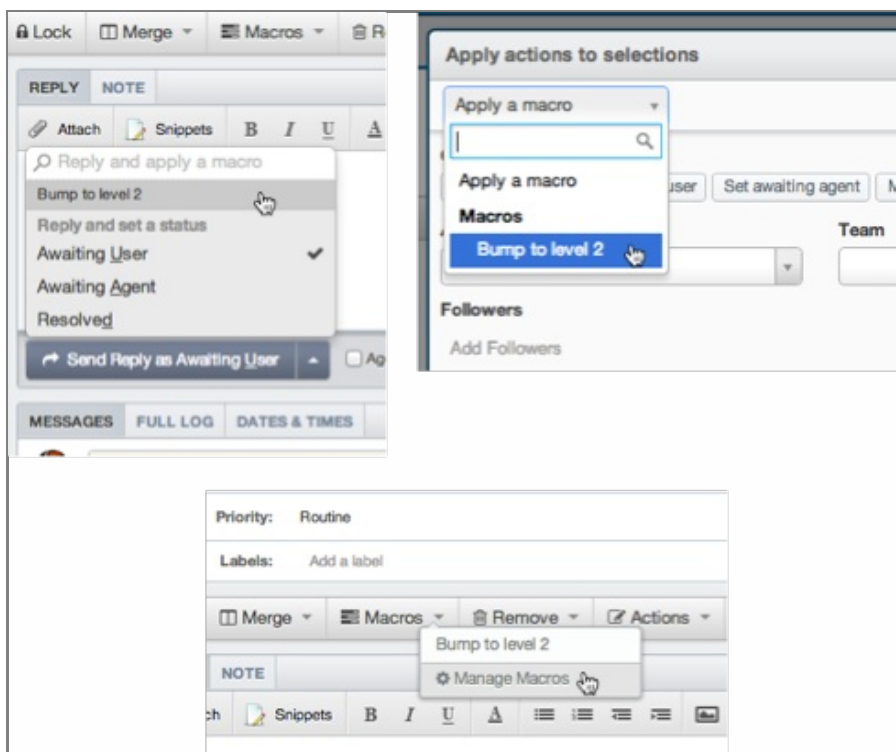
- **Apply to all tickets:** this applies the SLA to all tickets created *from now on* (not existing tickets)
- **Agents manually apply the SLA:** agents apply the SLA from the **SLAS** tab of the ticket
- **Apply to new tickets that match certain criteria:** you can enter conditions that tickets must meet for the SLA to apply.

7.3. Macros

Macros are stored sequences of actions which agents can apply from the agent interface. Macros are designed to automate sequences of steps that agents often have to repeat. An agent can apply a macro to a ticket with a couple of clicks. Using macros to perform commonly-repeated sequences of actions helps agents work more efficiently.

When a macro runs, the actions it contains are applied to the ticket it is run on.

Agents can apply macros from the content pane using the **Macros** button or the **Send Reply** control. They can also apply macros as a mass action from the list pane.



Macros can be shared (available to everyone), or limited to a specific agent. Note that if an agent tries to run a shared macro but lacks the permissions to run all its actions, the macro will not run.

Creating a macro

Agents can create macros in the **Preferences** area of the agent interface. You can create macros from **Tickets > Macros**, where you can also edit agent macros.

To create a macro:

- 1 Enter a descriptive **Title** that makes clear to agents what the macro does.
- 2 Select a **Permission** level; the macro can either be shared or limited to a specific agent.
- 3 Click the **+ Action** button and select an action.
- 4 Keep adding actions until the macro is complete.
- 5 Click **Save**.

7.4. Ticket escalations

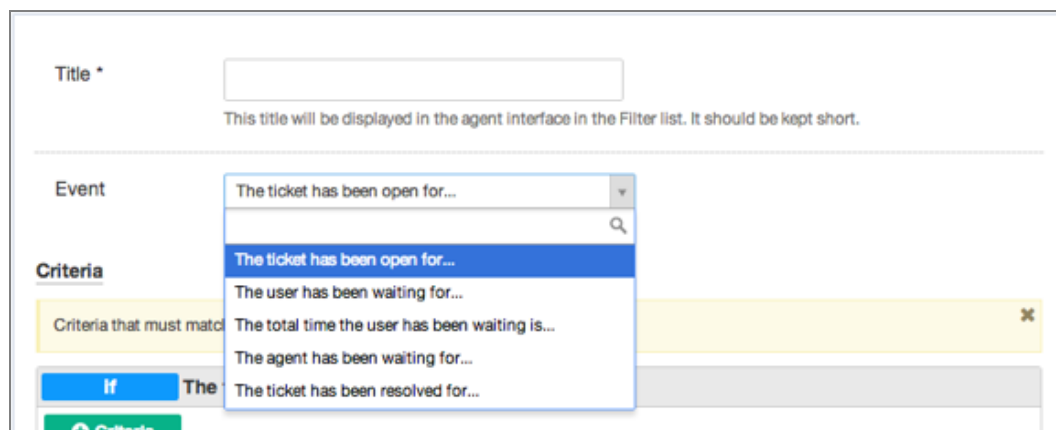
Ticket escalations run actions once a ticket has spent a set amount of time in a certain state.

They are useful to ensure that tickets are dealt with promptly. For example, you could use an escalation to increase the urgency of older tickets, or to automatically assign tickets that have gone unresolved for too long to a manager.

Creating an escalation

Create a new escalation by going to **Tickets > Escalations** and clicking **Add**.

When you create an escalation, you define what state a ticket must be in for elapsed time to count:



- **The ticket has been open for** total time the ticket has existed without being resolved.
- **The user has been waiting** total time the ticket has been *awaiting agent*, continuously since the most recent status change; for example, if the status is changed from *awaiting agent* and then back again, the clock resets.
- **The total time the user has been waiting** cumulative time the ticket has been *awaiting agent* over its whole life.
- **The agent has been waiting for** how long the ticket has been awaiting user, resets if the status changes; useful to detect tickets where the user stopped replying before the problem was resolved.
- **The ticket has been resolved for** time elapsed since the ticket was resolved.

You also define how long the ticket must spend in that state for the escalation to apply.

You can also set extra **criteria** that the ticket must meet for the escalation to run. For example, if you specify a department, the escalation will apply only to tickets in that department.

Criteria

Criteria that must match for the escalation to apply to a ticket.

if The following conditions are met:

Ticket Criteria

Status

Department

Agent

Agent Team

Product

Category

Priority

You then define the actions that run when the ticket has run up the correct elapsed time and meets any criteria.

Actions

then The followings actions will run:

Set Assigned Agent

then Set Priority

then Set Flag

+ Action

Save

7.5. Round robins

A common use for helpdesk automation is to assign tickets to agents. **Round robins** are a way to assign incoming tickets evenly to a queue of agents; each new ticket is assigned to the next agent in the queue.

When you create a round robin, an associated **Set Assigned Agent from Round Robin** action is available for use.

The first time this action runs, the relevant ticket is assigned to the first agent in the round robin. The next time the action runs, the ticket that caused it to run is assigned to the next agent in the queue, and so on. When the end of the queue is reached, assignment loops back to the first agent.

Note that you can have different triggers/SLAs/escalations that all use the same round robin. In this case, there is still only one queue, shared between the different processes.

The screenshot shows the 'Round Robin' configuration form. At the top, there is a 'Title *' field with the text 'Support round robin'. Below this field is a note: 'This is the title as it will appear throughout the agent and user interfaces.' Below the title field is a section for 'Agents *'. This section contains a list of agents with checkboxes: Abed Syrah, Annie Golding, Chris, Samuel Blix, Shirley Brown, Troy Molesworth, Maria Cama, Jeff Rosti, Temp 2, and Temp Exam. The first eight agents have their checkboxes checked. Below the list of agents is a section for bulk adding agents, with the text 'Bulk add agents that are members of teams, departments or permission groups'. This section includes a dropdown menu and an 'Add' button.

To create and use a round robin:

- 1 Go to **Tickets > Round Robin**
- 2 Ensure that **Enable Round Robin** is selected.
- 3 Click **Add** and enter a **Title** for the round robin and select the agents.
- 4 Optionally, you can bulk add the agents from a particular team or permission group or with access to a particular department (note that this does not create an ongoing link: i.e. adding a team to a round robin then later adding an agent to that team *does not* add the agent to the round robin).
- 5 Drag agents to order them in the queue, if desired.
- 6 Click **Save**.

There will now be a **Set Assigned Agent from Round Robin** action matching the title you entered which you can use in triggers, escalations and SLAS.

Actions ?

then The following actions will run:

Set Assigned Agent from Round Robin Round robin test

Action

For example, you could create a New Ticket trigger that runs the round robin action for incoming user-created tickets.

Title * Round robin
This title will be used throughout the admin interface to refer to this trigger.

Event When a new ticket is created
☒ By a user ☐ By an agent
– ☒ via the web ☐
– ☒ via email
– ☒ via the API

Criteria ?

when The following conditions are met:

Agent is Unassigned

Criteria

or The following conditions are met:

Criteria

Actions ?

then The following actions will run:

Set Assigned Agent from Round Robin Support round robin

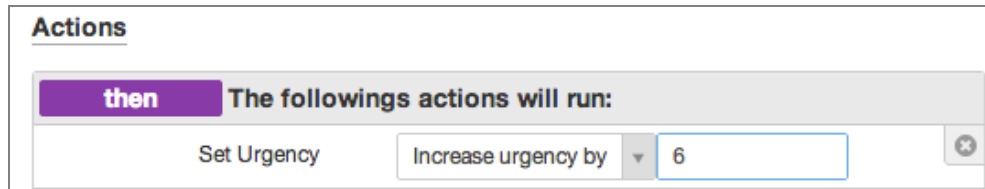
Action

7.6. Using urgency to prioritize tickets

A common use of automation is to set the **urgency** score on tickets, to help agents decide which ones to deal with first.

Urgency is a score from 1 to 10. For a newly created ticket, it defaults to 1. Agents can change the urgency score manually.

Actions can be used to increase or decrease urgency by a certain amount, instead of setting it to a particular value. This allows a number of different automated rules to be applied to a ticket, giving a combined urgency based on various factors.



Actions

then The followings actions will run:

- Set Urgency
- Increase urgency by 6

For example, suppose your organization wants to encourage agents to prioritize tickets from the VIP usergroup, and to prioritize support department tickets over sales department tickets.

You could set a trigger to increase the urgency for new tickets from the "VIP" usergroup by 6, and a second trigger to decrease the urgency for new tickets in the "Sales" department by 3.

A new "Support" ticket from a VIP would start with urgency 1, get +6 urgency from the first trigger, for an overall urgency of 7.

A new "Sales" ticket from a VIP user would start with urgency 1, get +6 from the first trigger, then -3 from the second trigger, for an overall urgency of 4.

Urgency can never go below 1 or above 10. If an action would decrease a ticket's urgency to below 1 or above 10, it will be set to 1 or 10 respectively.

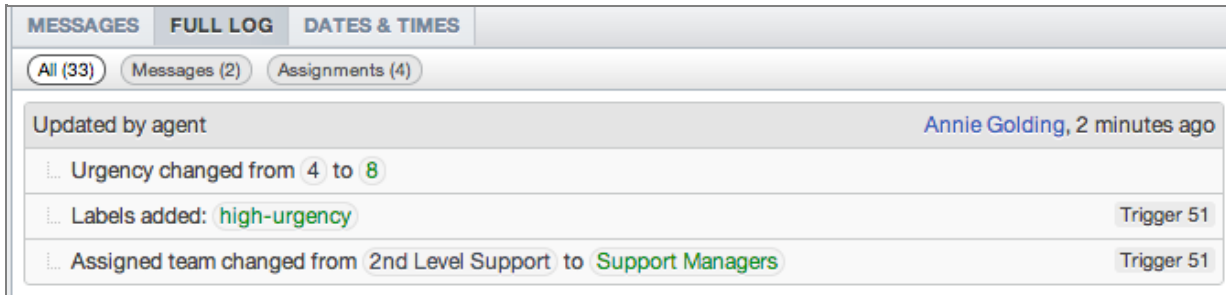
You can view the distribution of **urgency** values for your current unresolved tickets under **Tickets > Urgency**.



7.7. Displaying IDs for items

Every item within DeskPRO has an ID number. The most commonly-used example is the ticket ID, but there are IDs for most items you can create as an admin: triggers, SLAs, apps, custom fields, etc.

Knowing these IDs is useful when checking or troubleshooting your helpdesk automation by viewing the **full log** of a ticket. You can see which actions and processes have changed the ticket properties, but they are referred to by ID number.

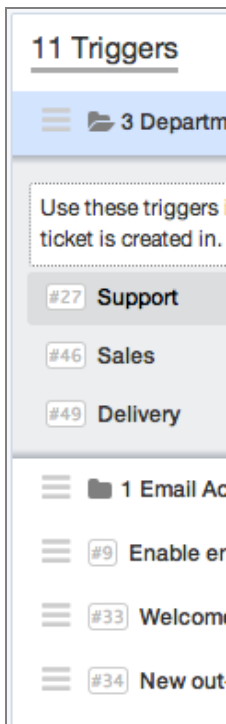


You can find out which ID belongs to which item from the admin page where you manage that item type.

Click the gear at the top right of the list of items, then enable **Show IDs**.



This is an example of how trigger IDs are displayed when **Show IDs** is enabled:



You will also need these IDs when you are referring to a particular item using a variable: for example, within a snippet (in the agent interface), when editing templates, or when you are creating a custom report in the Reports interface.

7.8. Sending SMS using automatic actions

You can automatically send SMS text messages to agents using a trigger/escalation/SLA. This is useful for alerting agents to particularly high priority tickets or SLA failures.

In order to do this you will need an account with either of the following third-party SMS services:

- Twilio
- Clickatell (Developers' Central account)

To enable SMS delivery, install the DeskPRO app for the service you wish to use from the **Apps** section of the admin interface.

Enter the required credentials from the service you are using.

For Twilio:

- Twilio Account SID (from **Account Settings** on the Twilio site)
- Twilio Auth Token (from **Account Settings** on the Twilio site) - click the padlock icon to reveal it
- From Number (must be a phone number registered with Twilio)

For Clickatell:

- Clickatell Username
- Clickatell Password
- API ID (from **APIs > Manage APIs** on the Clickatell site)

You will now have a **Send a Twilio SMS message**/**Send a Clickatell SMS message** action available for use in your triggers/escalations/SLAs.

Message:

To Agent(s):

☐ Assigned Agent

☐ Following Agents

☐ Ben Henley

☐ Chris Padfield

☐ Chris Tickner

☐ Christopher Nadeau

☐ Christopher Padfield

☐ David Lovell

Send SMS
via Clickatell SMS App

To Team(s):

☐ Assigned Team

☐ 1st Level Support

☐ 2nd Level Support

☐ Support

To Department(s):

☐ Sales

☐ Support

☐ Partnerships

☐ Media Requests

☐ General Contact

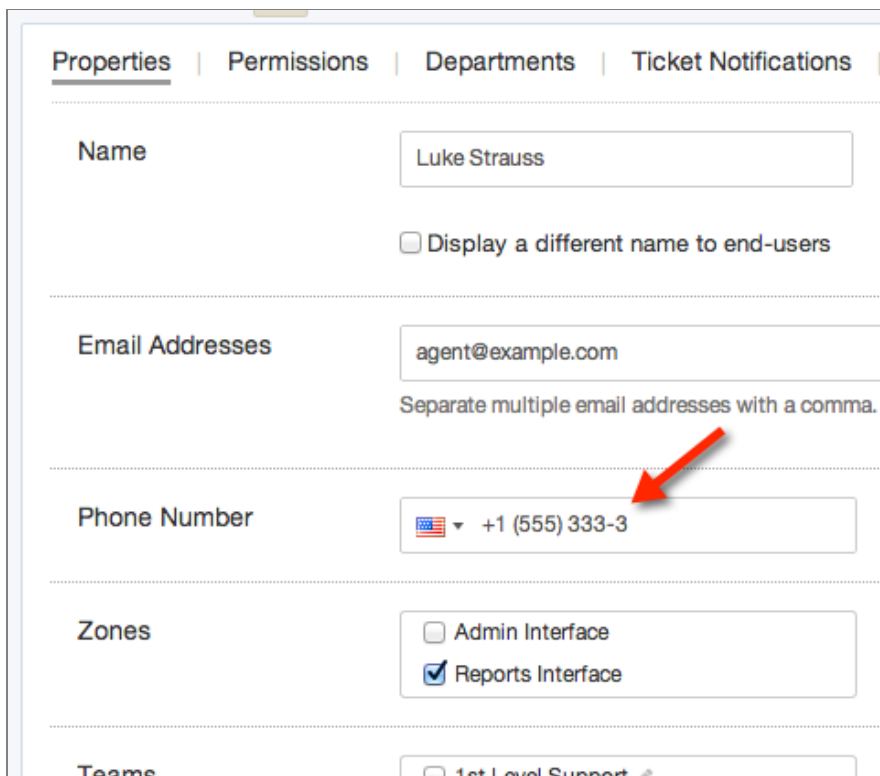
☐ Billing

To Number:

 ▼

➕ Action

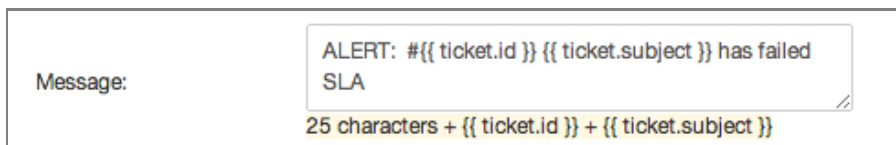
For an agent to receive SMS messages, ensure that there is a mobile phone number saved for them in the Properties tab of their details in **Admin > Agents**. Agents can also enter their mobile numbers through the agent interface in **Preferences > Profile**.



The screenshot shows the 'Properties' tab of an agent's configuration page. It contains several sections: 'Name' with a text field 'Luke Strauss' and a checkbox 'Display a different name to end-users'; 'Email Addresses' with a text field 'agent@example.com' and a note 'Separate multiple email addresses with a comma.'; 'Phone Number' with a dropdown menu showing a US flag and the number '+1 (555) 333-3', which is highlighted by a red arrow; and 'Zones' with checkboxes for 'Admin Interface' and 'Reports Interface' (which is checked). At the bottom, there is a 'Teams' section with a dropdown menu showing '1st Level Support'.

You can send an SMS to an individual agent, or all the agents within a team, a department (i.e. all agents with full permission to access that department) or to a specified number.

It is obviously a lot more useful if the SMS message contains details of the specific ticket. You can use the same variables as in snippets to automatically generate the message.



The screenshot shows the 'Message' field for an SMS. The message text is 'ALERT: #{{ ticket.id }} {{ ticket.subject }} has failed SLA'. Below the message field, it says '25 characters + {{ ticket.id }} + {{ ticket.subject }}'.

See the DeskPRO Knowledgebase article [Using variables in snippets](#) for full details of the available variables.

7.9. Actions guide

This section contains a complete list of the built-in actions you can use in triggers, SLAs, escalations and macros. Some actions are not available from every automation type.

Note that installing an app can provide you with extra actions: for example, installing the [Hipchat app](#) gives you an **Announce to HipChat** action, and installing an [SMS app](#) gives you an action to send an SMS message via the relevant provider.

Ticket assignment

- **Set Assigned Agent:**
 - assign the ticket to an agent you specify, or
 - **Unassign** the ticket, or
 - assign to **Current agent** (the agent who activated the trigger - if the trigger was activated by a user, this action will be ignored).
 - **Set Assigned Agent from Round Robin**
 - assign the next agent from the specified [round robin](#) queue .
 - **Set Assigned Team:**
 - assign the ticket to a team you specify, or
 - assign the ticket to **No Team**, or
 - assign to the **the Current Agent's Team:**
 - this means the agent who activated the trigger - if the trigger was activated by a user, this action will be ignored
 - if the agent has more than one team, the agent's **primary team** will be used (see [Setting primary team of an agent](#) for details).
 - **Set Agent Followers:**
 - add an agent you specify as a follower, or
 - assign to **Current Agent** (the agent who activated the trigger - if the trigger was activated by a user, this action will be ignored).

Ticket properties

- **Set Status:** set the ticket status from the built-in statuses
- **Set Department:** set the ticket's department
- **Set Product:** set the ticket's product
- **Set Category:** set the ticket's category
- **Set Priority:** set the ticket's priority
- **Set Workflow:** set the ticket's workflow
- **Set Urgency:**
 - **increase urgency** by an amount you specify
 - **decrease urgency** by an amount you specify
 - **set urgency** to a specified value
- **Set Subject:** set the ticket title to the specified value
- **Set Labels:** apply the specified labels to the ticket (specify them in a comma-separated list)
- **Set Flag:** apply the specified flag color to the ticket
- **Set Email Account:** set which ticket email account will send messages related to this ticket
- **Set CC'd Users:** enter user email addresses to CC:

Ticket SLAs

- **Set SLAs:** specify SLAs to apply or remove from the ticket
- **Complete SLAs:** complete the selected SLAs for the ticket; you can choose to apply SLA status as if the ticket passed, failed or is in warning, or keep the current status, or calculate the status based on the SLA criteria

Ticket actions

- **Set Ticket User:** set the user associated with the ticket (specify their primary email address)
- **Delete Ticket:** delete the ticket
- **Add Agent Reply:** send a reply from the selected agent (or, optionally, the assigned agent if there is one, otherwise the selected agent); you define the reply text

- **Add Agent Note:** works like Add Agent Reply except it adds an agent note to the ticket (for internal reference)
- **Require User Email Validation:** set the ticket status to "awaiting validation" and send the user a link to confirm their email address
- **Set Hold** - change the ticket's hold status
- **Call Web Hook** send an HTTP request to a web service; see the section on [Using web hook actions](#)

Send email

These actions send email. They all require you to specify:

- An email template
- From Name and From Email account

You can also add custom **Headers** to the email. The headers can affect how an email is processed by mail servers or other software.

Enter the header name on the left and the header value on the right (by convention, you should prefix the name of any custom email header with "X-").

The email actions available are:

- **Send Email To User** sends an email to the ticket's owner (i.e. the user); optionally include CC'd users
- **Send Email to Agents** sends an email to agents: either subscribed agents (i.e. those agents who have asked for ticket email notifications matching this event), or selected individual agents, or agents who match your set criteria:

- **Send Email to a specific email address** sends an email to a specific email address, or multiple email addresses separated by commas; this option can be useful if you want to alert someone who does not have a helpdesk account, or to use a third-party service that accepts input by email

When selecting an agent to send to when using **Send Email to Agents**, the "Agent Performer" means the agent who caused the action to run; the email will not be sent if the action is running in response to a user event, or in response to an SLA/escalation.

Trigger control actions

- **Stop Processing Triggers:** don't run actions from triggers below this one for the current event
- **Prevent Emails to User:** future actions which would result in sending an email to users about this ticket are ignored
- **Prevent Emails to Agents:** future actions which would result in sending an email to agents about this ticket are ignored
- **Force Agent Email Subscriptions** this action ensures that the selected agents receive updates about the ticket, overriding their notification preferences

For example, suppose you have a **New Ticket** trigger called "Set Foo" that *may* set the trigger variable `foo` to 1, based on a certain set of criteria. Suppose Ticket A meets those criteria; the value of `foo` is 1 when it's checked by any New Ticket triggers that are lower down the order than "Set Foo". But `foo` will not be 1 when a **New Reply** trigger later runs for Ticket A, or for Ticket B that did not meet the criteria.

You can access trigger variables when sending emails, either when using the **Enter a custom name** option to define the email **From Name**;; or from within the email template.

Use `{{ user_vars.foo }}` where `foo` is the name of the trigger variable.

Send User Email	Template:	New Ticket Auto-Response ▼
	Include CC's:	☐ CC in all additional user participants.
	From Name:	Enter a custom name ▼ {{ user_vars.fromname }}
	From Email:	The account set on the ticket ▼

Create task

This creates a task for an agent.

You have to specify:

- **Task Title**
- **Due Date**
- whether it is a **public** task (i.e. whether it can be seen by other agents).
- the agent who is the task's **Creator**.
- the agent who is the **Assignee** (i.e. is initially assigned to the task).

Custom ticket fields

- set values for any custom fields you have created

Custom person (user) fields

- set values for any custom user fields you have created

8. CRM

Usergroups control the [permissions](#) users have to view and interact with your helpdesk.

You can also set a [registration](#) policy which determines how users create an account on your helpdesk, and what they can do without registering.

You can configure the helpdesk so that can authenticate with credentials from outside DeskPRO, for example from a separate database, your network or using a social media account.

You can also set a password policy for users.

8.1. User permissions

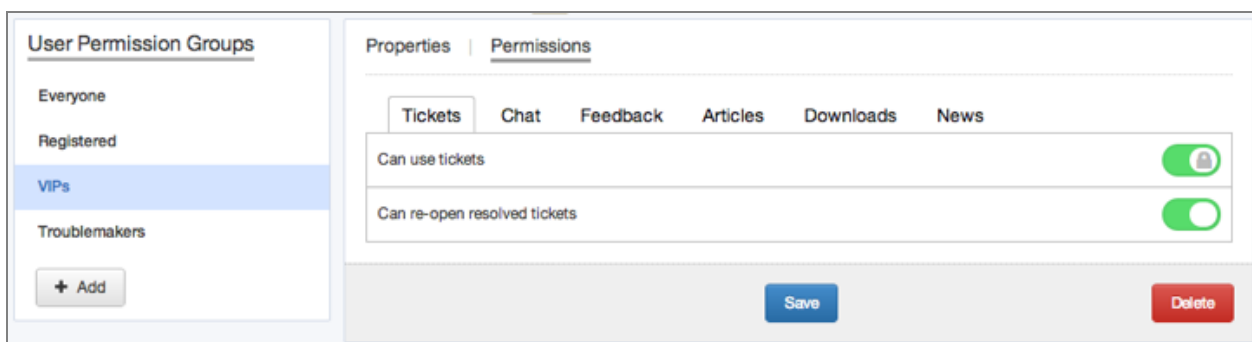
Usergroups determine what permissions users have. Usergroup membership grants user permissions, much like belonging to an agent permission group does for agents. Unlike with agents, there is no way to set permissions for an individual user.

Users can belong to multiple usergroups, and, as with agent permissions, the effect is additive; a user has all permissions that are granted by any of their usergroups.

Everyone usergroup

You may want users who aren't logged in (**guests**) to be able to access some portal content. There is a special **Everyone** usergroup which grants permissions to guests. All the permissions you grant to **Everyone** will be available to every other usergroup.

You can edit usergroups from **Users > Permission Groups**



Registered usergroup

The special **Registered** usergroup grants permissions to all registered users. Every usergroup (apart from **Everyone**) will have the permissions you grant to **Registered**.

Another way to think about this is that **Registered** inherits all the permissions from **Everyone**, and your custom groups inherit all permissions from **Registered**. Permissions that are inherited are marked with a padlock, and can't be changed.

You can grant extra permissions on top of inherited permissions, but you can't take permissions away. For example, suppose you want to make a "Basic Users" usergroup which can't access chat. You must remove chat access from **Everyone** and **Registered**, and make sure that users in "Basic Users" aren't members of other usergroups which grant chat access.

Portal access and usergroups

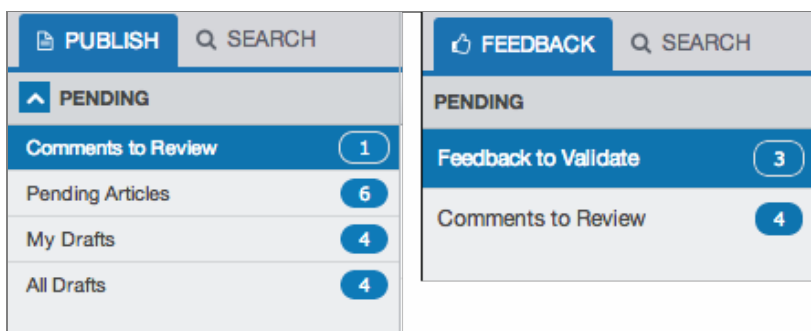
For each portal section, you can enable/disable access for the usergroup entirely.

Note that this is distinct from using the Portal Editor to enable/disable sections altogether. If you disable access to a portal section for the **Everyone** usergroup, it won't be seen by any users, but it will still be visible to agents.

For each section, you can enable/disable a usergroup's ability to rate content and submit content visible to other users, such as comments and feedback items.

For permissions that involve user-visible content, you can also choose whether or not to require agent validation. This means that agents have to approve content to check it is not abusive, spam etc.

Incoming comments and feedback in need of approval will be shown under **Comments to Review** and **Feedback** in the agent interface:



User permission details

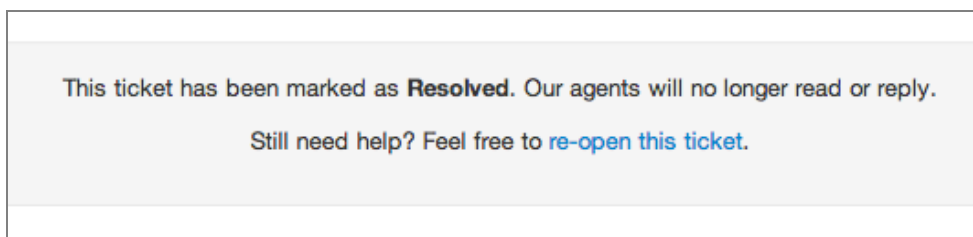
The user permission settings available are:

Tickets

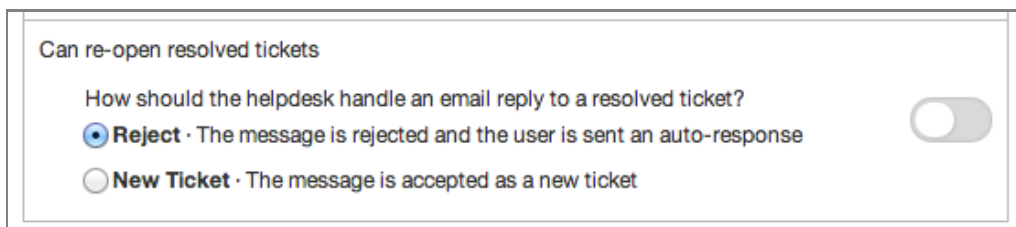
- Can use tickets (disabling this means the user can't see the Contact Us section)
- Can re-open resolved tickets

Enabling the "Can re-open resolved tickets" setting means that users can re-open a resolved ticket in the **Contact Us** section of the user portal, and by replying to an email about the ticket.

If re-opening tickets is enabled, users will see this option on the portal:



If you disable the option, you must select how your helpdesk deals with emails about resolved tickets:



You can choose whether to reject the message and inform the user with an automatic email response, or accept it as a new ticket.

Chat

- Can use chat (disabling this means the user can't see chat widgets)

Feedback

- Can use feedback (disabling this means the user can't see the Feedback section)
- Can submit new feedback
 - Do not require agent validation - this means feedback items from these users don't need to be approved from 'Feedback to Validate'
- Can vote on feedback
- Submit comments on feedback
 - Do not require agent validation - this means comments from these users don't need to be approved from 'Comments to Review'

Articles

- Can use knowledgebase (disabling this means the user can't see the Knowledgebase section)
- Can rate articles
- Submit comments on articles
 - Do not require agent validation

Downloads

- Can use downloads
- Can rate downloads
- Submit comments on downloads
 - Do not require agent validation

News

- News
- Can rate news posts
- Submit comments on news posts
 - Do not require agent validation

Users with the **Do not require agent validation** permission can post comments in the relevant portal section and have them go live immediately, without agent approval via the Publish/Feedback apps.

8.2. User registration and authentication

User registration settings govern how users create new accounts. This includes details like how the helpdesk checks that the supplied email address really belongs to the user, and your user password policy.

These settings interact with the user permissions system which determine what users (and unregistered guests) can do. Some of the registration settings override certain aspects of the permission system.

You can install **authentication apps** to let users or agents log in with credentials other than a DeskPRO account.

User registration settings

Go to **CRM > Registration** to change the helpdesk's settings for user registration and email validation.

The **Enable/Disable Registration** setting controls the ability for users to sign up for an account via the web portal. If you **disable registration**, agents must manually create an account for every user. Tickets submitted by email will only be accepted if the email From: address is already registered as a user's email address. Disabling registration gives you more control over who can access the helpdesk, at the expense of more time spent creating user accounts.

- **Require registration** hides any forms that a user can use to submit content: **Contact Us**, feedback and comments. If you enable this, even if the **Everyone** usergroup has the right permissions to submit content, the portal will not allow it.
- **Enable email validation for new registrations from the web portal** means that users who sign up for an account on your helpdesk must prove they own the associated email address before their account will work. This is done by sending a validation email with a special link that the user must click.
- **Enable email validation for new users submitting tickets from web** if this is not enabled, users can submit a ticket and provide any email address, even if it is not one they control; if it is enabled, the user must prove they own the email address by clicking a link in a validation email. Until the address is validated, the ticket is held in the **Awaiting Validation** filter in the agent interface. This option is useful if bots are submitting spam messages to your contact form. It also makes sure users have not made a mistake when entering their address.
- **Enable email validation for new users submitting tickets from email** it is easy for spammers to send email with a fake From: address. If this option is enabled, the user must prove their From: address is valid by clicking a link in a validation email. This option can help to prevent incoming spam email being turned into tickets.
- **Enable manual agent validation** means that your agents must approve every new account. Accounts that need approval will show up in the agent interface CRM app in the **Awaiting Agent Validation** filter:



While users are awaiting validation, they can still submit content, but it won't be shown on the portal until they are validated.

Restricting access to part of the portal

The registration and email validation settings apply to *every* part of the helpdesk.

You may instead want some sections of the helpdesk to be public while others have limited access. For example, suppose you want ticket submission to be locked down so it's available to approved users only, but you want to accept feedback from anyone without making them log in.

You can't apply the **Enable manual agent validation** setting just to Tickets, but the same effect can be implemented with the right combination of registration settings and [usergroups](#).

To implement this example, you'd check **Enable Registration** and uncheck **Require Registration**. You would need to keep the Everyone usergroup enabled, since it is required to accept feedback from users who aren't logged in.

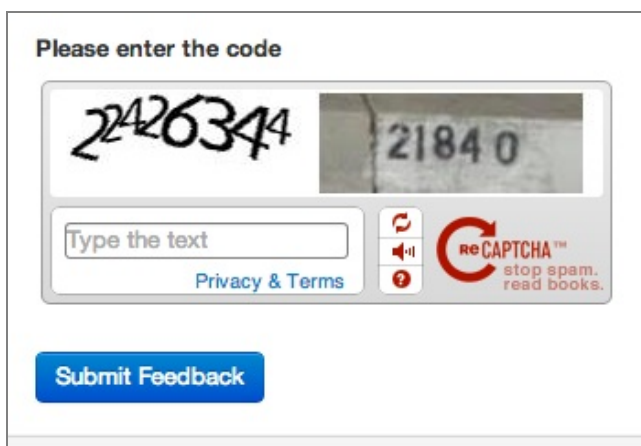
Then in **CRM > Permission Groups** you would set the Everyone group to have permission to use Feedback, but not Tickets. You don't want *every* registered user to be able to submit tickets, just ones you approve, so you would also remove Tickets permission from Registered and any other existing usergroups.

You would then create a new "Ticket Approved" usergroup with permission to use tickets. Users added to "Approved" could then use the ticket system as normal, while other users wouldn't see the **Contact Us** section of the portal at all.

CAPTCHAs

You can enable CAPTCHA tests on the portal. These require users to recognize text or audio to check that you are receiving input from a human and not an automatic spam-posting program.

DeskPRO uses the [reCAPTCHA](#) system.



Go to **Portal > Settings** to enable CAPTHCAs. You can have CAPTHCAs:

- for user registration
- for new feedback items and comments

If you require CAPTHCAs for feedback and comments, you can choose whether or not registered users need to pass a test for each comment.

Authentication apps

You can install **authentication apps** to enable users and agents to sign on with credentials from outside DeskPRO, instead of needing to create a DeskPRO account.

For example, you can allow agents to use their existing login information on your network, or allow users to sign in with their Facebook accounts.

See the section on [Extending DeskPRO](#) for more details.

8.3. Managing user passwords

Go to **CRM > Password Settings** to set the requirements for creating user account passwords.

By default, DeskPRO requires only that user passwords are 5 characters long. Select **Custom Password Policy** if you wish to specify more complex requirements to improve security.

☐ Standard Password Policy

☒ Custom Password Policy
Specify custom password requirements.

Password Policy

Minimum password length: characters
The minimum number of characters the password has to be.

Maximum password age: days
How many days a password can remain unchanged until DeskPRO requires the agent to change it.

☐ Forbid password reuse
When enabled, agents will not be able to change their password to a password they have used in the past.

Require this many lower-case characters:
Define the minimum number of lower-case (a-z) characters that must be used in the password.

Require this many upper-case characters:
Define the minimum number of upper-case (A-Z) characters that must be used in the password.

Require this many numbers:
Define the minimum number of numbers (0-9) that must be used in the password.

Require this many symbols:
Define the minimum number of symbols (!, @, \$, % etc) that must be used in the password.

The policy will only apply the next time each user sets their password. When you set a new password policy, users whose passwords do not meet the policy will not be forced to change them.

You can reset user passwords from the agent interface by going to the user's records, and from the **More** pull-down, selecting **Reset Password**.

8.4. Custom user and organization fields

You can define custom fields for users or organizations, just as you can for tickets.

These enable you to store extra information about users and organizations, beyond what

To create and manage custom user/organization fields, go to **CRM > Fields**. The options for fields are the same as for [custom ticket fields](#).

You can also [add a custom user field to your ticket forms](#), so that when a user or agent creates a ticket you can have them enter information about the user, such as an account number.

Adding custom fields to ticket forms

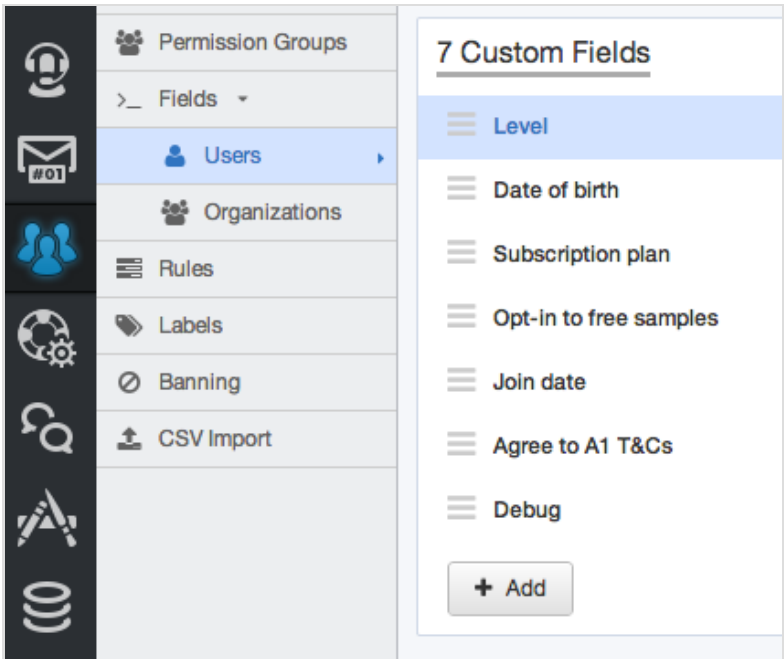
You can add custom user or organization fields to the ticket submission forms on the portal or embedded on your website.

This is useful to collect user-specific information which will help you solve tickets, and store it on the user-record for future use.

For example, suppose your agents often need to know a user's account number with your organization to solve their ticket. You could add a compulsory 'account number' field to your ticket form to ensure each user submits it.

Since the data from custom fields is stored as part of the user record, if a user submits a second ticket via the web, the account number field will already be prefilled for them.

Here's how to add a custom ticket field, using the example of an account number.

- 1 Go to **CRM > Fields > Users**

- 2 Click **+ Add**.
- 3 Select the type of input the field will take. For an account number, the simple Text option will work, so click **Configure a new Text Field**.

☒ Text

A text box is a single-line input box that the user types text into.

☐ Textarea

A textarea box is a multi-line input box that the user types text into.

☐ Choice

This is a set of predefined options presented as a drop-down select menu, radio buttons or checkboxes.

☐ Toggle (On/Off)

This field displays as a single checkbox.

☐ Date

This date field lets you select a date using a calendar widget.

☐ Display

This is a widget field that lets you enter any kind of HTML for display purposes. It does not take any input.

☐ Hidden

This is a hidden field in the new-ticket form with no visible UI for a user to interact with. The value can come from a cookie or a query string parameter, or you can use custom Javascript in your templates to set the field dynamically. The field is still editable in the agent interface via a text box.

Configure a new Text Field ↗

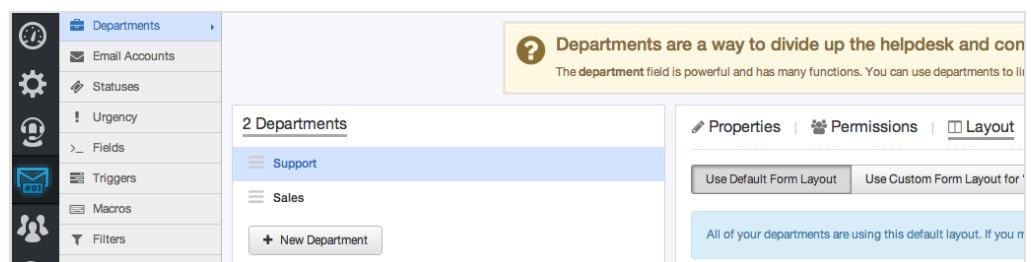
4

Enter the settings for the field. The **Title** and **Description** will be shown on the ticket form, so make sure they are phrased in a way that the user will understand.

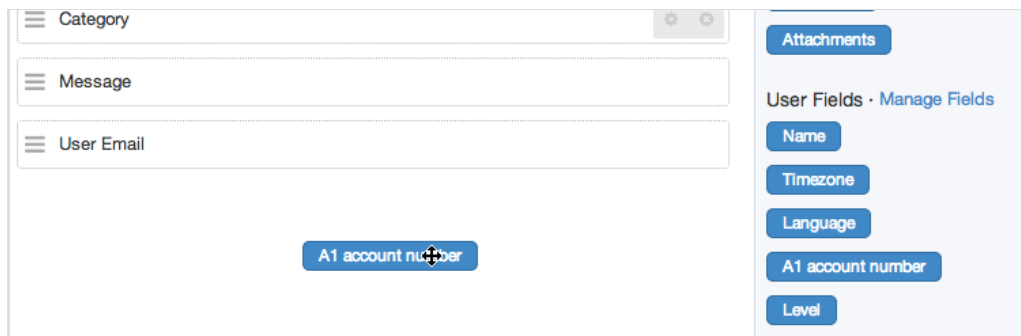
The screenshot shows a configuration panel for a field. At the top, there is a section labeled 'Enabled' with a checked checkbox 'Enable this field'. Below this are three input fields: 'Title *' containing 'A1 account number', 'Description *' containing 'Your 8 digit A1 account number', and 'Default value' which is empty. The 'User Validation' section has three radio buttons: 'No user validation', 'Require the user to provide a value', and 'Match regular expression' (which is selected). Below the selected option is a text input field containing the regular expression '\d{8}'. The 'Agent Validation' section also has three radio buttons: 'No agent validation', 'Require the agent to provide a value', and 'Match regular expression' (which is selected). Below the selected option is a text input field containing the regular expression '\d{8}'. At the bottom, there is a checkbox 'Only enable agent validation when the ticket is being resolved' which is unchecked. The 'Agent Only Field' section has a checkbox 'Only show this field to agents' which is unchecked.

You can define [validation](#) rules to make sure users/agents enter the account number correctly. In this example, a [regular expression](#) checks the account number has 8 digits. Make sure the field is set to be enabled, and click **Save**.

- 5 Now we have to edit the ticket forms. Go to **Tickets > Departments** and select the **Layout** tab. In this case, we want the user account number to be on every form, so we can just edit the default layout.



- 6 Drag the account number field into the user ticket form - the order of the fields is the order they will appear to the user.



Category

Message

User Email

A1 account number

Attachments

User Fields · [Manage Fields](#)

Name

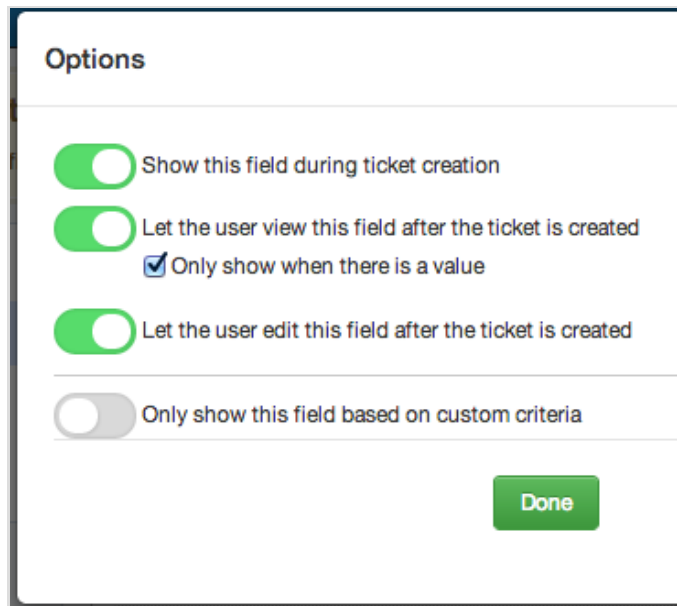
Timezone

Language

A1 account number

Level

- 7 Click the cog next to the field to edit the details.



Options

☒ Show this field during ticket creation

☒ Let the user view this field after the ticket is created

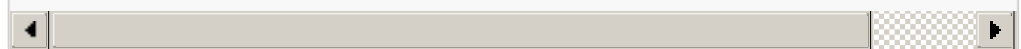
☒ Only show when there is a value

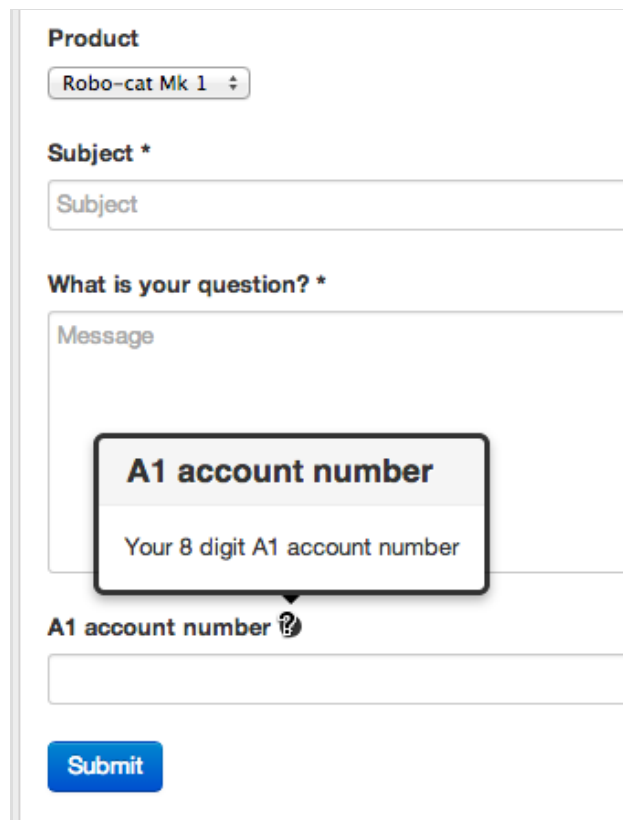
☒ Let the user edit this field after the ticket is created

☐ Only show this field based on custom criteria

Done

- 8 Disable "Only show when there is a value", so that the user is able to enter the account number on their existing tickets.
- 9 Click **Done** and **Save**.

- 10 The field is now shown on the user ticket form in the "Contact Us" section of the web
- 



The screenshot shows a web form for creating a ticket. It has the following fields and elements:

- Product:** A dropdown menu with "Robo-cat Mk 1" selected.
- Subject *:** A text input field with the placeholder text "Subject".
- What is your question? *:** A large text area with the placeholder text "Message".
- A1 account number ?:** A text input field. A tooltip is displayed over the question mark icon, containing the text "A1 account number" and "Your 8 digit A1 account number".
- Submit:** A blue button at the bottom left.

Note that mousing over the ? icon next to the field title shows the field description.

In this example, you would also want to add the account number field to the agent form, so agents could see the value entered by users, or add the value if they were creating a ticket on behalf of a user.

Per-user fields

In some situations, you may wish to add custom multiple-choice fields that offer different predefined values for each user.

For example, suppose you need a multiple-choice field to record which of a user's pets a ticket relates to. For one user, the possible values might be "Shadow" and "Felix" - for another, they might be "Fido" and "Tigger".

You couldn't use a normal custom field in this situation, because the values are the same for all users - you'd end up with a custom field that made the user pick from every pet name you have in your helpdesk.

The solution is to use a **per-user custom field**. This enables you to create a field with predefined choices, but with different values for each user. You can even allow the user to change the values through the portal.

If you added the **Pets** field to the portal contact form, here's what it would look like to the two example users we mentioned above:

To create a per-user custom field:

- 1 Go to **Admin > CRM > Fields > Users**
- 2 Under **Per-User Custom Fields**, click **+ Add**.

- 3 Enter the field details:

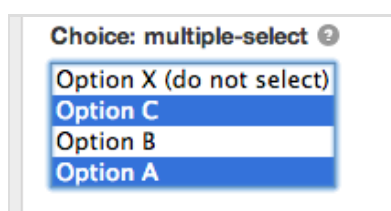
- **Title** and **Description** are displayed to users (if you add the field to the contact form), and agents. Be sure to explain the purpose of the field clearly.
- **Enabled** - select this to make the field active.
- Select the type of choice field you want:

- **Select box**



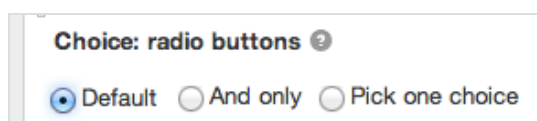
Only one can be selected at a time

- **Multiple-select box**



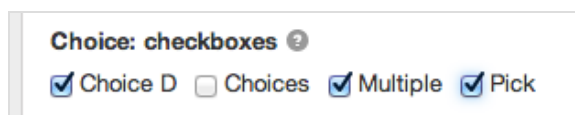
Allows multiple selections

- **Radio buttons**



Only one can be selected at a time

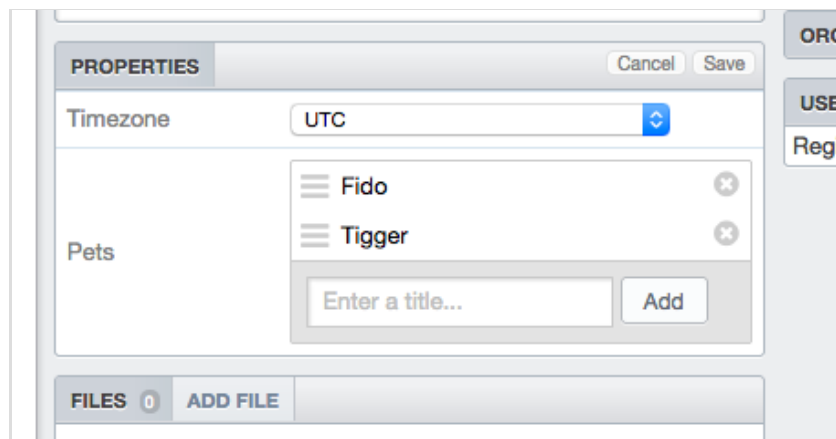
- **Checkboxes**



Allows multiple selections

- 4 Choose whether to enable **User can change values** - if enabled, the user will be able to edit the possible values for this field when logged in to the portal.
- 5 Click **Save** to store the field.
- 6 Go to **Tickets > Departments**, add the per-user field to the ticket form layouts you want (i.e. pick which departments, and whether it's for users and/or agents), and set display options. See the section on [Ticket form layouts](#) for details.
- 7 Now in **Agent > CRM** you will be able to go to each user's profile and edit their values for the field.

In the user's **PROPERTIES** section, click the small gear icon at the top right to see the values.



Click on a value to edit it. Click the X icon at right to remove a value.

You can drag values to reorder them using the handles at left.

- 8 Your users and/or agents will now see the new field according to the settings you chose in step 6.

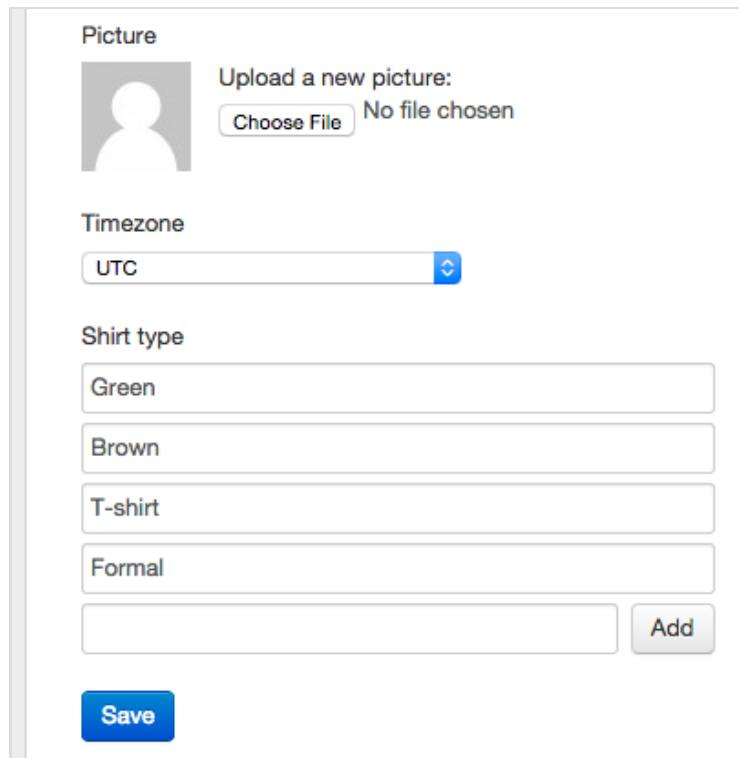
Note that by default, if a value isn't entered for the per-user/per-organization field after ticket creation, you should uncheck the **Only show when there is a value** option.

Editing per-user fields from the portal

If you want users to be able to change the values for a per-user field, make sure that in the field's settings in **CRM > Fields > Users**, you enable the **Allow users to edit values themselves** option.

When you enable this option:

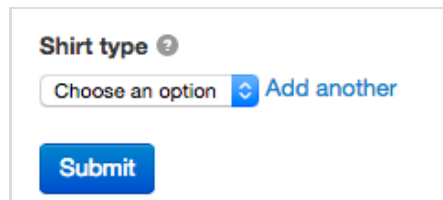
- 1 Users can add or change the available values from their profile on the portal. Deleting all the text and saving will remove a value altogether.



The screenshot shows a user profile form with the following sections:

- Picture:** A placeholder image icon, a text label "Upload a new picture:", a "Choose File" button, and the text "No file chosen".
- Timezone:** A dropdown menu currently showing "UTC".
- Shirt type:** A list of four buttons: "Green", "Brown", "T-shirt", and "Formal". Below these is an empty input field and an "Add" button.
- Save:** A blue button at the bottom left of the form.

- 2 When a user submits a ticket, they can choose to add a new value instead of selecting one of the existing options.



The screenshot shows a ticket submission form for the "Shirt type" field. It includes a question mark icon, a dropdown menu with "Choose an option", a blue "Add another" link, and a blue "Submit" button.

Per-organization fields

A per-organization field has different multiple-choice values for different organizations.

Per-organization fields work much like per-user fields, except that the values are edited on the organization profile.

Each user will see the values for their own organization (provided it's been added to the relevant layout). Users who are not members of an organization will not see the field at all.

Users cannot edit the values for a per-organization field from the portal.

9. Portal

The portal is the web-based interface to your helpdesk for users. They can use it to submit new tickets, view their previous tickets, access the help content created by agents within the agent interface **Publish** app, and submit feedback.

In addition, you can integrate portal functions within your existing website by [embedding](#) the following:

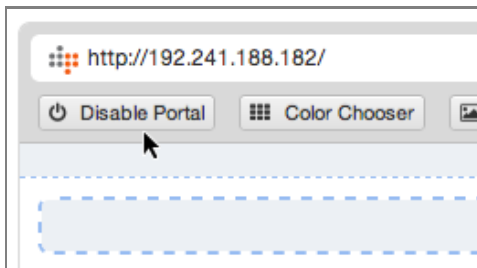
- The **Feedback & Support** tab widget enables users to interact with the helpdesk from an overlay without leaving your site
- The chat widget offers access to DeskPRO real-time chat from your website
- You can embed a contact form to submit a ticket on any web page.
- You can embed the entire portal within another site

9.1. Basic portal customization

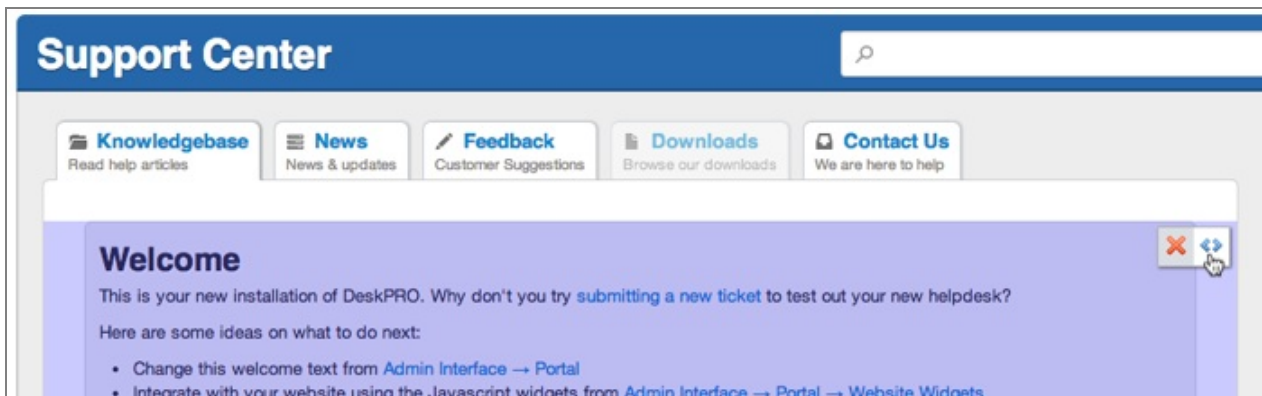
Click on the portal icon to configure it with the **Portal Editor**.



If you prefer not to use the portal, you can turn it off with the **Disable Portal** button.



If you do want to use the portal, you should replace the default Welcome message. Mouse over the **Welcome** section of the portal, and click the edit control.



You should change the welcome message to greet your users, and explain that this portal is the official way to get help from your organization. You can also mouse over and edit the words "Support Center".

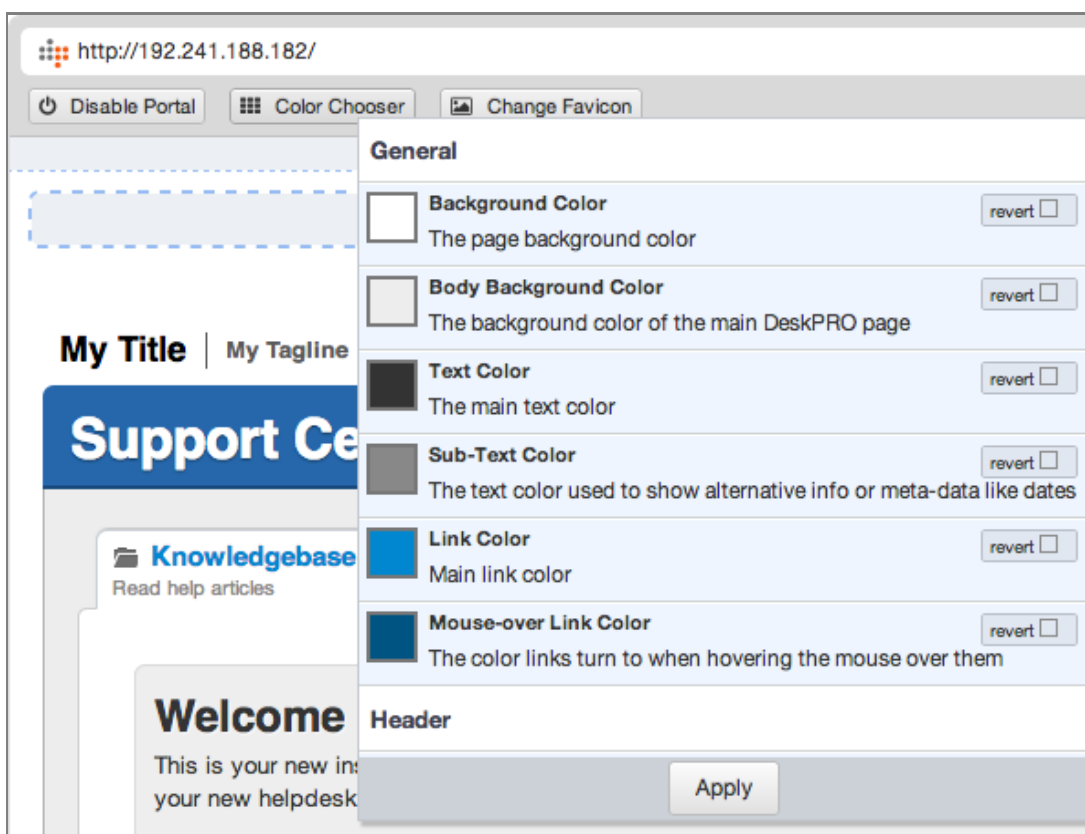
9.2. Portal branding

You can customize the look of the portal to make it closer to the design of your enterprise's website and other user-facing materials. You can make simple changes to the portal design through **Portal > Portal Editor**, or use the portal templates system to carry out more advanced customization.

Color chooser

You can pick custom colors for most of the elements of the portal.

Use the **Color Chooser** button at the upper left to edit the color scheme.

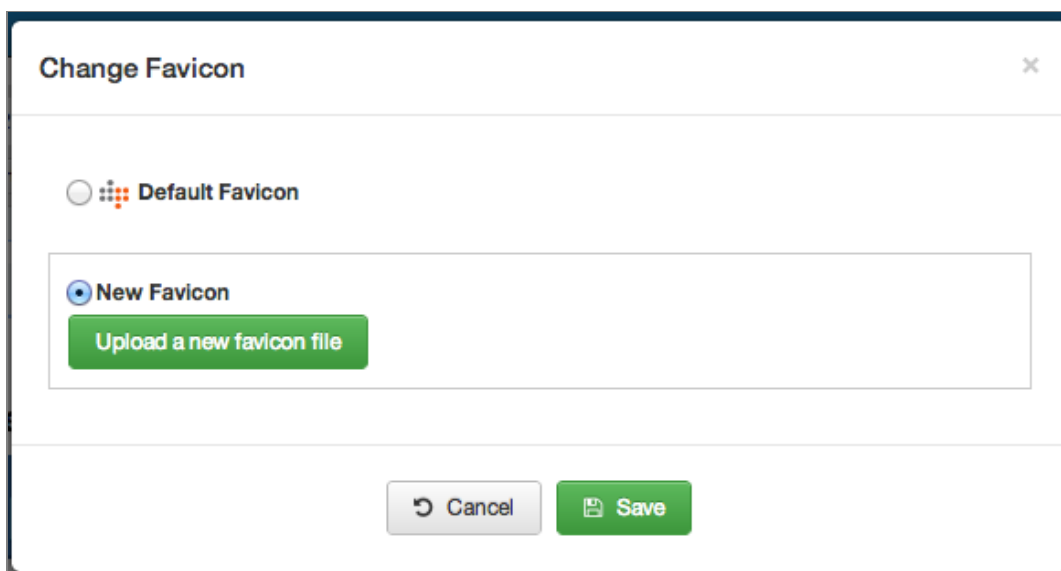


Custom favicon

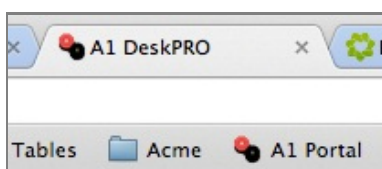
The favicon is the icon used by browsers next to the name of your site when it's opened in a tab, bookmarked, etc.

Use the **Change Favicon** button to replace the default DeskPRO icon with your own design.

The common ICO format for favicons is *not* supported. Instead, use a PNG or GIF image that is 16x16 or 32x32 pixels.



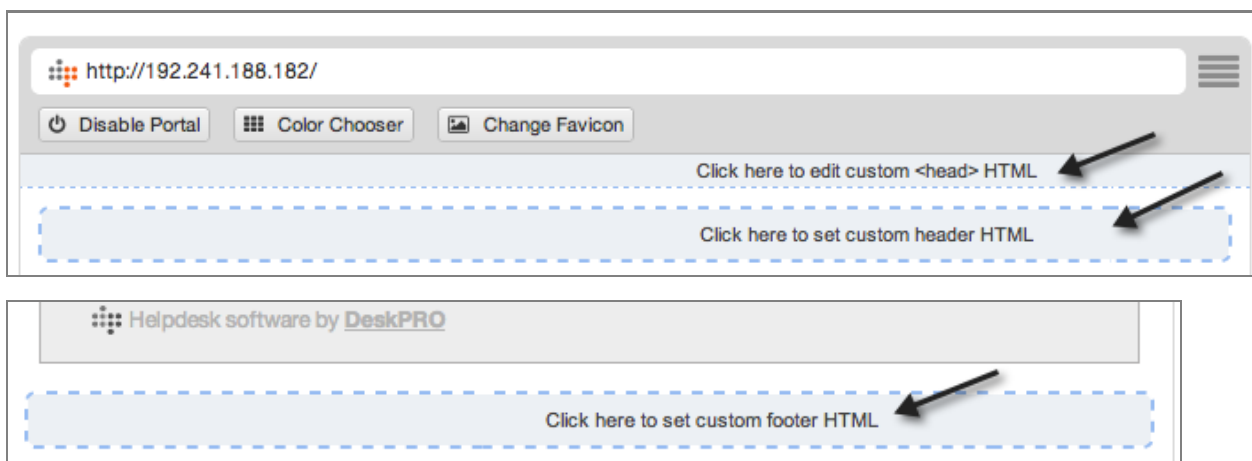
When you have uploaded the new favicon, you will see it immediately in the top left of the portal editor. It may take a while to update in your browser.



Adding custom HTML

You can insert a custom header and footer which will be displayed on all the pages of the web interface. You need to enter this as HTML code.

You can also enter code that will be inserted into the `<head>` element of the portal's HTML: this allows you to insert CSS or Javascript into the portal.

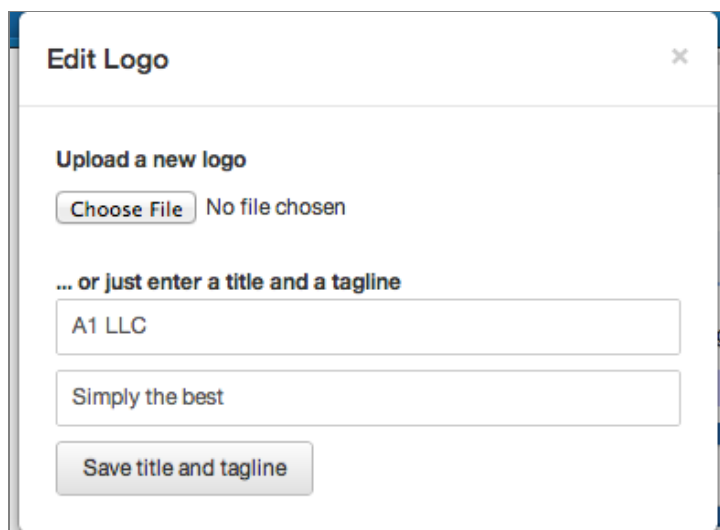


Adding a logo

You can add your logo to the portal.



Mouse over the title area and click the gear icon, then click **Choose file** to upload a logo image.



The image shows a dialog box titled "Edit Logo" with a close button (X) in the top right corner. Inside the dialog, there are two main sections. The first section is titled "Upload a new logo" and contains a "Choose File" button followed by the text "No file chosen". The second section is titled "... or just enter a title and a tagline" and contains two text input fields. The first input field contains the text "A1 LLC" and the second input field contains the text "Simply the best". Below these input fields is a "Save title and tagline" button.

Note that uploading a logo replaces any title and tagline you have entered.

9.3. Changing portal design and content with templates

If you want to customize the portal more fully, you can edit the HTML and CSS directly using **Portal > Templates**.

This lets you dramatically alter the design of the portal to make it look like part of your website.

You can also use the template system in combination with custom **phrases** to customize the text content of the portal.

See the [Editing Templates](#) section for more details.

9.4. Customizing portal content

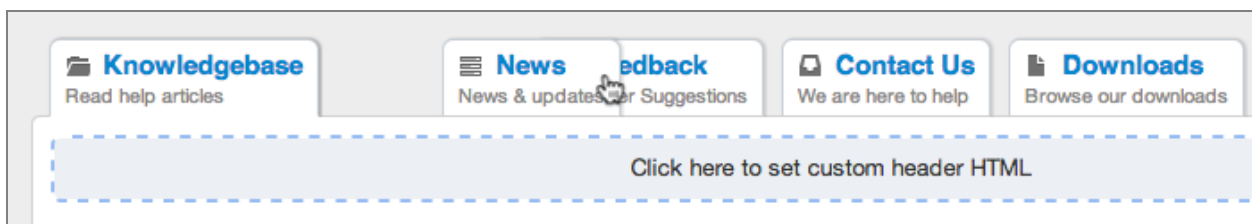
Most of the content displayed on the portal (Knowledgebase articles, news posts, downloads, and glossary terms) is created and managed from the **Publish** app in the agent interface. Feedback is managed from the agent **Feedback** app.

There are some aspects of portal content that can be customized from the admin interface.

Rearranging portal sections

You can change the order that the portal section tabs are displayed in.

Just click and drag a section (keeping the mouse button held down) to move it.

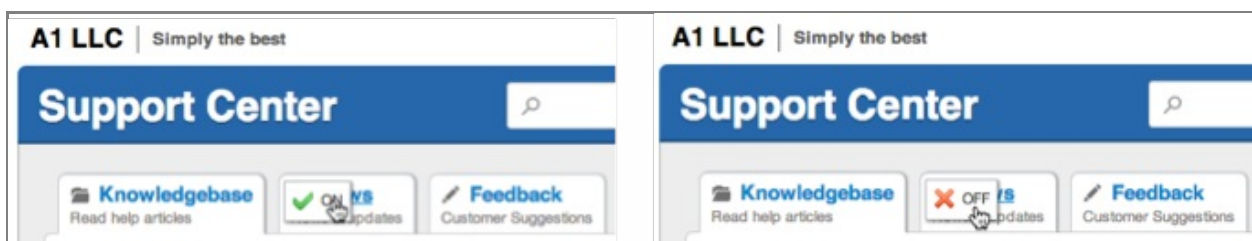


Note that the section at the very left will be the one that's displayed to users when they arrive at the portal.

Hiding and displaying portal sections

If you don't want to use a certain portal content type, such as news posts, you can disable the relevant section of the portal from **Portal > Portal Editor**. This avoids users seeing an empty area of the portal.

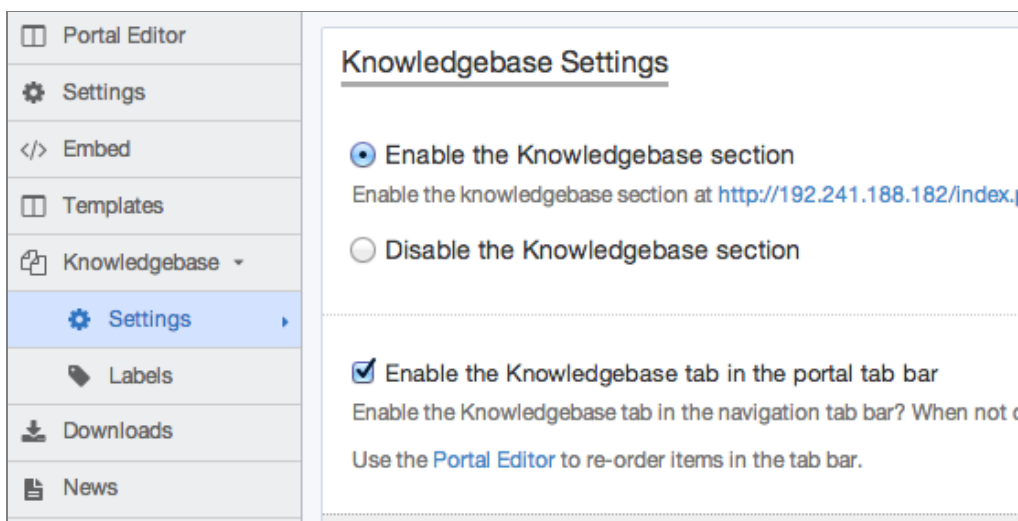
Mouse over a portal section, then click the **ON** control, to toggle the section off.



You can also drag the section tabs to reorder them.

Note that turning off a section in this way only prevents it from being displayed on the portal; any content isn't deleted, and remains accessible through **Publish/Feedback** in the agent interface. You can continue to create content even if the section isn't being displayed. So you could, for example, disable your Knowledgebase section while it's empty, write an initial set of articles, then enable it again.

In the individual settings for each portal section, you also have the option to disable display of the tab, while leaving the section active. This means the section is still available for users who know the correct web address, but users who come to the portal homepage won't be able to navigate to it.



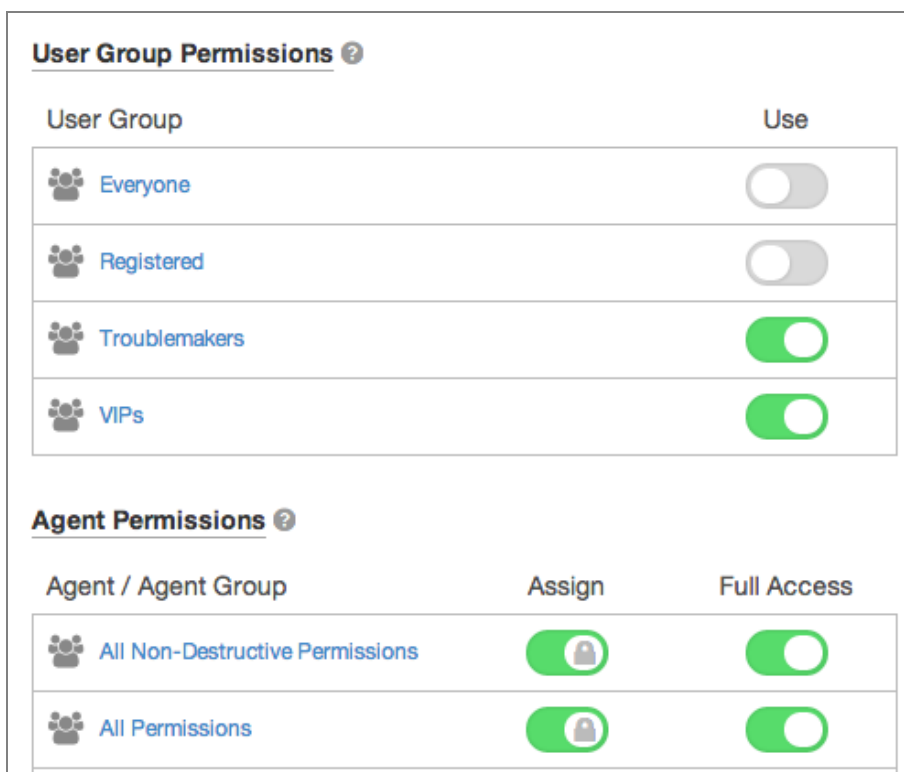
For example, suppose you have drivers in your Downloads section that should only be installed on the advice of an agent. You could disable the Downloads tab, so users can't find the section and experiment with drivers, but you could still easily send links to the drivers when needed.

Permissions will also affect the display of portal sections.

Sections are not shown if the user does not have:

- the correct *usergroup* permissions (set in **CRM > Permission Groups**)
- for Contact Us, sufficient *department* permissions to use at least one department available on the form (set in **Tickets > Departments** in the **Permissions** tab)

Note that the **Everyone** and **Registered** usergroups can affect what agents see on the portal. For example, suppose you have a single department, with **Everyone** and **Registered** permissions disabled. Your agents will not be able to see the **Contact Us** section on the portal, even if they have *agent* permissions to use that department in the agent interface.



Ticket deflection

Ticket deflection directs users who are about to submit a ticket to knowledgebase articles that might solve their problem.

If you enable this option from **Tickets > Settings**, when users start typing into the **Contact Us** ticket submission form on the portal, matching articles will be suggested automatically.

The screenshot shows a ticket submission form with the following fields and sections:

- Department ***: A dropdown menu with "Support" selected.
- Subject ***: A text input field containing "email problem".
- What is your question? ***: A large text area with the placeholder text "Message".
- We found the following content that may answer your question**: A section with a heading and a subheading "Do not worry, you can click a title and you will remain on this page." It contains two suggested articles:
 - Is there a debug mode for email processing?**: A link with a folder icon. Below it, text says: "Yes. You can run the following command from the command line: php cron.php --verbose -j process_ema..."
 - Manually run the email processing command**: A link with a folder icon. Below it, text says: "Sometimes you may want to manually run the email processing command. For example, to debug connectio..."

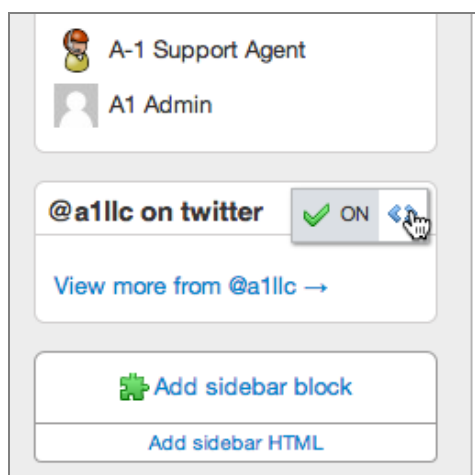
Portal contact form layout

You can edit the layout of the contact (ticket submission) form, including having different layouts for each department. See the section [Ticket form layouts](#) for details.

Showing your Twitter account on the portal

You can show recent tweets from a Twitter account you control in the portal sidebar with the Twitter block.

Mouse over the block and then click on the <> control to configure it.



Enter the Twitter account you wish to include, and how many recent tweets you want to show.

You must authenticate with Twitter by clicking the link and authorizing the portal to access the account.

Custom portal sidebars

You can add custom sidebar blocks to the portal.

If you want to include a simple sidebar that is the same for all users, click **Add sidebar block** and enter the block title, then its contents (marked up as HTML).

The **Add sidebar HTML** option is more powerful, because it allows you to use variables and conditional logic from the template system.

The following variables are useful for creating blocks:

{{ app.user.id }}	User ID - useful for conditional tests to see if user is logged in
{{ app.user.name }}	User's name
{{ app.user.primary_email_address }}	User email address
{{ app.user.organization.name }}	User organization name

Here's an example sidebar. It checks to see if the user is logged in (using `app.user.id`) and greets them by name. It then checks to see if they belong to an organization (using `app.user.organization.name`). If so, it displays a custom message about the organization.

```
{% if app.user.id %}

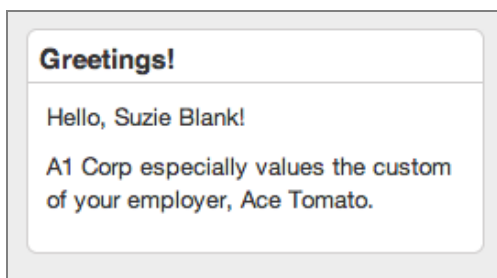
<p>Hello, {{ app.user.name }}!</p>

{% if app.user.organization.name %}

<p>A1 Corp especially values the custom of your employer,
{{ app.user.organization.name }}.</p>

{% endif %}
{% endif %}
```

This would display the following message for a user who was in the Ace Tomato organization.



See the section on [Editing Templates](#) for more about using the template system.

Custom ticket form

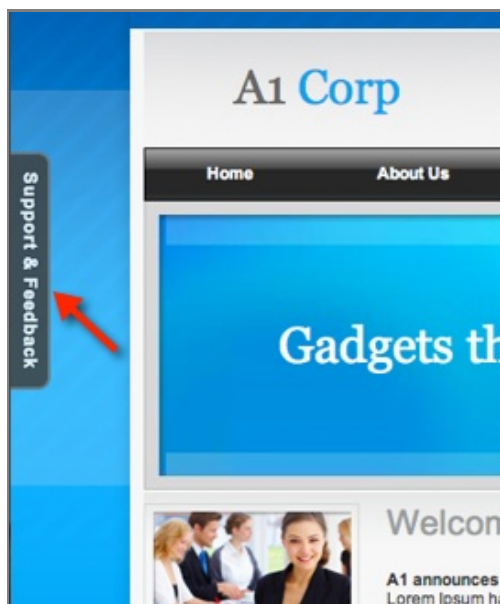
You can customize the layout of the Contact Us section ticket submission form from **Tickets > Departments**, using the **Layout** tab. See the section on [Ticket form layouts](#) for details.

9.5. Website integration

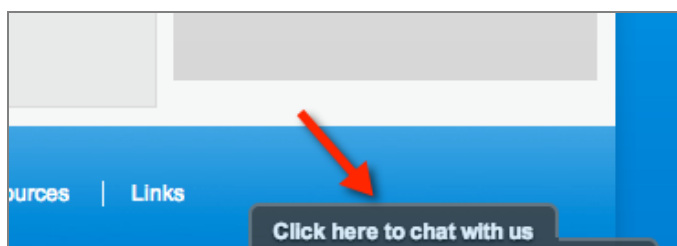
You can embed some DeskPRO functions, or your entire portal, into your existing website. This means that users don't have to navigate to your portal before using the helpdesk.

The available options for embedding are:

- a **Feedback & Support** tab at the left or right hand side of your site, giving users access to ticket submission, feedback, chat and the ability to search knowledgebase articles and news posts



- a **chat widget** at the bottom of your site

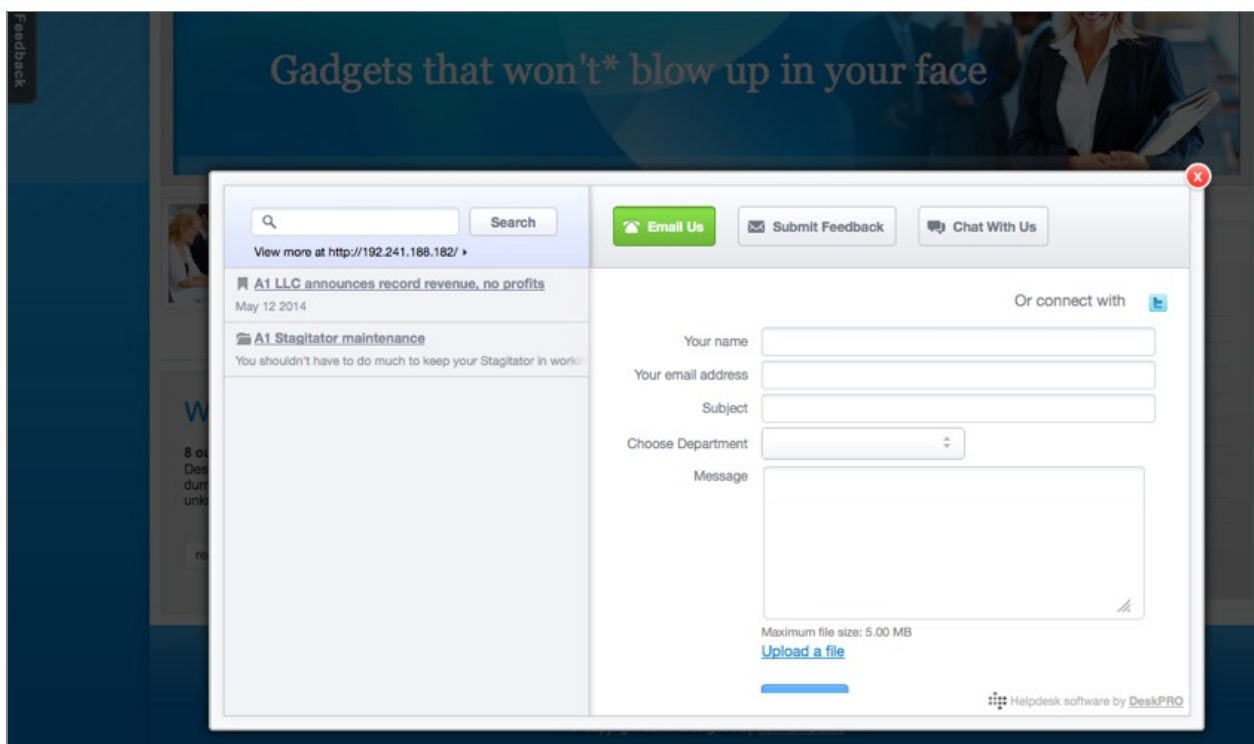


- a **contact form** for ticket submission, which can be set to submit to a particular department or allow the user to choose
- **embedding the whole portal** into a page on your site

Embedding any of these requires adding some Javascript code (which you can copy from the admin interface) to your website. It's simple, as long as you know how to edit the code of your website directly. Some "easy website builder" hosting services may not allow you to add the required code.

Embedding the Feedback & Support tab

When clicked, the Feedback & Support shows the following pop-up window over your website content:



The **Email Us** option displays a ticket submission form using your default layout from **Tickets > Departments**.

The **Submit Feedback** option enables the user to submit a feedback item, and the **Chat With Us** option enables the user to start a chat.

The search bar at the left searches through knowledgebase articles and news posts. You can also choose some articles or news items to display by default (see below).

To embed the Feedback & Support tab:

Feedback and Support Tab

The tab widget places a tab button on the left of your website. When the user clicks the tab

Tab title:

Tab position:

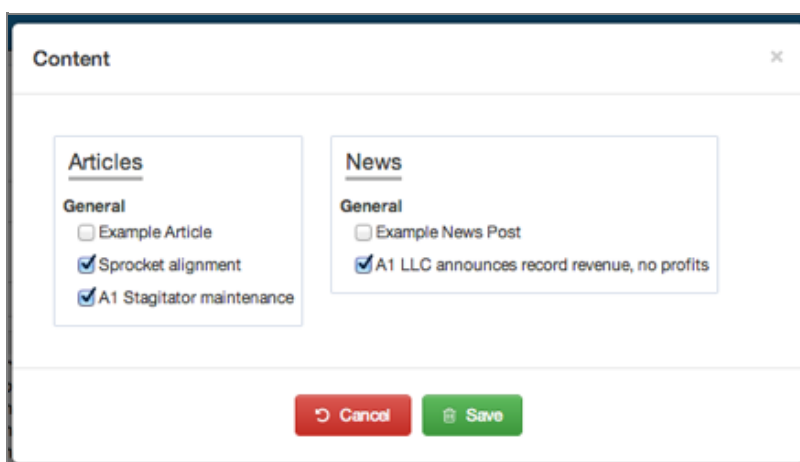
Language:

```
<!-- DeskPRO Widget -->
<script type="text/javascript">
  var DpOverlayWidget_Options = DpOverlayWidget_Options || {};
  DpOverlayWidget_Options.phrase = 'Support & Feedback';
  DpOverlayWidget_Options.tabLocation = 'left';
  DpOverlayWidget_Options.languageId = 0;
  DpOverlayWidget_Options.topPosition = '200px';
```

- 1 Go to **Portal > Embed**
- 2 In the **Feedback and Support Tab** section, set the options you want for your tab:
 - Edit the **Tab title** to change the text displayed on the tab.
 - Choose the **Tab position** (left or right edge of your site).
 - Set the **Language** displayed in the tab (note that unlike on the portal, the language cannot be changed by users).
- 3 Copy all the code in the text box.
- 4 Insert the code into the HTML of the page on your website where you want the tab to appear. We recommend putting it just before the `</body>` tag, i.e. after all the other page content.

Note that the settings for the above tab options are stored in the embed code. If you want to change them, you will need to copy and paste the Javascript again.

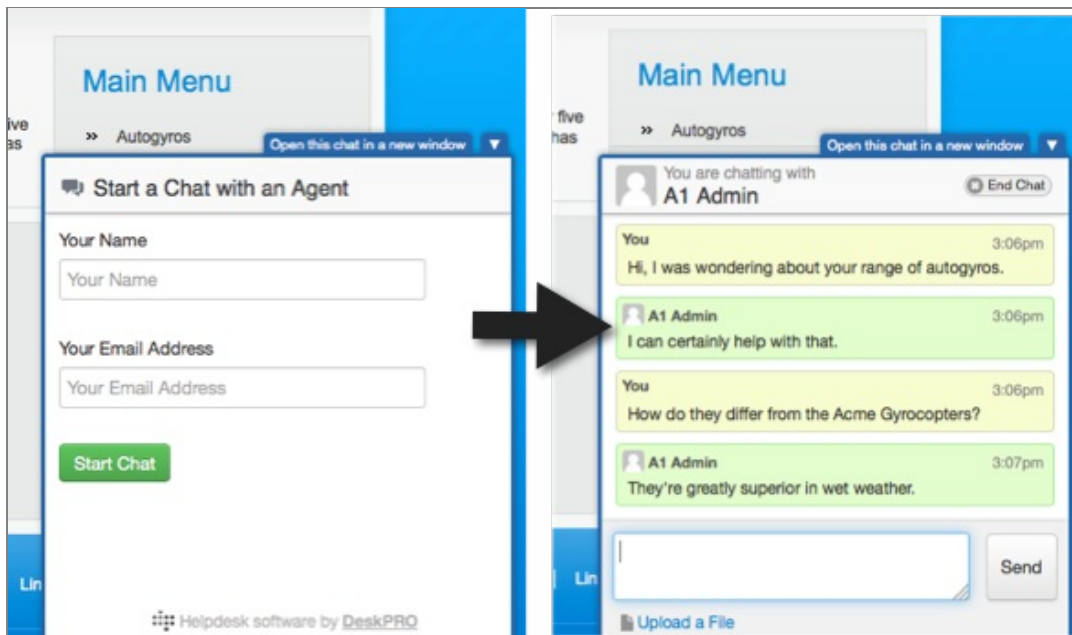
You can select the articles and news posts to display by default on the left of the overlay. Click the **Select content to show** button:



You can change this setting and the content displayed will update without you needing to re-paste the Javascript.

Embedding the chat widget

When clicked, the chat widget expands into a chat window.



Note that, depending on the chat permissions settings for the **Everyone** usergroup, users may have to log in before starting a chat.

To embed the chat widget:

Chat

The chat widget puts a small button on the bottom left of your website. When the user clicks the button, a chat to browse around your site while the chat is active.

Tab title:

Tab position:

Language:

☐ Show button even when chat is offline

```
<!-- DeskPRO Chat -->
<script type="text/javascript">
  var DpChatWidget_Options = DpChatWidget_Options || {};
  DpChatWidget_Options.startPhrase = 'Click here to chat with us';
  DpChatWidget_Options.tabLocation = 'right';

  DpChatWidget_Options.openInWindowPhrase = 'Open this chat in a new window';
```

- 1 Go to **Portal > Embed**
- 2 In the **Chat** section, set the options you want for the widget:
 - Edit the **Tab title** (text displayed on the minimized chat tab)
 - Choose the **Tab position** (bottom left or bottom right)
 - Choose the chat language.
- 3 By default, the chat widget is not shown if chat is disabled or there are no agents logged in. Check **Show button even when chat is offline** to change this; you can provide alternative text and a link where the tab will direct users when chat is offline (by default, this is to your contact form).
- 4 Copy all the code in the text box.
- 5 Insert the code on the page on your website where you want the chat widget to appear. We recommend putting it just before the `</body>` tag.

Using a custom button with the chat widget

If you want to use your own custom HTML element/button to launch the chat widget on a page, add the special classname `dp-chat-trigger` to it. For example:

```
<button class="dp-chat-trigger">Start chatting</button>
```

You can use custom CSS on your page to hide the default button, and make sure your button is hidden when agents are offline:

```
<style type="text/css">
#dpchat_wrap {
    display: none !important;
    visibility: hidden !important;
}

.dp-chat-trigger {
    display: none !important;
}

.dp-chat-enabled .dp-chat-trigger {
    display: block !important;
}
</style>
```

Embedding a contact form

You can embed a contact form to allow users to submit tickets from your website.

To embed a form:

```

<!-- DeskPRO Ticket Form -->
<div id="dp_newticket_form" style="display: none;"></div>
<script type="text/javascript">
var DpNewTicket_Options = DpNewTicket_Options || {};
DpNewTicket_Options.departmentId = 0;
DpNewTicket_Options.languageId = 0;

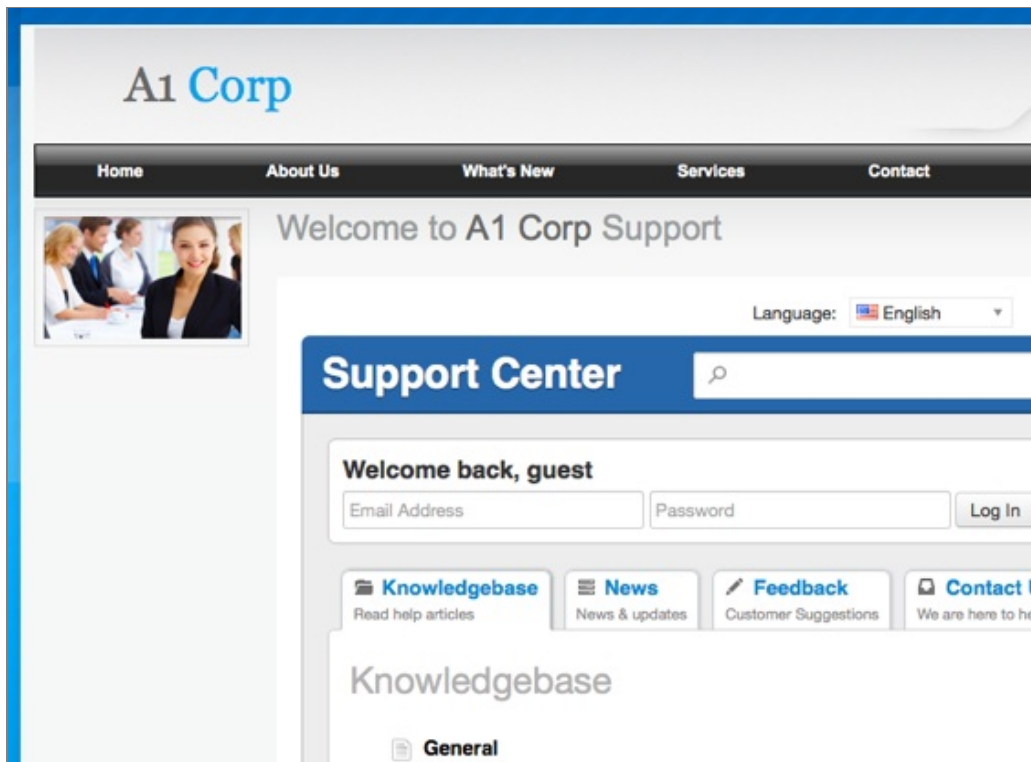
```

- 1 Go to **Portal > Embed**
- 2 In the **Contact Form** section, select the language for the form.
- 3 Choose whether to let the user select a department, or to make this form submit to a specific department only.
- 4 Copy all the code in the text box.
- 5 Insert the code where you want the form to appear.

You can use the ability to specify a department to put the most relevant forms on each area of your site. For example, you could put a form for your Sales department on a page about an upcoming product.

Embedding the whole portal

You can embed the whole portal within a page on your website.



- 1 Go to **Portal > Embed**.
- 2 Under **Embed your entire helpdesk**, set the embedding options.
 - **Hide the sidebar and border**(on by default)
 - Select which helpdesk section is displayed initially.
 - Select the initial language to display
- 3 Copy all the code in the text box.
- 4 Insert the code where you want the portal to appear.

To make the embedded portal look more like a part of your site, you could use the Portal Editor **color chooser** or edit the portal template.

9.6. Ticket ref codes

By default, a ticket's ID number is displayed to users in email notifications and the portal, and acts as the user's reference for that ticket.

The screenshot shows the 'Support Center' interface. On the left, a navigation pane includes 'Knowledgebase' and 'News'. The main content area shows a ticket titled 'Missing piece' with a 'Ticket Ref: 31' and a date 'Wed, 14th May 2014 9:35a'. Below this is a table with 'Assigned Agent' and 'Unassigned' columns, and a 'Department' row showing 'Support'. On the right, a 'Ticket' summary table lists details: ID 68, Status 'Awaiting Agent', Agent 'Acme Support Agent 1', Department 'Support Dept.', and Category 'Consumer'. Arrows highlight the ref code '31' and the ID '68'.

However, ticket ID numbers are assigned in sequence. That means they will reveal the total number of tickets you have had to your helpdesk, and let repeat users work out how many tickets you deal with. In addition, once you have a lot of tickets in your helpdesk, you will end up with long ID numbers that aren't readable or user-friendly.

You can choose to show **ref codes** to users instead of the ticket ID. A ref code is a string of numbers and letters that uniquely identifies a ticket, but isn't assigned sequentially, avoiding the problems mentioned above.

You can customize the ref code format to include information about the time the ticket was created; ticket reference `2014-01-26-AB379` is more meaningful than ticket ID `74983`.

To enable ref codes, go to **Tickets > Settings** and then the **Ticket Ref Codes** section, and check **Use ref codes**.

The 'Ticket Ref Codes' settings page includes a yellow informational box stating: 'Each ticket is given a unique ID number. You may also enable "ref codes" to show a reference string to users instead of the ID number. These ref codes are generally more readable and user-friendly. By default, ref codes are in the format "ABCD-1234-EFGH". You can customize the ref code format.' Below this, the 'Use ref codes' checkbox is checked, with a note: 'Show user the ref code instead of numeric ID's in email notifications, the web portal and URLs.' The 'Use a custom ref code' checkbox is unchecked, with a note: 'Enable to use a custom format for your ref codes.'

By default, ref codes are in the format `ABCD-1234-EFGH`.

If you check **Use a custom ref code** you can enter the format you want as a string of tokens, surrounded by angle brackets. For example, `<YEAR>-<MONTH>-<A><A><A><#><#>` would give you codes such as `2014-05-QZD45`.

Anything you enter that isn't in angle brackets will be the same in every code.

Here are the available tokens:

Token	Result
<A>	A random letter

<#>	A random number
<?>	A random letter or number
<YEAR>	Year as a four-digit number (e.g. 2013)
<MONTH>	Month as a two-digit number (01-12)
<DAY>	Day as a two-digit number (01-31)
<HOUR>	Hour in 24-hour time (00-24)
<MIN>	Minute as a two-digit number (00-59)
<SEC>	Second as a two-digit number (00-59)

You can also choose to have up to 8 digits added onto the end of the code. These are added sequentially.

Ticket Ref Codes

Each ticket is given a unique ID number. You may also enable "ref codes" instead of numeric ID's. These ref codes are generally more readable and user-friendly. By default, ref codes are in the format "ABCD-1234-EFGH". You can customize the format and the number of digits added to the end of the code.

☒ Use ref codes

Show users the ref code instead of numeric ID's in email notifications

☒ Use a custom ref code ?

Enable to use a custom format for your ref codes.

<YEAR>-<A><A><?><?><?>

and append 3 digits

0 digits
1 digit
2 digits
3 digits
4 digits
5 digits
6 digits
7 digits
8 digits

Here are some examples:

Format	Example codes
<YEAR>-<MONTH>-<DAY>- and append 3 digits	2014-10-02-001, 2014-10-02-002 (counting upwards)
<?><?><?><?><?><?>	1A45QR, 6BB73S
<YEAR>-<A><A><?><?><?>	2013-AB34Q, 2013-HXR81
ACME-<A><A><#><#><#><#>	ACME-AB2952, ACME-FV4541

10. Editing Templates

The most powerful way to customize the user-facing content of your helpdesk is to use DeskPRO's **template** system.

You can edit templates to change:

- the content of automatic notification emails
- the portal design and interface text

You can also add custom email templates.

Templates can include **phrases**: short, re-usable pieces of text, used to store something like the user greeting at the beginning of an email, or the name of a section of the portal.

A major advantage of storing text using phrases is that you can include the same phrase in many templates. DeskPRO's multi-language support is also based around phrases.

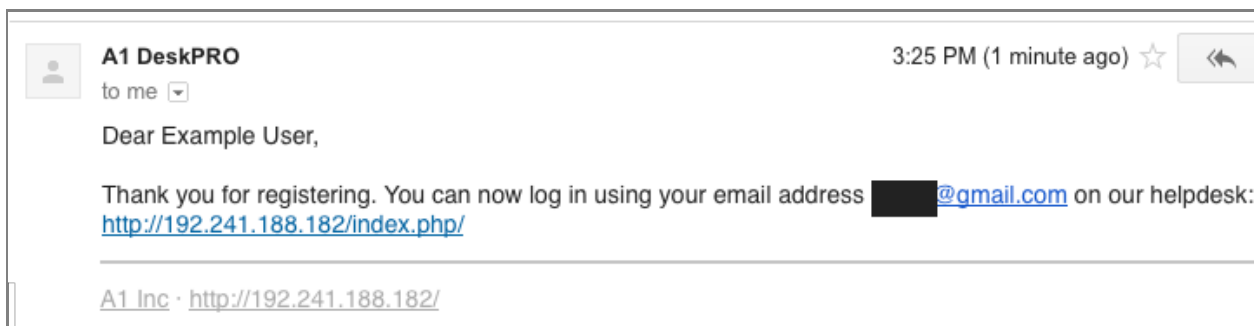
You can access **variables** from a template, enabling you to retrieve information about the user who is logged in, the ticket they're currently viewing, etc., and display that back to the user.

You'll need a basic understanding of HTML to edit templates. If you want to customize the portal design, you'll need a more advanced knowledge of HTML/CSS.

10.1. Introducing templates

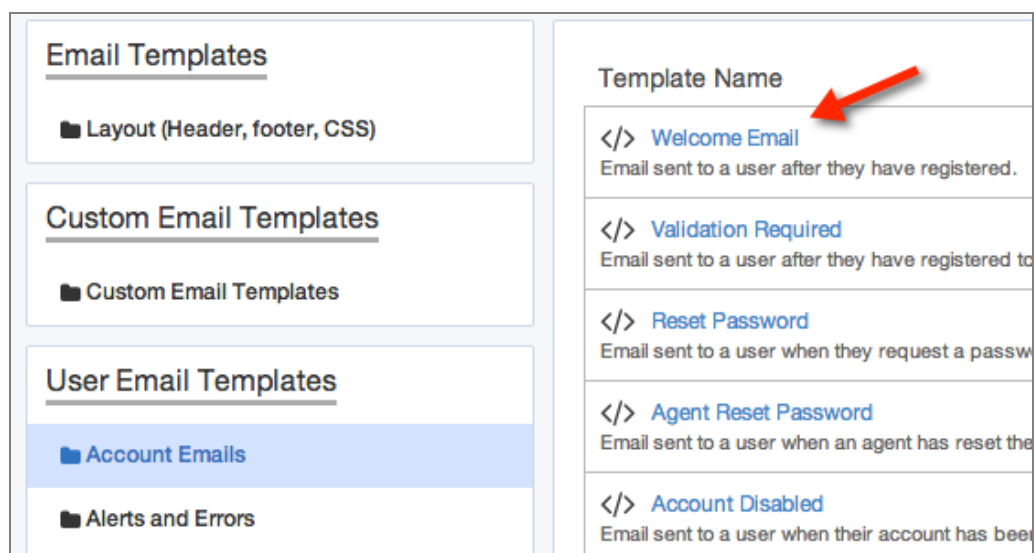
Here's an example of how to customize helpdesk content by editing a template and using phrases and variables.

Suppose you decide you want to edit the content of the automatic email users receive when they register an account. By default the email looks something like this:



As you'd expect, "Example User" is automatically replaced with the name of the user who's receiving the email. We'll see how this happens later.

Go to **Tickets > Email Templates** to find the corresponding template. Look under **User Email Templates** and you will see the **Welcome Email** template.



Click on the template to open the template editor window.

Welcome Email

Email sent to a user after they have registered.

Email Subject:

{{ phrase('user.email_subjects.register-welcome') }}

Email Body:

```
1 {{ phrase('user.emails.greeting') }}
2
3 <br /><br />
4
5 {{ phrase('user.emails.register-welcome') }}<br/>
6 <a href="{{ url_full('user') }}">{{ url_full('user') }}</a>
7
8 {% if not person.is_agent_confirmed %}
9 <br /><br />
10 {{ phrase('user.emails.register-agent-validation') }}
11 {% endif %}
```

Save

You'll see that the template contains code for the **Email Subject** and **Email Body**.

Let's look at the **Email Body**:

```
{{ phrase('user.emails.greeting') }}

<br /><br />

{{ phrase('user.emails.register-welcome') }}<br/>
<a href="{{ url_full('user') }}">{{ url_full('user') }}</a>

{% if not person.is_agent_confirmed %}
  <br /><br />
  {{ phrase('user.emails.register-agent-validation') }}
{% endif %}
```

You probably recognize that some of this is HTML markup: `<br />` for line breaks and `<a href=` to make a link.

The other parts, such as `{{ phrase('user.emails.greeting') }}` and `{% if not person.is_agent_confirmed %} ... {% endif %}`, are the template syntax.

`{{ phrase('X') }}` is the syntax to include a phrase in the email.

You can look up phrases by going to **Settings > Languages** Under **Installed Languages**, click on your default language, then **Edit Phrases**.

Default Language

English (English) ▼

Installed Languages

English (English) admin agent user

Title English
The language title as it a

Edit Phrases [Edit Phrases](#)

Under **User Interface Phrases**, click on **Emails** to see email phrases. The default content for `user.emails.greeting` is:

Dear {{to_name}},

When the email is sent, the content of the phrase is inserted into the email. Since the phrase contains the `{{to_name}}` variable, when the email is sent, the template system replaces this with the variable content, which is the name of the user.

If the email is being sent in another language, the translated version of `user.emails.greeting` from the corresponding language pack gets inserted instead. For example, here's the Spanish translation:

Phrase	Default	Translation	Custom
user.emails.greeting	Dear {{to_name}},	Apreciado/a {{to_name}}	Apreciado/a {{to_name}}

Let's suppose your marketing department has decided that user emails should start with "Hello ..." rather than "Dear ..."

You could replace `{{ phrase('user.emails.greeting') }}` with `Hello {{to_name}}`, but that would only change the greeting for this email type, and it wouldn't be translated into other languages.

A better solution is to go to **Setup > Languages** and use **Edit Phrases** to enter and save a custom version of the phrase.

user.emails.greeting Dear {{to_name}}, Dear {{to_name}},

[Reset](#)

All the other email templates which use the phrase will now use the custom version. You can also use **Edit Phrases** to enter a custom translation for any other languages you have installed.

user.emails.greeting Dear {{to_name}},preciado/a {{to_name}} Hola {{to_name}},

[Reset](#)

10.2. Twig syntax basics

The templating system DeskPRO uses is called [Twig](#).

Below are the features you'll need to know to do the most common forms of customization.

Variables

You insert a variable into a template with double curly braces:

```
{{ }}
```

There are dozens of variables available for DeskPRO.

Here are some examples:

```
{{ ticket.subject }}      - the subject line of the current ticket
{{ ticket.agent.display_name }} - the user-facing display name of the
                                agent assigned to the current ticket
```

You'll find a listing of common variables below.

Tags

Twig also includes **tags** which control the logic of the template.

For example, the `{% if %}` tag lets you have a section of your template that only displays *if* a certain condition is met.

This code on a portal page:

```
{% if app.user.id %}
  <p>You are logged in.</p>
{% endif %}
```

would display the text "You are logged in" if the template could retrieve a user ID - i.e. if the user is logged in.

Other useful tags you will see in templates:

- `{% else %}` - used in conjunction with `{% if %}` to display content if its condition is *not* met
- `{% for %}` - creates a loop while a certain condition is met; you'll see this used in portal templates for displaying a list of items
- `{% include %}` - includes the content of one template within another

See the official Twig documentation at <http://twig.sensiolabs.org/doc/templates.html> for more information about what tags can do.

Comments

You can leave comments to explain what your custom templates do. Comments will not affect how the template works; they are just for human reference.

```
{# this is a comment #}
```

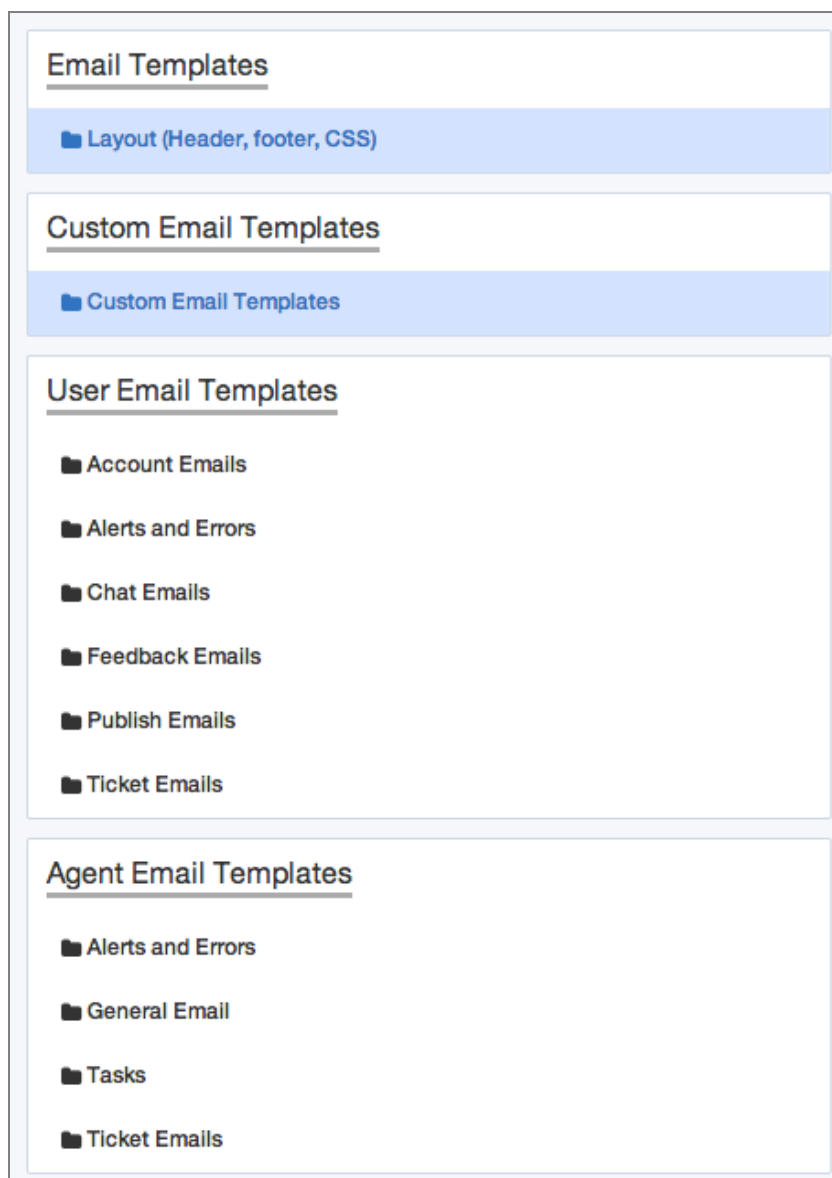
When you're troubleshooting a template, or trying to understand how the built-in templates work, it's useful to comment out a section of code to quickly disable it, and see how the template behaves without it. For example, here's

```
{# this comment wraps the code below  
{% for item in navigation %}  
    <li><a href="{{ item.href }}">  
        {{ item.caption }}</a></li>  
{% endfor %}  
#}
```

Don't use HTML comments within templates. If there are comments in a piece of inserted HTML, the two sets of comments can interact and cause mangled HTML display.

10.3. Customizing the email templates

You can access the email templates from **Tickets > Templates**.



The **Layout** templates affect the look of all emails by changing the HTML header/footer and CSS styling.

You can edit the built-in templates for the emails DeskPRO sends automatically.

If you're using a custom trigger to send an email for a particular situation, none of the default email templates may not be suitable. In that case, you can create a new **Custom Email Template**. Look at similar default templates to find variables and existing [phrases](#) you can use.

When creating a custom template, it's always best to store the user-facing text as phrases, rather than entering it directly into the template. Even if you only use one language in your helpdesk, using phrases can make it easier to maintain your templates.

For example, suppose your company slogan is "Simply the Best!" and you are going to include it in a lot of your outgoing emails. You should store it as a phrase or add it to the email header/footer. Then, when the slogan changes, you can change it in one place instead of having to update all your templates.

Templates example: custom email auto-response by department

Suppose your helpdesk is configured to send the default automatic email response when a user submits a new ticket.

The marketing department comes to you with another great idea: users who submit new tickets to the sales department should get a custom acknowledgement email, with a 10% discount code, to reward their interest in your company.

How can you implement this?

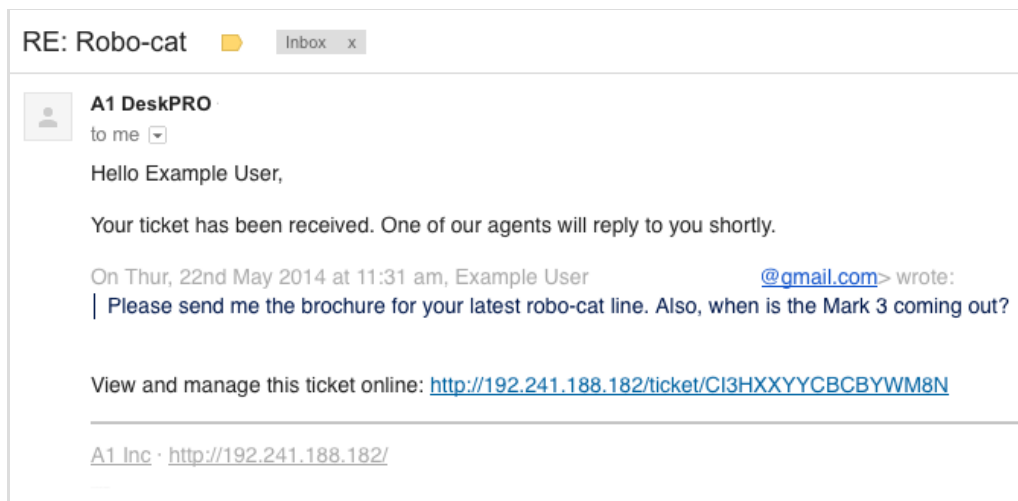
- 1 Got to **Tickets > Email Templates**, then click **Custom Email Templates**
- 2 Click **Create a New Email Template**
- 3 In the window that opens, enter a descriptive file name for the new template, e.g. `marketing-auto-response.html` . Save the new blank template.
- 4 You can use the template for the regular auto-response email as the basis for the new template. Under **User Email Templates**, click **Ticket Emails**, then **New Ticket Auto-Response**. Copy the code for the subject and body into a text editor.

Email Subject: `{{ phrase('user.email_subjects.tickets_re') }}`

Email Body:

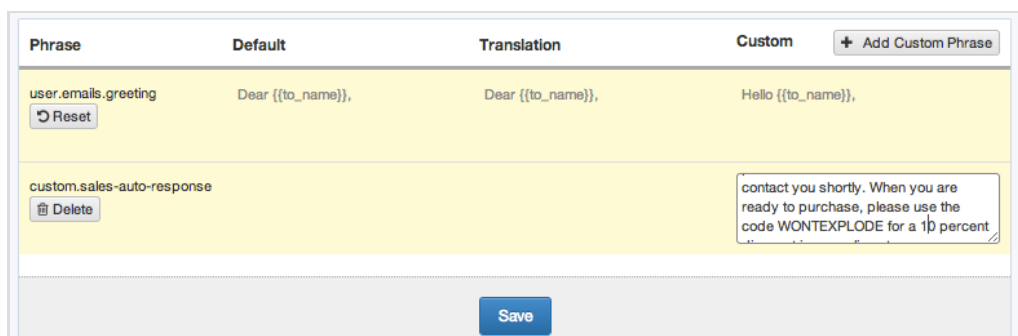
```
{{ phrase('user.emails.greeting') }}  
  
<br /><br />  
  
{{ phrase('user.emails.ticket_received') }}  
  
<br /><br />  
  
<dp:ticket-messages />  
  
{% if app.isPortalEnabled() %}  
  <br /><br />  
  
  {{ phrase('user.emails.ticket_access_ticket_online') }}  
  <a href="{{ ticket.link }}">{{ ticket.link }}</a>  
{% endif %}
```

This is an example of the output of the default template:



- 5 To change the user message, you can remove `{{ phrase('user.emails.ticket_received') }}` and replace it with a reference to a suitable custom phrase.

- 6 Go to **Setup > Languages**, click **All Custom Phrases** then **+ Add Custom Phrase**. Enter a suitable custom phrase, then save it as something descriptive, e.g. `custom.sales-auto-response` :



- 7 Back in **Tickets > Email Templates**, edit the custom `marketing-auto-response.html` and add the code from the default response.
- 8 Replace the default `{{ phrase('user.emails.ticket_received') }}` with `{{ phrase('custom.sales-auto-response') }}` :

```
{{ phrase('user.emails.greeting') }}  
  
<br /><br />  
  
{{ phrase('custom.sales-auto-response') }}  
  
<br /><br />  
  
<dp:ticket-messages />  
  
{% if app.isPortalEnabled() %}  
  <br /><br />  
  
  {{ phrase('user.emails.ticket_access_ticket_online') }}  
  <a href="{{ ticket.link }}">{{ ticket.link }}</a>  
{% endif %}
```

9

Now set your helpdesk so that it sends an email using the new template when a user submits a ticket to the sales department via email or the portal.

10.4. Customizing the portal templates

Editing portal and widget templates works along the same principles as editing email templates, but the templates tend to be more complicated.

If you are interested in redesigning the style of the portal, you will need HTML/CSS design skills. You can edit the **custom.css** template to override the default portal CSS.

10.5. Useful template variables

You may recognize that many variables used in templates are the same ones you use for snippets in the agent interface.

They're also used in the DeskPRO Query Language you use when writing custom reports.

Logged-in user information

These variables will only work in contexts where the user is logged: portal or widget, but not email.

```
{{ app.user.name }} user's name
{{ app.user.email_address }} user's email address
{{ app.user.organization.name }} the name of users organization
{{ app.language.id }} - ID of user's language
```

Language

```
{{ lang.id }}
{{ lang.flag_image }}
{{ lang.title }}
```

Search results

```
{{ res.type }}
{{ res.object.id }} .title
```

Agent

```
{{ agent.display_name_user }}
```

Tickets

```

{{ ticket.subject }}
{{ ticket.department.title }} & {{ ticket.department.full_title }}

{{ ticket.fieldX }} (for custom fields; replace X with the field id)

{{ ticket.product.title }} current product of ticket
{{ ticket.category.title }} current category of ticket
{{ ticket.workflow.title }} current workflow stage of ticket
{{ ticket.priority.title }} current priority of ticket
{{ ticket.id }} ticket ID number
{{ ticket.ref }} ticket ref code
{{ ticket.fieldX }} display a custom ticket field; replace X with field ID
{{ ticket.date_created }} time and date ticket was created e.g. 2014-03-14 11:38:17
{{ ticket.agent.display_name }} name of currently assigned agent (override name if set)
{{ ticket.person.display_name }} user's name
{{ ticket.person.fieldX }} display a custom user field; replace X with field ID

<dp:ticket-messages /> list of ticket messages - note this doesn't need to be surrounded with {{ }}

```

Ticket user information

```

{{ ticket.person.display_name }} name of user associated with ticket
{{ ticket.person.primary_email }} ticket's user's email
{{ ticket.person.organization.name }} ticket's user's organization
{{ ticket.person.organization_position }} ticket's user's position in organization
{{ ticket.person.fieldX }} for custom ticket fields; replace X with the field id

```

Ticket agent

```

{{ ticket.agent.display_name }}
{{ ticket.agent.primary_email }}
{{ ticket.agent_team.name }}

```

Ticket form

```

{{ f.title }}
{{ f.field_def.description }}

```

Article

```

{{ article.person.display_name_user }} (author)
{{ article.title }} title

```

Message

```
{{ message.person.display_name_user }}
```

News

```
{{ news.person.display_name_user }}  
{{ news.category.title }}  
{{ news.content }}  
{{ news.link }}
```

Download

```
{{ download.title }}  
{{ download.content_desc }}  
{{ download.filename }}  
{{ download.readable_filesize }}  
{{ download.person.display_name_user }} - author of download  
{{ download.date_created|date('full') }} - piping
```

Feedback

```
{{ feedback.status }}  
{{ feedback.person.display_name_user }}  
{{ feedback.message }}  
{{ feedback.rating }}
```

Category

```
{{ category.title }}  
{{ subcat.title }}
```

Participant

"Participant" means a CC'd user.

```
{{ part.person.id }}  
{{ part.person.email_address }}
```

For a complete listing of DeskPRO variables, see the [DPQL Field Reference](#) article on the DeskPRO Knowledgebase. Note that not all of these variables will be available from every template; the available variables depend on context. The best way to work out what variables are available is to look at the default templates.

11. Localization

We're committed to making DeskPRO suitable for international use.

There is currently support for 19 languages, with the ability to enter your own custom phrase translations.

You can also fully customize the date format to match how dates are written in your location.

11.1. Date & time settings

You can customize the format that DeskPRO uses to display dates and times in the **Dates and Times** section of **Settings**.

The formatting will apply to the agent interface and the user web portal.

Use the format characters from the table below to specify how DeskPRO should display dates.

For example, the date format `M j Y` means that when the interface needs to display a date, it will display something like "Jan 1 2015". If you wanted it to display "1st January 15", you would edit the Date format field to read `"jS F y"`.

Format character	Description	Example output
Day		
d	Day of the month, 2 digits with leading zeros	01 to 31
D	A textual representation of a day, three letters	Mon through Sun
j	Day of the month without leading zeros	1 to 31
l (lowercase 'L')	A full textual representation of the day of the week	Sunday through Saturday
S	English ordinal suffix for the day of the month, 2 characters	st, nd, rd or th. Works well with j
z	The day of the year (starting from 0)	0 through 365
Week		
W	ISO-8601 week number of year, weeks starting on Monday	Example: 42 (the 42nd week in the year)
Month		
F	A full textual representation of a month, such as January or March	January through December
m	Numeric representation of a month, with leading zeros	01 through 12
M	A short textual representation of a month, three letters	Jan through Dec
n	Numeric representation of a month, without leading zeros	1 through 12
Year		
Y	A full numeric representation of a year, 4 digits	Examples: 1999 or 2003
y	A two digit representation of a year	Examples: 99 or 03
Time		
a	Lowercase Ante meridiem and Post meridiem	am or pm

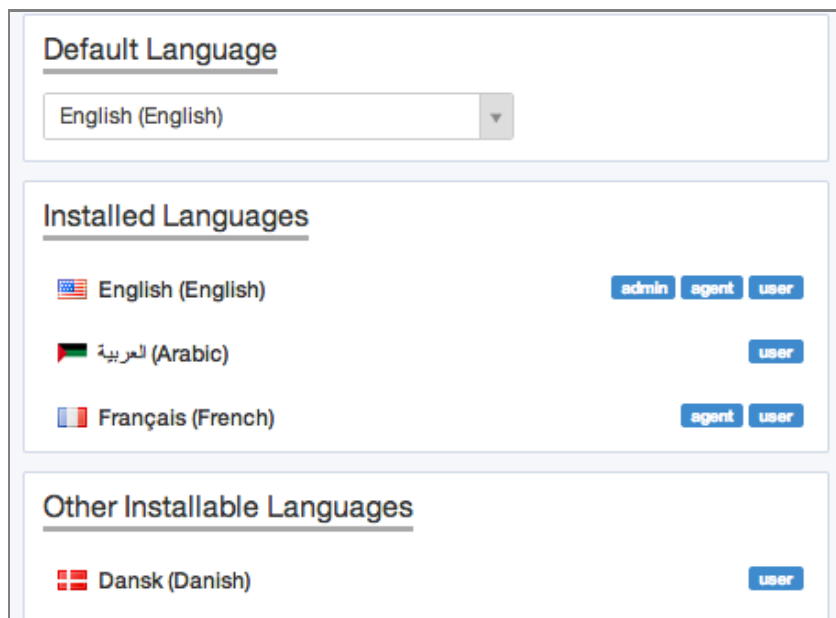
A	Uppercase Ante meridiem and Post meridiem	AM or PM
g	12-hour format of an hour without leading zeros	1 through 12
G	24-hour format of an hour without leading zeros	0 through 23
h	12-hour format of an hour with leading zeros	01 through 12
H	24-hour format of an hour with leading zeros	00 through 23
i	Minutes with leading zeros	00 to 59
s	Seconds, with leading zeros	00 through 59
Timezone		
e	Timezone identifier	UTC, GMT, Atlantic/Azores
I (capital i)	Whether or not the date is in daylight saving time	1 if Daylight Saving Time, 0 otherwise
O	Difference to Greenwich time (GMT) in hours	+0200
P	Difference to Greenwich time (GMT) with colon between hours and minutes	+02:00
T	Timezone abbreviation	EST, MDT
Full Date/Time		
c	ISO 8601 date	2004-02-12T15:19:21+00:00
r	RFC 2822 formatted date	Thu, 21 Dec 2000 16:01:07 +0200

11.2. Languages

DeskPRO has support for many international languages.

At **Setup > Languages** you can see:

- the default language of your helpdesk
- installed languages
- other installable languages



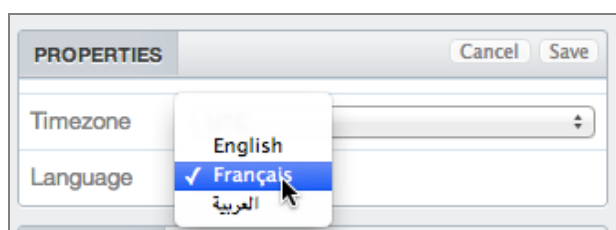
The level of translation support varies, as indicated by the blue labels on the right: most language packs currently have translations for **user** text (portal and emails) only. Some additionally translate the **agent** interface (and agent notification emails).

DeskPRO records the language of both tickets and users.

The language of a ticket is determined based on text character set. Users can pick their preferred language (from those you have installed) when they are logged in to the web portal, using the language selection control at the upper right.



Agents can change a user's language from within the agent interface, using the **Properties** section of the user record.



Changing a user's language will change the language in which they view the web portal, as well as the language of their notification emails. When a user's language changes, their tickets are switched to that language.

If the language for a ticket/user cannot be determined (e.g. the user has not yet logged in), the helpdesk default will be used.

Installing new languages

Installing a supported language is easy. Just go to the **Other Installable Languages** section, click on the language you want to install, then click **Install Language**.

After you've installed the language, you can click on it to view its settings.

The screenshot shows a form for configuring the English language pack. It includes a 'Title' field with 'English' and a description 'The language title as it appears to users.' Below this is an 'Edit Phrases' button. The 'Locale Code' field contains 'en_US' with the note 'This is the standard locale code for the language.' The 'Flag' field shows a US flag and 'us'. The 'Availability' section lists the areas covered by the language pack: 'The user portal and user emails', 'The agent interface and agent email notifications', and 'The admin interface and the reporting interface'. A 'Save' button is at the bottom.

If you want to make sure you support as many languages as possible, install every language pack, then click **Settings & Tools** at the bottom of the page and select **Automatically install new languages**.

You can select which flag to display for each language. This is useful if the language you've installed is used in multiple countries, but your customers who speak that language will mainly come from a particular country.

This screenshot shows the language settings form for Arabic. The 'Title' field contains 'العربية' and the 'Locale Code' is 'ar'. The 'Flag' dropdown menu is open, showing a list of countries with their flags and codes: 'le', 'il', 'in', 'io', 'iq', and 'ir'. The 'iq' option (Iraq) is currently selected. The 'Availability' section is partially visible on the right.

Translating names of custom values

For a full translation, you will need to provide the translated terms for any custom values you have entered:

- Department names (ticket & chat)
- Products
- Ticket priorities
- Feedback statuses
- Feedback types
- Knowledgebase categories
- Ticket fields
- Person (user) fields

For example, if you have enabled French, DeskPRO won't know the correct French name for your 'Returns' department, so you have to enter it.

- 1 Click on your newly-installed language to view detailed settings.
- 2 Click the **Edit Phrases** button, and on the following page, click on the first entry under **Resources**.
- 3 In each **Resources** section, click on each item in the **Translated** column, and enter a translation into the new language, then click **Save**.
- 4 Repeat until you have translated all your custom values.

Language: Français (French)	Default	Translated
Resources		
All Custom Phrases		
Ticket Departments		
Chat Departments		
Products		
	Support	Au secours!
	Sales	Prix réduit!
Save		

Editing user interface phrases

Almost all of DeskPRO's interface text (including text used in notification emails) is stored as **phrases** which are editable through **Setup > Languages**. This applies even if you only have one language installed.

You can enter a custom version of any phrase to change the user-facing text. You can use this to fix translation problems, or to rewrite user-facing text in your organization's style.

To change a phrase in any of our language packs:

- 1 Select the relevant language and click **Edit Phrases**.
- 2 Find the phrase you want to change in **User Interface Phrases**.
- 3 Enter a new version of the phrase in the **Custom** column.
- 4 Make sure you scroll down and click **Save**.

The **template** system, used to define the portal and notification emails, determines where each phrase is shown.

For example, look at **Tickets > Email Templates**. In all the templates, the body of the email begins:

```
{{ phrase('user.emails.greeting') }}
```

This tells DeskPRO to insert the `user.emails.greeting` phrase automatically at the beginning of every email.

If you wanted to change how you greet your users, you could enter a custom version of `user.emails.greeting`:

Phrase	Default	Translation	Custom
user.emails.greeting Reset	Dear {{to_name}},	Dear {{to_name}},	Howdy {{to_name}}!

See the chapter on [Editing Templates](#) for more information.

Changing default language

Simply select an installed language from the **Default Language** pulldown to change the default language.

From **Settings & Tools**, you can apply mass updates to tickets or users. This is useful if you want to apply a new language to users who have not explicitly picked a language.

For example, suppose you know that most of your users speak Russian and you have been evaluating DeskPRO with only English installed, during which time many users have registered.

Once you install Russian, instead of changing users one by one, you could just update all 'no language' users to Russian:

Mass update user language choice

Update all (no language) users

and set their language to Russian

Any users who haven't explicitly selected English as their language now have their language set to Russian.

Agent interface language setting

If a language has **agent** support, agents can select which language they wish to use from their **Preferences** in the agent interface.

You can't change an agent's language from the admin interface, except by logging in as the agent.

Crowdsourced translation

If the language you want is not available, or doesn't support all the interfaces you require, you can contribute a translation via our crowdsourcing system at <http://languages.deskpro.com>. You can use this system to fix any translation errors you find, so they will be corrected in future DeskPRO updates.

You will need to create an account or login via Facebook to contribute.

If you have other translation requirements, please contact us at support@deskpro.com.

12. Extending DeskPRO

There are several ways to integrate your DeskPRO helpdesk with other services, or add new functions.

We have provided a number of ready-made apps. Some of these are **authentication apps** which enable agents or users to log in using credentials from other services, rather than a DeskPRO account. Others provide more extensive integration with other services.

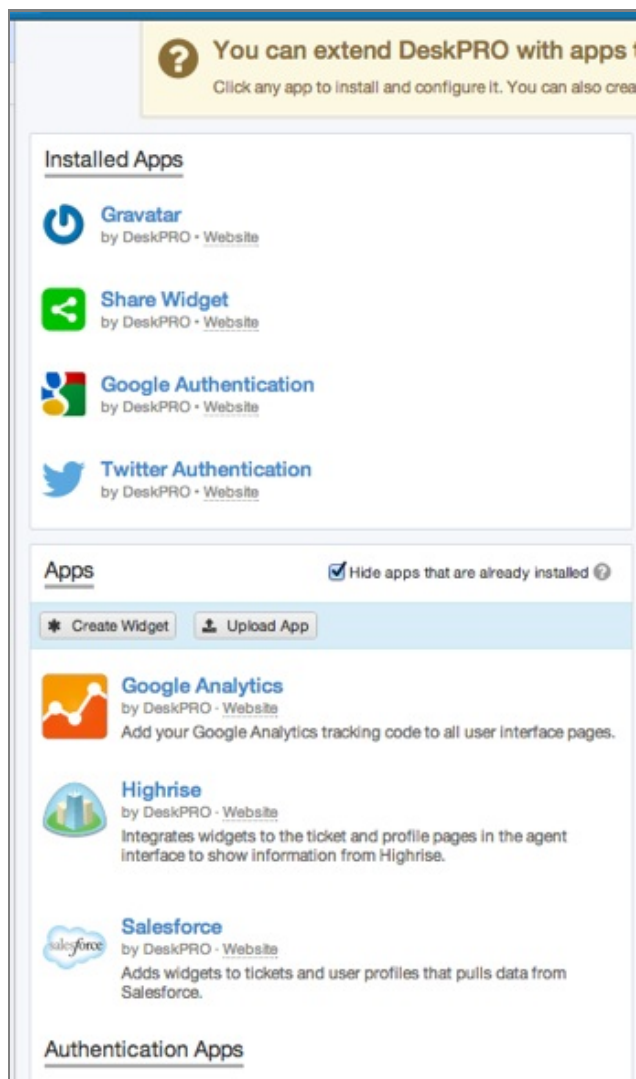
If the existing apps don't meet your needs, there are several ways to make custom integrations and extensions:

- [Web hook actions](#) enable DeskPRO automations (triggers, escalations, SLAs, macros) to make HTTP requests to external services. You can use this to integrate DeskPRO with any code that can accept HTTP requests; e.g. a web service with an API.
 - We offer a comprehensive [REST API](#) so that your code can access helpdesk data and carry out all the actions an agent can.
- **Widgets** are a simple way to customize the agent interface using HTML and Javascript.
- You can develop **full apps** using the same framework as the ready-made apps. These require more development skills, but give you more scope to change how the agent interface works.

12.1. Installing the ready-made apps

We have provided a variety of ready-made DeskPRO apps to provide new functionality and integration with other services.

Go to the **Apps** section of the admin interface to see the list of installed and available apps.



The [authentication apps](#) lower down the list allow users or agents to log in to DeskPRO with credentials from an external service (e.g. their Facebook account, a database you maintain, their network login at your company, etc).

Click on any app to install and configure it.

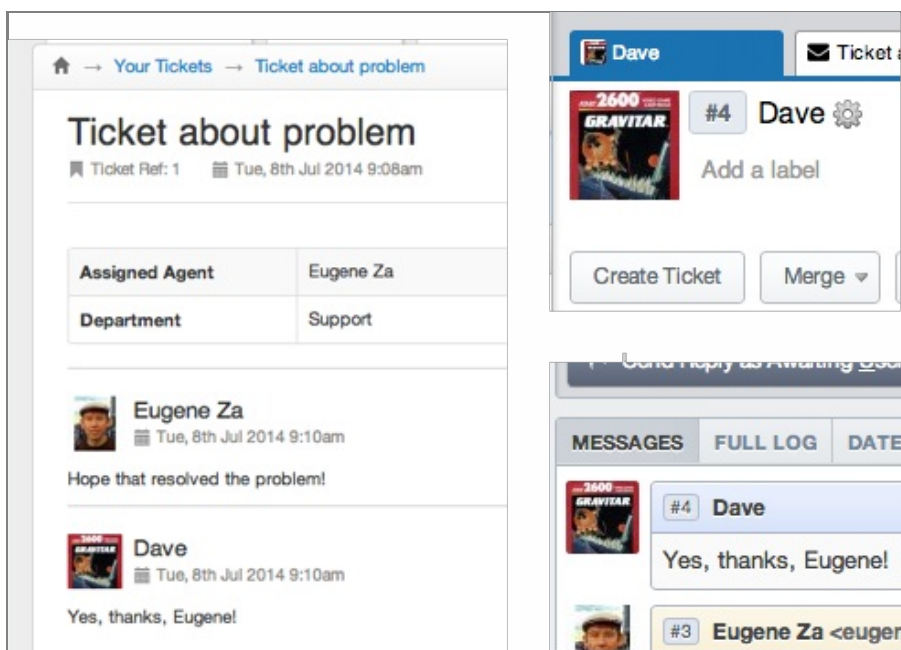
Most apps that operate with third-party services require you to provide an **API token** or some other authentication code from your account with the third-party service: if it is not clear where to get this from, consult the documentation for the service.

Gravatar

This app shows image avatars from the [Gravatar](#) service for any user or agent who has signed up for one.

The Gravatar app does not require any configuration. Simply select the app and click **Install** to add it to your helpdesk.

Gravatar is a shared profile service used on WordPress blogs and across many other sites. If a user or agent has a Gravatar image linked to their email address, it will be displayed next to their name in the agent interface and on the portal.

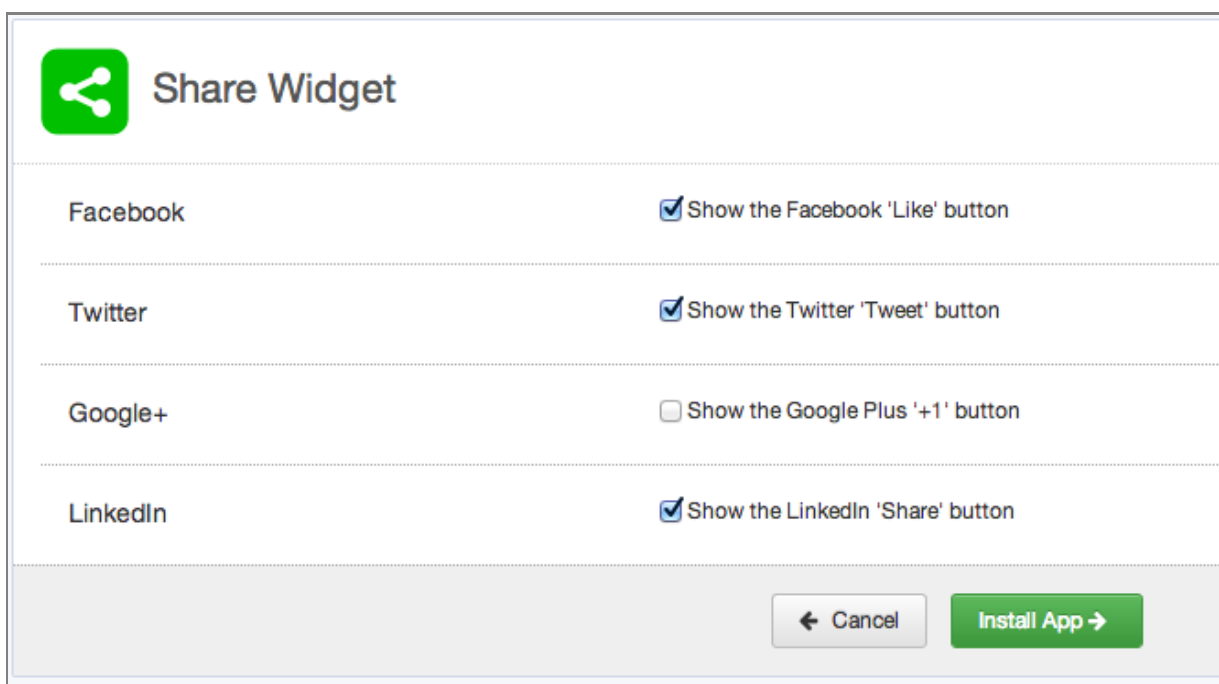


If the user or agent has already uploaded a profile picture, their Gravatar is not shown.

Share Widget

The share widget enables you to show buttons on the portal which let users quickly share selected categories of your portal content on social media sites (Facebook, Twitter, Google Plus and LinkedIn).

The only configuration the app requires is selecting which services you want to show buttons for.



The share buttons are shown on Knowledgebase articles, news posts and feedback items.

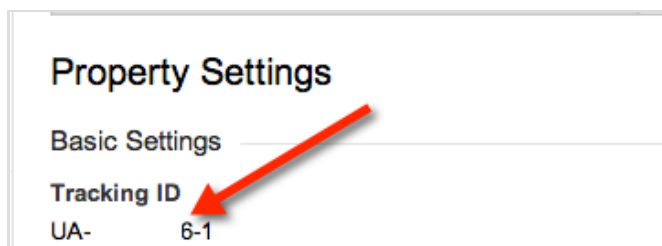
The screenshot shows a web page for an article titled "Gyrocopters". At the top, a breadcrumb trail reads: Home → Knowledgebase → Choosing a Product → Vehicles → Gyrocopters. Below the title, the author is listed as "Annie Edison", the date as "Thur, 12th Jun 2014", and there are "0 Comments". The main text of the article begins with "The Acme Gyrocopter is a perfectly cromulent vehicle, despite what you may have heard in the mass media or read in certain federally-mandated safety warnings." Below the text, there are two buttons: a green checkmark button labeled "This page was helpful" and a red X button labeled "This page was unhelpful". Underneath these buttons is a link that says "Subscribe to this article to get an email notification when it is updated." At the bottom, there is a "Share this:" section with four buttons: "Tweet" (with a count of 0), "Share" (Facebook icon, with a count of 0), "Share" (LinkedIn icon, with a count of 0), and "+1" (Google Plus icon, with a count of 0). This entire sharing section is highlighted with a red rectangular border.

Clicking one of the buttons takes the user to a screen where they can confirm that they want to Tweet/share/+1 a link to the item (if they are signed in to the relevant service - otherwise they will be prompted to sign in first).

Google Analytics

This app lets you use [Google Analytics](#) to understand traffic on your helpdesk portal.

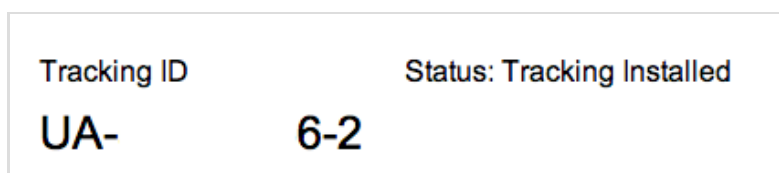
- 1 Go to <https://www.google.com/analytics/> and sign in to your existing Analytics account or sign up for a new account.
- 2 Create a new Analytics property for your helpdesk within Analytics: **Admin > Property pull-down > Create New Property**.
- 3 Enter the requested information and click **Get Tracking ID**.
- 4 The Tracking ID is shown on the following page. It's a code of the format UA-12345678-9. It's also shown next to the property e.g. **Admin > Property Settings**



- 5 In the DeskPRO admin interface, go to **Apps**, select Google Analytics and click **Install App**.
- 6 On the next screen, enter your Tracking ID.
Optionally, enable [Demographics & Interests reports](#) and [enhanced link attribution](#).

Click the **Install App** button.

- 7 The Google Analytics tracking code will now be added to the portal. To confirm it is installed, go to the Analytics interface, then in **Admin** select the property and go to **Tracking Info > Tracking Code**

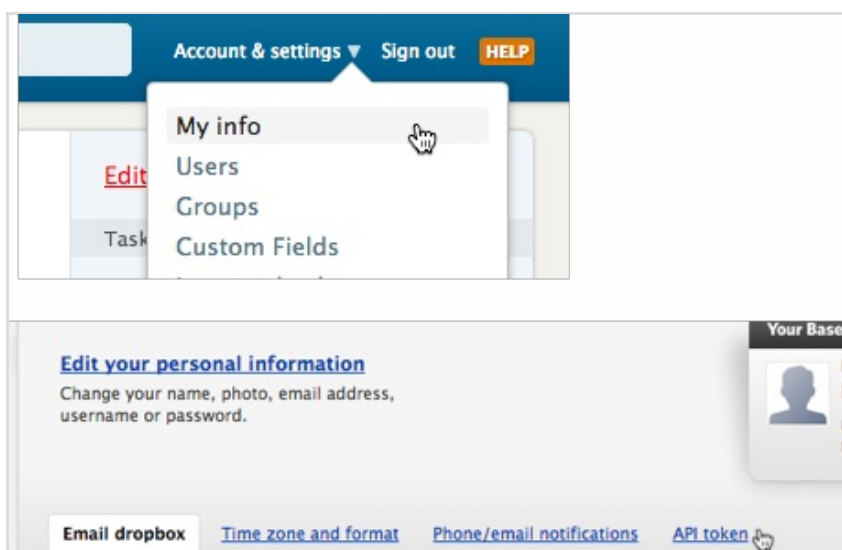


Once installed, you will be able to view traffic to the portal within Google Analytics.

Highrise

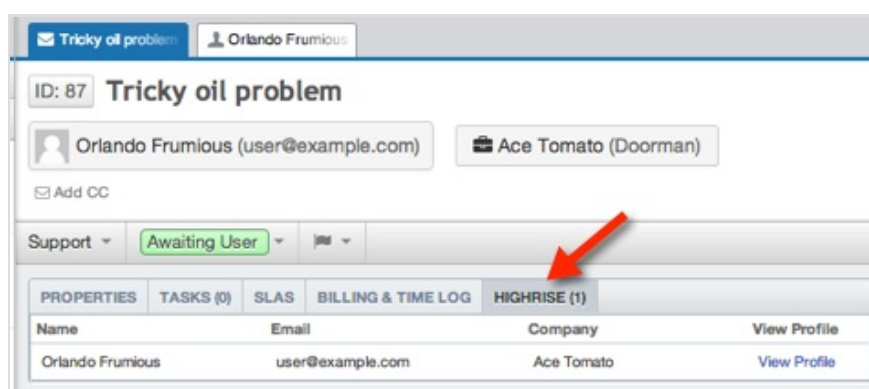
This app integrates information from the [Highrise](#) contacts tracking service into your agent interface.

- 1 Go to **Apps** and click Highrise, then click **Install**.
- 2 Enter the Highrise **API URL**: this is just the web address of your Highrise account, e.g. <https://acmel25.highrisehq.com>.
- 3 Enter the Highrise **API Token** from **Account & Settings > My Info > API Token**

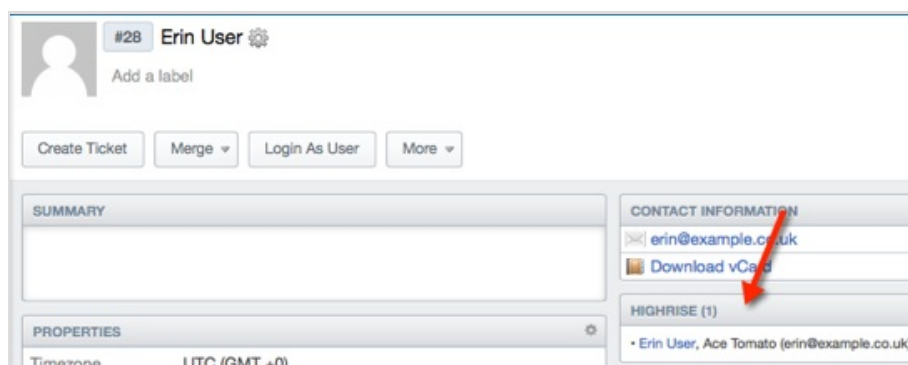


- 4 Select which of the two Highrise widgets you want to enable:

- The ticket widget adds a Highrise tab on tickets, providing a quick link to Highrise information.



- The profile widget adds a Highrise section when viewing a user's profile from the agent interface.



- 5 Click **Install App**.

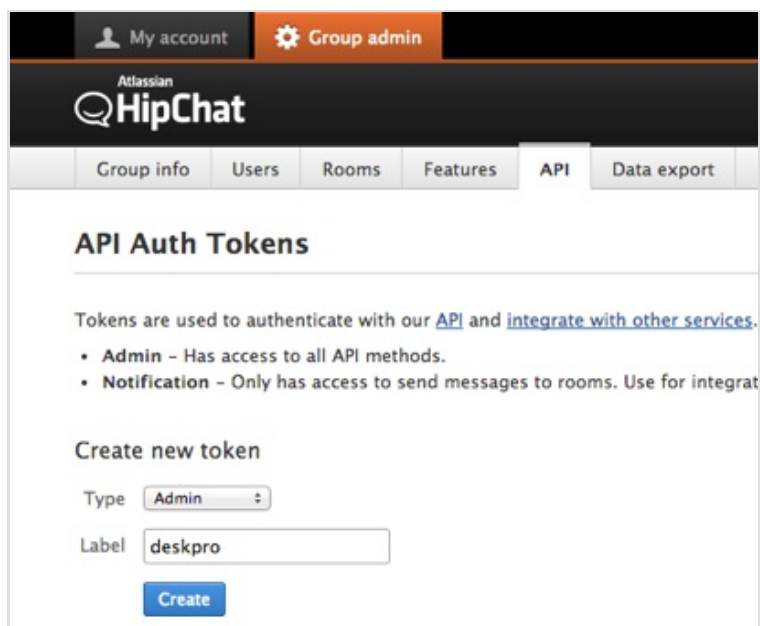
HipChat

Installing this app gives you a new [trigger](#) action which can send messages to a rooms on the [HipChat](#) IM service.

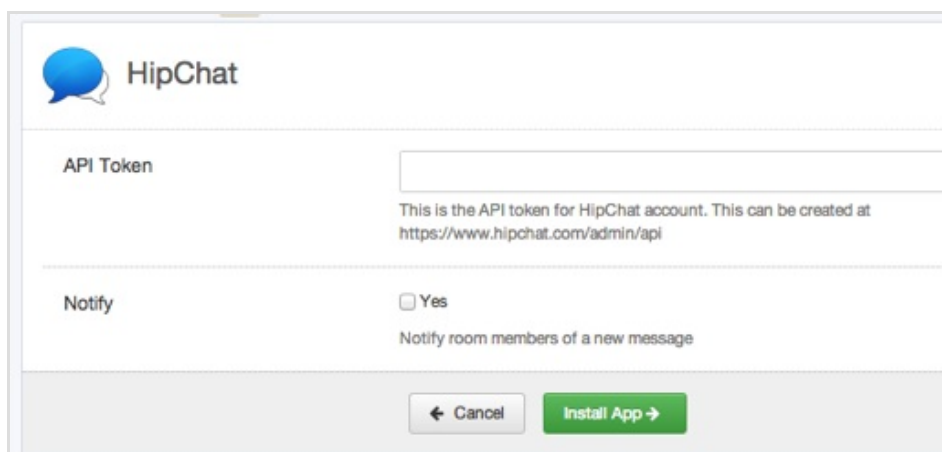
This enables you to set up useful IM alerts, for example, notification of important tickets.

You must sign up for a HipChat account first.

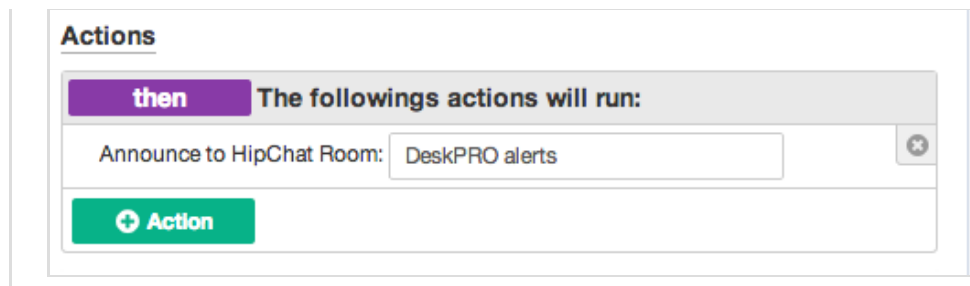
- 1 Log in to the HipChat web interface and go to **Group admin > API** or <https://www.hipchat.com/admin/api>.*
- 2 Under **Create new token**, make sure **Admin** is selected and enter a label so you can remember what the token is for.



- 3 Copy the API token.
- 4 In the DeskPRO admin interface, go to **Apps**, select **HipChat** and click **Install**.



- 5 Paste the **API Token** into the field. Check **Notify** if you want HipChat users who have joined the room to get an audio/pop-up notification when a new message is sent.
- 6 Click **Test Settings** to check that the API token is correct.
- 7 Click **Install App**.
- 8 You can now use the **Announce to HipChat Room** action in triggers.



The screenshot displays the 'Actions' configuration panel in DeskPRO. At the top, there's a purple 'then' button followed by the text 'The followings actions will run:'. Below this, there's a configuration row for the 'Announce to HipChat Room' action, with a text input field containing 'DeskPRO alerts' and a small 'x' icon to its right. At the bottom of the panel is a green button with a plus icon and the word 'Action'.

Make sure that you have entered the name of the HipChat room exactly as it appears in HipChat.

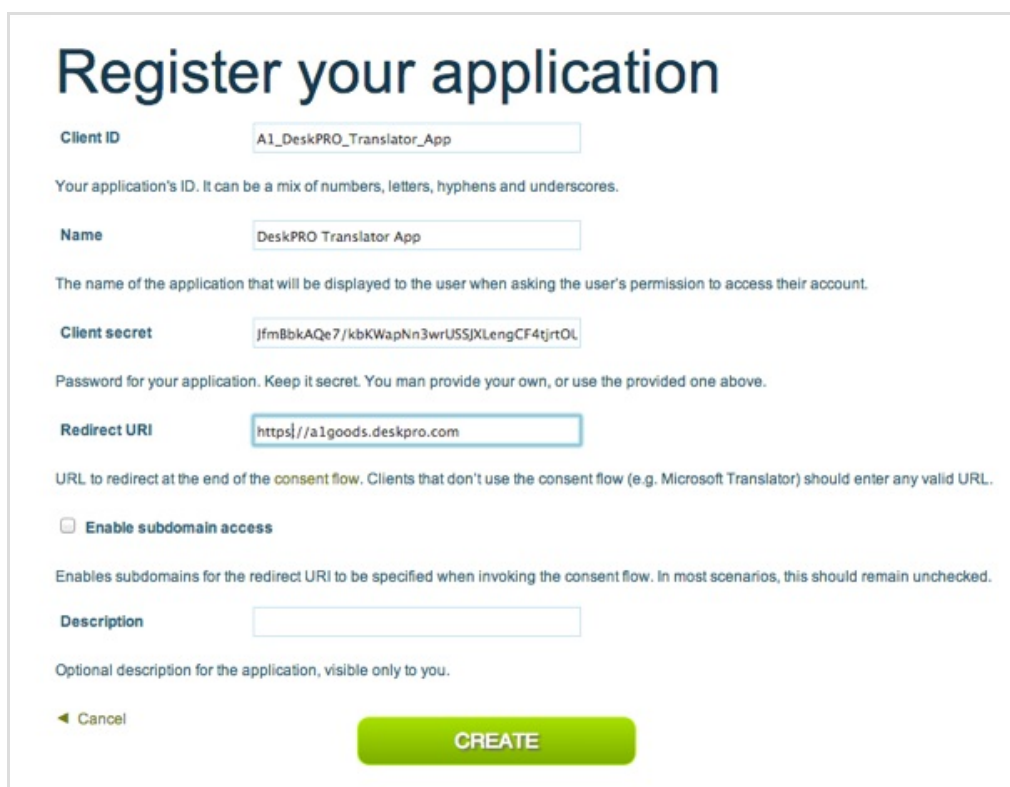
* Note, you must use an API token from the HipChat "Group admin" settings. The API access token for your individual account will not work. You may need to consult the person who administers your HipChat setup.

Microsoft Translator

Installing the Microsoft Translator app adds automatic message translation features to the agent interface.

To install this app, you first need to sign up with the Microsoft Translator service and get the needed credentials.

- 1 Sign up for a [Microsoft account](#), or log in to your existing Microsoft account.
- 2 Sign up for [Windows Azure Marketplace](#), or log in to your existing account.
- 3 Subscribe to the [Microsoft Translator API](#). You can select a free subscription which offers 2 million characters of translation per month.
- 4 In the lower navigation area of the [Windows Azure Marketplace](#), go to **Develop > Register Your Application**.
- 5 Enter the **Client ID** and **Name**. Copy the **Client Secret**. Under **Redirect URI**, put a valid web address such as the URL for your helpdesk or website.



Register your application

Client ID
Your application's ID. It can be a mix of numbers, letters, hyphens and underscores.

Name
The name of the application that will be displayed to the user when asking the user's permission to access their account.

Client secret
Password for your application. Keep it secret. You can provide your own, or use the provided one above.

Redirect URI
URL to redirect at the end of the consent flow. Clients that don't use the consent flow (e.g. Microsoft Translator) should enter any valid URL.

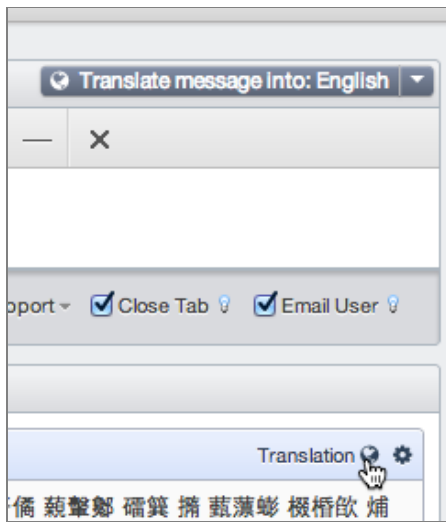
☐ **Enable subdomain access**
Enables subdomains for the redirect URI to be specified when invoking the consent flow. In most scenarios, this should remain unchecked.

Description
Optional description for the application, visible only to you.

[Cancel](#) [CREATE](#)

- 6 You can now install the app by going to **Apps** in the DeskPRO admin interface, selecting Microsoft Translator and clicking **Install**.
- 7 Enter the **Client ID** and **Client Secret** and click **Save**.

The Translator functions are now available from the agent interface. Agents can translate outgoing messages they are entering into the reply box, or translate messages from a ticket's history.



Remember that the Translator app can be useful to help agents understand messages in a language they don't speak, but they should be cautious about sending translated messages since machine translation is often far from perfect.

JIRA

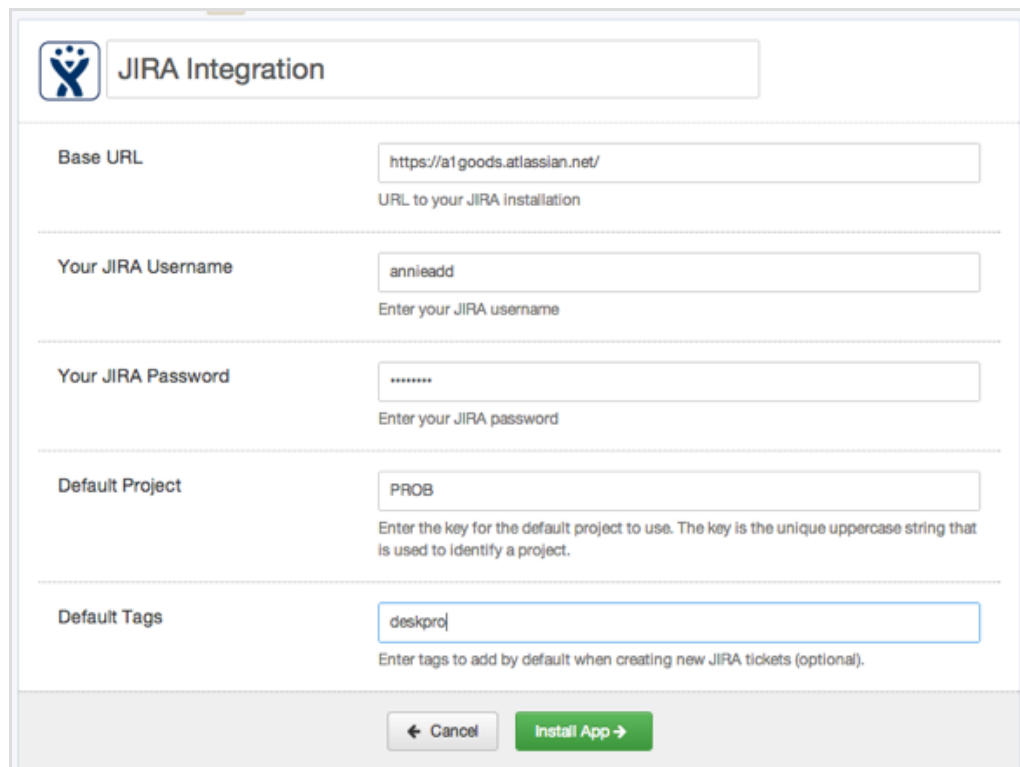
The JIRA integration app enables your agents to export tickets to issues in an installation of the [Atlassian JIRA](#) issue tracker software.

You can optionally set a default project, tag(s) or priority to use when creating the issue in JIRA.

The integration works with either the Download or OnDemand (hosted) versions of JIRA.

To install this app:

- 1 Go to **Apps** within the DeskPRO admin interface and click **JIRA Integration**.
- 2 Enter the requested information.



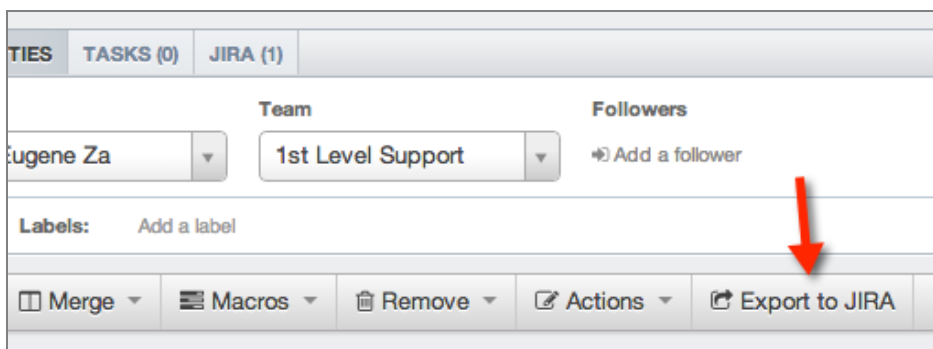
The screenshot shows the 'JIRA Integration' configuration form. It includes the following fields and instructions:

- Base URL:**
URL to your JIRA Installation
- Your JIRA Username:**
Enter your JIRA username
- Your JIRA Password:**
Enter your JIRA password
- Default Project:**
Enter the key for the default project to use. The key is the unique uppercase string that is used to identify a project.
- Default Tags:**
Enter tags to add by default when creating new JIRA tickets (optional).

At the bottom, there are two buttons: 'Cancel' and 'Install App'.

- 3 Click **Test Settings** to make sure that you entered the correct URL, username and password of your JIRA installation.
- 4 Click **Install App**.

The app adds an **Export to JIRA** button in the agent interface when you open a ticket in the content pane.



The screenshot shows the ticket content pane with the following elements:

- TABES:** TASKS (0) JIRA (1)
- Team:** Eugene Za (dropdown), 1st Level Support (dropdown)
- Followers:** Add a follower
- Labels:** Add a label
- Actions:** Merge, Macros, Remove, Actions, **Export to JIRA** (highlighted with a red arrow)

Click the button to create a linked issue within JIRA based on the ticket details.

Export to JIRA

Issue Title * [QDCR-4660-MWKC] Ticket about problem

Project * PROBLEMS Type * + New Feature

Priority * Blocker Assign To * Annie Spam [Admi...]

Description *

Hope that resolved the problem!

Dave(System) at 2014-07-08 09:10:40

Yes, thanks, Eugene!

Dave(System) at 2014-07-08 12:43:47

Please be sure to check the contents for any confidential information before exporting. You can also use the following "smart tags" to add DeskPRO relevant information to this issue.

1. @agent: The current agent
2. @department: The department in which the ticket is raised.
3. @ticket: The current ticket
4. @customer: The customer

Labels

deskpro

Add some tags/labels to better classify this issue

Reset Export

You can select the JIRA project, type and priority, as well as which JIRA user should be assigned the issue. The DeskPRO ID or ref code is added to the JIRA issue title. The issue description is prefilled with the ticket's full message history, which you can edit.

Once you've created a linked issue, you can view it in the ticket's JIRA tab.

Support ▾

Awaiting Agent ▾

4 ▾

 ▾

PROPERTIES

TASKS (0)

JIRA (1)

#	Issue Key	Status	Assigned To	Last Activity	Actions
1.	PROB-2	To Do	Nobody	a few seconds ago	 

The speech bubble icon in the **Actions** section of the tab adds a comment to the linked JIRA issue. Any comments added to the issue in JIRA are imported into DeskPRO as ticket notes.

The X icon unlinks the ticket from the JIRA issue.

Salesforce

The Salesforce app provides quick access to Salesforce customer profiles from the agent interface.

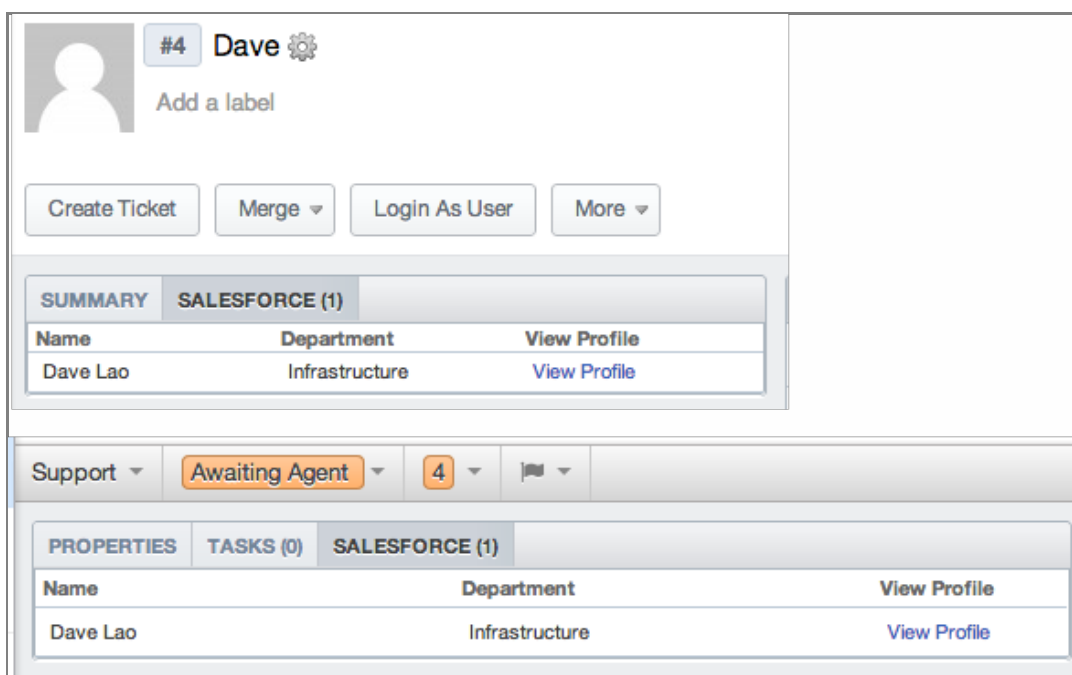
Note that your Salesforce account must include API access for this app to work. This is available to Enterprise or Performance accounts, and to Professional accounts for an extra fee; access may not be enabled for a trial account.

To install this app:

- 1 Go to **Apps** within the DeskPRO admin interface, select the Salesforce app and click **Install App**.
- 2 Enter your Salesforce username (usually an email address), password and API security token*.
- 3 Select whether you want a Salesforce widget to be displayed on tickets, user profiles, or both.

* The only way to get your API token from Salesforce is via email. If you don't have a token, go to **Your Name > My Settings > Personal > Reset My Security Token** from the Salesforce Web interface. Doing this will invalidate any previous tokens, so you will need to enter the new token for any other apps or services which use the API. Changing your Salesforce password also resets the token.

Once the app is installed, you will be able to see links to Salesforce customer profiles that match the DeskPRO user on tickets and user profiles (depending on which widgets you enabled).



Twilio SMS

The Twilio app enables you to send SMS text message notifications to agents' cellular/mobile phones as an action when automating the helpdesk.

To install this app you need to have a Twilio account and have registered a phone number with Twilio.

- 1 Go to your **Account Settings** on the Twilio site and copy down your Account SID and Auth Token (click the padlock icon to reveal it).
- 2 Go to **Apps** within the DeskPRO admin interface, select the Twilio app and click **Install App**.
- 3 Enter the SID, Auth Token and From number (the number that SMS messages will appear to originate from, which must be registered with Twilio). Be sure to use the flag icon to select the country of the phone number.
- 4 Click **Install App**.

You will now have SMS actions available for helpdesk automation. See the section [Sending SMS using automatic actions](#) for details.

Clickatell SMS

The Clickatell app enables you to send SMS text message notifications to agents' cellular/mobile phones as an action when automating the helpdesk.

To install this app you need to have a Clickatell Developers' Central account.

- 1 Log in to your account on the Clickatell site and copy down your API ID (from **APIS > Manage APIs**).
- 2 Go to **Apps** within the DeskPRO admin interface, select the Clickatell app and click **Install App**.
- 3 Enter your Clickatell username and password and the API ID.
- 4 Click **Install App** again.

You will now have SMS actions available for helpdesk automation. See the section [Sending SMS using automatic actions](#) for details.

Authentication apps

There are authentication apps for the following services:

- Microsoft's **Active Directory** network sign-on and the similar open **LDAP** standard
- **Database** - use credentials stored in an existing database: MySQL, PostgreSQL, SQLite, MsSQL and Oracle are supported
- **eZ Publish** and **Joomla** content management systems
- **Facebook**, **Google** and **Twitter**
- **phpBB**, **vBulletin** and **Xenforo** forum software

12.2. Using web hook actions

DeskPRO can use a **web hook action** (as part of a trigger, escalation or SLA) to interact with external services that accept commands using the web's [HTTP](#) protocol.

Many services offer HTTP APIs, so this provides a relatively simple way to have DeskPRO interact with other software you use (or custom software your organization has developed). For example, if a suitable API is available, you could post a warning on your intranet when a ticket fails an SLA.

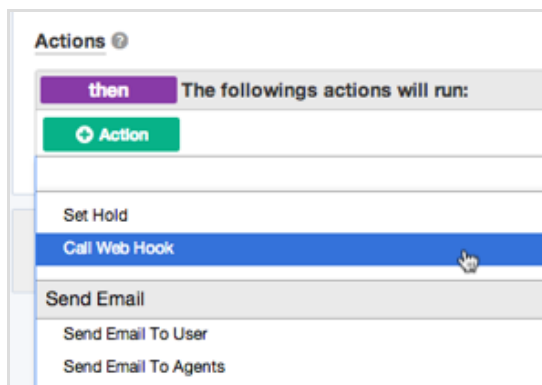
When the action runs, it sends an HTTP request using the HTTP method you specify from POST, GET, PUT and DELETE. A POST, PUT or DELETE request includes a representation of the relevant ticket in the request parameters. This enables the external service to use the information about the ticket. A GET request has an empty body.

See [this Knowledgebase article](#) for an example of the format.

Note that this option can only send data *from* DeskPRO *to* an external service. If you want an external service to change data within your helpdesk, you will need to use the DeskPRO API.

To create a web hook:

- 1 Select the **Call Web Hook** action type within a trigger, escalation, SLA or macro.



- 2 Fill in the requested settings:

- **URL** means the URL where the external service accepts requests
- **username and password** to authenticate with the external service (if required)
- **custom headers and data** that will be included in the request (consult the documentation for the API/service you are using to select appropriate values for these)
- **timeout** specifies how long in seconds DeskPRO will wait for the HTTP request to be acknowledged by the external service before giving up.

- 3 Click **Save**.

Whenever the action runs, DeskPRO will make the specified HTTP request.

To help with debugging, you can view the HTTP requests that an action is sending by using a tool such as [RequestBin](#), to collect and display requests in a readable way. You will need to temporarily change the URL for the action to the address provided by RequestBin.

Using custom headers and data

You can send custom values in the HTTP headers by entering them in the **Custom Headers** field of the web hook action.

The format is:

```
X-Custom-Header: value
X-Custom-Header2: value2
```

You can also send custom data in the body of the post, to be included in the **custom_data:** parameter.

You can use **variables** to send particular properties of the ticket that triggered the action in the custom headers or data.

For example, when creating the web hook action you could enter a Custom header of:

```
X-Deskpro-TicketId: {{ ticket.id }}
```

Then the HTTP headers sent due to an action on the ticket with ID 5 would include "X-Deskpro-TicketId: 5".

The syntax and possible values are the same as DeskPRO uses for snippets in the agent interface and when sending messages with actions.

Custom headers:	<pre>X-Custom-Header: acme1 X-Deskpro-TicketAgentTeam: {{ ticket.agent_team.name }}</pre>
Custom data:	<pre>{{ ticket.id }}, {{ ticket.subject }}, {{ ticket.department.title }}</pre>

Ticket variables

```
{{ ticket.subject }}
{{ ticket.department.title }}
{{ ticket.product.title }}
{{ ticket.category.title }}
{{ ticket.workflow.title }}
{{ ticket.priority.title }}
{{ ticket.id }}
{{ ticket.fieldX }} (for custom fields; replace X with the field id)
{{ ticket.ref }}
{{ ticket.date_created }} (time and date ticket was created e.g. 2014-03-14 11:38:17)
```

User variables

```
{{ ticket.person.display_name }} (user's name)
{{ ticket.person.primary_email.email }} (user's email)
{{ ticket.person.organization.name }} (user's organization)
{{ ticket.person.organization_position }} (user's position in organization)
{{ ticket.person.fieldX }} (for custom fields; replace X with the field id)
```

Agent variables

```
{{ ticket.agent.display_name }}  
{{ ticket.agent.primary_email.email }}  
{{ ticket.agent_team.name }}
```

Note that if you are sending a custom header using a variable which has no value, the header will not be sent at all (rather than a blank header being sent).

For example, if you have a custom header like this:

```
X-Deskpro-TicketTeam: {{ ticket.agent_team.name }}
```

but the ticket is not assigned to any team, no X-Deskpro-TicketTeam header will be sent.

12.3. Using the DeskPRO API

DeskPRO has a full REST API allowing access to the functions of the helpdesk using web protocols.

The API enables you or your developers to have your external code retrieve helpdesk information or act like an agent.

See the [API section of the DeskPRO Knowledgebase](#) for full details.

You can explore the DeskPRO API and quickly try out the available operations using our API browser. This gives you a web interface to interactively enter API requests and see the results, so you can test how the API works.

Go to `yourhelpdesk/api/api.html` to see the API browser. You will need to generate and enter an [API key](#) to use the browser.

Managing DeskPRO API keys

For external code to access the DeskPRO API, it will need to use a DeskPRO **API key**.

Bear in mind that anyone with access to one of your API keys can change data in your helpdesk, so you should be very careful about sharing or publishing them.

When you create an API key, you must select what privilege level it has:

- **Superuser Key:** can act as any agent, as specified in the API request - optionally, you can choose a default agent that will be used if no agent is specified. You can optionally give a superuser key access to use admin functions.
- **Normal Key:** API requests are treated as if they were performed by a specific agent.

If you enter a superuser key in the API browser, note that operations will always be carried out as the default agent; if you don't set a default agent for the key, it won't work in the API browser.

If it's not important for the app to be able to act on behalf of your human agents, you may want to create a dedicated agent account for it.

The screenshot displays the 'API Keys' management page. On the left, a sidebar shows '2 Api Keys' and a list with a key named 'Test'. The main area is the configuration form for the selected key. It includes a 'Description' field with the value 'Test', a note about its use, an 'API Key Code' field showing a long alphanumeric string, and a 'Regenerate API Key' button. The 'API Key Type' is set to 'Specific Agent' (selected with a radio button). Below this, a light blue box states 'Perform all API requests as a specific agent.' and a dropdown menu is set to 'Shirley Bennett'. At the bottom, there are 'Save' and 'Delete' buttons.

To generate a key:

- 1 Go to **Apps > API Keys**.
- 2 Click **+ New API Key**.
- 3 Enter a brief **Description** which explains what the key will be used for.
- 4 Set if you want the API Key to be a Superuser Key or a Specific Agent key.
- 5 Click **Save**. Click on the key in the list at the left to view the API key.

You should make a separate key for each external service/app that will be accessing your helpdesk through the API.

Note that you can test which API key has been used in the criteria for a trigger or other automated action.

To help with app development and monitoring API usage, you can see a log of the last 50 API requests in the **API Key Logs** tab.

If you think a key has become known to somebody you don't trust, you can **Delete** or **Regenerate** it. Regenerating a key means its description is kept, but the key code is changed and the old one will no longer work.

The screenshot shows the 'API Key' configuration page in DeskPRO. It features a form with the following sections:

- Description:** A text input field containing 'Intranet integration'. Below it, a note states: 'Used as note for this API key. You could use this field as a reminder of what this API key will be used for.'
- API Key Code:** A field displaying the code '3:XABDQQ2X3S6BWT2ZXWHHSCXK6'. To the right is a green button labeled 'Regenerate API Key'.
- API Key Type:** Two radio button options: 'Super User' (which is selected) and 'Specific Agent'.

A light blue informational box contains the text: 'A super user key is not associated with a particular agent. Every request must specify the agent performing the action.'

At the bottom of the form are two buttons: a blue 'Save' button and a red 'Delete' button.

12.4. Creating your own widgets and apps

See the [Apps Developer Guide](#) on the DeskPRO Knowledgebase for how to develop widgets. You will need a basic understanding of HTML and Javascript. We are working on expanding the Guide with details of how to develop full apps.

13. Chat

DeskPRO offers IM-style live chat to allow users to message your agents. Users can access chat from your portal or from a chat widget which you can embed into your website.

Use **Chat > Setup** to enable or disable the chat system altogether. This page also provides the code for embedding the chat widget - see the section on [Website integration](#) for details.

Note that by default, the chat widget (either on the portal or embedded) won't appear unless there is at least one agent signed into chat from the agent interface.

You can customize the information stored for a chat session using custom chat **fields**, under **Chat > Fields**. These work much like custom ticket fields.

You can also manage the [labels](#) that agents have applied to chats under **Chat > Labels**.

13.1. Chat departments

Chat departments are used to control access to chats in conjunction with usergroups and agent permission groups. See the section on [Department permissions](#) for more details.

Chat departments are a separate system from ticket departments - creating a ticket department will not create the corresponding chat department. Manage them under **Chat > Departments**

14. Labels

Agents can apply **labels** from the agent interface. A label is a text tag that can be applied to the following helpdesk content:

- Tickets
- Users
- Organizations
- Downloads
- News posts
- Feedback

Labels are not shared between different content types. Agents can see each others' labels.

Labels have the following benefits:

- 1 Searches from the agent interface and portal match labels. For example, a ticket labelled "engine" would show up in a search for "engine" even if the word "engine" wasn't mentioned in the ticket messages.
- 2 Agents can use labels to filter tickets from the agent interface filter pane.
- 3 Agents can view labels in the list and content pane. This gives them more information about tickets at a glance, especially if you apply [label colors](#).
- 4 Labels applied to Publish content can help users navigate the portal.

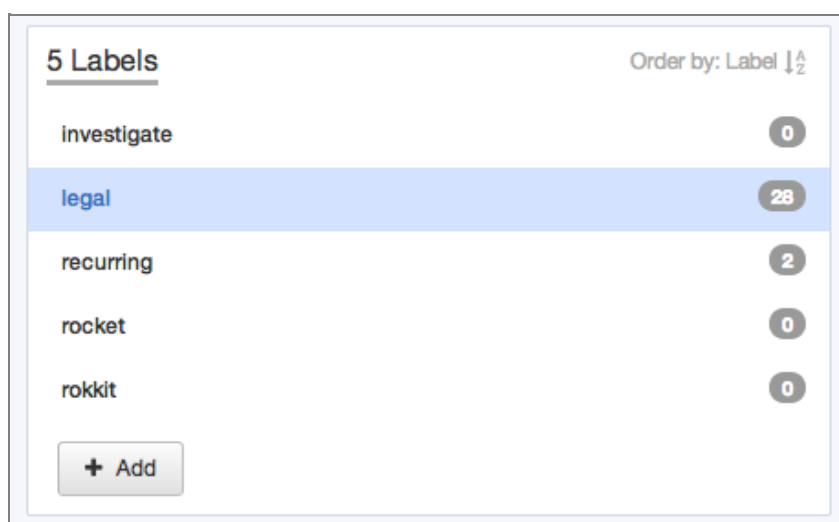
You can apply labels with automatic actions. For example, you might want to set up a **Ticket Update** trigger so that when a ticket is assigned to a certain **Product**, a label of the product name is applied. That way agents can quick search for the product to find matching tickets, even if they don't mention the full product name directly.

14.1. Managing labels and controlling agent label creation

To view and edit labels, go to the **Labels** section for the relevant type of content:

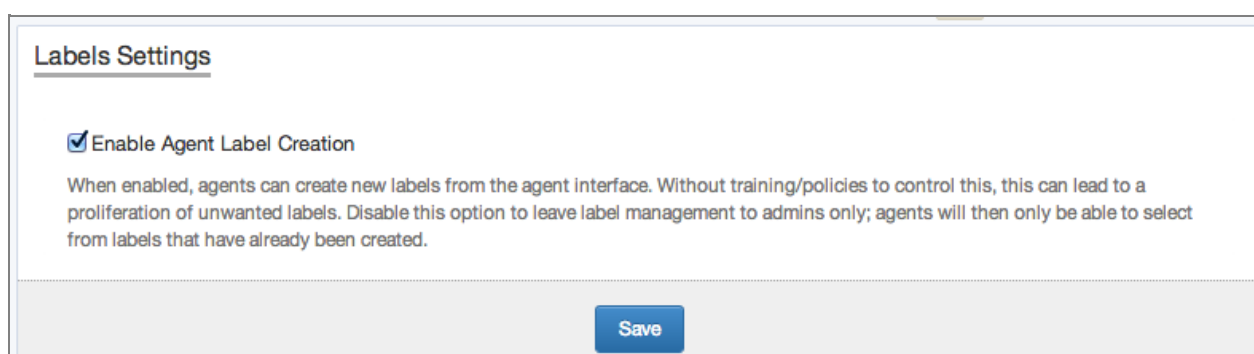
- **Labels > Tickets**
- **CRM > Labels > Users**
- **CRM > Labels > Organizations**
- **Portal > Knowledgebase > Labels**
- **Portal > Downloads > Labels**
- **Portal > News > Labels**
- **Portal > Feedback > Labels**

Each Labels section shows a list of labels that admins or agents have created.



By default, agents can add any label they want from the agent interface, but they will be prompted to choose from the list of existing labels.

If you find that agents are creating too many unwanted labels (such as variants of the same term), you can disable the ability for agents to create labels, so they can only apply existing labels.



You can create labels from the admin interface so they show up on the list of choices. Simply click the **+Add** button to add a label.

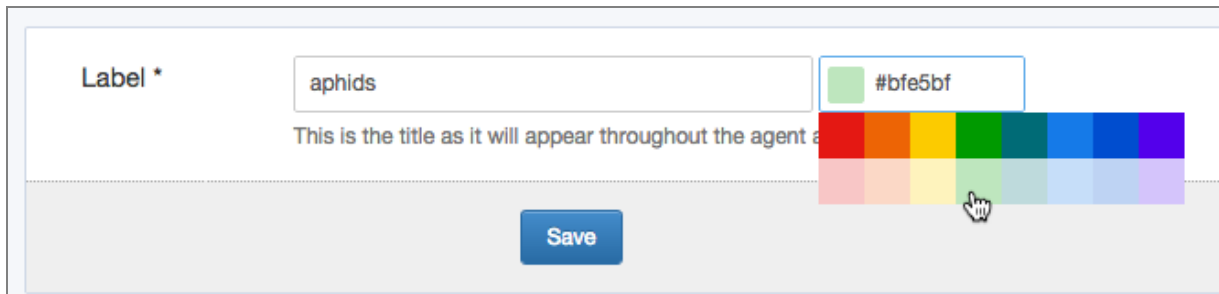
You can also [set a color for each label](#) that will be shown in the agent interface.

You can **delete** a label by selecting it and clicking the **Delete** button. This removes it from all items to which it has been applied.

14.2. Setting label colors

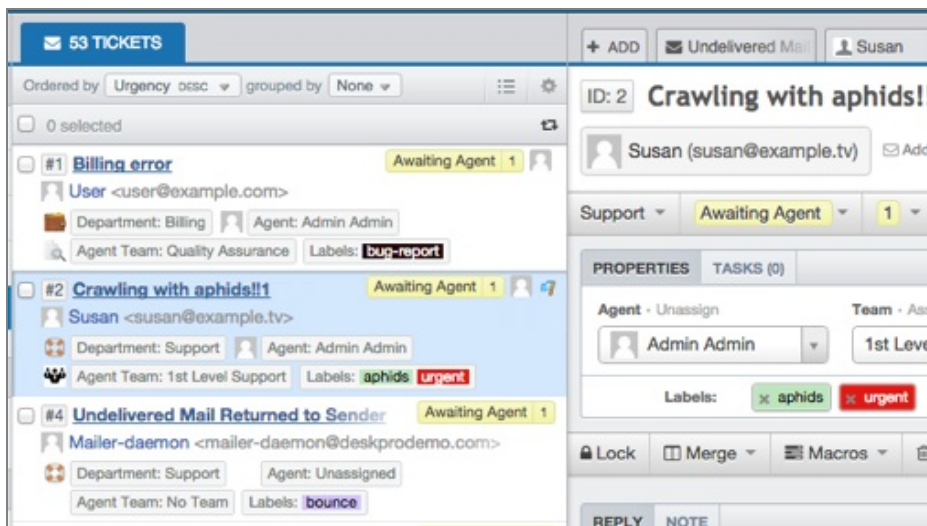
By default, labels are displayed as gray in the agent interface.

You can assign a color to each label from the **Labels** screen.



You can either specify a color by entering a [hexadecimal value](#), or just mouse over the color square and pick one of the 16 predefined shades.

These colors are used in the agent interface when the label is displayed on a ticket:



14.3. Merging labels

An alternative to deleting is to **merge** two labels. For example, suppose your helpdesk has two versions of a label, one correct and one misspelled.

rocket	6
rokkkit	3

Deleting the incorrect version would lose the information about which tickets it had been applied to. A better alternative is to merge both versions.

Click on the incorrect version, edit the label text so it's exactly the same as the correct version, then click **Save**.

5 Labels

Order by: Label ↓

investigate

0

legal

27

recurring

2

rocket

6

rokkkit

3

+ Add

Label *

rocket

This is the title as it will appear through

Save

All the tickets with either label are now labelled with the correct version:

rocket	9
--------	---

15. Agent Interface Options

You can change how the agent interface works to fit the needs of your agents.

The options available are:

- [Automatic ticket locking](#)
- [Ticket reply and message display defaults](#)
- [Time log and billing](#)
- [Custom filters](#)
- Ticket [archiving](#) and [purging](#) options

15.1. Automatic ticket locking

Confusion about who is working on a ticket is usually prevented by:

- agent assignment
- the ability for agents to see each other's draft messages

If an agent needs to prevent other agents changing a ticket, they can use the DeskPRO **locking** feature to manually lock it.

You can also enable **automatic locking** of tickets from **Tickets > Settings**, based on which agent has opened the ticket:

Ticket Locking ?

☐ Automatically enable lock

When enabled, a ticket will be automatically locked when an agent opens it in the agent interface.

☐ Automatically release lock

When enabled, the ticket will be automatically unlocked when the owner closes the ticket tab.

Lock timeout

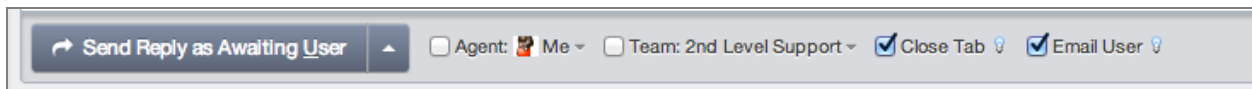
The time a ticket can be kept locked to one agent before the system automatically unlocks it.

Automatic locking might be useful if you find your agents are duplicating work, or getting into disagreements about who is replying to a ticket.

The **lock timeout** is the period of time before the lock is automatically removed. It applies to both automatic and manual locks. This is necessary to prevent locks being left on too long (for example, if an agent locks a ticket and then forgets about it).

15.2. Setting ticket defaults for agents

Under **Tickets > Settings > Ticket Defaults** you can configure the default values for the agent interface when the agent is creating or replying to a ticket.



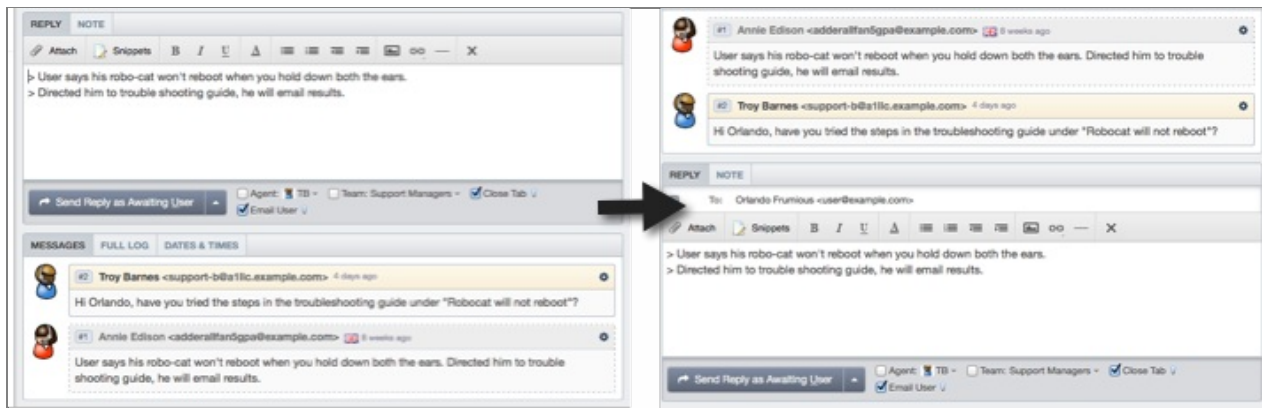
You can change these to reflect how your agents use DeskPRO. Pick defaults which reflect what agents do most often to increase their productivity.

For example, when an agent is replying to an unassigned ticket, the default is to assign it to that agent when the reply is sent, unless the **Agent** checkbox is cleared. If this default does not reflect how your agents use DeskPRO, you can change it.

The options available are:

- New Ticket defaults
 - Set status - choose the default status when creating a new ticket
 - Set agent (when ticket is unassigned) - choose whether to assign to self or leave ticket unassigned
 - Set team (when ticket has no team)
 - User notification - choose whether to check the "send user email notification" checkbox by default; if you disable this, agents must manually select for the user to receive an email notification of the new ticket
 - Ticket Reply defaults
 - Set status - choose the default status when replying
 - Set agent (when ticket is unassigned)
 - Set agent (when ticket is already assigned)
 - Set team (when ticket has no team)
 - Set team (when ticket is already assigned to a team)
 - User notification - choose whether to check the "send user email notification" checkbox by default; if you disable this, agents must manually select if the user receives their reply by email (not recommended unless users regularly check the portal)
 - Automatically set status - when changing the assigned agent from the reply box, choose whether to automatically change the status to **Awaiting Agent** (since normally if an agent is assigning the ticket to another agent, that agent is likely to be the next person to reply)
 - Automatically enable "close tab" - when changing the status to Resolved from the reply box, choose whether to automatically check the "close tab after reply" option.
- Note that these settings only affect defaults when agents reply via the agent interface, not if they reply via email. You may want to automatically assign tickets to the agents who reply to them by email. You can do this with [triggers](#triggers-example-- automatically-assign-agents-to-tickets-they-reply-to-by-email).
- Ticket View
 - Reverse message order - choose whether to show newest messages first with the reply box at the top.

This lets you select the default order that agents see the messages on a ticket, newest at top or oldest at top:



Agents can change this for themselves using the agent interface Preferences. The agent Preferences setting overrides the default you select here.

15.3. Time log and billing

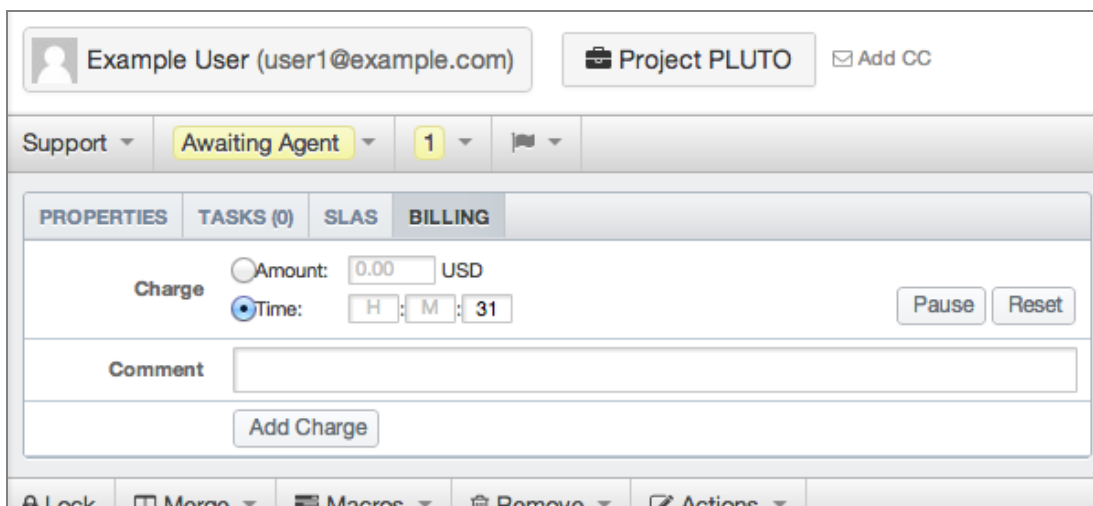
If your agents need to bill users based on the support they provide, you can enable the **time log** and **ticket billing** features.

The time log lets agents record the amount of time they spend on a ticket, while ticket billing lets them specify an amount of money to charge for support.

{This is not to be confused with the **Billing & Licensing interface** where you pay for your DeskPRO

license, accessible through the  icon at the lower left).

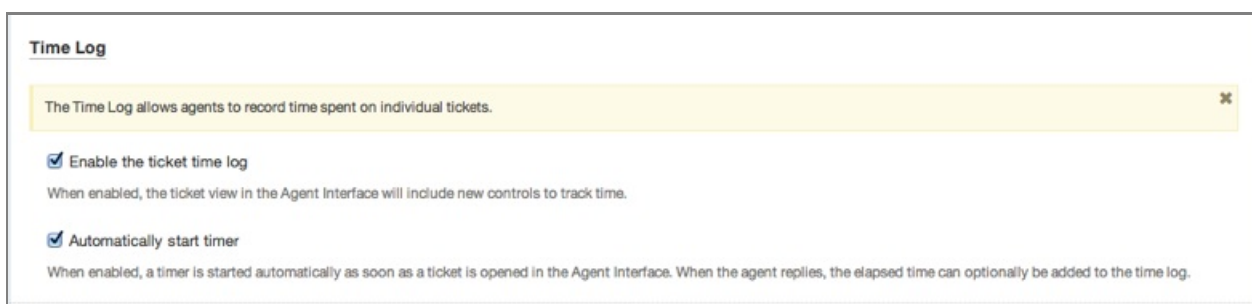
You can enable time logging and billing separately. Depending on which option is enabled, tickets in the agent interface will have a **Billing/Time Log/Billing & Time Log** tab.



The screenshot shows the DeskPRO agent interface. At the top, there's a header with a user profile (Example User, user1@example.com), a project name (Project PLUTO), and an 'Add CC' button. Below the header is a navigation bar with 'Support', 'Awaiting Agent', and a dropdown menu showing '1'. The main content area has tabs for 'PROPERTIES', 'TASKS (0)', 'SLAS', and 'BILLING'. The 'BILLING' tab is active, showing a 'Charge' section with radio buttons for 'Amount' (0.00 USD) and 'Time' (H: M: 31). There are 'Pause' and 'Reset' buttons. Below this is a 'Comment' text area and an 'Add Charge' button. At the bottom, there's a toolbar with 'Lock', 'Merge', 'Macros', 'Remove', and 'Actions'.

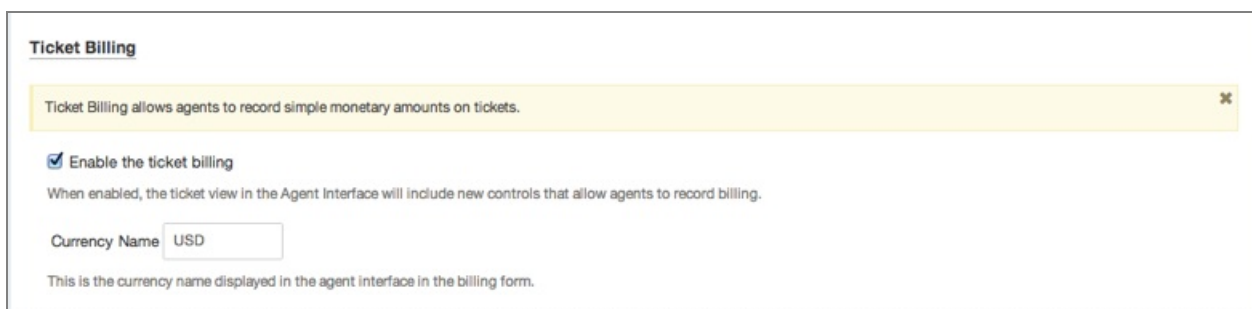
Go to **Tickets > Time Log & Billing** to change the settings.

The **Time Log** sections lets you enable the time log, and choose whether to **Automatically start timer** as soon as the ticket is opened by an agent. Note that the agent still has to press the **Add Charge** button to save the elapsed time.



The screenshot shows the 'Time Log' settings panel. It has a title 'Time Log' and a description: 'The Time Log allows agents to record time spent on individual tickets.' There are two checkboxes: 'Enable the ticket time log' (checked) and 'Automatically start timer' (checked). Below each checkbox is a brief explanation of what happens when it's enabled.

The **Ticket Billing** section lets you enable or disable monetary billing, and specify the currency.



The screenshot shows the 'Ticket Billing' settings panel. It has a title 'Ticket Billing' and a description: 'Ticket Billing allows agents to record simple monetary amounts on tickets.' There is a checkbox 'Enable the ticket billing' (checked). Below it is a text input field for 'Currency Name' with 'USD' entered. A note at the bottom states: 'This is the currency name displayed in the agent interface in the billing form.'

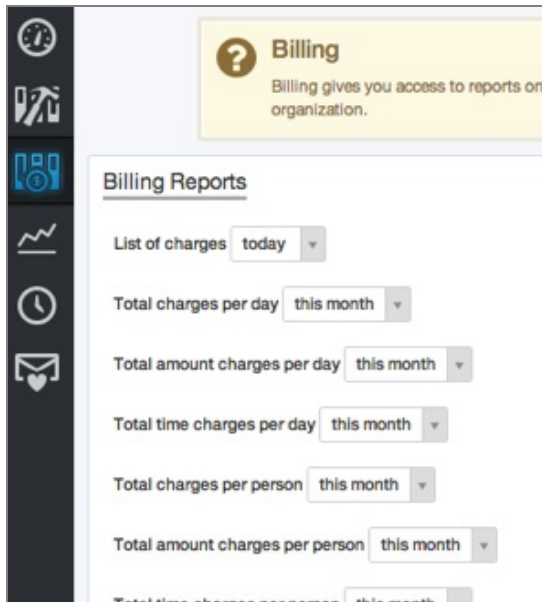
Modify billing/logs agent permission

The agent permission **Can modify billing and time log records** controls whether the agent can edit previously entered charges. An agent with this permission can also enter amounts of time directly, rather than running the timer.

Billing users

DeskPRO only records times and charges - users are not notified, and you will need to bill users outside of DeskPRO.

To facilitate billing, you can use the **Reports** interface to see summaries of all charges; note that you can generate reports by organization as well as by user.



Agents can also see the history of charges for a user and organization from profiles in **CRM**.

15.4. Custom filters

Ticket filters enable agents to view a list of all the tickets which meet a set of conditions. Agents can create their own filters from the agent interface; these are visible only to the agent who made them.

You can define custom filters for your helpdesk to be shown to all agents, to supplement the built-in filters like **My Tickets**. Filters make it easy for agents to track specific groups of tickets without repeated searching.

Create filters from **Tickets > Filters**.

The image shows two parts of the DeskPRO interface. On the left is the 'Create Filter' form, and on the right is a snippet of the agent interface showing the filter pane. An arrow points from the 'Criteria' button in the form to the filter list in the agent interface.

Create Filter Form:

- Title ***: Zenith robo-cat critical
This title will be displayed in the agent interface
- Permission**:
 - ☒ Everyone can use this filter
 - ☐ This filter belongs to a specific agent
 - ☐ This filter belongs to a specific team
- Search Terms ?**
 - If** The following conditions are met:
 - Priority is Critical
 - Department is not Sales
 - Product is Robo-cat Mk 1
 - Organization Name is Zenith Inc.
 - Criteria** (button)

Agent Interface Filter Pane:

- TICKETS** | SEARCH
- AWAITING AGENT**
- My Tickets** (11)
- Tickets I Follow (0)
- Unassigned Tickets (43)
- All Tickets (57)
- FILTERS** (gear icon)
- Zenith robo-cat critical (1)

Each filter has its own set of criteria and will only show tickets that match them all. Filters are shown in the agent interface filter pane. Filters you create from the admin interface can be visible to a specific agent, to a specific team, or to everyone.

15.5. Ticket archiving

Once a ticket has been marked resolved, most helpdesks can leave it that way. There is no need to delete resolved tickets.

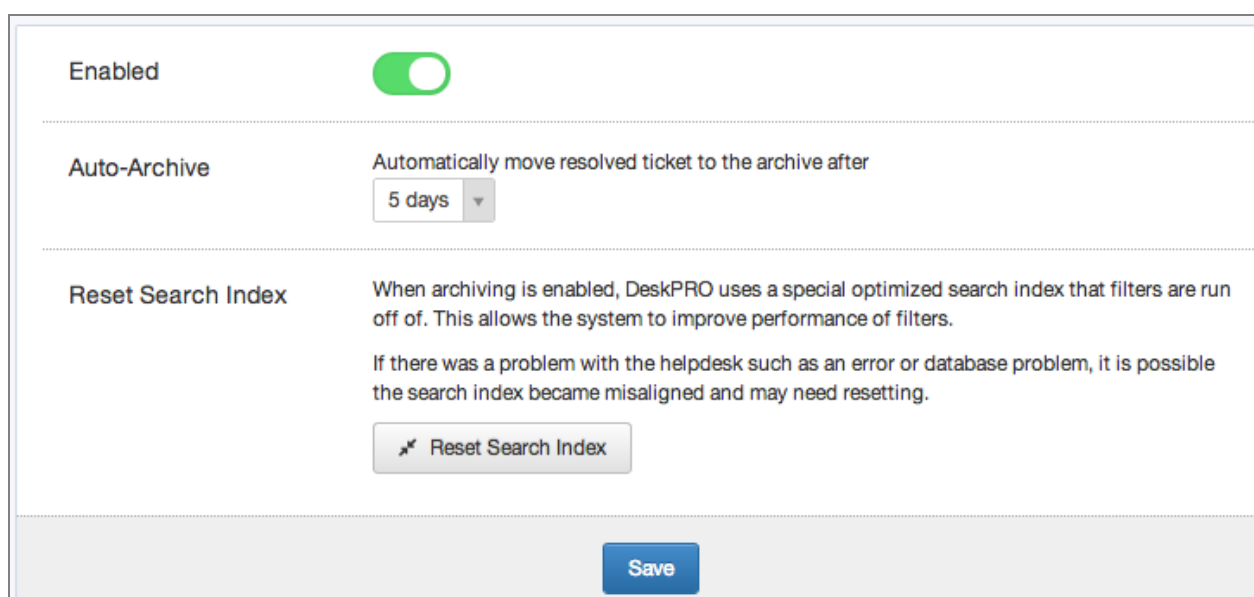
However, if your helpdesk has a very large number of old tickets (over a million), you can greatly improve agent interface performance by enabling **ticket archiving**.

Archiving automatically moves resolved tickets to a special **archived** status which means they don't show up in the main filters.

Agents can still view archived tickets using the **Archived** filter, and find them from the Quick Search bar or by carrying out an advanced search and including archived status.

Agents can change archived tickets back to another status if they have the correct permission.

To enable archiving, go to **Tickets > Statuses > Archived** and check **Enabled**.



The screenshot shows the 'Ticket Archiving' settings page. At the top, there is a toggle switch labeled 'Enabled' which is currently turned on (green). Below this, there is a section for 'Auto-Archive' with the description 'Automatically move resolved ticket to the archive after' and a dropdown menu set to '5 days'. Further down, there is a section for 'Reset Search Index' with a detailed explanation: 'When archiving is enabled, DeskPRO uses a special optimized search index that filters are run off of. This allows the system to improve performance of filters. If there was a problem with the helpdesk such as an error or database problem, it is possible the search index became misaligned and may need resetting.' Below this text is a button labeled 'Reset Search Index' with a small icon. At the bottom right of the form is a blue 'Save' button.

The **Auto-Archive** setting controls how long DeskPRO leaves a ticket as 'resolved' before setting it to 'archived'.

The **Reset Search Index** button is used to resolve certain very rare errors and database issues that can happen when archiving is enabled. There is no need to use it unless you are directed to do so by a DeskPRO technical support agent.

15.6. Ticket purging

When agents delete a ticket or mark it as spam, it is kept in the Recycle Bin/Spam for a set period of time before being permanently deleted by an automatic cleanup process. This gives your agents some time to retrieve tickets that are mistakenly deleted or marked as spam.

You can set the period until purging by going to **Tickets > Statuses > Hidden Statuses** and then clicking **Deleted** or **Spam**.

The default period until automatic cleanup is **1 month**.

You can use the **Run cleanup now** button to instantly delete all the tickets in the Recycle Bin or Spam. You might want to do this to purge your helpdesk of test tickets before you go into production. Bear in mind this cannot be undone.

16. Importing User Data

If you have data on users from another system, and you want to quickly create DeskPRO accounts for them, you can import data as a CSV (*Comma-Separated Values*) file. CSV is a simple comma-separated text format that can be exported from spreadsheet programs (e.g. Excel) and databases.

The exact format of the CSV file containing user data doesn't matter. The DeskPRO import function lets you select which column from the CSV file corresponds to which field in DeskPRO.

If you are using DeskPRO Download, you can use a more flexible data importer which can import all the data from [legacy helpdesk systems](#).

An alternative to importing account data is to enable users or agents to log in using credentials from a different system (for example, from an existing database, or their network login) by installing [authentication apps](#).

If you have other data import requirements, please contact us at support@deskpro.com.

16.1. Importing user data in CSV format

Comma Separated Values (CSV) is a simple format that is widely available as an export option from spreadsheet or database software. You can import user data in CSV into DeskPRO.

Here's an example of how a user data CSV file might look in your spreadsheet program:

	A	B	C	D
1	Brian Coyot	coyote1@example.com	Aardvark Inc.	enter
2	Maria User	maria1961@example.com	DreadCo	rando
3	Davey	daveyh@example.com		extra
4	Sue Cheam	suey333@example.com	Alpha	junk

and in a text editor:

```
Brian Wolf,coyote1@example.com,Aardvark Inc.,enter
Maria User,maria1961@example.com,DreadCo,rando
Davey,daveyh2@example.com,,extra
Sue Cheam,suey333@example.com,Alpha,junk
```

Some programs output CSV files with a semicolon instead of a comma as the field delimiter in between values:

```
Brian Wolf;coyote1@example.com;Aardvark Inc.;enter
```

In some CSV output, the fields are enclosed in quotes, e.g.:

```
"Maria User","maria1961@example.com","DreadCo","rando"
```

DeskPRO can handle both these variations in the CSV format.

These are the DeskPRO user fields which you can map to your CSV data:

- Basic Information
 - First Name
 - Last Name
 - Name
 - Title (Mr., Mrs., etc)
 - Primary Email (required)
 - Secondary Email
 - Password
 - Organization
 - Organization Position
- Contact Details
 - Phone Number
 - Website URL
 - IM
 - Twitter account
 - LinkedIn profile URL
 - Facebook profile URL
 - Address Line 1
 - Address Line 2
 - City
 - State
 - Post Code
 - Country
- Custom Fields
 - either existing custom user fields, or a new one you define (see [Importing into custom fields](#)).

All users must have at least one email address, so you must select a mapping to the "primary email" field.

- Users without a valid email address will not be imported.
- If an account already exists with a user's primary email address, no details from the user will be imported.

If you do not map a password field, a random password will be generated automatically for each user.

To import CSV:

- 1 In **CRM > CSV import**, click **Select CSV File** and select the CSV file with your data.
- 2 Set the **Field Delimiter** value (i.e. whether fields are separated by commas or semicolons).
- 3 Set the **Field Enclosure** value (i.e. whether the fields are enclosed by quotes).
- 4 Click **Prepare File**.

Import CSV File

The exact format of the CSV file does not matter. You will n

Field Delimiter Comma

Field Enclosure

Prepare File None Quotes

Once the file is uploaded, you can use the next screen to map the CSV columns to DeskPRO fields.

Import CSV File

Configure Importing:

Column Name	Example Data	Mapping Field
Brian Coyot	Maria User	First Name
coyote1@example.com	maria1961@example.com	Primary Email
Aardvark Inc.	DreadCo	Organization ☒ Create if does not exist
enter	rando	Do Not Map

☐ Skip the first row
 This should be selected if the "column name" values above contain labels rather than actual data.

☒ Import the first row
 This should be selected if the "column name" values above contain actual data.

[Click here to start Import](#)

By default, the importer assumes that the first row of your data consists of labels like "Name", "Email" etc. If that is *not* the case (as in this example) you should select the **Import the first row** option.

Once you have the configuration you want, you can start the import. The import will be automatically scheduled and done for you in the background. You do not need to keep your browser on the page except to get status updates as the import happens.

Importing into custom fields

If your data set has fields that you can't map to standard DeskPRO fields, you can import them into custom fields. If you haven't created the fields ahead of time, you can create them when configuring the import by selecting the **New Custom Field** mapping option.

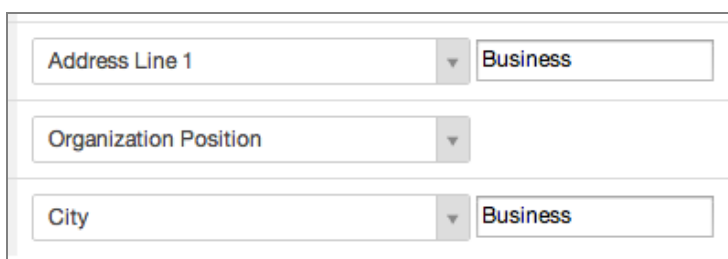
If you create a new choice-based field, new choices will be automatically created as new values are found for the field. If you import into an existing choice-based custom field, you can control whether new choices will be automatically created.

To import into a date-based custom field, the data you are importing must either be a UNIX timestamp or in the format YYYY-MM-DD (for example, 2012-10-24).

Importing contact data

The CSV importer supports importing contact data such as phone numbers and addresses and assigning a "label" to them. This label can be used to indicate, for example, a business phone versus a home phone. The label is optional in most cases.

However, when importing addresses, the label field is used to group related address parts together. If your data set has two or more sets of addresses, you need to enter a distinct label for each set of address fields that make up an address.



The screenshot shows a form with three rows of input fields. Each row has a dropdown menu on the left and a text input field on the right. The first row has 'Address Line 1' in the dropdown and 'Business' in the text field. The second row has 'Organization Position' in the dropdown and is empty in the text field. The third row has 'City' in the dropdown and 'Business' in the text field.

For example, if your CSV data has fields for "Business Street Address", "Business City", "Home Street Address" and "Home City", then you would need to use labels to group them correctly.

You could apply the label **Business** to "Business Street Address" and "Business City", and **Home** to "Home Street Address" and "Home City".

16.2. Importing data from legacy systems

DeskPRO Download offers a flexible data importer for transferring information from other systems into your helpdesk. It can read data in CSV and osTicket formats, and with PHP programming skills, you can extend it to handle other formats, e.g. data from your own internal systems.

See the [DeskPRO Importer Manual](#) for details.

If you are using DeskPRO Cloud, please contact us at support@deskpro.com to discuss data import.

Importing data from Zendesk

If you are using DeskPRO Download, you can import the data from a Zendesk helpdesk using the Zendesk API. Please refer to the knowledgebase article [Importing from Zendesk](#) for details.

If you are using DeskPRO Cloud, please contact us at support@deskpro.com to discuss importing Zendesk data.

Importing data with the DeskPRO API

You can also use the DeskPRO API to move data between an external source and your helpdesk. This is the most flexible method, but requires software development knowledge. See the Knowledgebase article [API Basics](#) for an introduction to the API.

17. Billing and licensing

You can view the status of your DeskPRO license in the **Billing** interface.



Your DeskPRO license entitles you to a certain number of agent accounts for a certain period of time.

In Billing you will be able to see your current number of agents, and when your license expires.

License

License ID	OIGW-1146
Expires	Your license expires on Wed, 1st Jan 2025.  Extend your license
Agents	5 Agents Add Agent Seats
License Code	<pre>T01HVy0xMTQ2LVdRQ1VFTk1NVk5JVUNRVzY0MTFXR01PWA1WLV tiA2BmXDUuVGvie08CQDAqWFZjUQoPKwEsNz90f1QnIzo0bwt5 JAENGRI7VC03Fzctd2YAFRADCAwBfx8pHmUIDghgDA1dFR1eQw c0eQx0BVoILwIvCS1KQ18qZXhsYm0GXXtUc2ArQgdbIyJAejlS ehwsfRIuCBi1WT0Za0EoIFc+eHE9SipeDH0YWyowD21DIQMCDw 41dGAJIB8NGg5jCm0djUQoPKwEsNz90f1Qne08CQDAqWFZjUQo PKwEsNz90f1QnIzo0bwt5JAENGRI7VC03Fzctd2YAFRADC</pre>  Update License Code

Click **Extend your license** to renew your existing license with the same number of agents.

If you want to increase your agents, click **Add Agent Seats**.

If you try to add more agents than your license allows from any part of the admin interface, you will also be prompted to add agents.

Note that DeskPRO Download agent licenses are sold in blocks of 5.

17.1. Deleting agents to free up agent seats

The licensing limit is on the number of active agents you have at any one time, *not* the total number of agents you have *created*.

You can [delete unwanted agent accounts](#) in order to increase the agents available.

For example, if you have a license for 10 agents and have hit the limit, but delete two accounts for agents who have left the company, you will then be able to create two more agents.

18. Managing DeskPRO Download

DeskPRO Download gives you more flexibility and control than the Cloud version, but requires some administration.

This chapter contains advice on maintaining and optimizing a DeskPRO Download installation.

18.1. Setting account credentials from the command line

Note that if you are using DeskPRO Download, for security reasons, admins need to use a command-line tool to reset their passwords.

To view a list of the admins and agents currently in your helpdesk:

```
php /path_to_deskpro/cmd.php dp:agents
```

replacing `path_to_deskpro` with the path to the directory where DeskPRO was installed.

Then execute this command:

```
php /path_to_deskpro/cmd.php dp:agents --reset-password
```

You will be prompted to enter the email address of the account you want to reset, and then to specify a password.

Note that you do not need to follow any password policy you have set on the helpdesk when setting a password this way.

If you need to promote an agent to an admin from the command line, you can use this command:

```
php cmd.php dp:agents --make-admin
```

Token login method

In some cases, you may be locked out and unable to access the normal login screen. For example, if you enable IP whitelisting but can't receive the email to verify your IP address, or if you have disabled the DeskPRO usersource and there is a problem with the authentication app you are using.

There is an alternative token-based access method for this situation.

Execute this command:

```
php cmd.php dp:login-token youremail@example.com
```

making sure you use the email of an admin account.

You will be given some Quick Login URLs which you will be able to follow to instantly log in to the different interfaces. They will be valid for the next 5 minutes.

18.2. Backing up or moving a DeskPRO server

You should turn off the helpdesk before moving your DeskPRO installation to a new server.

There are only two parts of DeskPRO with data that you need to copy when moving or backing up a server:

- 1 The MySQL database
To copy the MySQL database, use the [mysqldump](#) tool to dump the data (and then restore it if necessary). We recommend using the MySQL tools, not phpmyadmin.
- 2 The files on the filesystem.
Copy or move all the files within the root directory of your web server where you installed DeskPRO originally e.g. `/var/www` if you're using Apache.

When restoring a server, make sure you edit the `config.php` file to update any values that have changed in the new environment: e.g. location of your `/data/` directory. You will also need to recreate the `cron` jobs or Scheduled Tasks.

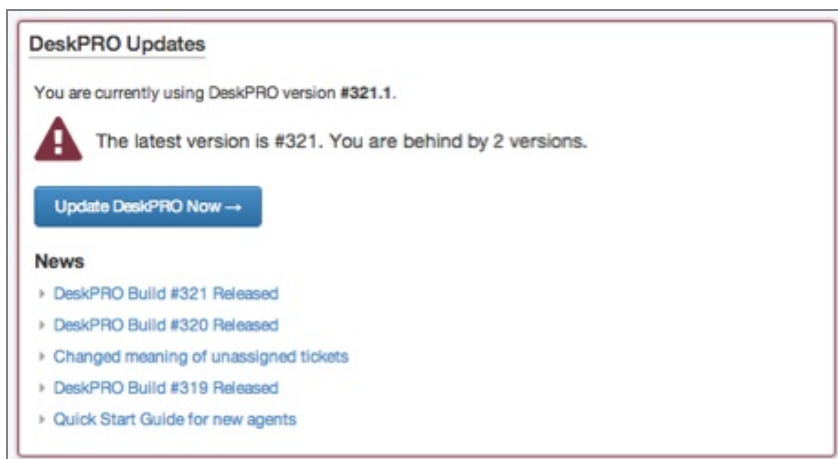
If you have moved a server and it is acting as if it is a fresh installation, check that the file `is_installed.dat` exists within your data directory.

18.3. Updating DeskPRO

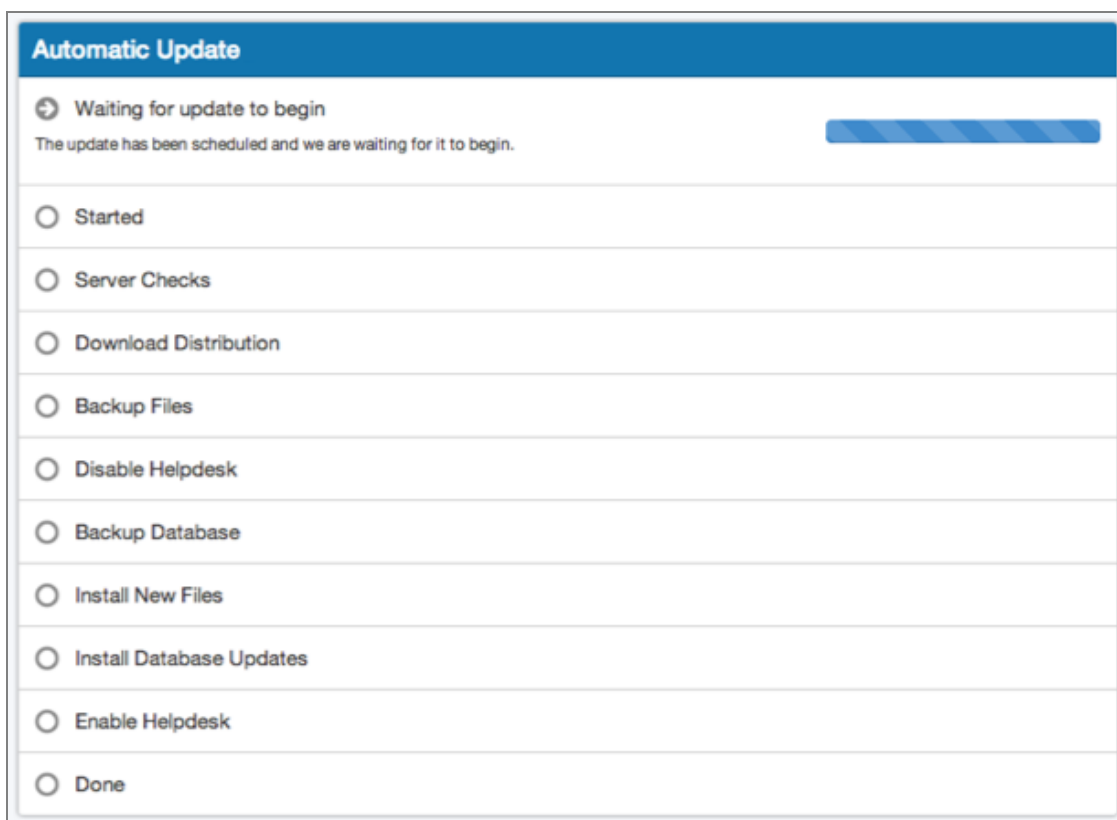
If you're using the Cloud version of DeskPRO, updating is automatic; you don't have to do anything to make sure that you have the latest version.

To update DeskPRO Download, you can use our automatic upgrader tools. They handle all the details of transitioning between versions, such as database schema changes.

When an update for DeskPRO Download is available, you will be notified on the Home page. The best way to update for most installations is to use the automatic updater by clicking **Update DeskPRO Now**.



The automatic updater uses the cron/Scheduled Task system. This means the cron/Scheduled Task user must have write permissions on all DeskPRO files for the update to work. If the updater gets stuck at an early stage of the process, as shown in the screenshot below, that probably indicates a permissions problem. To work round this, you can use the command-line updater instead.



The automatic updater gives you the option to back up the database and/or the DeskPRO source files, so you can revert to them in the unlikely event that you encounter a problem.

You can also choose to set a time delay before the update begins. This option will show an alert to any agents who are using your helpdesk, so they know to finish their work. You can use this to schedule the update for a time when there will be little usage of the helpdesk.

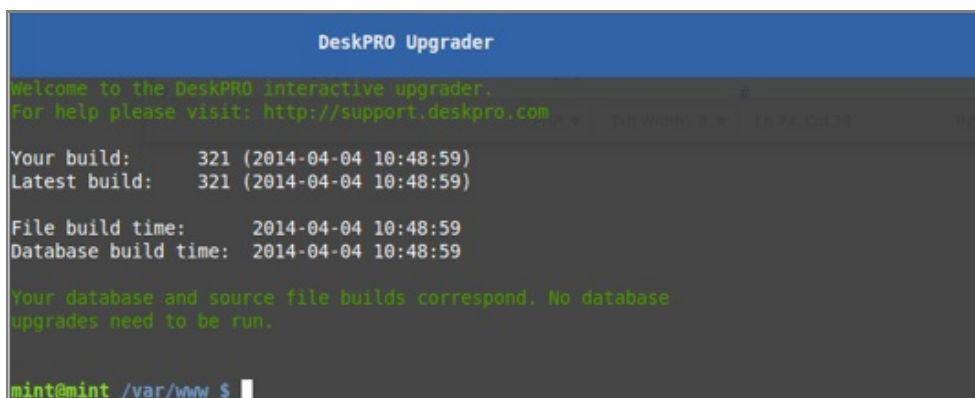
While the update happens, the helpdesk will be unavailable for a few minutes (depending on the size of your database). Under **User Offline Message** you can customize the notice that users will see on the portal during the update.

Command-line updater

If updating your configuration does not work from the admin interface, you can use an interactive command-line update utility.

From the directory where you installed DeskPRO, run:

```
php upgrade.php
```



```
DeskPRO Upgrader
Welcome to the DeskPRO interactive upgrader.
For help please visit: http://support.deskpro.com

Your build:      321 (2014-04-04 10:48:59)
Latest build:   321 (2014-04-04 10:48:59)

File build time: 2014-04-04 10:48:59
Database build time: 2014-04-04 10:48:59

Your database and source file builds correspond. No database
upgrades need to be run.

mint@mint /var/www $
```

The utility will guide you through the process of updating.

Manually installing updates

If the command-line updater does not work, you can manually update using the following steps:

- 1 Download the latest version of DeskPRO from <https://www.deskpro.com/downloads/deskpro.zip>.
- 2 Make a backup of the current DeskPRO files before you install updates in case you need to revert back.
- 3 Move the existing `data/` directory outside of the original DeskPRO install folder.
- 4 Copy your existing `config.php` to a safe place.
- 5 Extract the contents of `deskpro.zip` you downloaded in step 1, overwriting all existing files.
- 6 From inside the DeskPRO install directory, run the command-line upgrade tool:

```
php cmd.php dp:upgrade
```

The upgrade tool will ask if you want to produce a database backup before applying updates. We highly recommend you answer 'yes', in case you need to revert the upgrade.
- 7 The `data/` directory created in the install process may contain important log information, so copy it. Move the original `data/` directory from step 3 back into the DeskPRO install directory.
- 8 Move the original `config.php` file from step 4 back into the DeskPRO install directory.

Updating multiple servers

If you have multiple webserver (e.g. for load-balancing purposes), bear in mind that you will need to update each one.

18.4. Proxy servers and load-balancing

You can run DeskPRO with proxy servers. You may also choose to run it using multiple load-balanced nodes to increase availability. In these scenarios, you should enable proxy mode so that DeskPRO will trust IP addresses forwarded in HTTP headers.

To enable proxy mode, in the DeskPRO `config.php` edit the following line:

```
$DP_CONFIG['trust_proxy_data'] = false;
```

You must specify the IP addresses or address ranges of trusted proxies using [CIDR notation](#):

```
$DP_CONFIG['trust_proxy_data'] = array(  
    '192.168.1.83',      // whitelist this specific IP  
    '192.168.0.0/24'     // whitelist this range (192.168.0.0 - 255)  
    '@etc/proxy_list.php', // read addresses from file (as PHP array)  
);
```

If this is not configured correctly, it may appear that all users have the same IP address (the address of the proxy server).

For DeskPRO to function properly, the servers it interacts with must be sending proper headers with each request:

- **X-Forwarded-For:** The IP address of the user
- **X-Forwarded-Host:** The real hostname the user requested
- **X-Forwarded-Port:** The real port the user requested (usually 80 or 443 for SSL)
- **X-Forwarded-Proto:** The protocol (http or https) of the user request

18.5. DeskPRO Download performance tuning

For most helpdesks, DeskPRO is fast and reliable when properly configured, even on relatively modest hardware or budget virtual private server hosting.

If you are experiencing slow performance, consider the following:

- 1 Check **Server > Error Logs** for any repeated errors that are impacting performance. Note that while the DeskPRO Error Log is specific to DeskPRO, the PHP error logs will include PHP errors from any other PHP applications you are running on the server.
- 2 Make sure you have switched from storing binary files in the MySQL database to either filesystem storage or Amazon S3 cloud hosting. See the sections on [switching to filesystem storage](#) or [switching to Amazon S3 storage](#) for details.
- 3 If you are using a version of PHP below 5.5, either upgrade to 5.5, or install a supported **PHP accelerator** (software which caches compiled PHP code to improve performance):
 - [APC](#) (Alternative PHP Cache) for Apache or nginx
 - [Windows Cache Extension](#) for IIS
- 4 If you have a very large number of resolved tickets (over a million) and the agent interface is slow, enable [ticket archiving](#).
- 5 If you are running DeskPRO on a virtual machine, give the VM the same number of processors that the physical machine actually has.

18.6. Storing binary data

By default, uploaded files (such as email attachments) are stored in the MySQL database. However, databases are not designed to store binary data like this efficiently, so this may reduce performance/increase the required storage space.

DeskPRO supports two alternative storage methods:

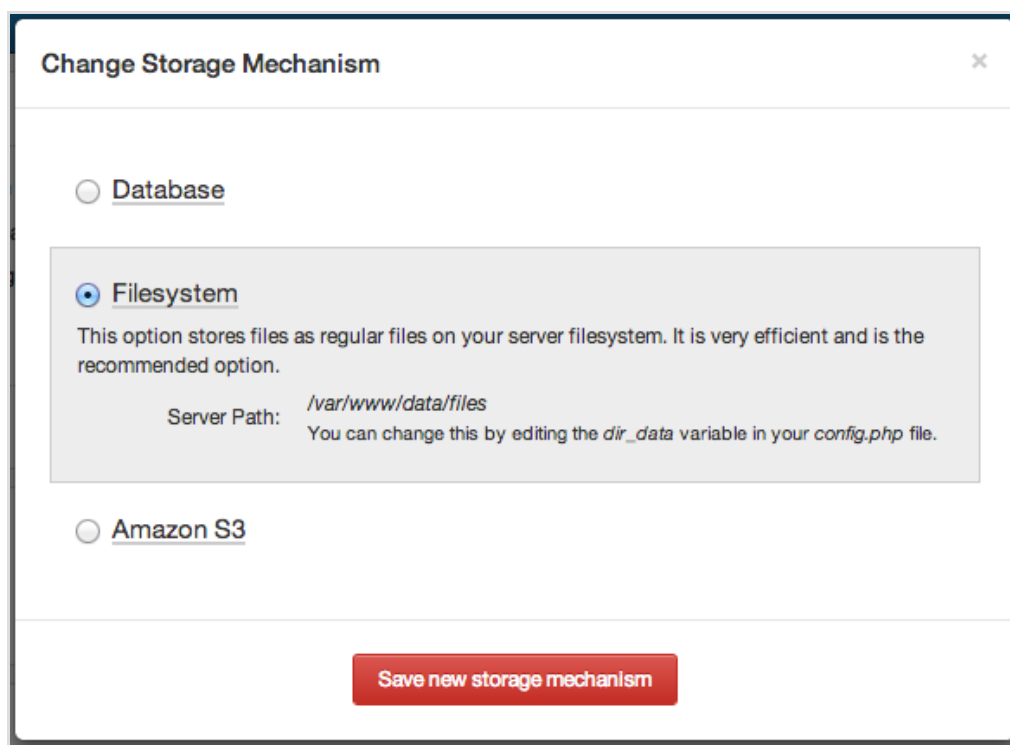
- Storing files in the [filesystem of your server](#).
- Storing files using [Amazon's S3 cloud storage service](#).

For better performance, we recommend switching from database storage to either of these two methods.

Switching to filesystem storage

To switch to filesystem storage:

- 1 Go to **Server > File Uploads**
- 2 Click **Change Storage Mechanism**.
- 3 In the pop-up window that opens, select **Filesystem**.



- 4 Click **Save new storage mechanism**.

Binary files will begin to be stored in `/data/files` within the directory where you installed DeskPRO. Any files already stored in the database will gradually be transferred to the filesystem in small batches (a few files will be moved each time the once-per-minute cron job/Scheduled Task runs).

Filesystem storage with multiple web servers

If you have multiple web servers, you must ensure that they are all accessing the same shared filesystem (e.g. using Network Attached Storage). If each web server is storing files to its own filesystem, then when web server A stores a file, web server B will not be able to access it, and you will experience errors.

You can set a custom location for filesystem storage by editing the `dir_data` variable in the DeskPRO `config.php` file. Use this to set the location of the shared filesystem.

Another way to deal with the issue of file storage with multiple servers is to use [Amazon S3](#). All the web servers will be storing and retrieving their files from a single S3 'bucket'.

Switching to Amazon S3 storage

Switching to Amazon S3 cloud storage has two main advantages:

- S3 storage is highly available and highly scalable. This reduces your server administration burden, since you don't have to ensure that your servers have disk space for all the files.
- If your DeskPRO installation uses multiple web servers, it's easy to have them share the same S3 storage. Using filesystem storage with multiple web servers requires extra configuration overhead.

To use Amazon S3, you must register with [Amazon Web Services](#) at <http://aws.amazon.com/s3/>.

- 1 Once you've registered, in the top right corner of the AWS site, you will see your name with a pull-down menu control. Use this to access the **AWS Security Credentials** screen.
- 2 Click the + icon next to the **Access Keys** section. Click **Create New Access Key** and save the Access Key ID and Secret Access Key. It's very important to save the Secret Access Key now; you will not be able to retrieve it once you close the window.
 - Note: You may wish to create a separate user with specific permissions instead of using your main AWS user (which has full AWS permission to do everything). Refer to Amazon's [Identity and Access Management Best Practices guide](#) for more information.
- 3 Open the **Services** menu in the top left corner of the screen and click on **S3**.
- 4 Click on **Create Bucket** and choose a bucket name; for compatibility the name should:
 - Have 3 to 63 characters.
 - Contain only lower case letters (at least one), numbers, `.` and `-`.
 - Not start with `.` and shouldn't follow or precede `.` with a `-`).
- 5 Select a location that is closest to the location of your DeskPRO Download webserver.
- 6 Select your new bucket in the list and click **Properties** tab in the right pane. Expand the **Permissions** section and click **Add Bucket Policy** and paste the following policy into the box:

```
{
  "Version": "2008-10-17",
  "Statement": [
    {
      "Sid": "AllowPublicRead",
      "Effect": "Allow",
      "Principal": {
        "AWS": "*"
      },
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::YOUR_BUCKET_NAME/*"
      ]
    }
  ]
}
```

(Replace "YOUR_BUCKET_NAME" with the name of the bucket you just created).

This policy makes the bucket public, so DeskPRO can serve binary objects or "blobs" directly from the Amazon S3 servers.

To enable S3 storage within DeskPRO, in the admin interface go to **Server > File Uploads**. Click the **Storage Mechanism Options** button and switch the mechanism to Amazon S3.

Enter the Access Key, Secret Key and S3 Bucket Name you configured above, then click **Save**.

DeskPRO will automatically move existing files to S3. This is done gradually, in small batches, as part of the default once-per-minute scheduled task (the same one that is responsible for processing email).

This process can take a long time, especially if you have a lot of data. If you would like to speed up the process, you can use the dedicated command-line tool:

```
$ cd /path/to/deskpro
$ php cmd.php dp:move-blobs      # Shows an overview
$ php cmd.php dp:move-blobs --run # Actually starts moving files
```

S3 security and reliability

It may seem insecure to have every file within your helpdesk (including email attachments) hosted publicly on S3. However, the URL for each file on S3 includes a very long string of random and unguessable characters that essentially acts as a password and ensures that people or bots browsing the web cannot find the 'public' files.

Only people who are authorized to see the file can get the URL. While a user or agent who is authorized to see the file could copy the URL and pass it to somebody else, they could also copy the file itself and pass it on. So enabling S3 doesn't make your files any more accessible than if you use filesystem or database storage.

Note that in the unlikely event that the S3 service becomes unavailable, links to existing files will not work. For example, if you had an image attachment saved to S3, that image would not load in the browser.

However, your helpdesk as a whole will remain operational. If S3 is unavailable, DeskPRO automatically switches back to database storage.

Server configuration and file size

Your server configuration determines the maximum file size DeskPRO can handle; factors include the amount of memory available to PHP and the values of some settings in your `php.ini` file. This determines the size of file that DeskPRO can handle as an upload from a web interface (agent, portal, chat widget etc) or as an email attachment.

See **Server Settings > File Uploads** for details of which settings affect the file size limit, and how to change it.

Raising the file size settings in **Setup > Settings** (under **User Attachments** and **Agent Attachments**) will not work if it exceeds the maximum file size determined by the server configuration.

18.7. Error reporting

Note that in the unlikely event that DeskPRO Download encounters a problem in its software, the error will be reported to the DeskPRO servers to enable rapid diagnosis and resolution of any problem. To disable this reporting, please add the following line to the bottom of your config.php file.

```
$DP_CONFIG['debug']['no_report_errors'] = true;
```

18.8. Elasticsearch

By default, DeskPRO uses MySQL to power searches made by agents and users. You can greatly improve search performance and quality by using the specialized [Elasticsearch](#) technology.

Elasticsearch indexes your helpdesk content and provides a much improved search experience from the agent interface and on the portal.

To use Elasticsearch, you need to set up a running instance of the Elasticsearch server software (called a cluster) and connect it to your DeskPRO installation.

The benefits of installing Elasticsearch are:

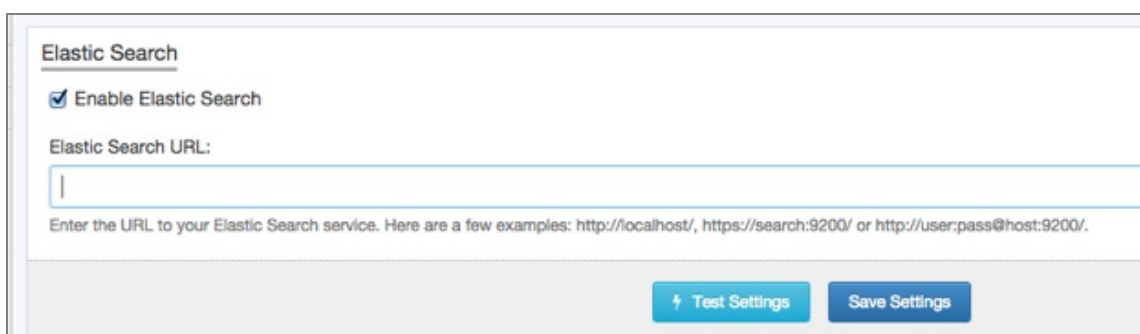
- Greatly increased search scope for agents, including the full text of ticket messages, agent notes and chat logs (for user chats, but not agent IM).
- Search results for agents and users are more accurate with better ranking.
- Search is much faster (especially if you install Elasticsearch local to your DeskPRO server).
- Search for a phrase by enclosing it in quotation marks e.g. a search for "audio feedback" will only return mentions of that exact phrase.
- Agents can choose how to sort search results (either **Best Match**, **Last Activity** or **Date Created**).
- User portal search includes full-text search of the user's tickets (or if the user is an organization manager, the organization's tickets).

Elasticsearch can run on the same server as DeskPRO, or a different server on your network, or you can use a hosted Elasticsearch service from a third-party provider (which may not deliver such a dramatic speed increase).

Installing Elasticsearch

See the [Installing Elasticsearch](#) article for a quick guide to setting up Elasticsearch on Linux or Windows, or using a hosted service. You should refer to the Elasticsearch documentation for how to configure your cluster for production.

Once you have set up your Elasticsearch cluster, go to **Server > Admin** and click **Enable Elasticsearch**, then enter the URL for the Elasticsearch service.



Click **Test Settings** to check that the service is accessible to DeskPRO and running correctly, then click **Save Settings**.

Elasticsearch will then index your helpdesk. When it is complete, you will benefit from the improved search.

18.9. Securing DeskPRO

The standard security precautions for any web-based software apply to DeskPRO Download. Here are some specific suggestions:

- 1 Enable [IP whitelisting](#).
- 2 Make DeskPRO application files non-writable by the web server. This is good practice in general, so DeskPRO is no different.
- 3 You can go a step further and make DeskPRO files non-writable by anyone. Updating to a new version using the admin interface requires write-access to files, so if you do this, you will need to install updates manually.

The `data/` directory must always be writable by both the cron user and the web server. It's where log info and temp files are stored.

- 1 Move your `data/` directory outside of the web root. You must specify the new path in `config.php` using the `$DP_CONFIG['dir_data']` variable.
- 2 Configure the web server not to execute or serve files in the `data/` directory. This ensures that even if a malicious PHP file is written to `data/`, it cannot execute.
- 3 Regularly monitor the PHP and MySQL error logs from **Server > Error Logs** and the `data/logs/error.log` file that DeskPRO writes.

19. DeskPRO Terminology

Admin: staff member within your company who has administrator access to configure and customize your helpdesk

Agent: staff member within your company who uses DeskPRO to provide help

Article: in the context of DeskPRO, an article refers to a piece of **knowledgebase** content

Assignment: a ticket can be assigned to an agent or a team to track who is dealing with that ticket

CC: a user who is CCed on a ticket will receive notifications about its progress. The equivalent for agents is **following**

Chat: real-time instant messaging between agents and users; may be available to users from the web portal or integrated into a website

CRM: the app used for managing users and assigning them to **organizations** and **usergroups**; short for *Customer Relationship Management*

Downloads: useful files you provide to users through your web portal; e.g., software drivers or product manuals; maintained in the **Publish** app

Feedback: suggestions from users submitted through your **web portal**; e.g. a feature request or new product idea

Following: an agent added as a follower of a ticket can get notifications and see it in the **Tickets I Follow** filter; useful for keeping track of a ticket without being assigned to it

Glossary: part of the **Publish** app used to maintain definitions of terms which are automatically displayed in **knowledgebase** articles

Knowledgebase: part of the **web portal** which contains rich-text help **articles**

News: part of the **web portal** which displays news posts chronologically; useful for product announcements

Organizations: used to track the company or other enterprise a user belongs to; you can assign users to organizations using the **CRM** app, or DeskPRO can automatically assign organizations based on user email addresses;

Permissions: settings which control what an agent or user can do with DeskPRO; your agent permissions are set by your admins and might, for example, prevent you deleting tickets; a user's permissions depend on their **usergroup**

Publish: the app used to maintain content for the web portal: knowledgebase articles, news posts, downloads and glossary entries

SLAs: short for *Service Level Agreements*; goals for dealing with tickets that your helpdesk aims to meet, e.g. that you must reply to a new ticket within 4 hours; shown in the **Tickets** app.

Snippets: short pieces of text which you can quickly insert into messages to users; useful to save time entering standard greetings, phrases, canned answers etc.

Status: a ticket's status is used to track who needs to act on a ticket next: **awaiting agent** means the helpdesk must do something related to that ticket; **awaiting user** means the helpdesk is waiting for the user to reply; **resolved** means the issue that ticket represents has been resolved

Tasks: an app used to track "to-do" items for agents

Teams: agents are grouped into teams to enable group ticket assignment

Ticket: represents a user question, problem or complaint; can be created automatically from incoming user email, or by a user from the **web portal**, or by an agent on a user's behalf

Ticket email account or just **email account:** an email account that is linked to DeskPRO so that incoming emails are turned into tickets e.g. `support@example.com`

User: one of the people the helpdesk is trying to help

Usergroup: users belong to usergroups which determine what they are allowed to do and see when interacting with DeskPRO; created by admins, assigned with the **CRM** app

Web portal: the user-facing web interface of DeskPRO - can be integrated into your own website; content on the portal is maintained using the **Publish** app